1 General

1.1 Summary

.1 The Work of this Section includes, but is not limited to the following:

.1 Administrative and procedural requirements for portions of the Work the design of which is delegated to the Contractor.

1.2 References

.1 Definitions:

.1 Delegated: Means transferred by the Consultant to the Contractor.

.2 Design: Means the complete planning, arrangement, and coordination of a discrete portion of the work, along with its graphic and written communication, including determination and engineering of the item's organization and structure in response to aesthetic requirements, functional requirements, dimensional and geometric limits, and the arrangement, performance, and other criteria indicated in the Contract Documents.

.3 Engineering services: Means structural engineering services performed for the design, fabrication, and installation of systems, assemblies, and components similar in material, design, complexity and extent to that indicated for this Project.

1.3 Administrative requirements

.1 Portions of the Contract Documents may delegate the design of certain items to the Contractor, or may otherwise specify "delegated design requirements" in individual Specification Sections.

.2 The Contractor is professionally liable for delegated design work, including design, engineering, and conformance to specified performance requirements.

.3 Drawings of delegated design portions of Work are diagrammatic; they do not identify or imply solutions to engineering issues, and are intended to only show:

.1 The design intent of finished materials, profiles, shapes and forms;

.2 Relationships between items;

.3 Location, identification, dimension and size of components, assemblies, accessories, and other items; and

.4 Schematic attachment details and diagrams of fasteners and connections.

.4 Specifications for delegated design portions of the Work are the performance type, and establish the minimum criteria for materials, fabrications, products, systems, assemblies, and methods of execution, along with the minimum performance requirements for indicated portions of the Work.

.5 The Consultant reviews and determines whether or not the Contractor's designs:

.1 Conform to the overall project design;

.2 Conform to the specified performance requirements, including subsequent modifications; and;

.3 Are appropriately integrated into the overall design of the project.

.6 In the event of a dispute regarding the Contractor's proposed delegated design solutions and the design intent of the Contract Documents, the decision of the Consultant is final.

1.4 Procedural requirements

.1 Design requirements: Proposed delegated design solutions must demonstrate conformance to the original design intent of the Contract Documents, as determined by the Consultant.

.1 Unless otherwise defined by the Contract Documents, the appearance of exposed elements, including member sizes, profiles, and alignment of components must be:

.1 Within the dimensional limits and section profiles indicated; and

.2 Consistent throughout the Project.

.2 Deviation from the profiles, layouts, dimensional locations, or arrangements indicated is not permitted without prior written consent from the Consultant.

.3 Contractor-proposed delegated design solutions that exactly follow the details indicated on the Drawings do not relieve the Contractor from liability for the design and performance of the delegated design portions of the Work.

.2 Engineering requirements: Engineer delegated design portions of the Work:

.1 Will meet or exceed the specified performance criteria;

.2 Will conform to the profiles indicated and to other requirements of the Contract Documents;

.3 Will satisfy the requirements of the authorities having jurisdiction; and;

.4 Will provide structurally sound, leak-proof, non-corroding, and weather tight assemblies, as applicable, that accommodate, resist, distribute, or transfer, as applicable, the minimum specified in-service loads, and thermal, seismic, and wind sway, or other types of movement, without incipient or catastrophic failure.

.3 Regulatory requirements: Delegated design items must be engineered in conformance with the latest edition of the OBC.

1.5 Submittals

.1 General: Coordinate and process submittals for delegated design portions of Work in same manner as for other portions of Work.

.2 Professional Engineer's qualifications:

.1 In all cases where delegated design services are specified to be performed under the Contract or where Work is specified to be performed by a “Professional Engineer” or “Engineer”, including but not limited to sealed shop drawings and quality verification services.

.2 The person providing such Work shall be a Professional Engineer, licensed by in the Province of the Work, providing such services under a certificate of authorization issued by the PEO and who carries errors and omissions (professional liability) insurance coverage.

.3 Design data: Submit structural engineering calculations demonstrating conformance to the requirements of the Contract Documents and of the authorities having jurisdiction.

.1 Calculations must be legible and incorporate sufficient cross-references to shop drawings to make calculations readily understandable and reviewable.

.2 At a minimum, structural calculations must contain:

.1 An analysis of framing members;

.2 Section property computations for framing members;

.3 An analysis of anchors, including anchors embedded in concrete; and

.4 The signature and seal of the Professional Structural Engineer, licensed in the Province of the Work, and responsible for their preparation.

.3 Test reports are not an acceptable substitute for calculations.

1.6 Quality assurance

.1 Professional Engineer's qualifications: Must be legally qualified to practice in the Province of the Work, with five (5) consecutive years' experience providing engineering services for projects similar in material, design, complexity, and extent to this Project.

2 Products

2.1 Materials

.1 Provide materials, fabrications, products, components, and accessories required for a complete design, whether or not such items are indicated on the Drawings or in the Specifications.

.2 Provide anchors, attachments, inserts, fasteners, clips, bracing, framework, and similar items as required to meet specified design and performance requirements, and to securely attach delegated design Work to adjacent supports, or to related adjoining work, whether or not such items are indicated on the Drawings or in the Specifications.

3 Execution

3.1 Design

.1 Unless otherwise indicated or specified, maintain the design intent and conform to the performance requirements indicated on the Drawings and in the Specifications, as determined by the Consultant.

.1 In the interest of certain fabrication or erection methods, minor dimensional changes and detailing adjustments to the original design communicated in the Contract Documents may become necessary.

.2 Obtain written approval from the Consultant for proposed changes and adjustments before procurement, fabrication, manufacture, assembly, or installation, as applicable.

.2 Engage a qualified professional structural engineer to design connection details and determine fastener types and sizes.

.1 Fasteners or connections may neither conflict with nor require revision to the finish profiles indicated or the supporting work.

.2 Connections may not impose eccentric loading, nor induce twisting or warping to the supporting structure.

.3 Connections must be designed to accommodate potential and actual misalignment of adjacent work within tolerances specified in other Sections.

3.2 Delegated design schedule

.1 Section 03 10 00 Concrete forming and accessories: For concrete formwork, falsework, reshoring and accessories.

.2 Section 04 20 00 Unit masonry: For unit masonry work, ties and anchors.

.3 Section 04 42 00 Exterior stone cladding: For hung stone work, anchors, dowels, and similar connections.

.4 Section 05 12 23 Structural steel for buildings: For connections related to structural steelwork.

.5 Section 05 50 00 Miscellaneous and metal fabrications: For metal fabricated items, such as but not limited to handrails and railings.

.6 Section 06 10 00 Rough carpentry: For structural wood studs and connections.

.7 Section 06 18 00 Glued-laminated construction: For engineered steel connectors for glued-laminated construction.

.8 Section 06 26 00 Mass timber soffit systems, ceiling and wall paneling: For soffit system including thermal spacer clips and girt support structure.

.9 Section 07 46 19 Metal siding: For metal siding system including thermal spacer clips and girt support structure.

.10 Section 07 61 00 Standing seam metal roofing: For metal roof system.

.11 Section 07 62 00 Flashing and sheet metal: For sheet metal flashing work to meet wind uplift requirements.

.12 Section 08 44 00 Aluminum work: For aluminum work systems and glazing.

.13 Section 09 21 16 Gypsum board: For light gauge metal framing of ***tall-spanning shaft walls as specified and*** gypsum board ***suspended*** ~~partitions and~~ ceilings.

.14 Section 10 73 00 Protective covers / pergolas: For pergola system.

.15 Section 11 82 00 Waste compactors and rubbish chutes: For rubbish chutes and connections.

.16 Section 26 56 13 Lighting poles and standards: For soiling testing and pole base for lighting poles.

.17 Section 31 00 00 Earthwork: For shoring, bracing and underpinning, if required.

.18 Section 32 31 13 Wood fences: For wood fences serving as a guard.

.19 Section 32 32 23 Segmental retaining wall: For retaining wall system.

End of section

1 General

1.1 Section includes

.1 The General Conditions and Supplementary Conditions of the Contract, Division 01 Specification Sections, and Contract Drawings apply to this Section.

.2 Work of this Section includes, but is not limited to the following:

.1 Brick veneer and lip brick.

.2 Stone veneer***,*** ~~and~~ sills ***and caps***.

.3 Concrete masonry units.

.4 Reinforcements and accessories.

.5 Mortar and grout materials.

1.2 Reference standards

.1 ASTM A167, Specification for Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet and Strip.

.2 ASTM A1064/A1064-M, Standard Specification for Carbon-Steel Wire and Welded Wire Reinforcement, Plain and Deformed, for Concrete.

.3 ASTM C91/C91M, Standard Specification for Masonry Cement.

.4 ASTM C207, Specification for Hydrated Lime for Masonry Purposes.

.5 ASTM C216, Standard Specification for Facing Brick (Solid Masonry Units Made from Clay or Shale).

.6 ASTM C615, Specification for Granite Dimension Stone.

.7 CAN/CSA A82, Fired Masonry Brick Made from Clay or Shale.

.8 CSA A165 Series, CSA Standards on Concrete Masonry Units.

.9 CSA A179, Mortar and Grout for Unit Masonry.

.10 CSA A370, Connectors for Masonry.

.11 CSA A371, Masonry Construction for Buildings.

.12 CAN/CSA A3001, Cementitious Materials for Use in Concrete.

.13 CSA G30.18, Carbon Steel Bars for Concrete Reinforcement.

.14 CSA S304, Design of Masonry Structures.

1.3 Design requirements

.1 Design unit masonry according to the following Climatic Design Data for Whitby contained in the Ontario Building Code:

.1 Design temperature: January 1%, July 2 1/2%.

.2 Hourly wind pressures: 1 in 50 year occurrence.

1.4 Submittals

.1 Product Data:

.1 Submit Product Data for masonry veneers, mortar and all accessories according to Section 01 33 00 indicating:

.1 Characteristics, performance criteria, and limitations.

.2 Preparation, installation requirements and techniques, Product storage, and handling criteria.

.2 Shop Drawings:

.1 Submit Shop Drawings according to Section 01 33 00 indicating:

.1 Wall sections and details, reinforcing and anchors, special detailing, patterning and locations of control joints.

.3 Samples:

.1 Submit samples according to Section 01 33 00:

.2 Submit samples of each type and colour of masonry unit used prior to placing order.

.3 Submit samples of coloured mortar to match masonry samples.

.4 Submit samples of masonry anchors, and ties.

.5 Submit 250 x 200 mm samples of dampproof course and flashing.

.4 Sustainability submittals: Submit applicable supporting documentation according to sustainability requirements for the approval of the Consultant.

.5 Certificates and reports:

.1 Submit manufacturer's certificates stating that materials supplied are according to this Specification.

.2 Submit masonry unit laboratory test reports certifying compliance with Specification and applicable CSA and ASTM standards.

1.5 Quality assurance

.1 Provide plain and reinforced masonry according to CSA A370, CSA A371, and CSA S304.

.2 Retain a licensed professional engineer, registered in province of Ontario, to perform following services for unit masonry work:

.1 Design of unit masonry work, ties and anchors.

.2 Review, stamp and sign shop drawings.

.3 Conduct shop and field inspections and prepare and submit inspection reports.

.3 Cold weather protection:

.1 To CAN/CSA-A371 and as follows:

.1 Maintain ambient temperature of masonry work and its constituent materials between 5 deg. C and 45 deg. C, until batch is used or becomes stable and protect masonry work from windchill.

.2 Maintain temperature of masonry above 0 deg. C for minimum of three (3) days, after mortar is installed.

.3 Preheat unheated wall sections in enclosure for minimum 72 hours above 10 deg. C, before applying mortar.

.4 Do not use scorched aggregate. Do not use salts or anti-freezes. Only use approved smokeless heaters.

.4 Hot weather requirements:

.1 To CAN/CSA-A371 and as follows:

.1 Plan in advance for hot weather construction. Protect freshly laid masonry from drying too rapidly, by means of waterproof, non-staining coverings.

.2 Avoid using dry masonry in hot weather conditions. Use pre-dampened masonry unit nominally saturated, but surface dry at time of laying. Do not dip masonry unit in bucket of water.

.3 Spread only enough mortar to permit soft setting of masonry units; do not over mix mortar materials; do not retemper mortar after two (2) hours of use; do not retemper pigment coloured mortar; do not spread more than 900 mm of mortar for placement of masonry unit.

.5 Mock-up:

.1 Construct one (1) 3 m^2 layered mock-up of unit masonry construction, to review all wall assembly components, in a location accepted by Consultant. Mock-up to include exterior reveal at brick to window/louvre head.

.2 Construct one mock-up of typical granite base installation, sized at 450 mm high by 1200 mm wide, to review all assembly components, in a location accepted by Consultant.

.3 Demonstrate use of reinforcement, ties, through-wall flashing, weep holes, jointing, coursing, coping and sills, mortar, bonding, control joints, and workmanship.

.4 Mock-up may form part of Work if accepted by Consultant. Mock-ups which do not form part of Work are to be removed from site during final cleanup, or when directed by Consultant.

1.6 Delivery, storage, and handling

.1 Deliver, store and handle Products according to the conditions of the Contract and as specified herein.

.2 Remove unacceptable materials from site and replace to acceptance of Consultant. Store materials off ground protected from wetting by rain, snow or ground water, or inter-mixture with earth or other materials. Store metal ties and reinforcement to prevent corrosion.

.3 Do not concentrate storage of materials on any part of structure beyond design load, take particular care not to overload unsupported portions of structure which may have not attained their full design strength.

.4 Comply with CSA A371. Do not use salt or calcium-chloride to remove ice from masonry surfaces.

.5 Deliver mortar materials in original unbroken and undamaged packages with the maker's name and brand distinctly marked thereon. Prevent damage to units.

.6 Keep masonry materials free from ice and frost. Keep units protected from concrete, mortar and other materials which could cause staining.

2 Products

2.1 Masonry units

.1 Clay face brick (BRK1):

.1 ASTM C216, Type FBA, grade SW or CAN/CSA A82 equivalent, exterior molded clay face brick, modular sized at 57 x 92 x 194 mm, installed with Flemish bond pattern.

.2 Provide lipped brick version of specified Product where shown on Drawings.

.3 Acceptable Product:

.1 ‘Lakeshore Blend’ with ‘Buff’ colour and having sandmold/smooth finish by The Belden Brick Company.

.2 Or approved equivalents by:

.1 Endicott.

.2 Glen-Gery.

.3 Or approved equivalent manufacturer.

.2 Clay face brick (BRK2):

.1 Exterior extruded face brick, DNF format sized at 54 x 108 x 228 mm, installed with running bond pattern.

.2 Acceptable Product:

.1 'D-Bricks Waterstruck' in colour 'D70' with rustic/buff finish by Petersen Tegl. Colour mix of brick to exclude very green variants.

.2 Or approved equivalents by:

.1 Endicott.

.2 Glen-Gery.

.3 The Belden Brick Company.

.4 Or approved equivalent manufacturer.

.3 Stone veneer (STN1 & 2) and stone sills ***and caps***:

.1 ASTM C615, granite fabricated by A. Lacroix Granit or approved equivalent.

.2 Colours and veining as selected by the Consultant.

.3 Stone veneer types (STN 1 & 2): 90 mm thick, granite with bush hammered finish in heights as indicated on Contract Drawings.

.4 Stone to be sound, hard and durable, well seasoned and of uniform strength, colour and texture, free of quarry sap, flaws, cracks, seams, sand holes, starts, iron pyrites or other mineral or organic defects that could impair structural integrity or function.

.5 Sills ***and caps*** (STN2): Provide slope in direction as indicated, complete with drips. Dowel and hook anchors to be stainless steel.

.6 Sealer:

.1 Factory seal stone units with two coats of recommended sealer.

.2 Stone sealer: Low VOC, penetrating sealer as recommended by stone fabricator.

.4 Concrete block units (CMU):

.1 Lightweight units: CSA A165 Series, size as indicated on Contract Drawing, classifications as follows:

.1 H/15/D/M.

.2 SS/15/D/M.

.3 SF/15/D/M.

.2 Normalweight units, CSA A165 Series, sizes as indicated on Contract Drawing, classifications as follows:

.1 H/15/A/M.

.2 SS/15/A/M.

.3 SF/15/A/M.

.5 Special shapes: Unless indicated otherwise, supply and install corner returns, bull-nosed or double bull-nosed units for exposed and external corners, bond beams, sash blocks for control joints, solid block where noted, concrete block lintels over openings in concrete block walls and any additional special shapes as indicated.

.6 Obtain each masonry unit type from same manufacturer. Supply and install units of uniform texture and colour for each kind required.

.7 Supply masonry units with exposed surfaces free of cracks, chips, blemishes, and broken corners.

2.2 Accessories

.1 Reinforcement: CSA A370, CSA A371, and ASTM A1064/A1064-M and ASTM A167, components to be hot dip galvanized and stainless steel, Type 304 as specified herein:

.1 Acceptable Products:

.1 Blok-Lok Limited.

.2 Or approved equivalent by:

.1 Hohmann & Barnard Inc.

.2 Or approved equivalent manufacturer.

.2 Type 1 (single wythe):

.1 Truss type; galvanized steel for single wythe partitions.

.2 Acceptable Product:

.1 ‘Blok-Trus BL30' by Blok-Lok Limited.

.2 Or approved equivalent by:

.1 Hohmann & Barnard Inc.

.2 Or approved equivalent manufacturer.

.3 Type 2 (back-up, typical – wood stud and cast-in-place concrete):

.1 Dual-diameter, stainless barrel and wire hook, with flame resistant plastic coated wings to create a thermal break, and factory-installed EPDM washers to seal air barrier and air, vapour and water resistive barrier.

.2 Acceptable Product:

.1 ‘Thermal Concrete 2-Seal Wing Nut Anchor’ by Hohmann Barnard, Inc.

.2 Or approved equivalent by:

.1 Blok-Lok Limited.

.2 Or approved equivalent manufacturer.

.4 Type 3 (screen wall, cast-in-place back-up):

.1 Anchors fabricated from 1.5 mm stainless steel plate x 50 mm wide with 25 mm bend with 4.76 mm stainless steel wire, complete with screws.

.2 Acceptable Product:

.1 ‘BL-5407' with ‘Flex-O-Lok tie’ by Blok-Lok Limited.

.2 Or approved equivalent by:

.1 Hohmann Barnard, Inc.

.2 Or approved equivalent manufacturer.

.5 Connectors: CSA A370 and CSA S304.

.6 Reinforcing steel: CSA G30.18, Grade 400, refer to Contract Drawings for number, size, and location.

.2 Cavity wall insulation: According to Section 07 21 00.

.3 Galvanized loose steel lintels, lateral support angles and shelf angles: Supplied as part of work of Section 05 50 00.

.4 Through wall flashings: Prefinished two-piece metal thru-wall flashings according to Section 07 62 00, continuous strips with a 25 mm folded drip edge.

.5 Dampproof course and flashing:

.1 Reinforced SBS rubberized asphalt compound laminated to cross-laminated polyethylene film, 40 mils thick, complete with primer and adhesive recommended by flashing manufacturer.

.2 Acceptable Products:

.1 ‘Blueskin TWF’ by Henry Company Canada Inc.

.2 ‘Sopraseal WFM TWF’ by Soprema Canada.

.3 'Airshield Thru Wall Flashing' by W.R. Meadows of Canada.

.4 Or approved equivalent.

.6 Compressible filler:

.1 75 x 6 mm thick preformed, polyurethane foam.

.2 Acceptable Products:

.1 ‘25V’ by Emseal Joint Systems Ltd.

.2 Or approved equivalent.

.7 Sealant: According to Section 07 92 00, such as at horizontal and vertical control joints and typical sealant work as required.

.8 Mortar mesh:

.1 250 mm high x thickness to suit cavity, 90% open HDPE mesh.

.2 Acceptable Products:

.1 ‘Mortar Break’ by Advanced Building Products Inc.

.2 ‘Mortar Net’ or ‘Mortar Trap’ by Hohmann & Barnard, Inc.

.3 ‘MortarNet with Insect Barrier’ by Mortar Net Solutions.

.4 Or approved equivalent.

.9 Weep hole vents:

.1 Flexible ultra-violet resistant polypropylene-copolymer plastic.

.2 Provide manufacturer recommended clear silicone adhesive for suspended applications.

.3 Colour: To be selected by the Consultant.

.4 Acceptable Products:

.1 ‘Mortar Maze Cell Vents’ by Advanced Building Products Inc.

.2 ‘Cell-Vent’ by Blok-Lok Limited.

.3 ‘CellVent’ by Mortar Net Solutions.

.4 Or approved equivalent.

2.3 Mortar materials

.1 Loadbearing masonry: CSA A179, Type S, proportion method.

.2 Interior non-loadbearing masonry: CSA A179, Type N, proportion method.

.3 Exterior non-loadbearing masonry: CSA A179, Premixed 1-1-6 Type N, portland cement/lime, proportion method.

.4 Cement: CAN/CSA A3001, normal Portland, Type GU. Provide white cement where required for white or light colour mortars.

.5 Masonry aggregate: CSA A179. Provide white aggregate where required for white or light colour mortars.

.6 Hydrated lime: ASTM C207, Type S.

.7 Water: Clean potable, free from deleterious elements and free from salts that can cause efflorescence.

.8 Mortar pigment:

.1 Mortar pigment colour to later selection by Consultant.

.2 Acceptable Product:

.1 'Bay Ferrox' by Bayer Inc.

.2 Or approved equivalent by:

.1 Huntsman Pigments and Additives.

.2 Solomon Colors, Inc.

.3 Or approved equivalent manufacturer.

.9 Concrete fill and grout: Minimum 12.5 MPa concrete according to CSA A179.

.10 Pointing mortar/grout:

.1 Use flush pointed grout at exterior stone base applications.

.2 Pointing mortar/grout consisting of high strength portland cement, graded aggregates and colour-fast pigments, conforming to ASTM C91/C91M and having a minimum compressive strength of 24 MPa.

.3 Mortar pigment: To match colour of stone and Consultant approved samples.

.4 Acceptable Product: 'MVIS Pointing Mortar' by Laticrete International or approved equivalent.

3 Execution

3.1 Examination

.1 Verify condition and dimensions of previously installed work upon which this Section depends. Report defects to Consultant. Commencement of work of this Section means acceptance of existing conditions.

3.2 Protection

.1 Supply and install temporary waterproof, non-staining coverings, secured against displacement, to extend over walls and down sides to protect masonry work from snow and wind driven rain, and from drying too quickly, until masonry work is completed and protected by flashings or other permanent construction.

.2 Supply and install non-staining, protective coverings on horizontal and vertical surfaces to protect work of this Section from damage, staining, marking, and mortar droppings.

3.3 Workmanship

.1 Perform masonry work according to CSA A371 and as indicated.

.2 Supply and install masonry work plumb, level and true to line, with vertical joints in alignment and horizontal courses level, uniform, and straight.

3.4 Masonry – general installation

.1 Construct masonry work as required by jurisdictional authorities.

.2 Before commencing masonry work, verify required limitations for wall heights, wall thicknesses, openings, bond, anchorage, lateral support, and compressive strengths of masonry units and mortars.

.3 Construct masonry fire protection and fire separations of the thickness indicated on Contract Drawings for fire resistant ratings as noted on Contract Drawings, and conforming to the Fire-Performance Ratings, Appendix ‘D’ to the National Building Code of Canada.

.4 Fire separations and fire separations with fire resistance ratings: Construct walls tightly to construction above and at perimeter, and without openings or voids. Do not reduce the thickness of walls to less than the thickness indicated on the Contract Drawings or for the required fire resistance rating where required.

.5 Do not butter corner units, throw mortar droppings into joints, or excessively furrow bed joints. Do not shift or tap units after mortar has taken initial set. If adjustment is necessary after mortar has started to set, remove and replace with fresh mortar.

.6 Do not use admixtures without Consultant's written acceptance.

.7 Tooling of mortar joints:

.1 Tool mortar joints slightly concave with non-staining tools unless indicated otherwise.

.2 Strike joints flush in non-exposed areas or where shown on Contract Drawings.

.3 Provide flush pointed grout joints for exterior stone base units as indicated.

.4 Use sufficient force to press mortar tight against masonry units on both sides of joints. Remove excess, remaining mortar material and burrs.

.8 Install masonry walls 25 mm clear of underside of building frames, roof or floor deck. Install masonry with a 19 mm space beneath shelf angles and install compressible filler. Sealant work according to Section 07 92 00.

.9 Cut masonry units with a wet saw to obtain straight, clean, even, unchipped edges. Cut units as required to fit adjoining work neatly or for flush mounted electrical outlets, grilles, pipes, conduit, leaving 3 mm maximum clearance. Use full-size units without cutting wherever possible.

.10 Reinforce veneer walls with adjustable wall lateral ties at maximum 400 mm o.c. vertically and 600 mm o.c. horizontally. Install lateral ties according to manufacturer's instructions. In veneer walls extend lateral ties from support wall, spanning cavity into exterior wythe. Place at maximum 75 mm o.c. each way around perimeter of openings, within 300 mm of openings.

.11 Reinforce block walls with continuous wire reinforcement in every second block course. Supply and install prefabricated L and T sections. Cut, bend and lap reinforcing units as per manufacturer's printed directions for continuity at returns, offsets, pipe enclosures, and other special conditions. Bending of masonry reinforcement is not permitted.

.12 Reinforce masonry walls with reinforcing steel as indicated on Contract Drawings. Vertical reinforcing shall be fully grouted in masonry cores with grout.

.13 At openings in block walls install extra reinforcement, so first and second courses above and below openings are reinforced. Extend extra reinforcement 600 mm beyond opening in each direction.

.14 Reinforce joint corners and intersections with strap anchors 400 mm o.c.

.15 Do not place reinforcement across masonry wythes at control joints.

.16 Install masonry with 10 mm thick joints unless indicated otherwise. Make vertical and horizontal joints equal and of uniform thickness.

.17 Build control joints in masonry walls at intervals and in locations shown. Form joints for block walls using sash block units according to details shown. Form joints for veneer walls by leaving head joints between stacked units void of mortar. Fill chase and joint with silicone sealant according to Section 07 92 00 for full height of control joints.

.18 Install control joints in masonry walls where indicated on Contract Drawings and at projections and changes in direction or between changes in supports. Where control joints have not been indicated provide joints at 6100 mm o.c. for exterior walls and 9150 mm o.c. for interior walls.

.19 If required, provide movement joints, similar to building control joints, installed between areas with different support conditions.

.20 Supply and install solid block or metal lath under block, and fill block cells solid for lintel bearing and as required to secure built-in anchor bolts and/or anchors shown.

.21 Do not tooth intersections of walls except as otherwise indicated.

.22 Weep vents:

.1 Install weep hole vents according to manufacturer's directions, in exterior wythe of masonry above dampproof courses and flashings and at tops of walls using adhesive.

.2 Space weep hole vents maximum 600 mm o.c. horizontally. Prevent weep hole vents from becoming plugged with mortar or debris.

.23 Coordinate installation of masonry with installation of air barrier and air/vapour barrier to ensure continuity of these systems.

3.5 Dampproof courses and flashing

.1 Install dampproof courses beneath first masonry bearing course on slabs-on-grade. Trim dampproofing to conceal it.

.2 Install flashings in masonry according to CSA A371.

.3 Install flashings under exterior masonry bearing on foundation walls, slabs, shelf angles, and steel angles over openings and elsewhere as indicated. Where flashings occur over openings in walls extend them past openings a minimum of 200 mm and turn up minimum 150 mm at each end to create a waterproof dam to prevent water draining into cavity.

.4 In veneer walls, install flashings continuously from front edge of masonry, under outer wythe, turn up backing minimum 200 mm and provide watertight seal against support wall.

.5 Lap dampproofing and flashing 150 mm and seal according to manufacturer's instructions.

.6 At bottom of cavity install mortar mesh to manufacturer's instructions. Apply additional mortar mesh layer as required to fill cavity thickness. Place mesh in continuous layer.

.7 Before masonry work begins, place specified dampproofing under first course of masonry. Install continuous dampproofing with ends lapped and cut flush with exterior face of wall. Place similar dampproofing over top course.

3.6 Mortar mixing

.1 Thoroughly mix mortar ingredients in proper quantities needed for immediate use to requirements of CSA A179.

.2 Measure and batch mortar materials either by volume or weight, to accurately control and maintain proportions. Do not measure materials by shovel.

.3 Mix mortar with maximum amount of water consistent with workability for maximum tensile bond strength within capacity of mortar.

.4 Do not use mortar which has begun to set. Use mortar within 2 hours after initial mixing. Re-temper mortar during two (2) hour period only as required to restore workability.

.5 Add mortar colour and admixtures to requirements of manufacturer's instructions.

.6 Provide uniformity of mix and colouration.

3.7 Block

.1 Lay blocks in running bond except as indicated otherwise. Align block webs vertically and install thicker ends of face shells up.

.2 Install a full bed of mortar for first courses of masonry, for masonry units 100 mm thick and less, and between solid units. For remaining courses bed face shells, including vertical end joints, fully in mortar.

.3 Install special shaped and sized concrete block units as indicated and as required for a complete and coordinated assembly and to minimize cut units.

.4 Supply and install two (2) courses of solid block beneath lintel bearing.

.5 Stagger end joints in every course. Align joints plumb over each other in every other course.

.6 Bond intersecting block walls in alternate courses. Where block work abuts concrete, anchor each block course to concrete.

3.8 Masonry veneer (exterior brick and stone)

.1 Work of this Section is to include typical cavity wall assembly and mortar-filled cavity wall as shown on Contract Drawings.

.2 Prior to installation of masonry veneer, coordinate installation of air, vapour and water resistive barrier with Section 07 26 00 as required for wall assembly conditions.

.3 Prior to installation of cavity insulation, examine air barrier and air, vapour and water resistive barrier membrane and make good damage. Install cavity wall insulation according to Section 07 21 00.

.4 Lay masonry veneer in bond patterns as specified and indicated, and in a full bed of mortar.

.5 Form angle corners with special shaped units; cutting of units is not permitted.

.6 Erect exterior cavity wall construction as shown on Contract Drawings.

.7 Typical: Install masonry veneer to prevent mortar droppings and protrusions from impeding drainage and pressure equalization of rainscreen cavities and drained walls.

.8 Apply sufficient mortar on end of stretchers to ensure end joints are compressed full when masonry unit is pressed into place.

.9 Mortar filled cavity (stone veneer):

.1 Where indicated on Contract Drawings, fill cavity with mortar.

.2 Work mortar into masonry cores, insulation and cavity extent to eliminate any voids

.10 Pointing mortar/grout:

.1 Provide pointing mortar/grout for stone base units.

.2 Where stone base unit has set, fill joints between units with coloured grout/pointing mortar taking care not to smear grout on masonry face.

.3 Fill joints to be without voids or cracks. Remove excess grout.

.4 When grout is firm or thumb-print dry, point joints to be flush with non-staining pointing tool.

.5 Clean and finish joints according to manufacturer's instructions.

3.9 Stone sills

.1 Install dampproofing or flashings continuous under full length of stone sill shapes.

.2 Install stone sill shapes in full mortar bed and secure units to each other with stainless steel dowels and to masonry units with stainless steel hook anchors, fully grouted.

3.10 Interior stone veneer, direct applied

.1 Install direct applied interior stone veneer where shown on drawings.

.2 Apply specified thin-set mortar bed to intended substrate to manufacturer recommended coverage rate with manufacturer recommended trowel. Back butter all stone units. Press units firmly into position in soft mortar bed, wiggle and apply slight pressure to unit to ensure firm bonding causing mortar to extrude slightly around edges on units.

.3 Ensure 100% mortar contact coverage to back of stones. Contact must be evenly distributed to give full support of the stone.

.4 Adjust joints between units uniform, plumb, straight, even, and true, with adjacent stone flush. Align grout joints in both directions unless indicated otherwise.

.5 Do not proceed with grouting until minimum 48 hours after stone has set, to prevent displacement of stones.

.6 When stones have set, fill joints between units with coloured grout/pointing mortar taking care not to smear grout on stone face. Fill joints to be without voids or cracks. Remove excess grout. When grout is firm or thumb-print dry, point joints slightly concave with non-staining pointing tool. Clean and finish joints according to manufacturer's instructions.

3.11 Lintels

.1 Install concrete block lintels over openings in masonry except where steel lintels are indicated.

.2 Set lintels with minimum of 200 mm uniformly distributed bearing at each end. Provide bond breaker under bearing ends.

.3 Install reinforcing steel and concrete fill in block lintels.

.4 Install loose steel lintels, as indicated in Contract Drawings. Centre over opening width.

3.12 Lateral support angles

.1 Where non-load bearing unit masonry partitions meet structural elements at top of partitions, install supplied lateral supports as required by the Ontario Building Code and according to structural details.

.2 In areas where ceilings are scheduled, use 150 mm lengths of steel angle located each side of partition at 1200 mm and staggered.

3.13 Built-in items

.1 Coordinate and locate build-in items required to be built into masonry or supplied under work of other Sections including hollow metal doors, windows, lintels, sleeves, inserts, and similar items. Build-in items to present a neat, rigid, true and plumb installation.

.2 Build wall openings, slots, and recesses required for ducts, grilles, pipes and other items.

.3 Coordinate installation of conduit, outlet boxes and other mechanical and electrical built-ins with work of Divisions 21, 22, 23 and 26.

.4 Prevent displacement of built-in items during construction. Check plumb, location and alignment frequently, as Work progresses.

.5 Brace door jambs to maintain plumbness. Set anchors between metal frames and masonry and fill voids between hollow metal frames and masonry walls with mortar.

3.14 Installation tolerances

.1 Install masonry work to a plane flatness and exposed end tolerance of 3 mm in 3000 mm.

.2 Variation in alignment from unit to adjacent unit: 1.5 mm maximum.

.3 Plumb within 6 mm in 3 m, or in 6 mm in 6 m at external corners, expansion joints, or other conspicuous lines.

.4 Level within 6 mm in any bay or 6 m maximum distance, and 12 mm in 12 m or more.

.5 Located from position shown, and from related position of columns, walls, and partitions within 12 mm in any bay or 6 m maximum distance, and 19 mm in 12 m or more.

.6 Opening sizes within 6 mm of designated dimension.

.7 Column and wall cross-section dimensions within minus 6 mm and plus 12 mm.

.8 Joint widths to dimensions indicated or specified herein, but in no case greater than 12 mm. Variation of mortar joint thickness: 1 mm every metre.

3.15 Repair and pointing

.1 Remove and replace masonry units which are loose, chipped, broken, cracked, marked, stained, discoloured, or otherwise damaged. Supply and install new units to match adjoining units and install in fresh mortar, and point to eliminate evidence of replacement.

.2 During tooling of joints, enlarge any cracks, holes, or other defects, point and completely fill with mortar.

.3 Point-up joints including corners, openings and adjacent Work for a neat, uniform appearance, properly prepared for application of sealant compounds.

3.16 Cleaning

.1 Obtain and follow unit masonry manufacturer's written instructions for cleaning of masonry.

.2 Clean exposed, masonry surfaces, removing excess mortar as work progresses. Allow mortar droppings to partially dry then dry brush with a stiff fibre brush.

End of section

1 General

1.1 Section includes

.1 The General Conditions and Supplementary Conditions of the Contract, Division 01 Specification Sections, and Contract Drawings apply to this Section.

.2 Work of this Section includes, but is not limited to the following:

- .1 Finish carpentry, including wood materials, high pressure laminate and solid surface materials.
- .2 Millwork hardware.
- .3 Accessories.

1.2 Reference standards

.1 ANSI A208.1, Particleboard.

.2 ANSI A208.2, Medium Density Fibreboard for Interior Use.

.3 ANSI/HPVA HP-1, Hardwood and Decorative Plywood.

.4 ANSI/NEMA LD 3, High-Pressure Decorative Laminates.

.5 APA - The Engineered Wood Association.

.6 ASTM E84, Standard Test Method for Surface Burning Characteristics of Building Materials.

.7 ASTM F1667, Driven Fasteners: Nails, Spikes and Staples.

.8 Architectural Woodwork Manufacturers Association of Canada (AWMAC).

.9 North American Architectural Woodwork Standards (NAAWS).

.10 CSA O121-M, Douglas Fir Plywood.

.11 CAN/CSA O141, Softwood Lumber.

.12 CSA O151-M, Canadian Softwood Plywood.

.13 CAN/ULC S102, Standard Method of Test for Surface Burning Characteristics of Building Materials and Assemblies.

.14 National Hardwood Lumber Association (NHLA) Rules for the Measurement and Inspection of Hardwood and Cypress.

.15 National Lumber Grades Authority (NLGA) Standard Grading Rules for Canadian Lumber.

1.3 Submittals

.1 Shop Drawings: Submit Shop Drawings of finish carpentry work according to Section 01 33 00 indicating:

- .1 Materials, thicknesses, sizes, finishes, wood species, grades, profiles, connection attachments, shop jointing, field jointing, reinforcing, anchorage, fastener types and sizes, location of exposed fastenings, mechanical and electrical service routes, service outlets, cut-out locations, and sizes.
- .2 Include erection drawings, plans, elevations, sections, and details as applicable.
- .3 Include all solid surface slab installation denoting seam layouts, joints, trim and edge.
- .4 Details, elevations, sections, materials, sizes, anchorage and connection attachments of wood slat wall.

.2 Samples: Submit samples of the following according to the requirements of Section 01 33 00:

- .1 Two representative pieces of each type of wood to receive a stained or natural finish.
- .2 Two representative pieces of each type of wood finished as specified.
- .3 Two of each colour, pattern, gloss, and texture of plastic laminate, in manufacturer's standard tag size.
- .4 Two samples of laminated high pressure laminate joints, edging, cutouts and postformed profiles.
- .5 Two samples of melamine surfaced board, edging and postformed profiles.
- .6 Two 300 x 300 mm samples of wood slat wall for Consultant’s approval of material, slat size and finish.
- .7 Two of each solid surface, in 100 x 75 mm samples, in each required thickness.
- .8 One of each item of finish carpentry hardware.

.3 Sustainability submittals: Submit applicable supporting documentation according to sustainability requirements for the approval of the Consultant.

1.4 Quality assurance

.1 Execute work of this Section by member of AWMAC, with five (5) years experience in finish carpentry work of comparable complexity and scope. Submit proof of experience upon Consultant's request.

.2 Fabricate finish carpentry work according to NAAWS, Premium Quality materials and installation unless otherwise indicated. Perform work according to the definition of Good Workmanship as defined in the NAAWS.

.3 Remove and replace finish carpentry work which does not conform to the NAAWS or as amended by these Specifications.

.4 Mock-up:

.1 Typical: Shop fabricate one (1) mock-up of a base cabinet, wall cabinet, and counter top for each type of surfacing specified, complete with hardware and shop applied finishes, installed in location acceptable to Consultant.

.2 Wood slat wall: Provide one (1) mock-up of wood slat wall, full height by 1200 mm wide, including fasteners, wood profiles, slat spacing and corner detail for Consultant's review.

.3 Wood slat surround: Provide one (1) full size mock-up of wood slat surround at typical entry suite door, including fasteners, wood profiles and slat spacing for Consultant's review.

.4 Arrange for Consultant's review and acceptance, allow 48 hours after acceptance before proceeding with work.

.5 When accepted, mock-up will demonstrate minimum standard for this work. Mock-up may remain as part of Work if accepted by Consultant. Remove and dispose of mock-ups which do not form part of Work.

1.5 Delivery, storage, and handling

.1 Deliver, store, and handle finish carpentry according to the NAAWS. Control the temperature and humidity according to the NAAWS recommendations, before, during, and after finish carpentry delivery, and also during storage and installation.

.2 Cover finished high pressure laminated work with heavy kraft paper or put in cartons during shipment. Protect installed surfaces by approved means. Do not remove until immediately before final inspection.

2 Products

2.1 Materials

.1 General:

.1 All materials under work of this Section, including but not limited to, adhesives and mastics, are to have low VOC content limits.

.2 Adhesives - Urea-formaldehyde-free glues.

.2 Concealed framing lumber and plywood:

.1 Eastern Spruce, Balsam Fir, or Jack Pine, to CAN/CSA O141, NLGA, and NAAWS Custom Grade, S4S, average moisture content 7% +/- 2% at installation.

.2 Softwood plywood: CSA O151-M; 19 mm unless indicated otherwise, (G2S).

.3 Marine grade plywood (sinks): APA, Grade A-A, T & G, composed entirely of Douglas Fir or Western Larch, laminated with structural waterproof adhesive, exterior grade. Thickness as indicated on drawings.

.4 Hardwood lumber and wood veneer (WD01 & 02):

.1 Wood veneer (WD01):

.1 Rift cut white oak, conforming to ANSI/HPVA HP-1 having finishes and meeting grades as follows.

.1 Transparent finish, Grade AA.

.2 Sizes, thickness, and shapes as indicated.

.2 Hardwood lumber (WD02): Rift cut white oak, conform to NHLA and NAAWS Premium Grade, S4S, average moisture content 7% +/- 2% at installation.

.3 Refer to Section 09 06 00 - Material and Finishes Schedule for additional information regarding wood veneer and hardwood lumber materials.

.5 Paint grade lumber: Solid poplar to NHLA and NAAWS Premium Grade, S4S, average moisture content 7% +/- 2% at installation. MDF material not permitted for painted millwork applications.

.6 High pressure laminate (PLAM): Provide high pressure laminates conforming to ANSI/NEMA LD 3 as follows:

.1 Flatwork face sheet: 1.2 mm thick typical and 0.7 mm for fingerprint resistant type, heavy wear resistance.

.2 Vertical interior face sheets: 0.8 mm thick.

.3 Postformed face sheet: 0.8 mm thick.

.4 Backing sheet: thickness to match face sheet, high pressure laminate, manufactured by same manufacturer as face sheet.

.5 Refer to Section 09 06 00 - Material and Finishes Schedule for high pressure laminate types, colours, finishes and additional information.

.6 Provide high pressure laminate material with fingerprint resistant technology, repairable with heat technology where shown and as scheduled.

.7 Edging to be done in 3 mm thick PVC to match laminate colour.

.8 Acceptable manufacturers:

.1 Fenix or Wilsonart as scheduled.

.2 Or approved equivalents by:

.1 Arborite.

.2 Formica.

.3 Forbo.

.4 Pionite.

.5 Nevamar.

.6 Or approved equivalent manufacturer.

.7 Veneer core plywood (substrate):

.1 CSA O121-M, sanded, douglas fir plywood (DFP), G2S, Grade A-A, for typical veneer core substrate.

.2 APA, Grade A-A, where hardwood plywood is required by NAAWS for millwork items.

.3 Sizes, thickness and shapes as indicated.

.8 Medium Density Fibreboard (MDF), core or panel (typical, NAUF):

.1 ANSI A208.2; no-added formaldehyde; meeting the following minimum criteria:

.1 Density: 753 kg/m^3.

.2 Internal bond: 1.17 N/mm^2.

.3 Modulus of rupture: 31.0 N/mm^2.

.4 Modulus of elasticity: 3103 N/mm^2.

.5 Moisture content: 4-6%.

.2 Acceptable Product:

.1 'Medite II' by Roseburg Forest Products.

.2 Or approved equivalent by Flakeboard Company Limited or approved equivalent manufacturer.

.9 High Density Fibreboard (HDF) core and panel:

.1 ANSI A208.2; omnidirectional, light coloured with uniform density throughout; high density fibreboard (HDF), meeting the following minimum criteria:

.1 Density: 905 kg/m^3.

.2 Internal bond: 1.7 N/mm^2.

.3 Modulus of rupture: 45 N/mm^2.

.4 Modulus of elasticity: 4500 N/mm^2.

.5 Face screw holding: 1700 N.

.6 Core screw holding: 1500 N.

.7 Moisture content: 4-6%.

.2 For use at stair stringers.

.3 Acceptable Product:

.1 Arauco.

.2 Marlite.

.3 Uniboard Canada Inc.

.4 Or approved equivalent.

.10 Melamine surfaced MDF (ML):

.1 Provide melamine surfaced board for inside facing of millwork, unless otherwise indicated.

.2 ANSI A208.2, MDF substrate board faced with a melamine impregnated decorative paper thermofused onto the surface, no urea-formaldehyde.

.3 Melamine faced MDF to meet the following criteria:

.1 Modulus of rupture: 3130 psi.

.2 Modulus of elasticity: 313, 000 psi.

.3 Internal bond: 78 psi.

.4 Linear expansion: 0.33% max.

.5 Formaldehyde emissions: 0.11 ppm.

.4 Melamine colour: Inside facing of millwork; refer to Section 09 06 00 - Material and Finishes Schedule for melamine type, colour, finish and additional information.

.5 Edge banding: Minimum 1 mm thick, unless otherwise indicated, PVC to match door/drawer front high pressure laminate colour.

.6 Acceptable Product: 'Panolam Thermofused Melamine Panels' by Panolam or approved equivalent.

.11 Particle board core:

.1 ANSI A208.1, no-added formaldehyde, FSC certified particleboard containing recycled content; meeting the following minimum criteria:

.1 Density: 720 kg/m^3.

.2 Internal bond: 0.76 N/mm^2.

.3 Modulus of rupture: 25.4 N/mm^2.

.4 Modulus of elasticity: 2861 N/mm^2.

.2 Acceptable Product:

.1 'Purekor Platinum Particleboard' by Panel Source.

.2 Or approved equivalent by Roseburg Forestry Products or approved equivalent manufacturer.

.12 Solid surfacing (SS):

.1 13 mm thick, as indicated on Drawings, acrylic polymer sheet stock, provide with eased edges and all cut-outs as required.

.2 Refer to Section 09 06 00 - Material and Finishes Schedule for solid surfacing types, colours and additional information.

.3 Installation and seam adhesives to be as recommended by solid surfacing manufacturer, colour matched to solid surfacing.

.4 Acceptable manufacturer: Formica or approved equivalent.

.13 Laminating adhesive: CSA O112 Series, water resistant type, low VOC content, selected by laminate manufacturer for intended end use.

.14 Draw bolts and splines: Type as recommended by fabricator.

.15 Nails and staples: Conforming to ASTM F1667; Size and type to suit application, galvanized for exterior work, interior humid areas and for treated lumber; plain finish elsewhere.

.16 Bolts, nuts, washers, blind fasteners, lags and screws: Size and type to suit application. Stapling is not acceptable.

.17 Adhesive and bituminous mastic: Selected by the millwork fabricator with low VOC content.

.18 Acoustic felt: Class A rated to ASTM E84 or CAN/ULC S102, black acoustic felt for use at slat wall, type as recommended by millwork fabricator.

.19 Miscellaneous metals: According to Section 05 50 00.

.20 Acoustic/fire rated insulation: According to Section 07 21 00.

~~.21 Glass and glazing materials: According to Section 08 80 00.~~

.21 Tile base: According to Section 09 30 00.

.22 Resilient base: According to Section 09 65 13.

.23 Insect repellant powder: Provide diatomaceous earth insect powder as specified herein. Refer to Section 01 57 16 for additional pest control requirements.

.24 Finishing: According to Section 09 91 00.

2.2 Hardware

.1 The following hardware is the minimum quality standard for the work of this Section. Equivalents may be considered provided they are approved by Consultant prior to ordering of products.

.2 Drawer slides:

.1 Full extension with soft-closing feature***, having a minimum load capacity of 45 kg (100 lbs).***

.2 Acceptable Product:

.1 '8400 Series' with '8450FM Soft-Close Drawer Slide' by Knape & Vogt.

.2 'Xpert Series' with soft-closing by Richelieu.

.3 Or approved equivalent by Accuride or approved equivalent manufacturer.

.3 Pilasters:

.1 Clear anodized aluminum recessed shelf standards with 12 mm divisions.

.2 Acceptable Product: 'Model 233' by Knape & Vogt or approved equivalent.

.4 Clips:

.1 Bright zinc plated, adjustable height shelf supports.

.2 Acceptable Product: 'Model 256' by Knape & Vogt or approved equivalent.

.5 Shelf pins:

.1 5 mm diameter, bright zinc plated, adjustable height shelf pin supports.

.2 Acceptable manufacturers: Richelieu or approved equivalent.

.6 ***Cabinet hinges: Heavy duty, concealed, 100 degree, clip, self-closing, 'Model MODUL' by Blum or approved equivalent.***

~~.6 Cabinet hinges and restrictor clips:~~

~~.1 Heavy duty, concealed, minimum 170 degree opening angle, self closing, complete with plastic hinge restrictor clip.~~

~~.2 Supply manufacturer's recommended number of hinges to suit door size and thickness.~~

~~.3 Acceptable Product: 'Blumotion' soft-closing feature with plastic hinge restrictor clip by Blum or approved equivalent.~~

.7 Drawer and cabinet pulls:

.1 8 mm dia. x 104 mm length, D-shaped steel pull with projection of 30 mm.

.2 Finish and colouring: To be selected by the Consultant from manufacturer's standard range.

.3 Acceptable Product: 'Modern Metal Pull - 2288' by Richelieu Hardware or approved equivalent.

.8 Magnetic catches: 'Model 918' by Knape & Vogt or approved equivalent.

.9 Locks:

.1 Cam locks/deadbolt locks complete with lock core, type to suit application and installation.

.2 Acceptable manufacturers:

.1 Hafele.

.2 Richelieu Hardware.

.3 Or approved equivalent.

.10 Cable grommets: round, one-piece chrome polished '429.94.258' by Hafele or approved equivalent.

.11 Closet rods and shelves:

.1 Closet rails, typical:

.1 27 mm diameter wardrobe rail stainless steel finish.

.2 Acceptable Product: 'Model 660 Round Tubing' and tubing flange 'Model 734' by Knape & Vogt or approved equivalent.

.2 Suite closet shelf and rod:

.1 Wire closet shelf with rod and white powder coated finish.

.2 Acceptable Product: 'Closet Shelves' by Alsta Architectural Systems or approved equivalent.

.12 Nylon beads: Clear nylon bead ***cabinet*** door and drawer stops; provided at each corner***.***

.13 Pocket door kit:

.1 Pocket door kit for use on doors up to 125 lbs. Kit shall be complete with the following:

.1 Steel jambs.

.2 Aluminum header.

.3 Track.

.4 Hangers.

.5 Guides.

.6 Floor plates.

.2 Acceptable Product: 'Crowderframe Type B Kit' by K.N. Crowder or approved equivalent.

.14 Pull-down shelving system:

.1 Two-tier pull-down shelf with gas assisted lifting/lowering mechanism, fabricated from heavy-gauge chrome plated metal wire construction, complete with side and bottom mount brackets and capable of locking in the down position.

.2 Acceptable Product: 'Rev-A-Shelf Pull-Down Shelf (Item No. 5PD24CRN)' by Richelieu or approved equivalent.

.15 Pull-out pantry system:

.1 Full height, bottom mount, pull-out pantry system, consisting of a sliding system ''Fulterer' EZ Close Slide System', top guide rail, natural maple frame, adjustable shelves and chrome rail accents. Width of 140 mm (5-1/2").

.2 Acceptable Product: 'Rev-A-Shelf Pull-Out Pantry with Slides – Wood (Item No. 448TPF5851)' by Richelieu or approved equivalent.

.16 ***Pull-out counter extension:***

.1 ***Pull-out counter extension system equipped with aluminum rails capable of accommodating a high-load capacity of 40 kg (88 lbs.), evenly distributed, dampening pistons for smooth opening and a trimmable front bar.***

.2 ***Top work surface and front panel to match finish, material and appearance of adjoining joining counter material.***

.3 ***Acceptable Product: 'Table Extension Mechanism OPLA +39 (Item No. 4117391010)' by Richelieu or approved equivalent.***

2.3 High pressure laminate work

.1 Perform high pressure laminate work according to NAAWS and ANSI/NEMA LD 3.

.2 Ensure adjacent parts of continuous laminate work match in colour and pattern.

.3 Laminate high pressure laminates to specified and indicated core substrates according to manufacturer's instructions.

.4 Cabinetry typical:

.1 Unless noted otherwise or specified, all casework to be constructed of veneer core plywood, minimum thickness 19 mm.

.2 Laminate plastic laminate work to specified laminate work in accordance with manufacturer's written instructions.

.3 Back panels with a minimum thickness of 13 mm.

.4 Cabinet doors and drawer fronts constructed of particleboard or MDF, minimum thickness of 19 mm.

.5 Provide exterior-grade veneer core/marine grade plywood for bottoms of sink boxes and areas that may come into contact with water.

.6 High pressure laminate for exposed laminate surfaces.

.7 Melamine interiors to cabinets and drawers.

.5 Laminate postformed laminates to particle board core or MDF according to manufacturer's instructions.

.6 Fabricate core surfaces and profiles with continuous support and bond over entire surface to receive high pressure laminate.

.7 Apply laminate backing sheets to balance shrinkage stresses induced by high pressure laminate face sheets.

.8 Joints:

.1 Install joints according to reviewed Shop Drawings.

.2 Jointing shall be placed at logical locations in intended millwork item and shall meet the overall aesthetic intent of the Consultant.

.3 Minimize joints in high pressure laminate work.

.4 Do not install joints in high pressure laminate work in less than 2400 mm o. c.

.5 Locate joints minimum 610 mm from cut-outs.

.6 Offset core and high pressure laminate facing joints.

.9 Form shaped profiles and bends as indicated, using postformed grade laminate to laminate manufacturer's instructions.

.10 Edging to be done using 3 mm thick PVC to match high pressure laminate colour to cover exposed edge of core material. Chamfer exposed edges uniformly at approximately 20 degrees. Do not mitre laminate edges.

.11 Apply laminated high pressure liner sheet to interior of cabinetry and where indicated.

.12 Fabricate units by solid surfacing manufacturer's certified or approved fabricator/installer. Fabricate built-up profiles as indicated.

2.4 Fabrication

.1 Finish carpentry work required by this Section is to include but not be limited to the following items:

.1 Casework.

.2 Countertops.

.3 Shelving.

.4 Window sills.

.5 Wood benches with solid wood edges.

.6 Door frames and casings.

.7 Mailbox cubbies.

.8 Walls slats, such as typical entry suite door surrounds and lobby area.

.9 Wood bases.

.10 Stair stringers/panels.

~~.11 Glazed wood screens.~~

.11 Wardrobe and tall storage cabinets.

.12 Millwork at waste bin.

.13 Millwork feature around mailboxes.

.2 Be responsible for methods of construction and for ensuring that materials are rigidly and securely attached and will not be loosened by the work of other sections.

.3 Coordinate locations of concealed supports and blocking with other parts of work. Provide cut-outs for outlet boxes and other fixtures.

.4 Fabricate work in a manner which will permit expansion and contraction of the materials without visible open joints. Conceal joints and connections in wherever possible.

.5 Set nails and countersink screws, apply wood filler to indentations, sand smooth and leave ready to receive finish.

.6 Mitre exposed corners, no end grain shall be visible in completed installation.

.7 Finish millwork according to Section 09 91 00. Finished millwork shall be free from bruises, blemishes, mineral marks, knots, shakes and other defects and shall be selected for uniformity of colour, grain and texture.

.8 Shelving to cabinetwork to be adjustable unless otherwise noted.

.9 Recess shelf standards, unless noted otherwise. Stagger recessed shelf standards on opposite sides of divider.

.10 Do not exceed maximum 760 mm unsupported span for 19 mm thick shelving. House fixed shelving into gables and divisions.

.11 Shop assemble finish carpentry to accommodate delivery and handling and to ensure passage through building openings.

.12 Shop install cabinet hardware for doors, shelves and drawers. Recess shelf standards unless noted otherwise.

.13 Window sills:

~~.1 Provide wood veneered wood sill, jambs and head, complete with solid wood edges for all punched windows.~~

.1 Provide solid wood ***(WD02)*** ~~and plastic laminate faced~~ window sills where shown on drawings.

.2 Conceal all fixings.

.3 Provide all solid continuous blocking and shims required for installation.

.14 Wood base (WB01):

.1 Fabricate base from MDF wood in thickness and profile as shown on drawings, finished according to Section 09 91 00.

.2 Refer to Section 09 06 00 – Material and Finishes Schedule for additional information regarding wood base.

.15 Wood slat wall, surrounds and panels:

.1 Provide wood slat wall/surrounds as shown on drawings, consisting of wood ***veneer (WD01) faced*** slats ***with MDF core***, black acoustic felt, acoustic/fire rated insulation, blocking/furring as required for complete installation and ***adjoining*** panel components with high pressure laminate finish on MDF core ***and painted wood casings*** as indicated.

.2 Insulation thickness as shown on drawings.

.3 Wall slat systems, surrounds and panels to be designed to meet requirements of authorities having jurisdiction and have a Class A rating to ASTM E84 or CAN/ULC S102.

.4 Provide cut-outs as required.

.16 Millwork feature around mailboxes: Coordinate with Section 10 55 23 as required for sizing and installation of millwork components around mailboxes.

3 Execution

3.1 Examination

.1 Verify condition and dimensions of previously installed work upon which this Section depends. Report defects to Consultant. Commencement of work of this Section means acceptance of existing conditions.

3.2 Installation

.1 Install finish carpentry work according to NAAWS and tolerances for architectural woodwork and reviewed Shop Drawings.

.2 Set and secure finish carpentry in place, rigid, plumb, square, and level.

.3 Provide insect repellant powder under all lower cabinetry behind kickplates and within wall framing cavities and according to pest control requirements as indicated in Section 01 57 16.

.4 Scribe and cut as required, fit to abutting walls, and surfaces, fit properly into recesses and to accommodate columns, fixtures, outlets, or other projecting, intersecting or penetrating objects leaving a 0.8 mm gap maximum.

.5 Coordinate cut-outs for plumbing fixtures, inserts, appliances, outlet boxes, and other fixtures, in finish carpentry. Round internal corners of cut-outs and seal exposed cores.

.6 Form joints to conceal shrinkage.

.7 Install draw bolts and splines in laminated plastic counter top joints at maximum spacing 450 mm o. c., and 75 mm from edge. Make joints flush, hairline butt joints.

.8 Install finishing hardware accurately and securely according to manufacturer's directions, adjust and clean.

.9 Install prefinished millwork at locations shown on Contract Drawings. Position accurately, level, plumb straight.

.10 Apply bituminous coating over wood framing members in contact with masonry or cementitious construction.

.11 Melamine panels: Assemble melamine millwork using dowelled/wafered-and-glue construction. Installed melamine panels shall not show any exposed fasteners on finished/exposed surfaces.

.12 Solid surfacing:

.1 Install solid surfacing according to manufacturer's instructions.

.2 Align work plumb and level.

.3 Seal perimeter of fabrication to adjacent construction according to Section 07 92 00.

.13 Wall slats, panels and surrounds:

.1 Install wood slats, panels and surrounds in locations indicated on drawings.

.2 Ensure that slats, panels and surrounds are securely fastened in true vertical and horizontal manner.

~~.14 Glazed wood screens:~~

~~.1 Coordinate installation of glazed wood screens with work of Section 08 80 00.~~

~~.2 Install wood screens in locations indicated on drawings. Ensure that screens are level and plumb when installed.~~

~~.3 Glazing work for wood screens to be installed under work of Section 08 80 00.~~

.14 Door frames:

.1 Butt and cope internal joints of door frames to make snug, tight, joint. Cut right angle joints with mitred joints.

.2 Fit backs of frames snugly to wall surfaces to eliminate cracks at junction of frame with walls.

.15 Sills: Install window sills level, plumb and even in locations as indicated and ensure that sills are securely fastened.

.16 Wood benches:

.1 Construct wood benches of sizes and details as noted, complete with solid wood edges.

.2 Anchor wood to supports in a concealed manner.

.3 Mitre joints at corners. Keep joints to a minimum.

.4 Round all corners, edges and ends.

.5 Install bench support structure supplied under work of Section 05 50 00.

.17 Mailbox cubbies: Set and secure cubbies in place, rigid, plumb, square, and level.

.18 Fastening:

.1 Coordinate wall securement, anchorage, and blocking for finish carpentry items.

.2 Position items of finished carpentry work accurately, level, plumb, true and fasten or anchor securely.

.3 Design and select fasteners to suit size and nature of components being joined. Use proprietary devices as recommended by manufacturer.

.4 Provide heavy duty fixture attachments for wall mounted cabinets.

.5 Set finishing nails to receive filler. Where screws are used to secure members, countersink screw in round cleanly cut hole and plug with wood plug to match material being secured.

.19 Remove and replace damaged, marked, or stained finish carpentry.

.20 Adjust millwork hardware for smooth and efficient operation.

End of section

1 General

1.1 Section includes

.1 The General Conditions and Supplementary Conditions of the Contract, Division 01 Specification Sections, and Contract Drawings apply to this Section.

.2 Work of this Section includes, but is not limited to the following:

.1 Flashing and sheet metal.

.2 Gutters, downspouts and fascias.

.3 Reveal-shaped flashing, fasciae, closures and trims.

.4 Sealant.

.5 Accessories.

1.2 Reference standards

.1 ASTM A653/A653M, Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process.

.2 ASTM C920, Specification for Elastomeric Joint Sealants.

.3 ASTM C1193, Standard Guide for Use of Joint Sealants.

.4 ASTM D4138, Standard Practices for Measurement of Dry Film Thickness of Protective Coating Systems by Destructive, Cross-Sectioning Means.

.5 CSA A123.21, Standard Test Method for the Dynamic Wind Uplift Resistance of Membrane-Roofing Systems.

.6 CSA A231.1/A231.2, Precast Concrete Paving Slabs/Precast Concrete Pavers.

.7 CRCA Roofing Manual, Canadian Roofing Contractors Association.

.8 CSA S136, North American Specification for the Design of Cold Formed Steel Structural Members.

.9 SMACNA, Sheet Metal & Air Conditioning Contractors' National Association, Architectural Sheet Metal Manual.

1.3 Design requirements

.1 Sheet metal flashing and trim assemblies, including cleats, anchors, and fasteners, are to withstand structural movement, thermally induced movement, and exposure to weather without failure due to defective manufacture, fabrication, installation, or other defects in construction. Completed sheet metal flashing and trim are not to rattle or loosen.

.2 Design flashing elements and fastenings to withstand wind loading and perimeter and corner uplift pressures for roof system according to CSA A123.21.

1.4 Submittals

.1 Shop Drawings:

.1 Submit Shop Drawings according to Section 01 33 00 indicating:

.1 Fabrication and installation layouts of sheet metal flashing and trim, including plans, elevations, expansion-joint locations, and keyed details. Distinguish between shop- and field-assembled work.

.2 Proposed method of shaping, forming, jointing.

.3 Application of flashing and sheet metal work.

.4 Interface of thermal spacer clip and girt support structure with adjacent construction, with Shop Drawings developed in coordination with Section 07 05 43.

.5 Flashing fastening method, fasteners, and fastener spacing to meet wind loading and uplift pressures.

.6 Details and components of gutter, downspouts and fascias.

.7 Details of reveal-shaped flashing.

.8 The following criteria and information:

.1 Identification of material, thickness, weight, and finish for each item and location in Project.

.2 Details for forming sheet metal flashing and trim, including profiles, shapes, seams, and dimensions.

.3 Details for joining, supporting, and securing sheet metal flashing and trim, including layout of fasteners, cleats, clips, and other attachments.

.4 Include pattern of seams.

.5 Details of termination points and assemblies, including fixed points.

.6 Details of edge conditions.

.7 Details of special conditions.

.8 Details of connections to adjoining work.

.9 Detail formed flashing and trim at a scale of not less than 1:10.

.2 Samples:

.1 Submit following samples according to Section 01 33 00:

.1 50 x 50 mm samples of sheet metal material, colour and finish.

.2 Representative sample section of prepainted metal flashing illustrating typical S jointing method, minimum 600 mm long.

.3 Representative sample section of prepainted metal flashing illustrating two-piece split jointing method, minimum 600 mm long.

.4 Representative sample section of prepainted reveal-shaped flashing, minimum 600 mm long.

.5 300 x 300 mm sample of high temperature waterproofing membrane.

.6 300 x 300 mm sample of EPDM gutter membrane.

.3 Sustainability submittals: Submit applicable supporting documentation according to sustainability requirements for the approval of the Consultant.

1.5 Quality assurance

.1 Retain a licensed professional engineer, registered in province of Ontario, to perform following services for sheet metal flashing work:

.1 Design sheet metal flashing work to meet wind uplift requirements and to include thermal spacer clip, girt support structure and complete anchorage devices.

.2 Review, stamp, and sign shop drawings.

.3 Conduct shop and field inspections and prepare and submit inspection reports.

.2 Mock-up:

.1 Gutters: Construct one (1) complete 1800 mm long mock-up of a typical gutter including one (1) 600 mm gutter corner condition, one (1) reveal condition and juncture with roof edge, in location acceptable to Consultant.

.2 Reveal-shaped flashing: Construct one (1) complete 1800 mm long layered mock-up of a reveal-shaped flashing condition, including thermal spacer clip, girts, insulation and complete anchorage devices, in location acceptable to Consultant.

.3 Demonstrate installation method, anchoring and workmanship.

.4 Arrange for Consultant's review and acceptance.

.5 Mock-up may remain as part of Work if accepted by Consultant. Remove and dispose of mock-ups which do not form part of Work.

.6 Upon acceptance, mock-up shall serve as a minimum standard of quality for the balance of the work of this Section.

2 Products

2.1 Materials

.1 All materials under work of this Section, including but not limited to, sealants and paints are to have low VOC content limits.

.2 Prepainted sheet steel:

.1 ASTM A653/A653M; Classification LFQ, Grade A, Z275 zinc coating designation, commercial quality, prefinished with coating system as specified herein.

.2 Minimum base steel thicknesses:

.1 Typical***, reveal-shaped flashing, fascias***: 0.8 mm (22 ga.).

.2 ***Gutters and downspouts/rain water leaders*** ~~Reveal-shaped flashing, fascias~~: 3 mm thick.

.3 Finish and colour:

.1 Factory applied, two (2) coat silicone modified polyester (SMP) coating system.

.2 Exposed coating thickness: Dry film coating system thickness not less than 22 micrometres and in compliance with ASTM D4138.

.3 Acceptable Products:

.1 'Perspectra Series' by ArcelorMittal Dofasco.

.2 'WeatherXL' by VicWest Steel.

.3 Or approved equivalent.

.4 Colour: To be selected by the Consultant from manufacturer's standard colour range.

.3 Thermal spacer clip system and girt support system: According to Section 07 05 43.

.4 High temperature waterproof membrane:

.1 Minimum 1.0 mm thick, high temperature composite sheet waterproof membrane, comprised of SBS modified bitumen with woven polyethylene reinforcement.

.2 Primer recommended by membrane manufacturer.

.3 Acceptable Products:

.1 'CCW 300 HT' by Carlisle Coatings and Waterproofing.

.2 'Blueskin PE200HT' by Henry Company Canada Inc.

.3 'Lastobond Shield HT' by Soprema Canada.

.4 Or approved equivalent.

.4 For use at parapet caps and any additional areas as indicated on Contract Drawings.

.5 Gutter membrane: 1.5 mm thick, non-reinforced, cured, synthetic single-ply EPDM. Adhesive as recommended by EPDM manufacturer.

.6 Plastic cement: Trowel grade asphalt mastic.

.7 Sealant:

.1 ASTM C920, Type S, Grade NS, Class 35; High-performance, medium-modulus, one-part, neutral-cure silicone sealant.

.2 Acceptable:

.1 ‘Dowsil CWS’ by Dow Consumer Solutions.

.2 ‘Sikasil WS-305 CN' by Sika Canada.

.3 ‘Tremsil 400' by Tremco Canada.

.4 Or approved equivalent.

.8 Cleats and starter strips: Starter strips to be continuous, of same material as flashing used, 1.2 mm thick.

.9 Fasteners:

.1 Provide Type 300 or 400 series stainless steel fasteners.

.2 Flat head roofing nails of length, type and thickness suitable for metal flashing application.

.3 Provide gasketed hex head fasteners for roof side of metal cap flashing.

.10 Washers: of same material as sheet metal, 1 mm thick with rubber packings.

.11 Touch-up paint: Same colour and material as prepainted sheet steel, as recommended by prefinished coating manufacturer.

.12 Plywood sheathing: According to Section 06 10 00.

.13 Tapered insulation: According to Section 07 53 24.

.14 Splash pad:

.1 To CSA A231.1/A231.2, minimum 87 kg/m^2, maximum 107 kg/m^2, 5% air-entrainment, 610 x 610 x 57 mm thick natural precast concrete with minimum 55MPa compressive strength and shotblast finish.

.2 Acceptable manufacturer: Brooklin Concrete Products Limited or approved equivalent.

2.2 Fabrication

.1 Fabricate copings, flashings, curb counter flashings, starter strips, gutters, downspouts, reveal-shaped flashing, closures, fasciae, trims and miscellaneous flashings according to CRCA, SMACNA and to details shown.

.2 Form prepainted sheet material at shop to shapes shown. Make end joints where adjacent lengths of metal flashing meet, according to jointing method specified.

.3 Form pieces in 2400 mm maximum practical lengths. Make allowance for expansion at joints.

.4 Hem exposed edges 13 mm minimum on underside for appearance and stiffness. Mitre and seal corners with sealant.

.5 Reglets and cap flashing: Form flashings of as detailed and according to CRCA. Provide slotted fixing holes and steel/plastic washer fasteners.

.6 Gutters and downspouts/rain water leaders (RWL):

.1 Provide gutter and downspouts formed from prefinished galvanized steel sheet metal and to CRCA requirements.

.2 Provide goosenecks, strainer baskets and necessary fastenings.

.7 Reveal-shaped flashing, closures, trims and fasciae:

.1 Form reveal-shaped flashing, closures, trims and fasciae from ~~minimum 3 mm thick~~ prefinished galvanized steel sheet metal. Profiles as indicated and to CRCA requirements.

.2 Provide necessary fastenings.

3 Execution

3.1 Examination

.1 Verify condition and dimensions of previously installed work upon which this Section depends. Report defects to Consultant. Commencement of work of this Section means acceptance of existing conditions.

3.2 Installation

.1 Install coping flashings, curb counter flashings, starter strips, scuppers, miscellaneous flashings, and fasteners to details indicated and according to reviewed Shop Drawings and CRCA and SMACNA requirements.

.2 High temperature waterproofing membrane:

.1 Install primer and waterproof membrane continuously over plywood sheathing, according to manufacturer's instructions.

.2 Overlap waterproof membrane 50 mm along sidelaps and 75 mm on end laps and lap in direction of waterflow.

.3 Use concealed fasteners. Exposed fasteners such as pop rivets are not allowed.

.4 Install continuous starter strips to present a true, non-waving, leading edge. Anchor to back-up for a rigid, secure installation.

.5 Counterflash bituminous flashings at intersections of roof with vertical surfaces and curbs. Flash joints using S-lock forming tight fit over hook strips.

.6 Typical S jointing method: Make end joints using an S lock joint. Execute by inserting end coping length in 25 mm deep S lock formed in end of adjacent length. Extend concealed portion of S lock 25 mm outwards and nail to substrate. Face nailing of joints will not be permitted.

.7 Split joint method: Provide two-piece split jointing method where indicated and in accordance with CRCA requirements.

.8 Seal where necessary to form weathertight seal between flashing and adjoining surfaces and between flashing and other work. Sealing work consists of bedding between members where possible. Tool sealant to concave profile where exposed. Install sealant according to ASTM C1193.

.9 Insert metal flashing under cap flashing to form weathertight junction.

.10 Caulk flashing at cap flashing with sealant.

.11 Gutters, downspouts and fascias:

.1 Fabricate and install gutters, downspouts and fascias according to reviewed shop drawing and manufacturer's written instructions.

.2 Longitudinal gutter joints not acceptable.

.3 Line gutter application with fully adhered EPDM. Ensure gutters are water tight.

.4 Slope gutters to downspouts as indicated. Seal joints watertight.

.5 Ensure gutters are installed without rattling, buckling or defects in appearance.

.6 Install downspouts and provide goosenecks back to wall. Secure downspouts to wall with straps at 1800 mm oc; minimum two straps per downspout.

.7 Install fascias without rattling, buckling or defects in appearance.

.12 Reveal-shaped flashing, closures, trims and fasciae:

.1 Install reveal-shaped flashing, closures, trims, fasciae, thermal spacers and supports according to reviewed shop drawings and according to Section 07 05 43.

.2 Install reveal-shaped flashing, closures, trims and fasciae without rattling, buckling or defects in appearance.

.3 Install reveal-shaped flashing, closures, trims and fasciae, carefully formed and profiled, complete with sealant to provide a weathertight installation.

.13 Install one splash pad for each downspout, angled away from building.

End of section

MATERIAL AND FINISHES SCHEDULE

Section	Code	Description	Product Details
06 20 00	ML01	Melamine (White - Interior of Cabinets)	Manufacturer: Wilsonart or approved equivalent Product: Melamine / Thermally Fused Laminate Colour: Designer White Code: D354 LRV: ##
06 20 00 & 08 14 16	PLAM01	Plastic laminate (Woodlook - Typical)	Manufacturer: Wilsonart or approved equivalent Product: White Oak PB Finish: 05 Timbergrain Code: 8281K Finish: 05 Timbergrain LRV: 33
06 20 00	PLAM02	Plastic laminate (Dark Brown)	Manufacturer: Formica or approved equivalent Product: Fenix Pattern: Grigio Aragona Code: J0793
06 20 00	PLAM03	Plastic laminate (White)	Manufacturer: Fenix or approved equivalent Product: Grade JL Standard Core 0.7mm thick Colour: Bianco Kos Code / Finish: J0793 LRV: 84

MATERIAL AND FINISHES SCHEDULE

Section	Code	Description	Product Details
06 20 00	SS01	Solid Surface (Resident Kitchen)	Manufacturer: Formica or approved equivalent Product: Everform Pattern: Bianco Mineral Code: 758 LRV: ##
06 20 00	SS02	Solid Surface (Resident WC)	Manufacturer: Formica or approved equivalent Product: Everform Pattern: Luna Concrete Code: 781 LRV: ##
06 20 00	SS03	Solid Surface (Public kitchen)	Manufacturer: Formica or approved equivalent Product: Everform Pattern: Luna Steel Code: 415 LRV: ##
06 20 00	WD01	Wood Veneer	Species: White oak or approved equivalent Cut: Rift cut Panel match: Book Match Finish: Clear Satin Finish LRV: TBD
06 20 00	WD02	Wood (solid)	Species: White oak or approved equivalent Cut: Panel match: Book Match Finish: Clear Satin Finish LRV: TBD

Note: Material and finishes have been included to represent the Basis of Design and are subject to change with similar or approved equal products.

MATERIAL AND FINISHES SCHEDULE

Section	Code	Description	Product Details
06 20 00	WB01	Wood Trims (Wood Base - Residential)	Manufacturer: As Per drawings Product: MDF Size and Profile as per drawings Finish: Painted - PT02 LRV: ##
09 30 00	CT01	Ceramic Tile - Wall (Amenity - Backsplash / Resident Suite - Kitchen & WC Sink Backsplash)	Manufacturer: Stone Tile or approved equivalent Product: CeSi Glazed Pattern: Bianco matte Size: 6 x 25 LRV: ##
09 30 00	CT02	Ceramic Tile - Wall (Common Spaces - Laundry / UNIV. Washrooms)	Manufacturer/Distributer: CE.SI. By stone tile or approved equivalent Product: Full body R10A+B Pattern: Fluoro Size: 20 x 20 LRV: ##
09 30 00	CT03	Ceramic Tile - Wall (Washroom shower walls)	Manufacturer: Stone Tile or approved equivalent Product: CeSi Glazed Pattern: Cotone matte Size: 20 x 60 LRV: ##

Note: Material and finishes have been included to represent the Basis of Design and are subject to change with similar or approved equal products.

MATERIAL AND FINISHES SCHEDULE

Section	Code	Description	Product Details
09 30 00	PCT01	Body though Porcelain Tile - (Ground floor lobby / washroom / laundry /amenity/Elevator Lobby)	Manufacturer: Stone Tile or approved equivalent Product: Lagom Oyster mat Pattern: Matte,60x120 LRV: ##
09 30 00	PCT02	Body though Porcelain Tile - (Suite WC floor)	Manufacturer: Stone Tile or approved equivalent Product: Lagom Smoke Pattern: Matte, 60x120 LRV: ##
09 30 00	PCT03	Body though Porcelain Tile - (wall base where PCT01 is used)	Manufacturer: Stone Tile or approved equivalent Product: Lagom Graphite Pattern: Matte,60x120 LRV: ##
09 30 27	TAI01	Tactile Attention Indicator (Exit Stairs)	Manufacturer: Access Tile Tactile Systems or approved equivalent Product: Access® FR Fire Resistant Tactile Surface Systems Size: Refer to Drawings Description: Tactile Walking Surface Indicator (TWSI) with Truncated Domes compliant with AHJ Colour: TBD LRV: ##

MATERIAL AND FINISHES SCHEDULE

Section	Code	Description	Product Details
09 51 00	ACT01	2’ x 4’ - Acoustical Ceiling Panel (Typical)	Manufacturer: CGC or approved equivalent Product: Mars High NRC Edge: Square Size: Panel size 2' x 4' NRC: .75 CAC: 35 Grid: DXT/DXLT - Flat White Standard Advantage Flat Lay in, 9/16" Track To be Mold- and mildew-resistant surface Washable, Scratch-resistant, Soil-resistant LRV: 90
09 65 13	RB01	Applied Rubber Wall Base -	Manufacturer: Johnsonite or approved equivalent Product: Millwork Wall Base System Profile : 3" OBLIQUE Code: Peppercorn LRV: ##
09 65 19	LVP01	Luxury Vinyl Plank - Woodlook (Resident Suite)	Manufacturer: Tarkett or approved equivalent Product: Event Wood - PECF Crafted Plank+ Size: 6 x 48 Colour: 11212 Savannah Installation: Glue down installation Installation pattern: TBD
09 65 19	RBF01	Rubber Floor (Stair & Stair Landings)	Manufacturer: Tarkett or approved equivalent Product: TBD Pattern: TBD Code: TBD LRV: ##

MATERIAL AND FINISHES SCHEDULE

Section	Code	Description	Product Details
09 65 19	RBST01	Rubber integrated stair tread and riser with contrasting strip	Manufacturer: Tarkett or approved equivalent Product: Anglefit Rubber Stair with Integrated Riser Colour: TBD Accesory: Grit Tape Insert Contrasting Strip: Textured Strip Black LRV (tread): ## LRV (strip): ##
09 67 23	EFC01	Epoxy Floor Coating (House Keeping)	Performance Specification refer to specifications
09 67 23	EFS01	Seamless Epoxy Floor System, with texture (Waste)	Performance Specification refer to specifications CB/EFS01 = Epoxy floor system coved base to match EFS01
09 67 70	WPC01	Water Containment Coating (Mechanical and Water Rooms)	Performance Specification refer to specifications
09 67 72	CONS01	Concrete floor sealer	Performance Specification refer to specifications
09 72 17	FRP	Fiberglass Reinforced Plastic (FRP) CladdinG (Waste Room & House Keeping)	Performance Specification refer to specifications

MATERIAL AND FINISHES SCHEDULE

Section	Code	Description	Product Details
09 83 00	ACB01	Acoustic Ceiling Baffles (Amenity Space)	Manufacturer: Armstrong Ceiling Solutions or approved equivalent Product: Feltworks Blades - VarAffix Finish: Frost Pattern: , 8"" x 96"" x 3/8"", 0.55 NRC, 6"" O.C. Spacing, no infill panels. 0.55 NRC Code: BPSM1069FSD LRV: ##
09 91 00	PT01	Paint - Typical (Walls & Ceilings)	Manufacturer: Benjamin Moore or approved equivalent Colour: Chantilly Lace - White LRV: ##
09 91 00	PT02	Paint - Door Frames	Manufacturer: Benjamin Moore or approved equivalent Colour: Black Tar LRV: ##
09 91 00	PT03	Paint - Painted Doors	Manufacturer: Benjamin Moore or approved equivalent Colour: Sweatshirt Grey LRV: ##
10 80 00	CG1	Corner Guards	Manufacturer: INPRO or approved equivalent Product: Stainless Steel Corner Guards Size: 1.5" wing size Height: Top of Base to underside of Ceiling Colour: SS LRV: ##

Note: Material and finishes have been included to represent the Basis of Design and are subject to change with similar or approved equal products.

MATERIAL AND FINISHES SCHEDULE

Section	Code	Description	Product Details
10 80 00	CG2	Corner Guards	Manufacturer: INPRO or approved equivalent Product: Tape-on Corner Guard Size: 1.5" wing size Height: Top of Base to underside of Ceiling Colour: TBD https://www.inprocorp.com/products/tape-on-corner-guards-11248-wall-protection LRV: ##
10 80 00	RCR	Crash Rails (Rubber) - Waste Room	Manufacturer: C/S Group or approved equivalent Product: Rubber Bump Guard Colour: TBD
10 80 00	EM01	Entry mat (Vestibules)	Manufacturer: C/S Group or approved equivalent Product: PEDIMAT Colour: TBD Entry Grilles and Carpet - drainless

End of section

1 General

1.1 Section includes

.1 The General Conditions and Supplementary Conditions of the Contract, Division 01 Specification Sections, and Contract Drawings apply to this Section.

.2 Work of this Section includes, but is not limited to the following:

.1 Gypsum board and associated materials.

.2 Exterior grade gypsum sheathing.

.3 Cement board.

.4 Metal CT and CH shaftwall studs.

.5 Metal framing.

.6 Suspension system components.

.7 Accessories.

1.2 Reference standards

.1 ASTM A653/A653M, Specification for Steel Sheet, Zinc-coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvanealed) by the Hot-Dip Process.

.2 ASTM C474, Standard Test Methods for Joint Treatment Materials for Gypsum Board Construction.

.3 ASTM C475, Specification for Joint Compound and Joint Tape for Finishing Gypsum Board.

.4 ASTM C645, Specification for Nonstructural Steel Framing Members.

.5 ASTM C754, Specification for Steel Framing Members to Receive Screw-Attached Gypsum Board.

.6 ASTM C840, Specification for Application and Finishing of Gypsum Board.

.7 ASTM C1002, Specification for Steel Self-Piercing Tapping Screws for the Application of Gypsum Panel Products or Metal Plaster Bases to Wood Studs or Steel Studs.

.8 ASTM C1177/C1177M, Standard Specification for Glass Mat Gypsum Substrate for Use as Sheathing.

.9 ASTM C1178, Specification for Glass Mat Water-Resistant Gypsum Backing Board.

.10 ASTM C1278, Specification for Fiber-Reinforced Gypsum Panel.

.11 ASTM C1280, Standard Specification for Application of Exterior Gypsum Panel Products for Use as Sheathing.

.12 ASTM C1396, Specification for Gypsum Board.

.13 ASTM C1658/C1658M, Standard Specification for Glass Mat Gypsum Panels.

.14 ASTM E84, Standard Test Method for Surface Burning Characteristics of Building Materials.

.15 CAN/ULC S102, Surface Burning Characteristics of Building Materials and Assemblies.

.16 CAN/ULC-S146, Standard Method of Test for the Evaluation of Encapsulation Materials and Assemblies of Materials for the Protection of Structural Timber Element.

1.3 Design requirements

.1 Design gypsum board wall and ceiling systems with a maximum deflection of l/360.

.2 Design ceiling suspension system according to manufacturer's printed directions and ASTM C754.

.3 Design ceiling system for adequate support of electrical fixtures as required by the current bulletin of the Electrical Safety Authority.

.4 Design hanger anchor and entire suspension system static loading not to exceed 25% of their ultimate capacity including lighting fixture dead loads.

.5 Design suspension system to support weight of mechanical and electrical items such as air handling boots and lighting fixtures, and with adequate support to allow rotation/relocation of light fixtures.

.6 Design subframing as necessary to accommodate, and to circumvent, conflicts and interferences where ducts or other equipment prevent the regular spacing of hangers.

.7 Design wall framing system and reinforce as necessary to accommodate and support items attached to and supported by wall framing system.

.8 Design shaft wall framing system for wall assemblies with a height greater than 3000 mm.

.9 ~~Design~~ ***P***artitions serving as guards to comply with OBC requirements and to withstand guard loads ***and engineered under Section 06 10 00***.

.10 Finish all vertical service spaces, exits and concealed spaces, such as within ceilings, bulkheads, chaises and shafts, where mass timber is exposed, with one layer of Type X gypsum board or Consultant approved other noncombustible material providing an encapsulation rating of 25 minutes and meeting CAN/ULC-S146. Finish joints using specified tape or by using standard methods of mudding and taping. No further finishing is required.

1.4 Regulatory requirements

.1 Provide fire separations and fire protection exactly as specified in test design specification that validates the specified rating. Verify that work specified in other Sections, as a part of the entire assembly, meets applicable validating test design specification.

1.5 Submittals

.1 Product data:

.1 Submit manufacturer's Product data according to Section 01 33 00 indicating:

.1 Performance criteria, compliance with appropriate reference standard, characteristics, and limitations.

.2 Product transportation, storage, handling and installation requirements.

.2 Shop Drawings:

.1 Submit Shop Drawings according to Section 01 33 00 indicating:

.1 Wall assemblies, suspension systems, adjacent construction, elevations, sections and details, dimensions, thickness, finishes and relationship to adjacent construction.

.2 Framing and blocking for items being supported of wall systems.

.3 Fire rated designs.

.3 Sustainability submittals: Submit applicable supporting documentation according to sustainability requirements for the approval of the Consultant.

.4 Certifications: Submit written certification stating that suspended ceiling system is designed for adequate support of electrical fixtures as required by the current bulletin of the Electrical Safety Authority.

1.6 Quality assurance

.1 Installer's qualifications: Execute the work of this Section by skilled, qualified, and experienced workers trained in the installation of the work of this Section.

.2 Retain a professional engineer, licensed in province of Ontario, with experience in work of comparable complexity and scope, to perform following services as part of work of this Section:

.1 Design of shaft wall systems with height greater than 3000 mm.

~~.2 Design partitions to serve as a guard where required.~~

.2 Design of suspended gypsum board assemblies.

.3 Review, stamp, and sign Shop Drawings and design calculations.

.4 Conduct shop and on-site inspections, prepare and submit written inspection reports verifying that this part of Work is according to Contract Documents and reviewed Shop Drawings.

1.7 Site conditions

.1 Do not begin work of this Section until:

.1 Mechanical and electrical work above the ceiling is complete.

.2 Substrate and ambient temperature is above 15°C.

.3 Relative humidity is below 80 %.

.4 Ventilation is adequate to remove excess moisture.

.2 Install temporary protection and facilities to maintain Product manufacturer's, and above specification, environmental requirements 24 h before, during, and 24 h after installation.

2 Products

2.1 Materials

.1 General: All materials under work of this Section, including but not limited to, sealants, adhesives, and primers are to have low VOC content limits.

.2 Wood studs, blocking, framing, furring, etc.: According to Section 06 10 00.

.3 Steel framing:

.1 ASTM C754; ASTM A653/A653-M, Z275; cold rolled, galvanized steel sheet.

.2 Acceptable manufacturer:

.1 Bailey Metal Products Limited.

.2 Corus Metal Profiles.

.3 Or approved equivalent.

.4 Main carrying channels: ASTM C645; Formed from galvanized steel sheet, 38 x 19 mm cold rolled, channels.

.5 Resilient channel: ASTM C645; 0.5 mm thick galvanized metal, 57 mm wide x 12 mm deep for walls and ceiling to reduce sound transmission.

.6 Furring channels: ASTM C645; Formed from galvanized steel sheet, 22 mm winged flange type, cold rolled.

.7 Furring channels (hat type): ASTM C645; 0.5 mm base steel thickness, galvanized. 70 mm wide x 22 mm deep hat shaped channel.

.8 Heavy duty furring channels: ASTM C645; 0.9 mm steel thickness, galvanized hat shaped channel with a wider and deeper size as required by manufacturers.

.9 Hanger wires: 4.1 mm minimum diameter galvanized pencil rod.

.10 Tie wire: 1.6 mm thick minimum diameter, soft annealed, galvanized steel wire.

.11 Corner bead, casing bead, and special shapes: Formed from 0.6 mm thick minimum, galvanized steel sheet, designed to be concealed by joint compound.

.12 Gaskets (acoustic partitions): Adhesive-backed, closed-cell vinyl foam strips that allow fastener penetration without foam displacement, 3.2 mm thick, in width to suit stud size.

.13 Control joint strip: Roll formed from galvanized steel sheet, with a tape protected recess, 6 mm wide x 11 mm deep.

.14 Screw fasteners: ASTM C1002 Type S; corrosion resistant.

.15 Anchors:

.1 Tie wire sleeve anchors suitable for anchorage to wood deck, manufactured by ITW Construction Products or approved equivalent.

.2 Concrete anchors: tie wire sleeve anchors, ‘Redi-Drive Anchors' by ITW Red Head or approved equivalent.

.16 Batt insulation: According to Section 07 21 00.

.17 Standard and acoustic sealants: According to Section 07 92 00.

.18 Fire rated sealants: Provide fire rated sealants for use at fire rated assemblies according to Section 07 85 00.

.19 Smart vapour control layer/retarder: According to Section 07 26 00.

.20 Vibration isolation ceiling hangers:

.1 Unit shall consist of a steel spring in series with a neoprene isolating element. Provide hangers to ensure that working load does not exceed 2/3 of solid load. Design hangers to tolerate 30 deg. misalignment.

.2 Acceptable Product:

.1 ‘Model W30N’ by Mason Industries Inc.

.2 Or approved equivalent by PAC International or approved equivalent manufacturer.

.21 Gypsum board:

.1 ASTM C1396; gypsum board 12.7 mm and 15.9 mm thicknesses of maximum practical lengths to minimize end joints, unless indicated otherwise.

.2 Acceptable manufacturers:

.1 Certainteed Gypsum Canada.

.2 CGC Inc.

.3 Georgia-Pacific Canada LP.

.4 Or approved equivalent.

.22 Fire rated gypsum board (Type X):

.1 ASTM C1396; gypsum board 15.9 mm thick of maximum practical lengths to minimize end joints, unless indicated otherwise.

.2 Acceptable manufacturers:

.1 Certainteed Gypsum Canada.

.2 CGC Inc.

.3 Georgia-Pacific Canada LP.

.4 Or approved equivalent.

.23 Cementitious board:

.1 ASTM C1177, high strength portland cement building panel with self adhesive glass tape; provide board with heavier mesh reinforcement for suspended applications.

.2 Acceptable Products:

.1 ‘Durock Cement Board Next Gen.’ by CGC Inc.

.2 Or approved equivalent by Certainteed Gypsum or approved equivalent manufacturer.

.24 Moisture and mould resistant board:

.1 15.9 mm thick, unless otherwise indicated, of maximum practical lengths to minimize end joints, unless indicated otherwise.

.2 Acceptable Products:

.1 ‘M2Tech Moisture and Mould Resistant' by Certainteed Gypsum Canada.

.2 ‘Sheetrock Mold Tough’ by CGC Inc.

.3 ‘DensArmor Plus High Performance Interior Panel’ by Georgia-Pacific Canada LP.

.4 Or approved equivalent.

.25 Abuse resistant panels:

.1 ASTM C1396; 12.7 and 15.9 mm thick, as indicated on drawings. For use up to 1200 mm high.

.2 Acceptable Products:

.1 ‘Abuse Resistant’ by Certainteed Gypsum Canada.

.2 ‘Sheetrock AR’ by CGC Inc.

.3 ‘ToughRock’ by Georgia-Pacific Canada LP.

.4 Or approved equivalent.

.26 Tile backer:

.1 Water resistant tile backer board meeting ASTM C1178 or ASTM C1278, thickness as indicated.

.2 Acceptable Products:

.1 ‘GlasRoc Tile Backer’ by Certainteed Gypsum Canada.

.2 ‘Fiberock Aqua-Tough Underlayment’ by CGC Inc.

.3 ‘Dens Shield’ by Georgia-Pacific Canada LP.

.4 Or approved equivalent.

.27 Exterior grade gypsum sheathing:

.1 ASTM C1177C1177M, 15.9 mm thick, exterior grade gypsum sheathing.

.2 Acceptable Products:

.1 ‘GlasRoc Brand Sheathing’ by CertainTeed Gypsum Canada.

.2 ‘Securock Glass-Mat Sheathing’ by CGC Inc.

.3 ‘Dens-Glass Sheathing’ by Georgia-Pacific Canada LP.

.4 Or approved equivalent.

.28 Shaftwall gypsum system (CT stud):

.1 Steel J-tabbed track profile:

.1 ASTM C645; rolled formed galvanized sheet steel, minimum 25 gauge.

.2 Acceptable manufacturer:

.1 Bailey Metal Products Limited.

.2 Or approved equivalent by:

.1 CGC.

.2 Certainteed Gypsum Canada.

.3 Gypsum Corporation.

.4 Or approved equivalent manufacturer.

.2 C-T stud:

.1 Hot-dipped galvanized, C-T steel stud.

.2 Acceptable Product:

.1 ‘CT Stud Profile’ by Bailey Metal Products Limited.

.2 Or approved equivalent by:

.1 CGC.

.2 Certainteed Gypsum Canada.

.3 Gypsum Corporation.

.4 Or approved equivalent manufacturer.

.3 Liner panel:

.1 ASTM C1396; gypsum wallboard panel, thickness: 25.4 mm, width: 610 mm. Type X shaftliner panel.

.2 Acceptable manufacturers:

.1 CGC.

.2 Certainteed Gypsum Canada.

.3 Gypsum Corporation.

.4 Or approved equivalent.

.4 Face panel:

.1 ASTM C1396; gypsum wallboard panel, 1 layer, thickness: 15.9 mm, width: 1219 mm. Unless otherwise indicated or required to meet fire rated design, Type X face panel.

.2 Acceptable manufacturers:

.1 CGC.

.2 Certainteed Gypsum Canada.

.3 Gypsum Corporation.

.4 Or approved equivalent.

.29 Shaftwall gypsum system (CH stud):

.1 Steel J-runner:

.1 ASTM C645; rolled formed sheet steel, 25 gauge.

.2 Acceptable manufacturers:

.1 CGC.

.2 Gypsum Corporation.

.3 Or approved equivalent.

.2 C-H stud:

.1 Hot-dipped galvanized, C-H steel stud.

.2 Acceptable manufacturers:

.1 CGC.

.2 Gypsum Corporation.

.3 Or approved equivalent.

.3 Liner panel:

.1 ASTM C1396; Gypsum wallboard panel, thickness: 25.4 mm, width: 610 mm.

.2 Acceptable Product:

.1 ‘M2Tech Shaftliner Type X’ by Certainteed Gypsum Canada.

.2 Or approved equivalent by:

.1 CGC.

.2 Gypsum Corporation.

.3 Or approved equivalent manufacturer.

.4 Face panel:

.1 ASTM C1396; Gypsum wallboard panel, 1 layer, thickness: 15.9 mm, width: 1219 mm.

.2 Acceptable Product:

.1 ‘GlasRoc Shaftliner Type X’ by Certainteed Gypsum Canada.

.2 Or approved equivalent by:

.1 CGC.

.2 Gypsum Corporation.

.3 Or approved equivalent manufacturer.

.30 Special trim pieces: to include, but not limited to, the following:

.1 J moulding: Formed from extruded aluminum alloy 6063 T5; manufactured by Fry Reglet, Gordon Trims or approved equivalent.

.2 Z-reveal: Formed from extruded aluminum alloy 6063 T5; manufactured by Fry Reglet, Gordon Trims or approved equivalent.

.31 Primer: Where indicated by board manufacturer, provide primer as required to achieve finishes as defined in ASTM C840.

.32 Latex fortified mortar: Of type recommended by cementitious board manufacturer to suit application.

.33 Joint reinforcing tape:

.1 Standard gypsum board: ASTM C475; 50 mm wide x 0.25 mm thick, perforated paper, with chamfered edges.

.2 Moisture resistant and tile backer boards: ASTM C475; fibreglass mat joint tape as recommended by board manufacturer to suit location.

.3 Fire rated gypsum board ceiling:

.1 Multi-layer, fiberglass reinforced joint tape with pressure sensitive adhesive tested to requirements for joint tape under ASTM C474, complying with requirements for joint tape under ASTM C475 for both paper-faced gypsum board (ASTM C1396) and glass-mat faced gypsum board (ASTM C1177 and ASTM C1658).

.2 Acceptable Product: ‘Flame Fighter Fire Tape’ by E-Z Taping Systems, or approved equivalent.

.4 Cement board: Mesh reinforcing tape recommended by cement board manufacturer.

.34 Bonding adhesive: Type for purpose intended and as recommended and approved by manufacturer.

.35 Joint and patching compound: ASTM C475; asbestos-free, supplied by manufacturer of gypsum board used.

.36 Fast setting patching compound:

.1 ASTM C475; asbestos-free, patching compound.

.2 Acceptable Products:

.1 ‘Sheetrock or ‘Durabond’ by CGC Inc.

.2 ‘Moisture and Mold Resistant Setting Compound’ with M2Tech’ by Certainteed Gypsum Canada.

.3 Or approved equivalent.

.37 Access doors: According to Section 08 31 00.

.38 Insect repellant powder/pesticide: Provide diatomaceous earth insect powder as specified herein. Refer to Section 01 57 16 for additional pest control requirements.

3 Execution

3.1 Examination

.1 Verify condition and dimensions of previously installed work upon which this Section depends. Report defects to Consultant. Commencement of work of this Section means acceptance of existing conditions.

3.2 Wood studs, blocking, framing and furring

.1 Coordinate with Section 06 10 00 as required for wood studs, framing, blocking and furring components.

.2 Provide diatomaceous earth under all interior walls. Refer to Section 01 57 16 for additional pest control requirements.

.3 Where possible, eliminate thru-bolts for reinforcing in partitions for door hardware, such as but not limited to exit devices, closer and closer arms, power operator arms, magnetic hold opens, magnetic locks, and similar door hardware items.

3.3 Suspension framing

.1 Install ceiling systems according to reviewed Shop Drawings and manufacturer's written instructions.

.2 Install hanger wires plumb and securely anchored to the building structural framing, independent of walls, pipes, ducts, and deck; install additional framing and hangers to bridge interference items.

.3 Install vibration isolation hangers at all locations where indicated according to manufacturer's printed instructions.

.4 Install hanger wires at 1200 mm maximum centres along carrying channels, not less than 25 mm, and not more than 150 mm from channel ends.

.5 Install additional hangers at lighting fixture and ductwork locations. Do not attach hanger wires to mechanical or electrical equipment. Do not support mechanical and electrical fixtures and fitting on ceiling without the ceiling manufacturer's written acceptance.

.6 Install main carrying channels transverse to structural framing members. Lap main carrying channels 200 mm minimum at splices and wire each end with two loops and prevent clustering or lining-up of splices.

.7 Install furring channels at 400 mm o.c., not less than 25 mm, and not more than 150 mm from perimeter walls, at openings, at interruptions in ceiling continuity, and at change in plane. Install furring channels to a tolerance of 3 mm maximum in 3600 mm.

.8 Install additional main carrying and furring channels to frame and to reinforce openings such as recessed lighting fixtures, access hatches, ceiling grilles, outlet boxes, ventilating outlets and similar items.

3.4 Fire rated assemblies

.1 Install Products in fire rated assemblies strictly according to reviewed Shop Drawings and applicable tested and approved designs required by authorities having jurisdiction.

.2 Install firestop fill material behind fire rated acoustical sealant and provide firestop identification tag.

.3 Where double layers of gypsum board are shown, and required for fire rating, screw first layer to studs and furring and laminate the second layer to the first using joint filler as an adhesive. Stagger joints between first and second layers.

3.5 Batt insulation

.1 Install non-rated and fire-rated/acoustic insulation as required for Work of this Project according to Section 07 21 00.

3.6 Acoustical partitions and sealant

.1 Install acoustical sealant to acoustically insulated partitions according to the manufacturer's instructions and Contract Drawings.

.2 Install acoustical sealant at entire partition perimeter on both sides and at both sides of openings, cut-outs, and penetrations, concealed from view in the final installation. Where larger gaps exist in acoustically sensitive partitions, provide a gypsum board patch.

.3 Smooth acoustical sealant with trowel prior to skin forming.

.4 Stagger gypsum board joints between layers in multi-layer assemblies.

3.7 Gypsum board

.1 Comply with ASTM C840. Install gypsum board according to reviewed Shop Drawings and manufacturer's written instructions.

.2 Install gypsum board vertically or horizontally, whichever results in fewer end joints. Locate end joints over supporting members.

.3 Install gypsum board in lightly butted contact at edges and ends and with 1.6 mm maximum open space between boards; do not force gypsum board into place. Do not install imperfect, damaged or damp boards.

.4 Install gypsum board butting paired tapered edge joints, and mill-cut or field-cut end joints; do not place tapered edges against cut edges or ends.

.5 Install vertical joints minimum 300 mm from the jamb lines of openings and stagger vertical joints over different studs on opposite sides of partitions.

.6 Do not locate joints within 200 mm of corners or openings, except where control joints occur at jamb lines or where openings occur adjacent to corners. Where necessary, place a single vertical joint over the centre of wide openings.

.7 Install gypsum board over concrete and concrete masonry units with adhesive as recommended by gypsum board manufacturer where indicated on Contract Drawings.

.8 Cut, drill and patch gypsum board as may be necessary to accommodate the work of other trades.

.9 Fire separations:

.1 Construct gypsum board assemblies, where located, according to tested assemblies to obtain required or indicated fire rated assemblies. As a minimum, fire separations shall consist of metal framing covered on both sides by fire-rated gypsum board.

.2 Apply diatomaceous earth under all interior walls, wood frame and gypsum board materials and according to requirements of Section 01 57 16.

.3 Install assemblies tightly to enclosing constructions to maintain integrity of the separations. Install casing beads at all perimeter edges.

3.8 Cement board

.1 Apply cementitious board to framing, with screw fasteners and taped joints according to manufacturers instructions.

.2 Pre-cut board to required sizes and make necessary cut-outs. Fit ends and edges closely but not forced together.

.3 Fasten board to steel framing with rust proof self-drilling, self-threading case hardened screws at 200 mm o.c for walls and 150 mm o.c for suspended applications.

.4 Apply mesh tape centred over all joints and corners but not overlapped.

.5 Apply 3 mm minimum thick skim coat of latex fortified mortar uniformly over entire cementitious board surfaces. Leave surface smooth and flat to receive subsequent finish.

3.9 Shaftwall liner (C-T studs)

.1 Install shaftwall assemblies including C-T stud and J-tabbed track according to Contract Drawings, reviewed Shop Drawings, manufacturer's written instructions and meeting requirements as specified herein.

.2 Vertical shaftwalls:

.1 Secure J-track as perimeter framing and plumb to ceiling, floor and sides, attached with suitable fasteners and spaced not more than 610 o.c. Apply a bead of non-hardening, flexible sealant to perimeter.

.2 Pre-plan stud layout 600 mm o.c. and adjust spacing at either end so end studs do not fall closer than 300 mm from end.

.3 Erect first 25 mm thick shaftliner panel, cut 19-25 mm less than total height of framed section. Plumb panel against web of J-track and bend out optional tabs in J-track to secure panels in place.

.4 Insert C-T Stud, cut 19 mm less than overall height, into top and bottom J-tracks and fit tightly over previously installed 25 mm thick panel. Allow equal clearance between top and bottom J-track.

.5 Install next 25 mm thick shaftliner inside J-track and within tabs of C-T stud.

.6 Progressively install succeeding studs and panels as specified above until wall section is enclosed. Final panel section may be secured with tabs from J-track at 300 mm o.c.

.7 Where wall heights exceed standard or available length of shaftliner panels, panels may be cut and stacked with joints occurring within top or bottom third points of wall. Alternately stagger joints of adjacent panels to prevent a continuous horizontal joint. Ensure gypsum panels engage a minimum of two (2) tabs.

.8 Install C-T Studs full height, one piece.

.9 For doors, ducts or other large penetrations or openings, install J-track as perimeter framing. Use 20-gauge track with a 75 mm back leg for elevator doors and block cavity with 300 mm wide gypsum filler strips for doors exceeding 2134 mm (7'-0") height.

.3 Horizontal shaftwalls/corridor ceilings:

.1 Secure J-track to all corridor walls or stair stringer framing around perimeter of ceiling or stair soffit to be protected.

.2 Position J-track with shorter 25 mm long leg of track facing floor.

.3 Place fasteners through web of J-track to supporting structure with manufacturer recommended spacing.

.4 Secure first liner panel to J-track with drywall screws spaced at 300 mm on-center and at each end through longer leg of J-track.

.5 Position C-T stud to secure free edge of first shaft liner panel within “T” portion of C-T stud.

.6 Do not splice C-T studs. Use only full length pieces of proper length.

.7 Secure both ends of C-T stud to both flanges of the J-track using recommended screws.

.8 Continue sequence of alternating placement of shaft liner panels and C-T studs with each C-T stud engaging long edges of the 25 mm thick shaft liner panels.

.9 Fasten the end of each 25 mm thick shaft liner panel within the perimeter J-track to the J-track flange with recommended screws at 300 mm o.c.

.10 Cut liner panels and butt factory ends of liner panels. Butt joints shall occur within outermost one third points of span. Alternately stagger joints in adjacent liner panels to prevent a continuous joint.

.11 Secure last liner panel to tabbed J-track with drywall screws spaced at 300 mm o.c. and at each end through longer leg of J-track.

.12 Install gypsum board layers as shown on Contract Drawings and as recommended by manufacturer to achieve intended fire rating.

.13 Caulk perimeter edges and abutments with dissimilar materials and any penetrations with a non-hardening flexible sealant.

.14 Tape and finish all joints on outer layer and finish fastener heads with joint compounds meeting ASTM C474.

3.10 Shaftwall liner (C-H studs)

.1 Plan and lay out metal framing components to ensure that all wall sections are plumb and properly aligned.

.2 Install J-track along the ceiling line and vertically at columns and abutting partitions, positioning the long legs closest to the shaft, using powder actuated fasteners or other approved method. Secure each piece with the appropriate fasteners spaced a maximum 610 mm O.C.

.3 Attach J-track to the floor with fasteners spaced at 610 mm O.C.

.4 Install shaftliner panels vertically. The leading edge of the first panel must be attached to the long leg of the vertical J-track with 41 mm Type S screws spaced 610 mm O.C. Secure the top and bottom edges using the same fasteners and spacing.

.5 Friction-fit C-H stud into the top and bottom tracks and slide it snugly against the shaftliner panel. Make sure the edge of the board is in full contact with the centre web of the stud and covered by all the tabs.

.6 Place the next shaftliner panel between the tabs and flange on the opposite side of the C-H stud and secure it to the top and bottom track with 41 mm Type S screws spaced 610 mm O.C.

.7 Install subsequent shaftliner panels and C-H studs in the same manner. Check periodically to ensure they are plumb.

.8 At the end of a partition run, cut the last shaftliner panel slightly narrower and shorter than the opening to facilitate installation.

.9 For walls exceeding 3.7 m in height, shaftliner panel end joints shall fall alternately in the upper and lower 1/3 of the partition. Use a C-H stud placed horizontally between panels to secure each joint.

.10 Frame all cut openings in the shaft side with J-track, providing adequate structural support for openings over 1219 mm.

.11 Anchor elevator door frames to shaftwall enclosures; however, they must remain independently supported by the building frame.

3.11 Exterior grade gypsum sheathing

.1 Install exterior grade gypsum sheathing materials according to ASTM C1280, reviewed shop drawings, and manufacturer’s written instructions.

.2 Install exterior sheathing with long dimension perpendicular to wood studs, offset joints and butt tight, centre edges of sheathing over wood studs, mechanically fasten with specified fasteners and washers according to manufacturer’s instructions.

3.12 Corner, casing beads and trim

.1 Corner reinforcing bead: Install along all external angles, erect plumb, level and with a minimum of joints. Secure with screws at 225 mm o.c. apply filler over flanges flush with nose of the bead and extending at least 75 mm onto surface of board each side of corner. When filler dries, apply a thin coat of topping cement and blend onto adjoining surfaces.

.2 Casing bead: Install where wallboard butts against a surface having no trim concealing the juncture and where shown on Contract Drawings. Erect casing beads plumb or level, with minimum joints, and secure with screws at 300 mm o.c. apply filler over flange flush with bead and extending at least 75 mm onto surface of board. When dry, apply a thin coat of topping cement and blend onto adjoining surfaces.

.3 Recess channels and trim: Install recess channels and special metal trim where shown. Secure to substrate. Provide casing beads full height on wallboard edges at recess channels and metal trim.

3.13 Joint taping and finishing

.1 Install reinforcing tape and a minimum of three (3) coats of joint compound over gypsum board joints, metal trim and accessories, and screw fasteners according to the gypsum board manufacturer's instructions.

.2 Fill gaps between, and any imperfections in, gypsum boards with joint compound, allow to dry, and sand smooth ready for painting.

.3 Install finished gypsum board work smooth, seamless, plumb, true, flush, and with square, plumb, and neat corners.

.4 Finish gypsum board according to ASTM C840 to the following grades:

.1 Level 0: No taping, finishing, or accessories required. Use above suspended ceilings and within other concealed spaces, unless the assembly is fire rated, sound rated, sound or smoke controlled, or unless the space serves as an air plenum.

.2 Level 1: At joints and interior angles embed tape in joint compound. Leave surface free of excess joint compound. Tool marks and ridges are acceptable. Use above suspended ceilings and within other concealed spaces if the gypsum board assembly is fire rated, sound rated, sound or smoke controlled, or the space serves as an air plenum.

.3 Level 2: At joints and interior angles embed tape in joint compound with one separate coat of joint compound applied over joints, angles, fastener heads, and accessories. Use for water resistant gypsum board indicated for use as a substrate for ceramic tile.

.4 Level 3: At joints and interior angles embed tape in joint compound with two separate coats of joint compound applied over all joints, angles, fastener heads, and accessories. Apply joint compound smooth and free of tool marks and ridges. Use where heavy grade wall coverings are the final decoration.

.5 Level 4: At joints and interior angles embed tape in joint compound with three separate coats of joint compound applied over all joints, angles, fastener heads, and accessories. Apply joint compound smooth and free of tool marks and ridges. Use for all locations except those indicated for other finish levels.

.6 Level 5: At joints and interior angles embed tape in joint compound with three separate coats of joint compound applied over all joints, angles, fastener heads, and accessories. Apply a thin skim coat of joint compound, or a material manufactured especially for this purpose, to the entire surface. Leave surface smooth and free of tool marks and ridges. Use where semi-gloss or gloss finish coatings are the final decoration.

3.14 Access doors

.1 Install access doors according to Section 08 31 00 and manufacturer's written instructions.

3.15 Site tolerances

.1 Install metal support systems to ensure that, within a tolerance of +3 mm and -1.5 mm for plaster thickness, finish surfaces will be flat within 3 mm under a 3 m straightedge, and with no variation greater than 1.5 mm in any running 300 mm, and that surface planes shall be within 3 mm of dimensioned location.

3.16 Repair

.1 Make good cut-outs for services and other work, fill in defective joints, holes and other depressions with joint compound.

.2 Make good defective work, and ensure that surfaces are smooth, evenly textured and within specified tolerances to receive finish treatments.

End of section

1 General

1.1 Section includes

.1 The General Conditions and Supplementary Conditions of the Contract, Division 01 Specification Sections, and Contract Drawings apply to this Section.

.2 Work of this Section includes, but is not limited to the following:

.1 Acoustic underlayment.

1.2 Submittals

.1 Product data: Submit manufacturer's Product data according to Section 01 33 00 indicating:

.1 Performance criteria, compliance with appropriate reference standard(s), characteristics, limitations, and trouble-shooting protocol.

.2 Product transportation, storage, handling and installation requirements.

.2 Samples:

.1 Submit following samples according to Section 01 33 00.

.1 Two 300 x 300 mm samples of underlayment.

.3 Sustainability submittals: Submit applicable supporting documentation according to sustainability requirements for the approval of the Consultant.

2 Products

2.1 Materials

.1 Acoustic underlayment ~~(Type 1)~~:

.1 25 mm thick dimpled acoustic rubber pad underlayment, manufactured by 94% recycled rubber content.

.2 Acceptable Product: ‘GenieMat FF25’ by Pliteq, Inc. or approved equivalent.

~~.2 Acoustic underlayment (Type 2, suites):~~

~~.1 Minimum 2 mm thick acoustic mat consisting of 92% recycled rubber. For use under floor finishes in suites as scheduled on drawings.~~

~~.2 Acceptable Product: 'AcoustiGuard QT-4002' by AcoustiGuard or approved equivalent.~~

.2 Tape: Joint tape as recommended by underlayment manufacturer.

3 Execution

3.1 Examination

.1 Verify condition of previously installed work upon which this Section depends. Report defects to Consultant. Commencement of work of this Section means acceptance of existing conditions.

3.2 Installation

.1 Review acoustic underlayment details with Consultant prior to installation.

.2 Unless otherwise specified, install materials according to manufacturer's printed instructions.

.3 Coordinate with applicable Sections as required for installation with intended floor applications.

.4 Loose lay underlayment and tape all joints. Secure butt joints and seams to prevent underlayment movement.

End of section

1 General

1.1 Section includes

.1 The General Conditions and Supplementary Conditions of the Contract, Division 01 Specification Sections, and Contract Drawings apply to this Section.

.2 Work of this Section includes, but is not limited to the following:

.1 Luxury vinyl plank.

.2 Rubber tile flooring (landings).

.3 Rubber stair tread with integral riser.

.4 Vinyl stair nosings.

.5 Transition strips, thresholds.

.6 Accessories.

1.2 Reference standards

.1 ASTM F710, Standard Practice for Preparing Concrete Floors to Receive Resilient Flooring.

.2 ASTM F1344, Standard Specification for Rubber Floor Tile.

.3 ASTM F1700, Standard Specification for Solid Vinyl Floor Tile.

.4 ASTM F1869, Standard Test Method for Measuring Moisture Vapor Emission Rate of Concrete Subfloor Using Anhydrous Calcium Chloride.

.5 ASTM F2169, Standard Specification for Resilient Stair Treads.

.6 ASTM F2170, Standard Test Method for Determining Relative Humidity in Concrete Slabs Using in-situ Probes.

.7 ASTM F3261, Standard Specification for Resilient Flooring in Modular Format with Rigid Polymeric Core.

1.3 Submittals

.1 Product data:

.1 Submit manufacturer's Product data according to Section 01 33 00 indicating:

.1 Performance criteria, compliance with appropriate reference standard, characteristics, and limitations.

.2 Product transportation, storage, handling and installation requirements.

.2 Methodology: Submit surface preparation methodology of stair substrate, treatment of joints, fasteners and penetrations for the Consultant’s review.

.3 Samples:

.1 Submit samples according to Section 01 33 00:

.1 Two 250 x 200 mm samples of each type of tile material and colour.

.2 Two 250 mm long samples of each base, accessory and colour.

.4 Sustainability submittals: Submit applicable supporting documentation according to sustainability requirements for the approval of the Consultant.

.5 Certificates and reports: Prior to proceeding with installation, submit the following:

.1 Manufacturer's certificate stating that moisture content of subfloor is according to manufacturer's requirements and is suitable for flooring installation.

.2 Report indicating test results of moisture content tests substantiating compliance with manufacturer's requirements.

.6 Sustainability submittals: Submit applicable supporting documentation according to sustainability requirements for the approval of the Consultant.

.7 Closeout submittals: Submit maintenance and cleaning data for incorporating into operations and maintenance manuals according to Section 01 78 00.

1.4 Quality assurance

.1 Mock-up:

.1 Luxury vinyl plank flooring: Construct one (1) 2 m^2 mock-up luxury vinyl plank flooring in location acceptable to Consultant.

.2 Resilient tread with integral risers: Construct one (1) mock-up resilient stair tread/risers over stair substrate, including minimum three (3) treads, in location acceptable to Consultant.

.3 Nosings: Construct one (1) mock-up stair nosings over stair substrate, including minimum three (3) nosings, in location acceptable to Consultant.

.4 Demonstrate surface preparation and workmanship of resilient stair accessories over wood stair substrate and lack of telegraphing of joints or fasteners in final installation.

.5 Arrange for Consultant's review and acceptance, allow 48 hours after acceptance before proceeding with work.

.6 Mock-up may remain as part of Work if accepted by Consultant. Remove and dispose of mock-ups which do not form part of Work.

.7 Upon acceptance, mock-up shall serve as a minimum standard of quality for the balance of the work of this Section.

1.5 Site conditions

.1 Maintain air temperature and structural base temperature at flooring installation area above 20 deg. C for 48 hr before, during and 48 hr after installation.

.2 Store materials for two (2) days prior to installation in area of Work to achieve temperature stability.

.3 Do not lay flooring in conditions of high humidity or where exposed to cold drafts. In hot weather, protect from direct sunlight.

.4 Provide adequate ventilation during installation.

1.6 Maintenance

.1 Submit extra 5% or to nearest full carton of each colour, pattern and type of flooring material required for maintenance use. Identify each carton. Store where directed. Submit maintenance material in one piece and of same production run as installed materials.

2 Products

2.1 Materials

.1 All materials under work of this Section, including but not limited to, primers and adhesives are to have low VOC content limits.

.2 Luxury vinyl plank (LVP):

.1 ASTM F1700, Class III, Type B, minimum 3 mm thick wood-look luxury vinyl plank, with ***glue-down*** ~~click/glueless~~ installation.

.2 Refer to Section 09 06 00 - Material and Finish Schedule for luxury vinyl plank type, size, colour and additional information.

.3 Stair treads with integral risers (RBST):

.1 ASTM F2169, one-piece rubber tread with integral riser with non-slip surface and nosing with depth to 50 mm, capable of accommodating various stair pan angles.

.2 Stair/riser unit includes visually impaired colour contrasting insert.

.3 Refer to Section 09 06 00 - Material and Finishes Schedule for integral rubber tread/riser type, colour and additional information.

.4 Rubber floor tile, landings (RBF):

.1 ASTM F1344, minimum 3 mm thick, rubber tile with non-slip surface.

.2 Refer to Section 09 06 00 - Material and Finishes Schedule for rubber floor type, colour and additional information.

.5 Stair nosing:

.1 Vinyl nosing with a 50 mm hinged square nose.

.2 Colour: To later selection of Consultant from manufacturer's full colour range.

.3 Acceptable Product: 'Flexible Vinyl Stair Nosing' by Tarkett or approved equivalent.

~~.6 Acoustic underlayment: According to Section 09 60 13, for use at floor finishes in suites as scheduled on the drawings.~~

.6 Bases:

.1 Wood base: According to Section 06 20 00.

.2 Resilient base: According to Section 09 65 13.

.7 Primers and adhesives: Low VOC, waterproof, of types recommended by flooring manufacturer for specific material on applicable substrate, above, on or below grade.

.8 Transition strips:

.1 Resilient transition trips:

.1 Reducing edge strips, thresholds: Nitrile rubber plasticized vinyl, 80-95 Shore A Durometer, adhesive as recommended by manufacturer.

.2 Manufactured by Roppe Corporation, Tarkett or approved equivalent.

.2 Metal transition strips: According to Section 09 30 00.

.9 Concrete skim coat compound: High-performance, rapid-setting cement based skim coating compound. 'Planiprep SC' by Mapei or approved equivalent for filling minor voids and leveling existing substrate.

.10 Sealer and wax: Type recommended by flooring manufacturer.

3 Execution

3.1 Examination

.1 Verify condition and dimensions of previously installed work upon which this Section depends. Report defects to Consultant. Commencement of work of this Section means acceptance of existing conditions.

.2 Defective work resulting from application to unsatisfactory surfaces will be considered the responsibility of those performing the work of this Section.

3.2 Preparation and subfloor treatment

.1 Prepare concrete substrates according to manufacturer’s written instructions and as specified herein.

~~.2 Coordinate with Section 09 60 13 as required for the installation of acoustic underlayment at intended areas.~~

.2 Moisture content testing, general:

.1 Test subfloor for moisture content to ensure that subfloor is suitable for flooring installation. Test methods shall be as recommended by flooring manufacturer and as specified herein.

.2 Obtain manufacturer's certificate of acceptance of substrate moisture content in writing and submit copy to Consultant.

.3 Moisture content must be below flooring manufacturer's requirements before any flooring is installed. Submit copy of test results to Consultant.

.3 Concrete subfloors/substrates:

.1 Flooring shall be installed over subfloors conforming to ASTM F710 for concrete.

.2 Ensure concrete floors meet the following minimum requirements and requirements of the flooring manufacturer. If there is a conflict between these requirements and those of the flooring manufacturer, the more stringent shall apply.

.1 Internal relative humidity test: Perform internal relative humidity testing according to ASTM F2170. Results shall not exceed 80% RH.

.2 Moisture test: Moisture emissions from concrete subfloors (cured for a minimum of 28 days) must not exceed 3 lbs per 1000sf per 24 hours (1.4 kg H2O/24 hr/93 m2) for acrylic adhesive and 5lbs for polyurethane adhesive via the Calcium Chloride Test Method (ASTM F1869).

.3 The pH level of the subfloor surface shall not be higher than 9.9. If higher, subfloor must be neutralized.

.4 Wood stair substrates:

.1 Prepare wood substrates according to manufacturer’s written instructions and requirements of ASTM F1482.

.2 Prepare wood stair substrate and screw fasteners and penetrations as recommended by flooring manufacturer and to Consultant approved methodology.

.3 Wood substrate must be free of any foreign material that may cause the discolouration of resilient flooring.

.5 Ensure that sub-floors have been provided as specified without holes, protrusions, cracks, depressions or other major defects.

.6 Remove sub-floor ridges and bumps on subfloors. Fill low spots, cracks, joints, holes and other defects with sub-floor filler as required for subfloors.

.7 Apply sub-floor filler to low spots and cracks to achieve floor level to a tolerance of 1:1000, allow to cure.

.8 Ensure that control joints have been filled and levelled.

.9 Clean and remove all deleterious materials from surfaces to receive this work according to the adhesive manufacturer's recommendations.

.10 Prime subfloors and substrates to flooring manufacturer's printed instructions.

3.3 Luxury vinyl plank flooring application

.1 Install luxury vinyl plank flooring according to manufacturer's written instructions, with ***glue-down*** ~~click~~ installation method.

.2 ***Apply adhesive uniformly using recommended trowel according to flooring manufacturer's instructions. Do not spread more adhesive that can be covered by flooring before initial set takes place.***

~~.2 Acoustic underlayment: Install acoustic underlayment according to Section 09 60 13.~~

.3 Install flooring parallel to long dimension of room. Locate centre axis so that no flooring unit is less than one-half size.

.4 Install floor planks with tongue and groove method, individually and in rows as recommended by flooring manufacturer.

.5 ***As installation progresses, and after installation, roll flooring in two (2) directions with minimum 45 kg minimum roller to ensure full adhesion.***

.6 ***Remove adhesive seepage at seams or surface while adhesive is still wet, according to manufacturer's recommendation.***

.7 Cut tile and fit neatly around fixed objects.

.8 Terminate flooring at centerline of door in openings where adjacent floor finish or colour is dissimilar.

.9 Ensure final installation of wood plank flooring is correctly ***adhered*** ~~clicked~~ in place, flat, seamless and meets the satisfaction of the Consultant.

.10 Transition strips:

.1 Install transition strips at unprotected or exposed edges where flooring terminates or where there are two finishes of different thicknesses, unless otherwise shown on Contract Drawings.

.2 Coordinate with applicable Sections as required for junctures of where luxury vinyl plank flooring meets other flooring materials.

3.4 Rubber tiles, stair treads and nosings

.1 Prepare adhesive and install materials according to manufacturer's written instructions.

.2 Stair treads with integral risers:

.1 Pre-cut and fit treads/risers prior to spreading adhesive. Fill back side of tread nose with a caulking bead; brush on adhesive on understeps and back of treads, as well as back of risers, and on receiving substrate. Allow to become tacky to touch before installing.

.2 Treads and risers shall be fully bonded to substrate, with tread nosing butted tight against stair tread nosing.

.3 Roll with hand roller after installation.

.3 Nosings:

.1 Pre-cut and fit nosing prior to spreading adhesive. Fill back side of stair nose with a caulking bead. Allow to become tacky to touch before installing.

.2 Ensure nosings are fully bonded to substrate, butted tight against stair.

.3 Roll with hand roller after installation.

.4 Rubber tile:

.1 Trowel adhesive evenly on substrate any lay tile with tight joints. Work from off the tile. If necessary to work on the tile, avoid shifting by using a kneeling board and by cutting tile to butt tightly at wall junctions.

.2 Roll tile both directions with 68 kg sectional roller when adhesive is tacky to touch. Use hand roller in areas inaccessible with large roller.

.5 Remove adhesive seepage at seams or surface while adhesive is still wet, according to manufacturer's recommendations.

3.5 Resilient base application

.1 Resilient base: According to Section 09 65 13. Coordinate with noted Section as required for installation of resilient base with resilient tile flooring.

3.6 Cleaning and waxing

.1 Forty-eight hours after installation, clean resilient tile surfaces with a mild soap solution approved by finish manufacturer. Rinse clean, dry and apply two (2) coats of wax. Polish thoroughly.

3.7 Protection of finished work

.1 Protect floors from time of final ***set of adhesive*** ~~installation~~ until final inspection.

.2 Prohibit traffic on floor for forty-eight (48) hours after installation.

.3 Cover waxed and polished surfaces with fibre reinforced, clean, non-staining kraft paper. Secure in position with gummed tape to prevent drifting. Remove covering when directed by Consultant.

End of section

1 General

1.1 Submittals

.1 Submit shop drawings/product data sheets for all products specified in this Section except pipe and fittings.

1.2 Closeout Submittals

.1 Submit a copy of plumbing inspection certificate prior to application for Substantial Performance of the Work.

.2 Record Drawings: Indicate inverts of new below grade sanitary and storm piping on as-builts drawings.

2 Products

2.1 Pipe, Fittings, and Joints

.1 PVC Sewer

.1 DR35 rigid, green PVC hub and spigot pattern sewer pipe and fittings to CAN/CSA B182.2, with gasket joints assembled with pipe lubricant.

.2 DR35 rigid, PVC sewer pipe and fittings, with solvent weld joints, all certified to CSA B182.1 and colour-coded as per local governing codes, regulations and standards.

.2 PVC - DWV

.1 For Low Buildings: Equal to IPEX System 15 drain, waste and vent pipe and fittings to CAN/CSA B181.2, complete with a flame spread rating not more than 25 when tested to CAN/ULC-S102.2, with solvent weld joints or MJ Grey mechanical joint couplings, and, for fire barrier penetration, approved firestop conforming to CAN/ULC S115.

.2 For High Buildings and Plenums: Equal to IPEX System XFR drain, waste and vent pipe and fittings to CAN/CSA B181.2, complete with a flame spread rating not more than 25 and a smoke developed classification not more than 50 when tested to CAN/ULC-S102.2, with solvent weld joints or MJ Grey mechanical joint couplings, and, for fire barrier penetration, approved firestop conforming to CAN/ULC-S115.

.3 Copper - Solder Joint

.1 Equal to Great Lakes Copper Ltd. Type DWV hard temper to ASTM B306, with forged copper solder type drainage fittings and 50% lead - 50% tin solder joints.

.4 Cast Iron

.1 Equal to Bibby St. Croix Class 4000 cast iron pipe, fittings, and mechanical coupling joints to CAN/CSA B70.

.5 Copper - Victaulic Coupling Joint

.1 For copper piping sizes 75mm (3”) to 150mm (6”), equal to Great Lakes Copper Ltd. Type DWV hard temper to ASTM B306, with factory or site rolled grooved ends (with grooving rolls designed for copper) and Victaulic "Copper Connection" wrought copper or cast bronze fittings and Style 606 gasket type couplings.

.6 Galvanized Steel - Victaulic Coupling Joint

.1 Schedule 40 mild steel, galvanized, ASTM A53, factory or site rolled grooved, complete with Victaulic galvanized ductile iron grooved end fittings and, unless otherwise specified, Victaulic Style 77 hot dip galvanized mechanical joint couplings with Grade M gaskets.

.7 PVC Weeper Piping

.1 Perforated solid PVC pipe with an integral geodesic sock, supplied in coils.

.8 Stainless Steel – For Storm Drainage

.1 Schedule 10S type 304 stainless steel, ASTM A312, factory or site roll grooved, complete with Victaulic or equal type 304 stainless steel roll grooved end fittings and, unless otherwise specified, Victaulic Style 807, 877 or 889 couplings and coupling gaskets equal to Victaulic Grade P fluoroelastomer.

.9 Stainless Steel – For Sanitary Drainage

.1 Schedule 10S type 304 stainless steel, ASTM A312, factory or site roll grooved, complete with Victaulic or equal type 304 stainless steel roll grooved end fittings and, unless otherwise specified, Victaulic Style 807, 877 or 889 couplings and coupling gaskets equal to Victaulic Grade P fluoroelastomer.

2.2 Shut-Off and Check Valves

.1 Shut-off Valves

.1 Class 600, 4140 kPa (600 psi) WOG rated full port ball valves, each complete with a forged brass body, blowout-proof stem, chrome plated solid brass ball, solder or screwed ends as required, and removable lever handle.

.2 Manufacturers:

.1 Toyo Valve Co. Fig. 5049A or Fig. 5044A;

.2 Milwaukee Valve Co. #BA-155 or #BA -125;

.3 Kitz Corporation Code 58 or Code 59;

.4 Victaulic Co. of Canada Ltd. Series 722;

.5 Apollo Valves # 77-100 or # 77-200;

.6 Watts Industries (Canada) Inc. #FBVS-3C.

.7 Or approved equivalent.

.2 Check Valves

.1 Class 125, bronze, 1725 kPa (250 psi) WOG rated vertical lift check valve with solder or screwed ends as required, and, for horizontal piping, Class 125, bronze 1380 kPa (200 psi) WOG rated swing check valves with solder or screwed ends.

.2 Manufacturers:

.1 Toyo Valve Co. Fig. 231 or Fig. 236 or Fig. 237;

.2 Milwaukee Valve Co. #1510 or #510;

.3 Kitz Corporation Code 36 or Code 22 or Code 23.

.4 Or approved equivalent.

2.3 Vent Stack Covers

.1 Equal to Lexcor Model "Flash-Tite" seamless, spun aluminum, insulated vent stack covers with caps and a factory applied asphalt primer coating on top and bottom of flange.

2.4 Cleanouts

.1 Horizontal and Vertical Piping

.1 “BARRETT" type cleanout tees, each gas and water-tight and complete with a bolted cover to match piping material.

.2 Urinal(s)

.1 Wall access cleanout assemblies, each complete with a tapered plug, threaded brass insert, urethane rubber seal, and polished stainless steel access cover with vandal-proof stainless steel securing screw.

.2 Manufacturers:

.1 Watts Industries (Canada) Ltd. #CO-590-RD.

.2 Jay R. Smith #SQ4-1819;

.3 Zurn #ZSS-1666-1;

.4 Mifab #C1440-RD;

.5 Or approved equivalent.

2.5 Floor Cleanout Terminations

.1 Factory finished cast iron terminations, each adjustable and complete with a cast iron body with neoprene sleeve, solid, gasketed, polished nickel-bronze scoriated top access cover to suit floor finish, a seal plug, and captive, vandal-proof, stainless steel securing hardware.

.2 Manufacturers:

.1 Watts Industries (Canada) Ltd. # CO-200-R-6.
.2 Jay R. Smith #4020-F-C-U Series;
.3 Zurn # ZN-1602-SP-VP Series;
.4 Mifab # C1100-XR-1 or #C1000-R-3;
.5 Or approved equivalent.

.3 Cleanout terminations in areas with a tile or sheet vinyl floor finish are to be as above but with a square top in lieu of a round top.

2.6 Floor Drains, Funnel Floor Drains, and Hub Drains

.1 Unless otherwise specified or indicated, floor drains are to be vandal-proof drains in accordance with drawing symbol list, each complete with a cast iron body and a trap seal primer connection. Cast iron components are to be factory finished with latex based paint coating.

.2 Floor drains in areas with a tile finish are to be as above but with a square grate in lieu of a round grate.

.3 Manufacturers:

.1 Watts Industries (Canada) Ltd.;
.2 Jay R. Smith Manufacturing Co.;
.3 Zurn Industries Ltd.;
.4 Mifab Inc.
.5 Or approved equivalent.

2.7 Roof Drains

.1 Unless otherwise specified or indicated, roof drains are to be cast iron body drains with aluminium domes, in accordance with the drawing symbol list. Cast iron components are to be factory finished with a latex based paint coating.

.2 Manufacturers:

.1 Watts Industries (Canada) Ltd.;
.2 Jay R. Smith Manufacturing Co.;
.3 Zurn Industries Ltd.;
.4 Mifab Inc.
.5 Or approved equivalent.

2.8 Drainage Trench Frames and Grating

.1 Welded, hot dipped galvanized, 45 mm x 45 mm x 6.4 mm (1-¾" x 1-¾" x ¼") carbon steel angle frame, 300 mm (12") wide, with anchor straps and lengths as required, and baked epoxy coated cast iron slotted grating in 600 mm (24") long sections.

.2 Manufacturers:

.1 Watts Industries (Canada) Ltd. #TD-910-B1-4;
.2 Jay R. Smith #2971VP.
.3 Zurn # Z796VP;
.4 Or approved equivalent.

2.9 Trench Drains

.1 Modular, pre-sloped, polyester fibreglass construction interlocking sections of drainage channel with overlapping joints, drain pipe connection outlets as required, end caps and covers to suit the application, integral anchor tabs for grate anchoring and trench levelling, heavy-duty coated steel angle top frames, and heavy-duty coated cast iron slotted grate supplied in 600 mm (24") long sections.

.2 Manufacturers:

.1 Watts Industries (Canada) Inc. “Dead Level” Series;
.2 Jay. R. Smith #9810 Series.
.3 Zurn "Flow-Thru" System;
.4 ACO Systems Ltd. "ACOdrain";
.5 Or approved equivalent.

2.10 Elevator Pit Scupper Drains

.1 Shallow-profile, heavy-duty elevator pit drain designed for limited clearance applications. Corrosion-resistant cast iron body, a removable stainless-steel or nickel-bronze slotted grate, bottom or side outlet as required, and a removable sediment bucket. A 1/2" NPT trap primer connection with threaded plug and membrane clamp for waterproofing integration.

.2 Backwater valve to be provided as specified on drawings.

.3 Manufacturers:

.1 Zurn;
.2 Watts Industries (Canada) Ltd.
.3 Or approved equivalent.

2.11 Laundry Trench Lint Trough Drain

.1 Manufacturer

.1 Zurn
.2 Or approved equivalent.

2.12 Interior Catch Basin Frames and Covers

.1 Heavy-duty, 508 mm (13") square, baked epoxy coated cast iron, non-removable, hinged slotted grate with coated steel frame with concrete anchors.

.2 Manufacturers:

.1 Watts Industries (Canada) Ltd. #FD-410;
.2 Jay R. Smith #8915FC;
.3 Zurn #Z-799-1;
.4 Mifab #F1570.
.5 Or approved equivalent.

2.13 Backwater Valves

.1 Heat bonded powder epoxy coated cast iron in-line type, each complete with a bolted and gasketed cover, bronze flapper, stainless steel extension, and stainless steel hardware.

.2 Manufacturers:

.1 Watts Industries (Canada) Inc. BV-230-R Series;
.2 Jay R. Smith #7022-CAN.
.3 Zurn #Z-1095-15-MJ;
.4 Or approved equivalent.

2.14 Extendable Backwater Valves

.1 Equal to “Mainline Adapt-a-valve” ABS/PVC backwater valve cassette with ABS/PVC flush cap cleanout for use with ABS or PVC pipe. Provide extension piece cut to length to extend valve access to finished floor level. Provide flush cap cleanout cover to match backwater valve size and material.

2.15 Exterior Catch Basins

.1 Pre-cast reinforced concrete catch basins manufactured to ASTM C478 and Municipal standards, each sized and arranged to suit drainage pipe size and arrangement, and complete with:

.1 cast iron frame and cover to Municipal standards;
.2 required masonry work to raise top of catch basins flush with finished grade or pavement surfaces.

.2 Masonry work is to consist of cement mortar and clay or shale bricks to ASTM C32 Grade M5, or Oaks Precast Industries "MODULOC" pre-cast interlocking concrete members and accessories.

3 Execution

3.1 Underground Municipal Service Connection

.1 Make required arrangements with Municipality for installation of drain service piping mains from Municipal main to property line.

.2 Pay charges levied by Municipality for service connection work.

.3 Municipal charges for underground street service connection work will be paid out of a prime cost allowance. Submit original copies of invoices issued by Municipality for street service connection work.

3.2 Drain and Vent Piping Installation Requirements

.1 Provide required drainage and vent piping. Pipe, unless otherwise specified, as follows:

.1 for underground pipe inside building and to points 1.5 m (5') outside building lines – rigid PVC sewer pipe, minimum 75 mm (3") dia.;
.2 for pipe inside building and aboveground in sizes less than or equal to 65 mm (2-½") dia. – type DWV copper;
.3 for pipe inside building and aboveground in sizes greater than or equal to 75 mm (3") dia. – Class 4000 cast iron;
.4 for storm piping draining to a stormwater management tank located below ground floor – Schedule 40 galvanized steel or stainless steel with Victaulic fittings and couplings.
.5 for pipe inside building and aboveground in lieu of type DWV copper and cast iron, at your option and where permitted by governing Codes and Regulations – rigid PVC DWV;
.6 for drainage pump discharge pipe connections from pump to and including shut-off and check valve connections – Type "DWV" copper

with Victaulic "Copper Connection" fittings and couplings, or Schedule 40 galvanized steel with Victaulic fittings and couplings.

.2 Unless otherwise specified, slope horizontal drainage piping aboveground in sizes to and including 75 mm (3") dia. 25 mm (1") in 1.2 m (4'), and pipe 100 mm (4") dia. and larger 25 mm (1") in 2.4 m (8').

.3 Install and slope underground drainage piping to inverts or slopes indicated on drawings to facilitate straight and true gradients between points shown. Verify available slopes before installing pipes.

.4 Unless otherwise specified, slope horizontal branches of vent piping down to fixture or pipe to which they connect with a minimum pitch of 25 mm (1") in 1.2 m (4').

.5 Extend vent stacks up through roof generally where shown but with exact locations to suit site conditions and in any case a minimum of 3 m (10') from fresh air intakes. Terminate vent stacks a minimum of 330 mm (13") above roof (including roof parapets) in vent stack covers. Where not shown on drawings, route vent piping from source to building exterior as required in order to satisfy local governing codes and authority. Coordinate vent routing with other building services and ensure there is no architectural impact.

.6 Provide cast brass dielectric unions at connections between copper pipe and ferrous pipe or equipment.

3.3 Installation of Shut-Off and Check Valves

.1 Provide a shut-off valve and a check valve in discharge piping of each drainage pump.

.2 Locate valves so they are easily accessible without the use of ladders or other such devices.

3.4 Supply of Vent Stack Covers

.1 Supply a properly sized vent stack cover for each vent stack penetrating roof.

.2 Hand vent stack covers to roofing trade at site for installation and flashing into roof construction as part of roofing work. Coordinate installation to ensure proper locations. Provide waterproofing caps over vent stacks.

3.5 Installation of Cleanouts

.1 Provide cleanouts in drainage piping in locations as follows:

.1 in building drain or drains as close as possible to inner face of outside wall, and, if a building trap is installed, locate cleanout on downstream side of building trap;

.2 at or as close as practicable to the foot of each drainage stack;

.3 at maximum 15 m (50') intervals in horizontal pipe 100 mm (4") dia. and smaller;

.4 at maximum 30 m (100') intervals in horizontal pipe larger than 100 mm (4") dia.;

.5 in the wall at each new urinal or bank of urinals in a washroom;

.6 wherever else shown on drawings.

.2 Cleanouts are to be same diameter as pipe in piping to 100 mm (4") dia., and not less than 100 mm (4") dia. in piping larger than 100 mm (4") dia.

.3 Where cleanouts in vertical piping are concealed behind walls or partitions, install cleanouts near floor and so cover is within 25 mm (1") of the finished face of the wall or partition.

3.6 Installation of Floor Cleanout Terminations

.1 Where cleanouts occur in horizontal inaccessible underground piping, extend cleanout TY fitting up to floor, and provide a cleanout termination set flush with finished floor.

.2 In waterproof floors, ensure each cleanout termination is equipped with a flashing clamp device. Cleanout terminations are to suit floor finish.

.3 Where cleanout terminations occur in finished areas, confirm locations prior to rough-in and arrange piping to suit.

.4 Ensure cleanout termination covers in tiled floor are square in lieu of round.

3.7 Installation of Floor Drains, Funnel Floor Drains and Hub Drains

.1 Provide floor drains, funnel floor drains and hub drains.

.2 Coordinate location of floor drains, funnel floor drains and hub drains with equipment provided by Mechanical Division and Owner's supplied equipment. Install in accordance with manufacturer's instructions.

.3 Equip each drain with a trap.

.4 In equipment rooms and similar areas, exactly locate floor drains to suit location of mechanical equipment and equipment indirect drainage piping. In washrooms, exactly locate floor drains to avoid interference with toilet partitions.

.5 Confirm exact location of drains prior to roughing in. Where floor drains occur in washrooms coordinate locations with toilet partition installations.

.6 Temporarily plug and cover floor drains during construction procedures. Remove plugs and covers during final clean-up work and when requested, demonstrate free and clear operation of each drain. Replace any damaged grates, and refinish any areas of the drain where cast iron finish has been damaged or removed, including rusted areas.

3.8 Installation of Roof Drains

.1 Supply roof drains and place roof drain bodies in position for flashing into roof construction as part of roofing work. Roof drain installation to be coordinate with architectural slab edge drawings. Connect with piping and provide accessories.

.2 Protect roof drains from damage and entrance of debris until roofing work is complete, and refinish any areas where cast iron factory finish has been damaged or removed, including rusted areas.

3.9 Installation of Drainage Trench Frames and Grating

.1 Supply frame and grating sections for drainage trench. Provide piping connections, traps, etc., as required.

.2 Hand frames to concrete trade forming and pouring trenches. Ensure frames are properly and accurately installed.

.3 Be present during concrete pour to ensure frames are not dislodged or damaged and remain straight and true. Immediately report any problems.

.4 Install grates and secure in place. Temporarily cover grates during construction procedures. Clean trenches when work is complete.

3.10 Installation of Trench Drains

.1 Provide pre-sloped sections of drainage channel and install so top frames are level and plumb in relation to floor finishes. Provide accessories, traps, etc., as required.

.2 Be present during concrete pour to ensure trench drainage is not dislodged or damaged and remains straight and true. Immediately report any problems.

.3 Install grating and secure in place.

.4 Temporarily cover trench drainage openings during construction procedures. Clean trenches when work is complete.

3.11 Installation of Interior Catch Basin Frames and Covers

.1 Supply frames and hinged grates for interior catch basins and provide sump inlet and outlet piping and accessories.

.2 Hand frames to concrete trade pouring concrete sump, and coordinate installation of sump piping with the formwork installation.

.3 Install grates and secure in place. Clean sumps when work is complete.

3.12 Installation of Backwater Valves

.1 Provide backwater valves in drainage piping and connect with piping.

.2 Set backwater valve assembly such that cover is flush with finished floor. Provide an extension piece if required due to depth of piping.

3.13 Installation of Exterior Manholes

.1 Provide pre-cast concrete manholes. Properly bed each unit and set to required invert.

.2 Provide a reinforced pre-cast concrete base slab and bottom section for each manhole, or provide a poured-in-place concrete base. Ensure each manhole is sized to suit pipe size and arrangement. Conform to Municipal installation standards.

.3 Provide masonry work required to raise top of each assembly flush with finished grade level.

.4 When work is substantially complete, clean out each manhole.

3.14 Installation of Exterior Catch Basins

.1 Provide pre-cast concrete catch basins. Properly bed each unit and set to required invert.

.2 Ensure each catch basin is sized to suit pipe size and arrangement. Conform to Municipal installation standards.

.3 Provide masonry work required to raise top of each assembly flush with finished grade level.

.4 When work is substantially complete, clean out each catch basin.

END OF SECTION

1 General

1.1 Submittals

.1 Submit shop drawings/product data sheets, complete with control components, and piping and wiring schematics.

.2 Submit a start-up and certification letter from equipment supplier as specified in Part 3 of this Section.

1.2 Quality Assurance

.1 Packaged terminal air conditioning equipment and installation of equipment are to be in accordance with requirements of following:

.1 all applicable Provincial Codes and Standards;

.2 ANSI/AHRI Standard 210/240, Performance Rating of Unitary Air-Conditioning and Air-Source Heat Pump Equipment.

.1 Unit must have AHRI certification at time of installation.

.2 Packaged terminal air conditioning system installation tradesmen are to be journeyman refrigeration mechanics.

2 Products

2.1 Manufacturers

.1 Innova;

.2 Inspiron Air;

.3 Or approved equivalent.

2.2 Packaged Terminal Air Conditioning Unit

.1 Equal to Innova .2.0. Factory assembled and tested, heat pump type package terminal air conditioning unit consisting of galvanized steel case construction with polystyrene foam insulation and baked-on enamel finish, electric resistance heating system, sealed refrigerant system programmed frost control sequence and internal condensate removal. Unit sizing and performance in accordance with drawing schedule, CSA or ETL listed and labelled, AHRI rated and certified and with a minimum system efficiency of 12.1 SEER.

.2 2kW electric resistance heater.

.3 Condensate drip tray

.4 Unit complete with heavy-gauge galvanized steel wall sleeve with insulation, , remote thermostat controls.

2.3 Packaged Terminal Air Conditioning Unit with Integrated Energy Recovery Ventilator

.1 Equal to Innova HRA-I PLUS. Concealed ceiling hung ducted packed air source heat pump unit with integrated energy recovery ventilator. Factory assembled and tested, heat pump type package terminal air conditioning unit consisting of galvanized steel case construction with polystyrene foam insulation and baked-on enamel finish, electric resistance heating system, sealed refrigerant system (rotary compressor, condenser, evaporator, R-32 refrigerant and connective tubing), programmed frost control sequence and internal condensate removal. Unit sizing and performance in accordance with drawing schedule.

.2 Air Filters:

.1 Cartridge filter on intake, exhaust and return air streams

.3 Fans:

.1 Fans equipped with DC motors

.4 Energy Recovery Heat Exchangers

.1 Polypropylene static enthalpy counterflow heat exchange

.5 Compressor

.1 Cased in acoustically isolated compartment, separated from air streams.

.6 12 HP DC Inverter with 2kW electric resistance.

.7 Unit complete with heavy-gauge galvanized steel wall sleeve with insulation, electronic LED display, remote thermostat controls, aluminum architectural rear grille - colour to be confirmed with architect.

3 Execution

3.1 Installation of Packaged Terminal Air Conditioning Unit

.1 Provide packaged terminal air conditioning unit and inspect unit prior to installation. Install per manufacturer's recommendations.

.2 Install wall sleeve and exterior grille.

.3 Prepare unit by removing packaging and room cover.

.4 Install unit into wall sleeve.

.5 Connection power cord or use direct connect kit as required by project.

.6 Reinstall room cover.

.7 Install internal drain per manufacturer's installation instructions.

.8 Allow unit to run for 48 hours to optimize performance.

.9 Refer to Section entitled Basic Mechanical Materials and Methods for equipment/system manufacturer certification requirements.

.10 Refer to Section entitled Basic Mechanical Materials and Methods for equipment/system start-up requirements.

.11 Include for 2 hours of on-site operation demonstration and training session. Training is to be a full review of all components including, but not limited to, a full operation and maintenance demonstration, with control set-up and abnormal events.

END OF SECTION

1 General

1.1 Section Includes

.1 Termination Blocks, Patch Panels, and Connectors.

1.2 Related Requirements

.1 Section 27 13 13 – Communications Copper Backbone Cabling.

.2 Section 27 13 23.00 - Communications Optical Fiber Backbone Cabling

1.3 Submittals

.1 Product Data: Provide manufacturers catalogue information.

2 Products

2.1 Copper Patch Panels and Termination Blocks

.1 Copper patch panel to be 19" rack mount, 24/48 port, Category 6A, 8 pin modular jack (RJ45) with labels and icons, icon color to match field outlet color. Patch panel to have 2 types, angled type for the cabling rack and flat type for equipment cabinets and remote wall mounted telecom cabinets.

2.2 Category 6A 1U-48 Port Flat Patch Panels

.1 Suitable for mounting to any standard 19" or 23" EIA rack or cabinet.

.2 Horizontal termination for data circuits shall consist of category 6A flat patch panels, which shall be 1U high and provide either 48 modular jack ports, as indicated on the drawings. Patch panels shall be configured with individually replaceable jacks.

.3 Each modular jack shall be provided with a bend-limiting strain relief. The strain relief shall provide cylindrical support to limit the bend radius at the point of termination.

.4 The front of each 6-port module shall be capable of accepting labels to TIA requirements.

.5 Each port shall be capable of accepting an icon to indicate its function. Patch panels shall meet the Category 6 performance requirements of ANSI/TIA-568-C.2.

.6 Panduit, Belden, Commscope, or approved equivalent.

2.3 Fibre Patch Panels and Splice trays

.1 Fiber patch panel to be 19” rack mount enclosure (in all 19” rack/cabinet) or wall mount enclosure, supporting optical fiber connector adapter panels (duplex LC Type), rear cable retention loops, entry cable strain relief supports, cable strength member clamps, D- grommets at cable entry, removable rear panel cover, accommodation and support for splice trays, removable front access cover, front space for patch cable storage, patch cable bend radius controls and identification labeling.

.2 Panduit, Belden, Commscope, or approved equivalent.

3 Execution

3.1 Installation

.1 Refer to Section 27 13 13 Copper Backbone Cabling.

.2 Refer to Section 27 15 13 Copper Horizontal Cabling.

END OF SECTION

1 General

1.1 Summary

.1 This Section includes:

.1 The supply, delivery, supervision, coordination, and installation of equipment items specified herein and shown on the Drawings.

.2 The testing, documentation, and instructions for completing the Fiber Backbone Cabling System.

.3 Products supplied but not installed under this section, including loose equipment specified herein, which is to be turned over to the Owner at the completion of this project.

.2 Examine the contract documents in their entirety (including drawings and specification sections in the other divisions) for requirements or work which may affect work under this section, regardless of whether such requirements or work are specifically indicated in this section.

.3 Contractor Shall Provide and Install

.1 The Contractor shall furnish and install telecommunications passive equipment, including:

.1 Fiber backbone cable.

.2 Splicing and terminations.

.3 Testing.

.4 Administration.

.2 Although such work is not specifically mentioned herein or on the Drawings, the Contractor shall furnish and install all miscellaneous items, accessories, appurtenances, and devices incidental to or necessary for a sound, secure, and complete installation, without claim for additional payment.

.3 The Contractor shall provide system testing and demonstration, system documentation, and instruction of Owner personnel, without claim for additional payment.

.4 Errors or Omissions in Drawings or Documentation:

.1 If any errors or omissions appear in Drawings, Specifications, or other documents, the bidding Contractor shall notify the Consultant no later than ten (10) days prior to submitting the bid.

.2 Should conflict occur in or between Drawings and Specifications, the bidding Contractor is deemed to have estimated the more expensive

way of doing the work, unless the bidding Contractor has asked for and obtained written decision (addendum) before submission of the bid as to which method or materials will be required.

1.2 Related Requirements

.1 Section 27 11 19 – Communications Termination Blocks and Patch Panels.

.2 Section 27 05 28 - Pathways for Communications Systems Section.

.3 Section 27 08 11 – Testing and Record Documentation for Communications.

1.3 Definitions

.1 ANSI – American Northern Standards Institute

.2 AWG – American Wire Gauge

.3 BICSI – Building Industry Consulting Service International

.4 BCT – Bonding Conductor for Telecommunications

.5 BD – Building Distributor – A distributor in which the building backbone cables terminate and at which connections to the campus backbone cables may be made

.6 CP – Consolidation Point – A connection facility within Cabling Subsystem 1 for interconnection of cables extending from building pathways to the equipment outlet

.7 EDA – Equipment Distribution Area – A space allocated for end equipment, including computer systems and telecommunications equipment

.8 EF – Entrance Facility – An entrance to a building for both public and private network service cables, including wireless, that includes the entrance point of the building and continues to the entrance room or space

.9 EIA – Electronics Industry Alliance

.10 ER – Equipment Room – An environmentally-controlled, centralized space for telecommunications equipment that serves the occupants of the building, considered distinct from a Telecommunications Room (TR) because of the nature or complexity of the equipment.

.11 ESD – Electro Static Discharge – The sudden flow of electricity between two electrically-charged objects caused by contact, an electrical short, or dielectric breakdown

.12 ETL – Intertek Certification Services

.13 IEC – International Electrotechnical Commission

.14 IEEE – Institute of Electrical and Electronic Engineers

.15 IDC – Insulation displacement contact

.16 ISO – International Standards Organization

.17 HC – Horizontal Cross-connect – A group of connectors, such as patch panels or punch-down blocks, that allow horizontal, backbone, and equipment cabling to be cross-connected with patch cords or jumpers

.18 HDA – Horizontal Distribution Area – A space in a computer room where a Horizontal Cross-connect (HC) is located, and which may include LAN switches, Storage Area Network (SAN) switches, and Keyboard/Video/Mouse (KVM) switches for the end equipment located in the Equipment Distribution Areas (EDAs)

.19 IC – Intermediate Cross-connect – A facility enabling the termination of different levels of backbone cabling and interconnection between them or equipment

.20 MC – Main Cross-connect – A facility enabling the termination of backbone cables and their connection to incoming services, other backbone cabling or equipment

.21 MDA – Main Distribution Area – The central point of distribution for the structured cabling system, which includes the Main Cross-connect (MC) and, when equipment areas are served directly from the MDA, may also include Horizontal Cross-connect (HC)

.22 MM – Multi-Mode Fiber

.23 NECA – National Electrical Contractors Association

.24 NFPA – National Fire Protection Agency

.25 NRTL – Nationally Recognized Testing Laboratory

.26 TIA – Telecommunications Industry Association

.27 SM – SingleMode Fiber

.28 UL – Underwriters Laboratory

.29 Provide: Furnish, install, terminate, label, test and certify a complete operating cabling system.

.30 Contract Documents (CD): Design drawings, specifications, sketches and schedules provided by the Engineer as they directly relate to this scope of work and this project.

.31 Structured Cabling Systems (SCS) wiring is defined as all required equipment and cabling including hardware, termination blocks, cross connect wire or cordage, patch panels, patch cords, telecommunication outlets, work area cords, UTP and fiber cable installed and configured to provide computer data and voice connectivity.

.32 Point–of–Entry (POE): Unmarked Manholes/Vaults at property line

.33 NET–POP Rooms/MPOE (Main Point of Entry): The area where the outside plant media/carrier services appear in the facility. The NET–POP contains equipment used by owner or carrier to hand–off/transition cable from outside plant into inside plant type.

.34 Network Center/Main Distribution Frame (MDF) Areas: This technology space houses Layer 2/3 network switching gear and other main network distribution equipment and acts as the mid–connection point between the Core/Network and the TR/IDF/access zones for all connections.

.35 Telecommunications Room (TR)/Intermediate Distribution Frame (IDF): is the location for the termination of backbone cables and for termination of horizontal cables, and for the interconnection of each. The space also hosts access–layer switches and user network connections within each floor.

.36 Active Equipment: electronic equipment used to develop various WAN, LAN, and voice services, e.g., digital multiplexers, RS-232 controllers, Ethernet hubs, switches, routers, PBX, etc.

.37 Campus Backbone: cabling system consisting of media and termination hardware interconnecting POE, Net–Pop’s and Future onsite buildings.

.38 Building Backbone: cabling system consisting of media and termination hardware interconnecting MDFs to IDFs.

.39 Horizontal: cabling system consisting of media and termination hardware interconnecting the Telecommunication Outlets (TOs) and the TRs.

.40 Bonding: permanent joining of metallic parts to form an electrically conductive path which will assure electrical continuity and the capacity to conduct safely any current likely to be imposed on it.

.41 Basket Cable Tray: A cable support and management system fabricated of continuous, rigid, welded steel wire mesh and available in many sizes with attachment hardware suiting multiple installation methods

.42 Cable Tray: vertical or horizontal open supports, usually made of aluminum or steel, which are fastened to the building structure. Cables are laid in and fastened to the trays.

.43 Cabinet: free standing, floor–mounted or wall-mounted modular enclosure designed to house and protect rack–mounted electronic equipment and passive terminations.

.44 Channel: The end–to–end transmission path between two points at which application specific equipment is connected; encompasses all the elements of the horizontal cabling link, plus the equipment cords in the telecommunications spaces and work area.

.45 Cross–Connect: equipment used to terminate and tie together communications circuits.

.46 Cross–Connect Jumper: a cluster of twisted–pair conductors without connectors used to establish a circuit by linking two cross–connect termination points.

.47 Grounding: a conducting connection to earth, or to some conducting body that serves in place of earth.

.48 Jack: receptacle used in conjunction with a plug to make electrical contact between communications circuits, e.g., eight–position/eight–contact modular jacks.

.49 Ladder Cable Tray: A fabricated structure consisting of two longitudinal side rails connected by individual transverse members (rungs).

.50 LAN: Local area network.

.51 Link: Horizontal cabling link encompassing all components of the horizontal cabling (TO, patch panels, blocks, jumpers and patch cords that join them in the horizontal cross–connect). It is distinguished from a channel because it does not include the equipment cables/cords at the telecom spaces or work area.

.52 Media: twisted–pair, and fiber optic cable or cables used to provide signal transmission paths.

.53 Mounting Frame: rectangular steel framework, which can be equipment rack or wall mounted to support wiring blocks, patch panels, and other communications equipment.

.54 Outside Plant (OSP): generally, any and all portions of the cable system that runs outside of an environmentally enclosed structure and/or building with each end terminated at different buildings. This specifically includes inter–building cables, conduits, manholes, hand–holes, and innerduct.

.55 UTP: Unshielded Twisted Pair.

.56 FO: Fiber Optic

.57 Passive Equipment: non–electronic hardware and apparatus, e.g., equipment racks, cable trays, electrical protection, patch panels, wiring blocks, fiber optic shelves, etc.

.58 Patch Cords: a length of wire or fiber cable with connectors on one or both ends used to join communications circuits at a cross–connect.

.59 Patch Panel: system of terminal blocks or connectors used with patch cords that facilitate administration of cross–connect fields.

.60 Pathway: facility for the placement of communications cable. A pathway facility can be composed of several components including conduit, wireway, cable tray, surface raceway, under floor systems, overhead systems, raised floor, ceiling support wires, etc.

.61 Protectors: electrical protection devices used to limit foreign voltages on metallic communications circuits.

.62 Raceway: an enclosed channel designed expressly for holding wires or cables; may be of metal or insulating material. The term includes conduit, tubing, wire ways, under floor raceways, overhead raceways and surface raceways; does not include cable tray.

.63 Racks: An open, freestanding, floor–mounted structure, typically made of aluminum or steel, used to mount equipment; usually referred to as an equipment rack.

.64 Riser Backbone: The Riser Backbone subsystem links the main cross connect (MDF) in the equipment room to the distribution rooms (TRs).

.65 Structured Cabling System (SCS): A SCS is defined as all required cabling including hardware, termination blocks, cross connect wire or cordage, patch panels, patch cords, telecommunication outlets, work area cords, UTP and fiber optic cable installed and configured to provide computer data and voice connectivity from each data or voice device to the network file server or voice network/switch designated as the service point of the local area network.

.66 Telecommunication Outlet (TO): Connecting device mounted in a work area used to terminate horizontal cable and interconnect cabling with station equipment.

.67 Trough or Ventilated Cable Tray: A fabricated structure consisting of integral or separate longitudinal rails and a bottom having openings sufficient for the passage of air and using 75 percent or less of the plan area of the surface to support cables.

.68 Work Area Subsystem: The connection between the telecommunications outlet and the station equipment in the work area is provided by the Work Area Subsystem. It consists of cords, adapters, and other transmission electronics.

.69 Wireless Access Point (WAP): Telecom outlet designated for use with wireless network devices. Such outlet shall be mounted above ceiling.

.70 Contractor – The successful bidder engaged to provide the work of this specification

1.4 References

.1 Most recent editions and addenda of the following documents:

.2 ANSI/TIA 568 series, most recent revisions, addenda and systems bulletins. All applicable

.3 ANSI/TIA–569 Telecommunications Pathways and Spaces, most recent revision including all relevant addenda and systems bulletins

.4 ANSI/TIA–606 Administration Standard for Telecommunications Infrastructure, most recent revision including all addenda and systems bulletins

.5 ANSI/TIA–607 Generic Telecommunications Bonding and Grounding (Earthing) for Customer Premises, most recent revision including all addenda and systems bulletins

.6 ANSI/TIA–862 Structured Cabling Infrastructure Standard for Intelligent Building Systems, most recent revision including all addenda and systems bulletins

.7 ANSI/TIA–942 Telecommunications Infrastructure Standard for Data Centers, most recent revision including all addenda and systems bulletins

.8 ANSI/TIA–1179 Healthcare Facility Telecommunications Infrastructure Standard, most recent revision including all addenda and systems bulletins

.9 ANSI/TIA–4966 Telecommunications Infrastructure Standard for Educational Facilities, most recent revision including all addenda and systems bulletins

.10 TIA–TSB–162 Telecommunications Cabling Guidelines for Wireless Access Points, most recent revision including all addenda and systems bulletins

.11 TIA-526 Series – Standard Test Procedures for Fiber Optic Systems

.12 TIA-942 – Telecommunications Infrastructure Standard for Data Centers

.13 Telecommunications Distribution Methods Manual, most recent edition

.14 Information Transport Systems Installation Methods Manual (ITSIMM), most recent edition

.15 National Electric Codes (NEC) – all applicable

.16 National Electric Codes (NEC) – all applicable

.17 NECA/FOA 301– Installing and Testing Fiber Optic Cables

.18 OSHA Standards and Regulations – all applicable

.19 Local Codes and Standards – all applicable

.20 UL444 – Standard for Safety of Communications Cable

.21 UL 1666 – Standard for Safety of Flame Propagation Height

.22 Local Authority Having Jurisdiction (AHJ)

.23 Anywhere cabling standards conflict with one another or with electrical or safety codes, Contractor shall defer to the NEC and any applicable local codes or ordinances, or default to the most stringent requirements listed by either.

.24 Manufacturers' Recommendations - Install all cabling and termination devices per the manufacturers' recommended installation practices for the applications warranties.

.25 Any violations of applicable standards or codes committed by the Contractor shall be remedied at the Contractor's expense.

1.5 System Description

.1 The Contractor will provide, install, and test a complete structured cabling system for the project's voice and data communications systems from the Telecommunications Outlet (TO) to the Telecommunications Room (TR), and between telecommunications spaces. The Contractor will provide and install all required components, as identified below.

.2 Fiber Backbone Cabling

.1 Vertical and horizontal backbone cabling will consist of Muilti-mode and Singlemode optical fiber cable installed from each TR to the ER.

.3 Backbone Cabling Termination

.1 Provide fiber distribution enclosures at each end, sized for the number of fibers to be installed.

.2 Terminate with field-installable connectors or with fusion splicing of factory cable assemblies and modular connector panels or splice cassettes.

.4 Typical Equipment Room (ER)

.1 Open racks and/or enclosures with vertical and horizontal wire management.
.2 Fiber Distribution Enclosures (FDEs).
.3 Fire-resistant plywood installed on at least one (1) wall, at a height of 96 inches Above the Finished Floor (AFF) on which to install wall-mounted equipment.
.4 A Grounding and bonding system connected to the building's main grounding electrode system.
.5 A cable runway system, installed above the racks and enclosures, to support and manage the cabling that runs from the racks and enclosures to the equipment in the space, which shall be fitted with all accessories required to adequately support the installed cabling, such as waterfalls, support components, and bonding components.

.5 Typical Telecommunications Room (TR)

.1 Open racks and/or enclosures with vertical and horizontal wire management.
.2 Fiber Distribution Enclosures (FDEs).
.3 Fire-resistant plywood installed on at least one (1) wall, at a height of 96 inches Above the Finished Floor (AFF) on which to install wall-mounted equipment.
.4 A Grounding and bonding system connected to the building's main grounding electrode system.
.5 A cable runway system, installed above the racks and enclosures, to support and manage the cabling that runs from the racks and enclosures to the equipment in the space, which shall be fitted with all accessories required to adequately support the installed cabling, such as waterfalls, support components, and bonding components.

.6 Pathways and Raceways

.1 Pathways and raceways are the support system for the infrastructure. All pathways and raceways shall conform to the standards referenced in this Section.
.2 All horizontal and backbone cable shall be properly supported every 48 to 60 inches. Infrastructure support systems include, but may not be limited to the following:

.1 Properly-supported cable trays and cable runway.
.2 Properly-supported conduits, inside or outside, above ground or underground.

.3 Non-continuous cable supports, which shall be spaced no more than 60 inches apart.

.4 Surface raceway systems that may consist of metallic or non-metallic raceways and boxes.

.7 Using a Combination of Cable Supports

.1 The preferred method for providing pathways is to use a combination of cable tray and non-continuous cable supports.

.2 Cable trays shall be used for main horizontal cable pathways on all levels from the ER and TR locations.

.3 Cable trays shall be installed in the main corridors.

.4 In areas of low cable density, use independently-supported non-continuous cable supports in lieu of the cable tray system.

.5 All backbone cable shall also follow these cable tray pathways.

.6 To allow for future maintenance and access, the primary cable routes shall be located over corridors.

.7 To protect cable from damage and to provide a suitable aesthetic appearance, in areas where the cable may be exposed, such as in open ceiling rooms, conduit or surface raceway must be used instead of non-continuous cable supports.

1.6 Submittals

.1 Consultant's Review

.1 The Consultant's review of shop drawings or samples shall not relieve the Contractor of responsibility for any deviation from the contract documents.

.2 With the shop drawings, the Contractor shall include an index sheet detailing all deviations from the contract documents, and will be held responsible for all deviations, unless the Contractor has received written approval from the Engineer for the specific deviation, separate from general shop drawing approval.

.3 The Consultant's review shall not relieve the Contractor from responsibility for errors or omissions in the shop drawings or samples.

.2 General Component Data

.1 For all products covered under this Section, the Contractor shall submit the following data for each component:

.1 A Specification Section

.2 The Manufacturer's name.

.3 The Manufacturer's model and part number

.3 Fiber Cable

.1 In addition to the general requirements above, the Contractor shall submit the following additional data:

.1 Cable identification numbers.

.2 Cable specifications including quantity of fibers, material, insulation, jacket, wavelength, attenuation, diameter, bend radius, core, cladding, coating, buffering, weight, and color.

.4 Splicing and Terminations

.1 In addition to the general requirements above, the Contractor shall submit splicing and terminating tools, materials, and methods.

.5 Testing and Test Results

.1 The equipment serial number.

.2 A graphic diagram documenting the test procedure, including all connectors, the light source (as applicable), the origin, and the destination of each cable tested.

.3 Refer to Section 27 08 11.

1.7 Quality Assurance

.1 Standards for Materials and Equipment

.1 The Contractor shall provide all materials, equipment, and installation in compliance with the latest applicable standards from ANSI, FCC, ASTM, EIA/TIA, IEEE, NEC, NFPA, NEMA, OSHA, REA, and UL.

.2 Installer Qualifications

.1 Refer to Section 27 05 00.

1.8 Delivery, Storage, and Handling

.1 To prevent damage, theft, soiling, and misalignment, protect equipment during transit, storage, and handling.

.2 The contractor shall coordinate the secure storage of equipment and materials on site, or, if no on-site storage is available, shall provide their own secure storage at the Contractor’s expense.

.1 Do not store equipment where conditions fall outside the manufacturer's recommendations for environmental conditions.

.2 Do not install damaged equipment. Remove environmental conditions from the site and replace damaged equipment with new equipment.

.3 If off-site storage of materials is necessary, this shall be at the Contractor's expense.

1.9 Coordination

.1 The Contractor shall coordinate with all other trades. The Contractor will submit a schedule for the installation within 10 days of contract award

.1 The schedule shall include delivery, installation, and testing for conformance to specific job completion dates.

.2 At minimum, the schedule shall provide dates for the start of demolition, the completion of demolition, the installation start date, the completion of copper cabling, the completion of backbone cabling, the completion of testing and labeling, cutover, the completion of the final punch list, final inspection, and acceptance.

.2 Meeting Attendance and Schedule Adherence

.1 The Contractor must attend all project-related meetings and adhere to schedule set by the Project Manager.

.3 Final Inspection

.1 The Contractor is required to notify the Engineer of a proposed appointment for Final Inspection at least 72 hours before the appointment.

.2 Within five working days after the final inspection, the Contractor shall send final project documentation and warranty information to the Owner and Engineer. The final project documentation shall include, but may not be limited to:

.1 As-Built Drawings, in an AutoCAD format, with legible outlet address and cable paths.

.2 Outlet location spreadsheets.

.3 Warranty paperwork.

.4 A copy of the Final Inspection and Acceptance Signoff Sheet.

.5 Photos of each ER and TR.

1.10 Project Conditions

.1 Project Environmental Requirements

.1 Fiber Optic Cable Safety

.1 The following warnings shall be posted on the job site:

.1 WARNING: PERMANENT EYE DAMAGE CAN RESULT FROM LOOKING DIRECTLY INTO A LIGHT BEAM

GENERATED BY AN LED OR LASER SOURCE OR INTO THE END OF A CABLE FIBER CONNECTED TO ONE OR THESE SOURCES.

.2 CAUTION: LIGHT GENERATED BY THESE SOURCES MAY NOT BE VISIBLE YET REMAIN HAZARDOUS TO THE EYE. LOOK FOR WARNING LABELS ON SOURCE DEVICES.

.2 Observe all warning signs on equipment and all written safety precautions in the equipment instruction and technical manuals.

.3 Always handle cable carefully to avoid personal injury. Care should be taken with individual fibers to prevent injury to the eyes or penetration of the fibers into the skin.

.2 Hazardous Materials Prohibition

.1 The Contractor shall ensure that all materials used in the project are asbestos-free, unless specifically authorized in writing by the Owner.

.3 Existing Conditions

.1 Verify that all conditions on the project site are acceptable for the Work specified in this Section. Prior to bid opening, notify the Consulting Engineer, in writing, of any discrepancies, conflicts, or omissions. Otherwise, correct these issues at no additional cost to the Owner.

.2 Continue to monitor the project site. If conditions develop that require a variance from the Specifications or Drawings, then immediately notify the Owner in writing. Otherwise, make recommendations, submit drawings showing how the Work may be installed, and, upon approval, proceed with the necessary changes without additional cost to the Owner.

.2 Record Drawings

.1 Keep a complete set of all telecommunications drawings in the job site office for demonstration of the actual installation work specified in this Section.

.2 Use this set of drawings for no other purpose.

.3 Where any material, equipment, or system components are installed differently than what is shown on the drawings, indicate the differences clearly and neatly using ink or indelible pencil.

.4 Upon completion of the project, submit the record set of drawings.

1.11 Use of the Site

.1 Where the Owner deems it necessary to place restrictions, use the site as directed by the Owner.

.2 When proceeding with the work, do not interfere with the ordinary use of streets, aisles, passages, exits, or operations of the Owner. During the day, set up cones and barriers in hallways and walkways. Do not string cable down the hallways during normal hours.

.3 Request a hazardous materials worksheet that identifies potentially-hazardous locations. Do not proceed with any work in locations where hazardous materials are known to be. Obtain instructions from the Contractor's Project Manager on and when to work in these areas.

.4 Multiple times each day, each contractor shall remove all trash and debris from the site. Before leaving the room each day:

.1 The Contractor shall replace all ceiling tiles that they have removed.

.2 The Contractor shall place all furniture and equipment that they have moved back into its original location.

.3 The Contractor shall return any equipment that they have disconnected to working order.

.4 The Contractor's Job Foreman shall inspect all work locations to ensure that the rooms are clean and that all of the tasks described above have been done.

.5 It is recommended that the Contractor inspect the site and take pictures to document the condition of the ceilings and walls.

1.12 Continuity of Services

.1 Take no action that will interfere with or interrupt existing building services, unless previous arrangements have been made with the Owner's representative. Arrange all work to minimize shutdown time.

.2 The Owner's personnel shall perform shutdown of operating systems. When shutdown of systems is required, the Contractor shall give three (3) days advance notice.

.3 Should building services be inadvertently interrupted:

.1 The Job Foreman shall immediately notify the Project Manager of the accidental disruption of services, the remedy, and how long it will take to restore services.

.2 The Contractor shall immediately furnish the labor, including overtime, the material, and the equipment necessary to promptly restore the interrupted service at no cost to the Owner.

1.13 Warranty

.1 Refer to Section 27 05 00.

2 Products

2.1 Manufacturers

.1 Manufacturer List:

.1 Panduit (basis of design).
.2 Corning.
.3 Belden.
.4 Commscope.
.5 Or approved equivalent.

.2 Substitutions: In accordance with Section 27 05 00.

2.2 General

.1 Refer to Section 27 05 00 for General Requirements

.2 All materials and products shall be:

.1 Appropriate for the intended use.
.2 Recognized as such by a Nationally Recognized Testing Laboratory (NRTL) such as Underwriters Laboratories (UL), ETL SEMCO (ETL), the Canadian Standards Association (CSA) or the American National Standards Institute (ANSI).
.3 Permitted by the Authority Having Jurisdiction (AHJ).

.3 All products shall be new, of the latest version at time of bid, and brought to the job site in original manufacturer's packaging. Used equipment and damaged material will be rejected.

.4 Any modifications to equipment to suit the intent of the specifications shall be performed in accordance with these requirements.

.5 Cable lubricants specifically designed for installing communications cable may be used as needed to reduce pulling tension when pulling cable into conduit.

.6 Take care during installation to prevent scratches, dents, chips, etc. Equipment with significant or disfiguring cosmetic flaws will be rejected.

.7 All components will be approved by the Engineer and shall have the most aesthetic value possible while maintaining specified functionality. Hardware shall:

.1 Be in compliance with the Construction Documents.
.2 Have fit and finish compatible with the existing surrounding structure.
.3 Be unobtrusive.
.4 Provide the required functionality.

.8 All work area termination hardware, including mounting boxes, faceplates, and outlets, shall match the existing wall surface color as closely as possible.

.9 All copper and fiber products shall be from a single manufacturer so that a single performance warranty covers all applications on vertical and horizontal links.

.10 Fabricate custom-made equipment with careful consideration given to aesthetic, technical, and functional aspects of the equipment and its installation.

.11 Provide products that are suitable for the intended use, including, but not limited to environmental, regulatory, and electrical factors.

2.3 Optical Fiber Cables

.1 Optical Fiber Strands

.1 All optical fibers shall:

.1 Be usable and shall meet required specifications.
.2 Be sufficiently free of surface imperfections and inclusions to meet the optical, mechanical, and environmental requirements of this specification.
.3 Consist of a doped silica core surrounded by a concentric glass cladding and have a matched clad design.
.4 Be proof tested by the fiber manufacturer at a minimum of 100 kpsi (0.7 GN/m2).
.5 Be coated with a dual layer acrylate protective coating that is in physical contact with the cladding surface.
.6 Have a maximum attenuation value for each cabled fiber at 23˚C ± 5˚C on the original shipping reel.

.2 Multimode OM4

.1 Single-mode fibers shall meet TIA 492AAAD, “Detail Specification for Class IVa Dispersion-Unshifted Multimode Optical Fibers with Low Water Peak,” and ITU recommendation TG.652,

.3 Fiber Optic Cable Fire Ratings

.1 Only OFNP listed optical fiber backbone cable is acceptable for use on this project.

.4 Fiber Optic Cable Termination

.1 Where cables are installed, the 900 µm buffer or 250 µm coated fibers contained in these cables may be terminated either by:

.1 Fusion splicing of factory-polished splice-on connectors or factory-terminated cable assemblies (“pigtails”).

.2 Cam-style mechanical splice connectors using a tool that provides calculated insertion loss at the point of termination

.3 Individual fibers secured in a protective covering, such as an aramid-reinforced tube with connectors mated to the resulting assembly

.5 Fiber Optic Cable Features

.1 The size and configuration of fiber optic cables shall be as shown on the Drawings.

.2 The buffered fibers shall be grouped in 12 fiber subunits.

.3 The fibers shall be stranded around a dielectric central member in the subunit.

.4 Layered aramid yarns shall serve as the tensile strength member of the subunit.

.5 To facilitate jacket removal, a ripcord may be applied between the aramid yarns and the subunit jacket.

.6 For physical and environmental protection, the subunit jacket shall be extruded over the aramid yarns. The jacket shall:

.1 Be continuous and free from pinholes, splits, blisters, or other imperfections

.2 Have a consistent, uniform thickness

.3 Be smooth, as is consistent with the best commercial practice

.7 The subunits shall be stranded around a dielectric central member, a ripcord shall be inserted beneath the outer jacket to facilitate jacket removal, an outer jacket shall be extruded around the subunits.

.8 An overall helically-wound interlocking metallic armor shall be provided, to surround the outer cable jacket, to which a listed outer jacket shall be applied.

.9 The individual fibers shall be color-coded for identification and shall meet these requirements:

.1 The optical fiber color coding shall be in accordance with EIA/TIA 598, “Optical Fiber Cable Color-Coding.”

.2 The coloring material shall be stable over the temperature range of the cable, shall not be susceptible to migration, and shall not affect the transmission characteristics of the optical fibers.

.3 Color-coded buffered fibers shall not adhere to one another.

.10 The overall jacket for graded index cables as specified herein shall be aqua for distribution cables, or black for indoor-outdoor rated cables.

.11 The overall jacket for single-mode cables as specified herein shall be yellow for distribution cables, or black for indoor-outdoor rated cables.

2.4 Optical Fiber Connectors and Terminations

.1 All terminations must be suitable for and meet the required standards and specifications as the fiber they intended for.

.2 Where fiber optic cables are installed fibers contained in these cables may be terminated either by:

.1 Fusion splicing of factory-polished splice-on connectors

.2 Factory-terminated cable assemblies (“pigtails”).

.3 All strands must be terminated with LC female connector on fiber patch panel as provided by the contractor.

2.5 Optical Fiber Cassettes

.1 Factory-manufactured modular fiber optic cassettes must comply with IEEE 802.3ae 10GbE and ANSI T11.2 Fibre Channel requirements.

.2 Optical fiber strands within the cassettes shall be equal to those used in the cable between cassettes.

.3 To meet the IEEE 802.3ae channel loss specification, optimized cassettes shall have a maximum insertion loss of 0.5 dB, and standard cassettes shall have a maximum insertion loss of 1.0dB.

.4 The adapter housing colors shall follow the color identification scheme suggested by ANSI/TIA 568.

.5 Cassettes shall interconnect with high-density small-form-factor pre-terminated cable assemblies.

.6 Terminations must be LC.

2.6 Optical Fiber Distribution Enclosures and Connector Panels

.1 Fiber patch panel to be 19” rack mount enclosure (in all 19” rack/cabinet) or wall mount enclosure, supporting optical fiber connector adapter panels (duplex LC Type), rear cable retention loops, entry cable strain relief supports, cable strength member clamps, D- grommets at cable entry, removable rear panel cover, accommodation and support for splice trays, removable front access cover, front space for patch cable storage, patch cable bend radius controls and identification labeling.

3 Execution

3.1 General

.1 Upon completion of work, a Registered Communications Distribution Designer (RCDD) shall submit as-built drawings to the Owner and Engineer.

.2 The Contractor shall input the cabling data into the cable management software.

.3 Install all cables in accordance with project Drawings.

.4 Provide any screws, anchors, clamps, tie wraps, distribution rings, miscellaneous grounding and support hardware, etc. needed to facilitate the installation of the cable plant system.

.5 Furnish any special installation equipment or tools required to properly complete the installation.

.6 Do not roll or store cable reels without an appropriate underlay.

.7 Failure to follow the appropriate guidelines may require the installer to provide the additional material and labor required to bring the installation back into alignment with the guidelines. This shall also apply to any and all damages caused to the cables by the installer during the implementation.

.8 Provide fire blocking at all fire-rated ceiling, wall, and floor penetrations.

.9 Plug conduits where cabling has been installed in the main equipment room, backbone and other cable entrance locations with re-enterable duct seal of flame-retardant putty.

.10 Provide bushings on all conduit ends.

.11 All wiring, materials, and equipment must be listed and labeled by an NRTL. To certify that performance characteristics, meet ANSI/TIA 568 Standards, provide all Original Equipment Manufacturer (OEM) documentation to the Owner.

.12 All techniques and fixtures used in the installation must minimize complexity must allow for easy maintenance of, and ready access to, all components for test measurements.

.13 No self-tapping screws shall be used.

.14 All parts shall be made of corrosion-resistant material, such as plastic, anodized aluminum, or brass.

.15 All materials used in the installation shall be resistant to fungus growth and moisture deterioration.

.16 To avoid corrosion caused by electrolysis between dissimilar metals under the environmental operating conditions specified, separate dissimilar metals with an inert dielectric material.

.17 All empty innerduct or conduit shall include a non-corrosive pull-rope.

.18 All of the pathways shown on the drawings are suggested routes for the Contractor to use as guidelines. Prior to construction, the Contractor shall coordinate in the field with other trades to determine the exact feeder, tie, and riser backbone cabling pathways. In any case where the communication pathway must be removed and re-routed, due to conflicts with other trades with which the Contractor did not previously coordinate, the Contractor is responsible for all costs associated with the removal and relocation.

3.2 Optical Fiber Cable

.1 Install the optical fiber backbone in a continuous length from the FDE in the MC to an FDE within each TR.

.2 Throughout its length, run the backbone cable in appropriate, listed raceway.

.3 Leave a 3 m long maintenance loop at each end of the link, neatly contained in the integral management rings and saddles in a “figure 8” loop at the rear of the FDE.

.4 Throughout the length of the cable, maintain the minimum bend radius and pulling force recommended by the manufacturer and required by industry standards, both during installation and after termination and testing.

.5 On each end, remove all outer jacket and strength member materials to expose the individual 900-micron buffering of the individual strands for a length of 0.5 m (18 inches).

.6 On each end, hold the cable ends securely in place with the cable clamping accessories in each FDE.

.7 Route individual strands in the rear of the FDE in a neat and orderly fashion and place them so as not to create undue stress or micro bending of the strands.

3.3 Cable Bundling Materials

.1 Secure all cable bundles with proper bundling or securing materials so as to ensure that the cable runs are securely held in place both vertically and horizontally.

.2 Do not tighten bundling materials or securing devices so tightly that they deform the inherent cable geometry or construction.

.3 Do not use cable ties or hook-and-loop tape to secure cable runs to other building systems such as electrical conduit, Electric Metallic Tube (EMT), sprinkler pipes, ceiling suspension members.

.4 In environmental air-handling spaces, only use appropriately-listed materials.

3.4 System Administration

.1 Uniquely identify all components of the installed system by location, function, unit, and sub-unit.

.2 Identify each location with a unique alphanumeric identifier.

.3 Assign a unique alphanumeric identifier for each equipment enclosure in the building.

.4 Identify each adapter module in each distribution or interconnect enclosure with an alphanumeric identifier.

.5 Identify all conduits, trays, and pathways with a unique alphanumeric identifier.

.6 Identify optical fiber cables by a textual label that indicates its type, strand count, point of origin, and termination.

.7 Supply a Cable Identification Matrix

.8 Supply all records in compliance with ANSI/TIA 606.

3.5 Identification

.1 Before installing or terminating cable, confirm all specific labeling requirements with the Owner or the Owner's Engineer.

.2 Mark each backbone cable at each endpoint and at all intermediate pull and access points and junction boxes with a label that indicates the origination and destination identifiers, the sheath identifier, and the strand or pair range.

.3 Fiber Distribution Enclosures (FDEs)

.1 Mark each FDE with an adhesive label that indicates the range of circuits installed within it.

.2 Label each port with the origination and destination grid identifier and the individual strand ID.

.3 At each end of each cable, within 75 mm (3 inches) of the end of the sheath, place a self-laminating label that surrounds the outermost jacket and bears the appropriate cable identifier.

.4 On each equipment enclosure, affix self-adhesive labels, bearing the enclosure's identifier in block characters, at the top center of the front and rear doors or faces.

.5 In all enclosures, place a label directly adjacent to the shortest side of each adapter that bears that adapter's identifier. Rotate the characters on the labels to maintain a left to right, top to bottom orientation.

.4 Refer to Section 27 05 53

3.6 Field Quality Control

.1 General Testing.

.2 Refer to Section 27 08 11 for complete testing specifications.

END OF SECTION

1 General

1.1 Summary

.1 This Section covers the supply and installation of a complete Communications Copper Horizontal Cabling system. The scope of work includes, but is not limited to, the following tasks, as well as all tasks specified by the Consultant or the Owner, stipulated in the Contract Documents, or required for proper completion and commissioning of the system.

.1 Supply, delivery, supervision, coordination, installation, testing and commissioning of a complete horizontal copper cabling system, as specified in the present document, the Contract Documents, and the Drawings.

.2 Submittal of complete documentation, including, certifications, test reports, progress reports, warranty certificates, manuals, as-built drawings, as specified in the present document and the Contract Documents, and as requested by the Consultant and the Owner for the purpose of proper completion of the project.

.2 Contractor shall examine, and is responsible for complying with, the Contract Documents in their entirety, including all relevant Drawings and Specification presented in other Sections and Divisions, regardless of whether such Drawings and Specification are referenced in this Section.

.3 Contractor shall provide, install and test a complete Communications Copper Horizontal Cabling system as follows.

.1 Contractor shall furnish and install all horizontal cabling system equipment, including:

.1 Horizontal Cables (Permanent links, patch cords and equipment cables).

.2 Termination Hardware.

.3 Telecom Outlets.

.4 Patch Panels.

.5 Device Connection.

.6 Testing (Including cable and channel verification, qualification and certification).

.7 Administration and Identification.

.2 Contractor shall furnish and install all miscellaneous items, accessories, appurtenances, and devices incidental to or necessary for

a sound, secure, and complete installation, without claim for additional payment, even if such work or components are not specifically mentioned in the Specification or the Drawings.

.3 Contractor shall provide system testing and demonstration, proper commissioning, system documentation, as-built drawings, training and installation manuals, and adequate training for the Owner's personnel, without claim for additional payment.

.4 Errors, Omissions and Conflicts:

.1 Contractor shall notify the Consultant no later than ten (10) days prior to submitting the bid of any errors or omissions encountered in the Drawings, Specification, or other Contract Documents.

.2 Contractor shall use the more expensive material or method whenever a conflict is encountered between or within Drawings and Specification, unless Contractor has requested received a written response (addendum) as to which material or method is suitable.

.5 Related Requirements:

.1 Section 07 84 00 – Firestopping.
.2 Section 26 05 26 – Grounding and Bonding for Electrical Systems.
.3 Section 27 08 11 – Testing of Structured Cabling Systems.
.4 Section 27 11 16 – Communications Cabinets, Racks, Frames, and Enclosures.
.5 Section 27 11 19 – Communications Termination Blocks and Patch Panels.
.6 Section 27 11 23.13 – Communications Cable Management.
.7 Section 27 11 26 – Communications Rack Mounted Power Protection and Power Strips.

1.2 Acronyms and Abbreviations

.1 ANSI - American Northern Standards Institute.

.2 AWG - American Wire Gauge.

.3 BICSI - Building Industry Consulting Service International.

.4 TDMM - Telecommunications Distribution Methods Manual (BICSI).

.5 EIA - Electronics Industry Alliance.

.6 IEEE - Institute of Electrical and Electronics Engineers.

.7 ISO - International Standards Organization.

.8 OESC – Ontario Electrical Safety Code.

.9 NFPA - National Fire Protection Agency.

.10 NRTL - Nationally Recognized Testing Laboratory.

.11 TIA - Telecommunications Industry Association.

.12 UL - Underwriters Laboratories.

1.3 References

.1 Most recent editions and addenda of the following documents:

.1 ANSI/TIA-568 - Commercial Building Telecommunications Cabling Standard Set
.2 ANSI/TIA-569 - Telecommunications Pathways and Spaces
.3 ANSI/TIA-606 - Administration Standard for Telecommunications Infrastructure
.4 ANSI/TIA-607 - Generic Telecommunications Bonding and Grounding (Earthing) for Customer Premises
.5 ANSI/TIA-862 - Structured Cabling Infrastructure Standard for Intelligent Building Systems
.6 ANSI/TIA-942 - Telecommunications Infrastructure Standard for Data Centers.
.7 ANSI/TIA-1179 - Healthcare Facility Telecommunications Infrastructure Standard.
.8 ANSI/TIA-4966 - Telecommunications Infrastructure Standard for Educational Facilities.
.9 BICSI/TDMM - Telecommunications Distribution Methods Manual.
.10 TIA-TSB-162 - Telecommunications Cabling Guidelines for Wireless Access Points.
.11 CSA C22.1 - Canadian Electrical Code Part 1.
.12 UL 444 - Standard for Safety of Communications Cable.
.13 UL 1666 - Standard for Safety of Flame Propagation Height.
.14 Local Authority Having Jurisdiction (AHJ).
.15 Anywhere cabling standards conflict with one another or with electrical or safety codes, Contractor shall defer to the OESC and any applicable local codes or ordinances, or default to the most stringent requirements listed by either.

1.4 System Description

.1 The Contractor shall provide, install, test, label and document a complete horizontal structured cabling system, spanning from a centralized telecom room(s) (TR) to the telecom outlets (TO), in a star network topology. The main components of the system are specified below.

.2 Horizontal Cabling

.1 The horizontal cabling system extends from the telecom room (TR) or telecom enclosure (TE) to the work area telecom outlet (TO). It includes horizontal cable, hardware terminations, jumpers, patch cords, and may incorporate multiuser telecommunications outlet assemblies (MUTOAs) and consolidation points (CPs).

.2 Maximum horizontal cable length shall be 90 m (295 ft.), independently of media type. Where an MUTOA is deployed, the maximum horizontal copper cable length shall be adjusted accordingly.

.3 Telecom Room / Enclosure (TR / TE)

.1 Telecom Room (TR) is a dedicated and controlled space on each building floor, used for housing telecommunications equipment, cable terminations, and cross-connect hardware serving that particular floor. TR constitutes the recognized connection point between horizontal and backbone cabling systems. The telecom standards (ANSI/TIA, BICSI TDMM, etc.) require a minimum of one telecom room, closet, or enclosure per floor.

.2 Telecom Enclosure (TE) usage is project-specific rather than a standard requirement. TE is intended to serve relatively smaller floor area, and is often used to extend the space and reach of a TR.

.4 Pathways and Raceways

.1 Pathways and raceways are conveyance and routing systems for distributing, supporting and protecting telecom infrastructure cabling systems. Pathway system type and installation method shall be suitable for the media being supported and the environment being traversed.

.2 Pathway and raceway systems shall conform to the standards specified and referenced in Section 27 05 28 and Section 27 05 36.

.3 Pathway design shall combine continuous cable trays and non-continuous cable supports. Cable trays should, nonetheless, form the main horizontal cabling pathway, and should be located over the main corridors, to facilitate access and maintenance tasks.

.4 In low cable density areas, non-continuous supports shall be used in lieu of a cable tray system, unless otherwise specified in the design documents.

.5 In open-ceiling spaces where the cable is exposed, a conduit or a surface raceway shall be used in lieu of non-continuous supports, to enhance cable protection and aesthetic appearance.

1.5 Submittals

.1 Provide submittals in accordance with Section 01 33 00.

.2 Shop Drawings

.1 Contractor shall submit shop drawings for all components and equipment used in the project, including, the items listed below.

.1 Modular Connectors.

.2 Telecom Outlet Faceplates.

.3 Horizontal Cable.

.4 Patch Panels.

.5 Patch Cables and Equipment Cords.

.6 Telecommunications Enclosures and Racks.

.7 Cable Management Components.

.8 Pathways and Raceways (Cable Trays, Cable Supports, Conduits, etc.).

.9 Firestopping Material.

.3 General Component Data

.1 Contractor shall submit the information and documents outlined below, with respect to all components and products covered under this Section or installed as part of communications copper horizontal cabling:

.1 Product Name, Manufacturer Name, Model Number and Part Number.

.2 Product Datasheets and Specifications.

.1 Complete specification, including but not limited to, standards compliance, Category type, fire rating, environmental rating, number and color code of pairs, conductor size and material, insulation type and material, performance and attenuation parameters, near-end cross talk (NEXT), cable diameter and weight, jacket type and material, cable color, etc.

.3 Product Manuals (Installation Instructions, User Manuals, Maintenance Manuals).

.4 Deviations

.1 Contractor shall be responsible for any deviation from Contract Documents, regardless of whether such deviation or associated drawings and samples have been reviewed by the Consultant, unless

deviation has been approved in writing by the Consultant and the Owner.

.2 Contractor shall submit an indexed sheet detailing all deviations from Contract Documents, and shall be responsible for all such deviations, unless these have been expressly approved in writing by the Consultant and the Owner.

.3 Contractor shall not assume that an approval of an item's shop drawing or sample is equivalent to an approval of the item itself. Deviations shall be treated specifically, and shall be approved separately as such, in writing by the Consultant and the Owner.

.4 Contractor shall be responsible for all deviations, errors and omissions in the submittal documents and process, regardless of whether such deviations, errors and omissions have been reviewed by the Consultant or the Owner.

1.6 Closeout Submittals

.1 Testing and Test Results

.1 Refer to Section 27 08 11.
.2 Specification and calibration sheets for all test equipment.
.3 Contracting firm qualifications and certifications.
.4 Installation team qualifications (by person).
.5 Test results (Intermediate and final test results, as specified in Part 3 - Execution).

.2 Contractor shall submit final closeout documents as required by the Consultant and the Owner. These shall include, but are not limited to, the following items.

.1 Test results covering all cable segments and links, in native tester and PDF formats, or as otherwise specified by the Consultant.
.2 Manufacturer and contractor warranties.
.3 Complete project manual set, including, product datasheets, material submittals, training manuals, user manuals, maintenance manuals and preventive maintenance plan.
.4 Spare parts recommendation list.

.3 Contractor shall submit a complete set of as-built drawings in an accessible AutoCAD format, as specified by the Consultant and the Owner. As-built drawings shall cover the following.

.1 Precise location and identification (labels) of all equipment, including, all data outlets, cables, cable trays, patch panels, racks, telecom

rooms and enclosures, and all other equipment which is part of the horizontal cabling system.

.2 Detailed floor plans of all telecom rooms and spaces.

.3 Detailed layout of all racks, patch panels, cable trays, pathways, conduits.

.4 Detailed layout of all pathways, cable trays, and cable routes.

.4 Contractor shall submit and mount on the wall of each telecom space or room, detailed as-built drawings depicting the floorplan, floor area, TO points, and cabling layout, for that space or room. Drawings shall be printed on ANSI C paper size or larger, shall be laminated, and shall be approved and wall-mounted in coordination with the Consultant and the Owner.

1.7 Quality Assurance

.1 Standards for Materials and Equipment

.1 Contractor shall provide all materials, equipment, and installations, in compliance with the latest applicable standards.

.2 Installer Qualifications

.1 Refer to Section 27 05 00 – Common Work Results for Communications.

1.8 Warranty

.1 Contractor shall ensure each piece of equipment installed including wiring is warranted by the manufacturer to be free of defects in operation, material, and workmanship for a period of 15 years from date of Substantial Completion.

2 Products

2.1 Manufacturers

.1 CommScope

.2 Panduit,

.3 Belden,

.4 Or approved equivalent.

2.2 Cable Colour

.1 All Access control and CCTV/Surveillance Camera cabling must be **RED** Cat6a.

.2 Cable other than CCTV/Surveillance Camera may be blue, white, black etc.

2.3 Category 6A Data Communications Horizontal Cable

.1 Category 6A cable shall meet ANSI/TIA-568.2-D Category 6A component standard. Cable conductors shall consist of 23 AWG copper wires and shall be twisted in pairs. In addition, Category 6A cable shall comply with the following specifications and standards:

.2 General Requirements

.1 Cable shall consist of 4 unshielded twisted pairs with no outer shielding (Cat 6A U/UTP), unless otherwise specified in the design and expressly approved by the Owner.

.2 Cable used in data networking systems shall have a blue color jacket.

.3 Cable conductor insulations shall be color coded in compliance with the TIA-568 standard.

.4 Cable individual conductors shall be 23 AWG copper wires. Special applications requiring the use of the thinner 24 AWG conductor in patch cords (short length, max 2 m) shall be specified and expressly approved by the Owner. Note: Only Category 6A cable with 23 AWG solid conductors is permitted in Permanent Link.

.5 Cable deployed as horizontal permanent link extending from the telecom enclosure/room to the telecom outlet shall have solid conductors.

.6 Equipment cable and patch cords shall have stranded conductors.

.7 Cable shall be properly rated for its use application and its installation environment (Outdoor/OSP, Plenum/CMP, Riser/CMR, LSZH, Industrial/Armored, Direct Burial, etc.), as required by the local and national codes.

.8 Cable weight shall not to exceed 30 lbs./1000 ft.

.3 Standards Compliance and Technical Specification

.1 Category 6A cable shall satisfy the following compliance and technical specification.

.2 Cable shall exceed requirements of ANSI/TIA-568.2-D Category 6A and ISO 11801 2nd Edition Class EA channels standard.

.3 Cable shall exceed requirements of ANSI/TIA-568.2-D and IEC 61156-5 Category 6A components standard.

.4 Cable shall meet requirements of IEEE 802.3af, IEEE 802.3at and IEEE 802.3bt for Power over Ethernet (PoE) applications up to 100 watts (IEEE 892.3bt Type 4, 4PPoE).

.5 Cable shall meet TIA TSB-184-A / IEEE 802.3bt cable requirements, and UL listed CMP-LP (0.7A).

.6 Cable shall be characterized up to 500 MHz or higher, as specified in the ANSI/TIA-568.2-D, Category 6A standard.

.7 Cable shall be third-party tested and verified for compliance with ANSI/TIA-568.2-D, Category 6A.

.8 Cable shall support IEEE 802.3 for 10GBASE-T (10 Gigabit Ethernet), 1000BASE-T (Gigabit Ethernet), 100BASE-T (Fast Ethernet), and 10BASE-T applications up to 100 metres.

.4 Safety Standards

.1 Category 6A cable shall strictly meet the following specification in order to achieve compliance with the local and national (CSA) safety codes and standards.

.2 Cable deployed in the plenum and/or raised floor space (whether partly or entirely) shall be plenum CMP rated and shall comply with the CSA FT6 or NFPA 262 (UL-910) fire test standards.

.3 Cable running between floors in non-plenum area shall be riser CMR rated and shall comply with the CSA FT4 and UL-1666 fire test standards.

.4 Plenum CMP rated cable (CSA FT6 or NFPA 262) may be used in riser applications, if needed. Riser CMR rated cable, however, is not approved for use in plenum space and shall not be deployed therein.

.5 Temperature Rating

.1 Installation Temperature Range: 32°F to 122°F (0°C to 50°C)

.2 Operating Temperature Range: -4°F to 167°F (- 20°C to 75°C)

3 Execution

3.1 General

.1 Comply with the design requirements and industry standards, and with the applicable local and national codes.

.2 Follow best practices guidelines and applicable industry standards for the installation of the cable and all horizontal channel components.

.3 Ensure that patch panels, connectors, and all structured cabling components are compatible with the Category rating of the cable. For example, if Category 6A cable is used; then, all structured cabling components shall be Category 6A compatible.

.4 Ensure that all materials, components and equipment being installed are properly rated (Plenum or Riser), as required by local and national codes.

.5 Install all cables and horizontal channel components (connectors, outlets, faceplates, jacks, etc.) as instructed in the manuals and specified in the Drawings.

.6 Provide all installation and mounting accessories and components, such as, screws, anchors, clamps, hooks and loops, grounding and support hardware, etc., as required to carry out the installation of the cabling system.

.7 Furnish all special installation equipment or tools, as needed to properly complete the installation.

.8 Provide appropriate underlay for storage of cable reels. Rolling of cable reels is not permitted.

.9 Provide all additional material and labour required to rectify a situation arising from their failure to comply with applicable standards and best practices guidelines. This equally applies to all damage inflicted to cables by Contractor during the installation process.

.10 Provide fire blocking material at all fire rated penetrations, as required by design, standards, and codes.

.11 Seal all conduits where cabling has been installed with re-enterable duct sealing system, as required by design, standards, and code.

.12 Provide and install bushings on all conduit ends.

.13 Ensure means and methods promote ease of access to cables for maintenance and testing purposes.

.14 The pathways shown on the Drawings are an approximate representation, and should not be interpreted as an absolute and accurate layout. Conduct site surveys, and coordinate with other trades, as required, in order to determine the exact final pathway layout.

3.2 Horizontal Cabling Guidelines

.1 Install all patch panels and network switches (if applicable) inside the data racks, at the specified telecom spaces.

.2 Perform all cabling tasks in compliance with best practices guidelines, and applicable industry standards, particularly, BICSI’S TDMM and ANSI/TIA-568.

.3 Ensure that all horizontal cables, including any required service loops, do not extend beyond 90 m (295 feet). Identify any areas that cannot be reached within the 90 metre constraint, and report them to the Consultant. Contractor

shall strictly adhere to this rule, unless a written approval has been issued by the Consultant.

.4 Provide cable slack (service loop) at both ends of the cable for maintenance or future cabling system changes. The recommended lengths (BICSI TDMM) are:

.1 Minimum of 3 metres in the telecommunications closet for both twisted-pair and fiber cable

.2 Minimum 1 metre for fiber at the outlet.

.3 Minimum 30 cm for twisted-pair cable at the outlet.

.5 Ensure that all equipment installed in the horizontal channel including but not limited to, cable, pathways, conduits, connectors, is properly rated in compliance with local and national codes. All cables and components installed partly or entirely in the plenum space shall be plenum rated. All cables or components installed partly or entirely in the riser space shall be riser rated. All cables or components installed partly or entirely outdoors shall be weatherproof and OSP (outside plant) rated.

.6 Install all cable pathways and conduits perpendicular or parallel to the ceiling structure, unless otherwise specified in the Drawings.

.7 Exercise care and follow proper guidelines during cable installation in order to prevent damage to cable and equipment. Use grommets to protect cable where conduit or chase nipples are not installed around cut-outs or knockouts.

.8 Contractor shall not expose cable to water, paint overspray, paint removal products, water-based pulling lubricants, or any other substances or environments which can damage or compromise the integrity of the cable.

.9 Provide metallic isolation barriers to isolate different wiring groups, where cables with different classifications share a common box or enclosure.

.10 Coordinate as needed with the electrical and mechanical trades in order to implement a proper earthing and bonding system, and an effective Electromagnetic Compatibility (EMC) scheme.

.11 Mitigate electromagnetic interference (EMI) by maintaining the recommended separation between telecom and power (EMI) sources, as specified by ANSI/TIA-569 and shown in the table below.

Minimum Separation Distance Between Power and Data Cables (ANSI/TIA-569)			
	Minimum Separation Distance		
Condition	**< 2 kVA in (mm)**	**2 - 5 kVA in (mm)**	**> 5 kVA in (mm)**
Unshielded power lines or electrical equipment in proximity to open or nonmetal pathways	5 (127)	12 (305)	24 (610)
Unshielded power lines or electrical equipment in proximity to a grounded metal conduit pathway	2.5 (64)	6 (152)	12 (305)
Power lines enclosed in a grounded metal conduit (or equivalent shielding) in proximity to a grounded metal conduit pathway		3 (76)	6 (152)
Electrical motors and transformers			48 (1,220)

.12 Do not install telecom cables and power cables in the same pathways, cable trays, or conduits.

.13 Provide vertical cable support for all cables installed in vertical cable trays, using properly sized vertical cable supports on every third floor. Use of nylon cable ties is not permitted.

3.3 Cable Installation in Conduit and Duct

.1 Apply best practices when pulling cable inside conduits and ducts.

.2 Ensure that pulling tension never exceeds the manufacturer’s specification.

.3 Ensure that cable bending during installation conforms to the manufacturer’s specifications.

.4 Use a pull mandrel one size smaller than the conduit.

.5 Use cable pulling lubricant to reduce friction and tension during cable pulling.

.6 Use a dynamometer to measure pulling tension in the case of long or difficult installations.

.7 Use pulling grips suitable for copper cable installations. Contractor shall consult the cable manufacturer to determine the suitable pulling grip and method of attachment.

.8 Seal loose cable ends using cable caps (heat-shrinking type) to protect them until they are terminated.

.9 Use corner blocks to facilitate the bending of cable. Contractor should use 45° cable block for bends between 5° and 45°, and 90° corner block for bends between 45° and 90°.

3.4 Cable Bundling Material

.1 Contractor shall use the appropriate cable bundling and securing material, in line with industry and best practices standards, to ensure cable runs are securely held in place, both vertically and horizontally.

.2 Contractor shall not tighten bundling and securing ties and straps around cables, to avoid deforming cable geometry, and as a result, cable performance.

.3 Contractor shall only use safe and soft cable fasteners to bundle or secure cable, such as, “hook and loop” cable fasteners. The use of regular plastic cable ties is strictly forbidden.

.4 Contractor shall not attach or secure cable runs to other building systems (such as, electrical or mechanical conduits, HVAC ducts, sprinkler pipes, ceiling suspension members, etc.).

.5 Contractor shall strictly use plenum rated cable bundling components inside suspended ceilings, raised floors, and air-handling spaces.

3.5 System Administration

.1 Refer to Section 27 05 53 – Identification for Communications Systems.

.2 Conform to the system administration guidelines of ANSI/TIA-606, Administration Standard for Telecommunications Infrastructure.

.3 Propose a consistent and comprehensive system administration scheme in coordination with the Consultant, the Owner and all stakeholders. Contractor shall seek the Consultant’s or the Owner’s written approval of the proposed scheme ahead of implementation.

.4 Uniquely identify all components of the installed system by location, function, unit, and sub-unit.

.5 Identify each location, telecom space, telecom room or enclosure in the building with a unique alphanumeric identifier.

.6 Identify each system component, including but not limited to, adapter modules, connectors, cables, patch panels, patch and equipment cords, cable management devices, grounding system components, power cords, faceplates, outlets, cable trays and pathways, racks, enclosures, etc. with a unique alphanumeric identifier.

.7 Contractor shall supply all the required records and documentation in compliance with ANSI/TIA 606.

.8 Contractor shall provide an ODBC-compliant database (Open Database Connectivity), for administration of the cabling system described in this Section.

3.6 Identification

.1 Contractor shall seek the =Consultant's and the Owner's approval of the project's labeling plan, ahead of cable installation.

.2 Labeling Cables

.1 Mark each cable at each endpoint and at all intermediate pull and access points, and junction boxes with labels that indicate the origin and destination identifiers, the sheath identifier, and the strand or pair range.

.2 Contractor shall mark each horizontal cable on the sheath at each end with the TR, patch panel, and panel port where cable is terminated.

.3 Labeling Conduits and Pathways

.1 Contractor shall label conduits and pathways within 0.5 m (18 inches) of each end, on an exposed and accessible spot.

.2 Contractor should apply additional labeling, where possible, at every 3 m (10 feet) of exposed length.

3.7 Cable Testing

.1 Refer to Section 27 08 11 for cable testing procedures.

.2 Ensure all testing procedures and field-test instruments comply with the requirements of:

.1 ANSI/TIA-1152-A, Requirements for Field Test Instruments and Measurements for Balanced Twisted-Pair Cabling

.2 ANSI/TIA-568.0-D, Generic Telecommunications Cabling for Customer Premises

.3 ANSI/TIA-568.1-D, Commercial Building Telecommunications Cabling Standard

.4 ANSI/TIA 568.2-D, Balanced Twisted-Pair Telecommunications Cabling and Components Standards

.3 Provide all labor, materials, tools, field-test instruments and equipment required for the complete testing of the work.

.4 Test all horizontal UTP cables as instructed below.

.5 Perform tests using a level IV accuracy cable tester, updated with latest firmware.

.6 Provide all test records in both “native” and pdf formats, as part of the as-built package.

.7 Unless otherwise specified by the Consultant or the Owner, all horizontal links and channels shall be tested for the following:

.1 Wire Map.
.2 Length.
.3 Propagation Delay.
.4 Delay Skew.
.5 DC Loop Resistance.
.6 Insertion Loss.
.7 NEXT (Near-End Crosstalk).
.8 PS NEXT (Power Sum Near-End Crosstalk).
.9 ACR-N (Attenuation to Crosstalk Ratio Near-End).
.10 PS ACR-N (Power Sum Attenuation to Crosstalk Ratio Near-End).
.11 ACR-F (Attenuation to Crosstalk Ratio Far-End).
.12 PS ACR-F (Power Sum Attenuation to Crosstalk Ratio Far-End).
.13 Return Loss.

.8 All horizontal links and channels shall pass the test requirements for respective cable Category, as specified in ANSI/TIA-568.2-D. Category 6 and 6A cables shall be tested up to 250 MHz and 500 MHz, respectively.

.9 Any cable or link that fails the requirements shall be identified and corrected.

3.8 Firestopping

.1 Comply with the requirements of Section 07 84 00 – Firestopping.

.2 Comply with the requirements of the latest TIA/EIA-569, Annex A, Firestopping.

.3 Comply with the requirements of the latest BICSI TDMM, Chapter 7, Firestop Systems.

.4 Firestop systems shall meet the requirements of ULC Standard CAN/ULC-S115.

END OF SECTION

1 General

1.1 Section Includes

.1 Twisted Pair Patch Cords.

.1 Schedule of copper patch cords to be supplied.

.2 Fibre Optic Patch Cords.

.3 Related Cross-connect components.

.4 Cross-connection and patching.

1.2 Related Requirements

.1 Section 27 13 23 – Communications Optical Fiber Backbone Cabling.

.2 Section 27 13 13 – Communications Copper Backbone Cabling.

.3 Section 27 11 19 – Communications Termination Blocks and Patch Panels.

1.3 Submittals

.1 Product Data: For each type of product indicated.

.1 For Category-6A patch cords, include the following installation data for each type used:

.1 Nominal OD.

.2 Minimum bending radius.

.3 Maximum pulling tension.

.2 For Fiber Optic patch cords, include the following installation data for each type used:

.1 Nominal OD.

.2 Minimum bending radius.

.3 Maximum pulling tension.

.2 Source quality-control reports.

.3 Field quality-control reports.

1.4 Quality Assurance

.1 Electrical Components, Devices, and Accessories: Listed and labeled by a qualified testing agency, and marked for intended location and application.

.2 Warranty: see Section 27 05 00.

1.5 Delivery, Storage, and Handling

.1 Test cables upon receipt at Project site.

.1 Test optical fiber cables to determine the continuity of the strand end to end. Use optical loss test set.

.2 Test each pair of UTP cable for open and short circuits.

1.6 Project Conditions

.1 Environmental Limitations: Do not deliver or install cables and connecting materials until wet work in spaces is complete and dry, and temporary HVAC system is operating and maintaining ambient temperature and humidity conditions at occupancy levels during the remainder of the construction period.

2 Products

2.1 Manufacturers

.1 Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include the following:

.1 Panduit, Belden, Commscope.

.2 Or approved equivalent.

2.2 Patch Cables

.1 General Requirements: Comply with TIA/EIA-569-C.

.2 Category 6A Copper Patch Cords:

.1 Factory-made, four-pair cables terminated with eight-position modular plug at each end in lengths as indicated in pricing sheet.

.2 Patch cords shall have bend-relief-compliant boots

.3 UTP Patch Cords will be supplied in the minimum following lengths:

Quantity	Length	Colour	Use	
25	5'	White	Desks	
25	7'	White	Cabinet	
25	10'	White	Cabinet	
25	7'	Yellow	Internet	
25	10'	Yellow	Internet	
1 per CCTV Camera	5’	Red	CCTV	

.3 1 5’ and 7’ patch cord must be supplied per installed telecom outlet.

.4 Fibre Optic Patch Cords: Factor-made, dual-fibre cables with LC duplex connectors.

.1 Fibre patch cords will be available in the following lengths.

.5 Estimated Quantities

.1 The estimate of cable counts and lengths is given for bid purposed only; the final count and lengths will be provided in the integration phase of the project.

.2 Pricing should include single cable pricing and quantity discount pricing.

.6 Cable Connecting Hardware:

.1 Comply with Optical Fiber Connector Intermateability Standards (FOCIS) specifications of TIA/EIA-604-2, TIA/EIA-604-3-A, and TIA/EIA-604-12. Comply with TIA/EIA568-C.3.

.2 Quick-connect, simplex and duplex, Type LC connectors. Insertion loss not more than 0.75 dB.

2.3 Identification Products

.1 Comply with TIA/EIA-606-A and UL 969 for labeling materials, including label stocks, and inks used by label printers.

.2 Comply with requirements in Section 26 05 53 "Identification for Electrical Systems."

2.4 Source Quality Control

.1 Factory test UTP cables according to TIA/EIA-568-C.2.

.2 Factory test multimode optical fiber cables according to TIA/EIA-526-14-A and TIA/EIA-568-C.3.

.3 Provide test and inspection reports.

3 Execution

3.1 Field Quality Control

.1 Perform tests and inspections.

.1 Visually inspect UTP and optical fiber cable jacket materials for NRTL certification markings. Inspect cabling connections for compliance with TIA/EIA-568-C.1.

.2 Visually confirm Category 6a, marking of patch cables.

.3 Visually confirm Fiber patch coble marking.

.4 Visually inspect cable placement, and patch cords, and labeling of all components

3.2 Supply of Patch Cords

.1 Patch cords included in contract; all patch cords to be supply only, installation by the Owner.

.1 Supply two (2) copper patch cords per each telecom field outlet.

.1 7’-0” length at telecom room.

.2 10’-0” length at workstation.

.2 Supply one (1) duplex Fiber patch cord for every 12 strand fiber backbone cable.

.3 Supply one (1) duplex Fiber patch cord for every 2 strand fiber horizontal distribution cable.

.2 Patch cord colour to match colour of system cabling.

END OF SECTION

1 General

1.1 Related Requirements

.1 Section 26 05 00 – Common Work Results for Electrical.

.2 Section 27 05 00 – Common Work Results for Communications.

.3 Section 27 05 28 – Pathways for Communications Systems.

.4 Section 27 15 13 – Communications Copper Horizontal Cabling.

1.2 References

.1 American National Standards Institute (ANSI/TIA/EIA) 568 - Commercial Building Telecommunications Cabling Standard.

1.3 System Description

.1 The Intercom System shall consist of field and termination equipment necessary to provide a fully automated system to control and monitor authorized traffic in and out of controlled areas of the Works.

.2 The system shall be an IP based, non–proprietary, redundant solution. The system shall be a complete solution that is scalable from the existing scope of this project. This future expansion capacity shall not require significant replacement or upgrading of hardware or equipment provided as part of the initial solution and communicate using a PoE network.

.1 Power Source: Power over Ethernet (802.3af).

.2 Network Interface: 10 BASE-T / 100 BASE-TX Ethernet (RJ-45).

.3 Network Protocols: IPv4, IPv6, TCP, UDP, SIP, HTTP, HTTPS, MJPEG, RTSP, RTP, RTCP, IGMP, MLD, SMTP, DHCP, NTP, DNS.

.4 Bandwidth Usage:

.1 G.711: 64 kbps x 2 per video call.

.2 64 kbps per monitor.

.3 H.264: 24 kbps ~ 2,048 kbps.

.5 Communication: Hands-free (VOX), push-to-talk (simplex), or handset (full-duplex).

.6 Video Display: 7-inch color LCD.

.7 Camera: Type:

.1 1/3-inch color CMOS. 1.23 Megapixels.

.2 View Area at 0-degree camera angle mounted at 4 feet 11 inches (1500 mm) AFF: 2 feet 3 inches (700 mm) vertical x 3 feet 9 inch (1150 mm) horizontal at 19 inches (500 mm).

.8 Video Stream: ONVIF Profile S.

.9 Door Release: Programmable Form C dry contact, 24V AC/ DC, 1A (use RY-1824L for larger contact rating, which requires a 24V DC power supply) or use IXW-MA with 10 multipurpose relays.

.10 Wire Type: min CAT5e.

.11 Distance: Any station to Network Node: 330 feet (100 meters).

.3 The Intercom system shall be installed such that no delay, distortion, or degradation of signal is experienced on any sub or master station during normal operation of the system.

.4 Intercom system shall be interfaced with the telephone system and provide the ability rollover and to forward calls to internal and external phone numbers and cellular phones.

.5 The Intercom system shall be AODA/ADA compliant and provide Induction loops as well as raised lettering and Braille signage where required by code.

.6 As part of this project the successful Contractor will connect IP Call Stations, Master Stations, and servers / exchanges to the client supplied network.

.7 The Intercom System shall include but not be limited to the following elements:

.1 Supply and install new devices as per the specifications and drawings, but not limited to:

.1 Door Stations.

.2 Master Stations.

.3 Duress Stations.

.4 System Exchange / Servers.

.8 Intercom door stations shall be provided at identified locations in Facility and departmental entrances as shown on the drawings.

.1 Intercom door stations shall be provided with discrete call button at the primary and employee entrances.

.9 Monitoring Locations

.1 Provide master intercom stations at Main Reception, and the second floor east Workstation.

.2 System shall allow forwarding of multiple simultaneous calls from this location to telephone numbers. At a minimum, three simultaneous phone calls should be supported.

.10 Equipment shall not carry any logos or text which identifies the manufacturer or Contractor.

.11 Network

.1 IP addressing schemes used for the intercoms shall be coordinated, reviewed, and approved by the Owner and their network integrator to facility network programing and commissioning of the intercom system.

.12 Power

.1 All intercoms shall be powered via PoE.
.2 120 VAC power (on centralized UPS with generator back-up) shall be provided to all other system devices.

1.4 Submittals

.1 Product Data: Manufacturer's data sheets on each product to be used, including:

.1 Preparation instructions and recommendations.
.2 Storage and handling requirements and recommendations.
.3 Installation methods.

.2 Shop Drawings: Submit the following:

.1 Wiring Diagrams: Indicate wiring for each item of equipment and interconnections between items of equipment.
.2 Include manufacturer's names, model numbers, ratings, power requirements, equipment layout, device arrangement, complete wiring point-to-point diagrams, and conduit layouts.

.3 Installation and Operation Manuals:

.1 Submit manufacturer's installation and operation manual, including operation instructions and component wiring diagrams.
.2 Provide detailed information required for Owner to properly operate equipment.

.4 Warranty: Submit manufacturer's standard warranty.

1.5 Closeout Submittals

.1 Operation and Maintenance Data: submit operation and maintenance data for intercommunications systems for incorporation into manual.

.2 Include parts list using component identification numbers standard to electronics industry.

.3 Training records including training attendance list.

2 Products

2.1 General Requirements

.1 The contractor shall be fully responsible for the ultimate design and implementation of the system topology (physical and logical) best suited for the project, given identified and recognized physical infrastructure and constraints.

2.2 Manufacturers

.1 Entry Stations,

.1 Akuvox Corp.

.2 Area of refuge Devices,

.1 Viking Electronic,
.2 Honeywell.

2.3 Technical Requirements

.1 Door Stations

.1 Door Stations shall incorporate two-way voice communication.
.2 Shall be IP based and support PoE.
.3 Chassis and face plate shall be constructed of stainless steel.
.4 Shall contain one discrete anodized aluminium button labelled “Call”.
.5 Signage shall be constructed of cast metal with raised lettering and Braille.
.6 Speaker and microphone shall be protected by non-ferrous metal screen to provide a barrier against vandalism.
.7 All Stations shall be flush mounted where possible.
.8 Should back boxes be required, the unit shall be stainless steel in color.

.1 Product: Akuvox X916

.2 Master Stations

.1 Master stations shall be IP based and support PoE.
.2 Master stations shall have been developed specifically for Reception, Comm Centres and the Security Workstation.

.3 Shall contain an alphanumeric display that allows scrolling of messages and calls and volume control.
.4 Shall provide high quality speech from a distance up to 7 m (23 ft).
.5 Shall support “loudspeaker” functions and contain a “gooseneck” microphone.

.3 Duress Stations

.1 Duress Stations shall be IP based and support PoE.
.2 Shall have a vandal resistant design that is poke protected and fitted with special security screws.
.3 The intercom shall contain a large raised and/or illuminated emergency call button.
.4 Temperature range from –40°C to +65°C
.5 Signage shall be constructed of cast metal with raised lettering and Braille.

.4 Digital Intercom Servers / Exchange

.1 New Intercom Servers shall be rack mounted in equipment cabinets located in remote telecommunication rooms and Data Centres.
.2 All configuration programming for the system is the responsibility of contractor, all network requirements shall be coordinated with the Owner’s network integrator.
.3 The configuration of the subscriber features shall be programmed and stored on internal non-volatile memory– no batteries shall be required to maintain configuration programming. All software licensing required for programming the system shall be supplied and installed by Contractor.
.4 Updates, programming and patching of the system should be done while the system is on and functioning (no downtime).

.5 UPS and Emergency Power

.1 Provide Centralized UPS backup power for all intercom servers and associated equipment, distributed throughout communication rooms.

.6 Area of refuge

.1 AOR devices must be a complete and functional system with all major components from the same system
.2 Must meet;
.1 NFPA 72 (2022) §24.10
.2 IBC (2021) §1009.8
.3 ICC A117.1 (2017) Ch. 7

.4 ADAAG (2010) Ch. 7

.5 AOD and AODA requirements

.3 Built-in battery backup (24-hours non-active, 4-hours active),

.4 Support at least 10 call stations,

.5 Include a celluar gateway.

3 Execution

3.1 Examination

.1 Examine areas prior to installation, notify Consultant of conditions that would adversely affect installation or subsequent use.

.2 Do not begin installation until unacceptable conditions are corrected.

.3 Product

3.2 Preparation

.1 Verify the following compliance before starting installation.

.1 Keep the intercom wires at least 1 foot (300 mm) away from strong electrical wiring (AC 100-240 V) including, in particular, wiring for inverter electrical appliances. Noise and malfunction could result.

.2 If a strong light shines on the main unit screen, the picture may turn white or only silhouettes will be visible.

.3 Other manufacturer's devices (such as sensor, detectors, door releases) used with this system, comply with the manufacturer's installation requirements.

3.3 Installation

.1 Install integrated security and communication system in accordance with manufacturer's instructions at locations indicated on the Drawings.

.2 Mount equipment plumb, level, square, and secure. For video entrance stations and video door stations, comply with manufacturer's design requirements to provide optimum picture quality of station monitoring.

3.4 Set-up and Adjusting

.1 Adjust integrated security and communication system for proper operation in accordance with manufacturer's instructions.

3.5 Programing

.1 All associated cost to program the system and ensure all configuration and naming conventions are reviewed and approved by the owner shall be included.

.2 All associated cost related to database programming and coordination wit the Owner's network integrator shall be included.

.3 The intercom contractor shall attend pre-installation meetings as required with the construction manager and/or the owner and their representatives to identify the specifics of the system programming as required.

.4 The contractor is to document all decided on software configuration parameters and submit for approval.

.5 Once approved the contractor shall implement and test all system programming required.

.6 All approved and commissioned software configuration parameters shall be submitted in final revisions shop drawings and included in as built documents.

.7 If building door numbers are not final, they are to be changed prior to final system programming and handover. The contractor is responsible to ensure final programming configuration and system documentation match the final as built door numbers.

3.6 Commissioning

.1 System Acceptance Tests (SAT)

.1 Inspection and functional testing shall be conducted on the basis of preapproved SAT procedures.

.2 The tests are to validate infrastructure, performance, and functional requirements and/or specifications compliance of the intercom system by the Owner or Owners Representative.

.3 On the date of Site Acceptance, the Contractor shall be present to perform testing. The Owner, and the Owners Representative shall be present to witness all of the tests performed including tests performed to confirm system installation and functionality.

.2 As part of SAT the contractor shall provide the following checklists to verify Intercom System Tests:

.1 Intercom Station Checklist.

.2 Intercom Exchange Checklist.

.3 Intercom Commissioning List.

.3 As part of SAT the contractor shall provide the following checklists to verify Intercom System Integrated Functionality Tests

.1 Intercom System – Phone System.
.2 Intercom System – Card Access System.
.3 Intercom System – Video Surveillance System.

3.7 Training

.1 The Training Phase of the project begins upon approval of the Shop Drawings. This phase is made up of two (2) milestones.

.1 Training Course Outline Approval
.2 Operator/Administrative Training

.2 Prepare and submit a digital copy of the training course outlines for review. As a minimum, this outline shall list all material to be covered, the time required to cover it, the organization and personnel conducting the training, any training aids or equipment to be used for demonstration.

.3 Course outlines shall be approved when all comments by the Owner and/or Owners Representative have been satisfactorily addressed.

.4 Training shall be provided on site and conducted by both the Contractor and/or system manufacturer. All associated cost shall be carried by the Contractor under his cost.

.5 Training shall consist of the following minimum requirements:

.1 The Contractor shall provide technical system training by the contractor and/or the system manufacturer’s training personnel to outline the complete operation of the system to relevant staff and operators.
.2 Administrators training conducted by the contractor and/or the system manufacturer’s training personnel to outline the infrastructure, programming, and maintenance requirements of the system.

3.8 Documentation

.1 Refer to Section 27 05 00.

END OF SECTION

1 General

1.1 Section includes

.1 Cuts and Removals

.1 Curb Cutting.

.2 Removal of curb and gutter.

.3 Removal of concrete barrier.

.4 Removal of concrete sidewalks.

.5 Removal of fencing (all types).

.6 Removal of asphalt pavement.

.7 Removal of driveway pavement.

.2 Subgrade and Fine Grading

.1 Fine grading

.2 Granular B, Type I (Modified).

.3 Granular A subgrade for roadways and high-density residential driveways.

.4 Granular A subgrade for sidewalks and concrete walkways.

.3 Hot Mix Asphalt

.1 HL 8 - binder (PGAC 58-28).

.2 Tack coat.

.3 HL 3 - surface (PGAC 58-28).

.4 Pavement Joint Treatment.

.5 Temporary asphalt curb at catch basins, HL 3 (PGAC 58-28).

.6 HL 8 asphalt infill (300 mm width) for concrete curbs completed after base asphalt.

.4 Concrete Curbs, Sidewalks, walkways, refuse area ~~and~~ other concrete appurtenances

.1 Concrete curb and gutter, including curb terminations (machine placed and hand placed).

.2 Concrete Barrier curb.

.3 Concrete sidewalk.

.4 Concrete walkways and building entrances.

.5 Concrete refuse area.

.6 Tactile walking surface indicator plates (760 mm x 610 mm).

.5 Pavement Marking

.1 Solid white (10 cm).

.2 Symbols – Handicap parking.

.3 Symbols – EV parking.

.6 Surface Restoration

.1 Topsoil and sod.

1.2 Reference standards

.1 Contract Drawings:

.1 SG-1

.2 SS-1

.2 Town of Whitby Standard Drawings:

.1 Standard Drawing 208

.2 Standard Drawing 404.40

.3 Standard Drawing 405.50

.3 Regional Municipality of Durham Core Specifications (RMDCS):

.1 Section 02700

.2 Section 02749

.3 Section 02905

.4 Regional Municipality of Durham Standard Drawings (RMDSD):

.1 S-301.010

.5 Ontario Provincial Standard Specifications (OPSS/OPSS.MUNI/OPSS.PROV):

.1 OPSS.PROV 308

.2 OPSS.MUNI 102

.3 OPSS.MUNI 310

.4 OPSS.MUNI 311

.5 OPSS.MUNI 312
.6 OPSS.MUNI 314
.7 OPSS.MUNI 351
.8 OPSS.MUNI 353
.9 OPSS.MUNI 501
.10 OPSS.MUNI 506
.11 OPSS.MUNI 510
.12 OPSS.MUNI 710
.13 OPSS.MUNI 802
.14 OPSS.MUNI 803
.15 OPSS.MUNI 804
~~.15~~.16 OPSS.PROV 905
~~.16~~.17 OPSS.MUNI 1001
~~.17~~.18 OPSS.MUNI 1003
~~.18~~.19 OPSS.MUNI 1010
~~.19~~.20 OPSS.MUNI 1101
~~.20~~.21 OPSS.MUNI 1103
~~.21~~.22 OPSS.MUNI 1150
~~.22~~.23 OPSS.MUNI 1350
~~.23~~.24 OPSS.MUNI 1716

.6 Ontario Provincial Standard Drawings (OPSD):

.1 OPSD 310.010
.2 OPSD 310.020
.3 OPSD 310.030
.4 OPSD 310.033
.5 OPSD 310.039
.6 OPSD 310.040
.7 OPSD 310.050
.8 OPSD 351.010
.9 OPSD 600.010
.10 OPSD 600.040

.11 OSPD 600.110

.12 OPSD 601.010

.13 OPSD 608.010

1.3 Submittals

.1 The Submit manufacturer's product data for granular materials, hot mix asphalt, concrete curb, concrete sidewalk, concrete walkways and appurtenances, pavement markings, and other roadworks.

1.4 Quality assurance

.1 Perform work of this section using personnel qualified for all associated roadworks.

.2 Should the contractor not complete the base asphalt in one day of paving they are required to leave a clean traverse edge to match into the following day of pacing as a part of the cost for placing base asphalt.

.3 Should the contractor not complete the surface asphalt in one day of paving they are required to provide a grinded lap joint leave a clean traverse edge to match into the following day of paving as a part of the cost for paving surface asphalt.

.4 Should surface asphalt be placed on a different day the base asphalt is placed the contractor is required to ensure the base surface is cleaned free of debris and apply tack coat prior to placing the surface course asphalt. This shall be included in the surface asphalt price.

.5 All sodding and seeding restoration must be performed by an experienced specialist Subcontractor using workers skilled in this trade.

.6 Granular materials shall meet the gradation, physical property, and quality requirements of OPSS-314 and OPSS-1010.

.7 Provide PGAC QA samples for every 500 tonnes of hot mix (minimum 1L per grade).

.8 All concrete shall be tested for slump, air, temperature, and strength as required by OPSS.MUNI 1350.

.9 The removal and replacement of defective materials, asphalt, sidewalk bays, or workmanship shall be executed at no additional cost to the Owner.

.10 Disturbed or damaged portions of existing elements not designated for removal shall be corrected or repaired at the Contractor's expense.

1.5 Site Conditions

.1 Ensure asphalt is clean prior to placement of subsequent lifts; cleaning of base asphalt is part of the scope.

.2 The Contractor shall not pave on soaked/wet granular or road base. Wet is considered by excess moisture as determined by the Geotechnical Engineer. The Contractor is responsible to coordinate paving based on favourable weather conditions.

2 Products

2.1 Acceptable manufacturers and approved materials

.1 Tactile Walking Surface Indicators material shall be as per OPSD 310.039 and compliant with the Town of Whitby Standard Drawing 208.

.2 Paint materials shall be water-borne products selected from the Ministry of Transportation (MTO) Designated Sources for Materials (DSM) List number 5.85.40 conforming to OPSS.MUNI 1716. No organic solvent paints are permitted.

2.2 General system requirements

.1 Tack coat shall be applied using self-propelled or tow-along pressure distributors equipped with a volume-determining device sensitive to the nearest five litres. Hand-held wands are permitted for small, irregular areas only.

.2 Provide expansion joints where sidewalk or concrete walkway abuts other concrete structures, walks, building foundations, curbs, utility poles, hydrants, maintenance holes, and any other location specified by the Contract Administrator.

.3 All cold joints, grinded lap joints, and daily match joints shall be clean, properly constructed, and treated as per RMDSD S-301.010 as part of the HMA placement.

.4 Base course asphalt left exposed prior to surface course placement shall be fully cleaned and receive a full application of tack coat at no extra cost to the Owner.

2.3 Materials

.1 Granular Materials:

.1 Granular A conforming to OPSS.MUNI 1010.

.2 Granular B, Type I (Modified) conforming to OPSS.MUNI 1010.

.2 Hot Mix Asphalt:

.1 HL 8 binder (PGAC 58-28)

.2 Tack Coat in accordance with OPSS.MUNI 310 and OPSS.MUNI 1103.

.3 HL 3 (PGAC 58-28).

.4 Pavement Joint treatment as per S-301.010

.5 Temporary Asphalt Curb - HL 3 (PGAC 58-28).

.6 HL 8 binder for 300 mm wide asphalt infill behind curbs completed after base asphalt placement.

.3 Concrete Curbs:

.1 Concrete mix design and materials conforming to OPSS.MUNI 353 and 1350.

.2 Concrete Curb and Gutter as per OPSD 600.040.

.3 Concrete Barrier Curb as per OPSD 601.110.

.4 Curb Terminations as per OPSD 608.010.

.5 Monolithic curb returns at entrance curbs where new barrier curb transitions to existing curb and gutter.

.4 Concrete Sidewalks:

.1 Concrete mix design and materials conforming to OPSS.MUNI 351 and 1350.

.2 Concrete sidewalk and granular base as per OPSD 310.010 and 310.020.

.3 Tactile walking surface indicator plates (760 mm x 610 mm) as per Town of Whitby Standard Drawing 208 and OPSD 310.039.

.5 Concrete Walkways and Building Entrances:

.1 Concrete mix design and materials conforming to OPSS.MUNI 351 and 1350.

.2 Concrete walkways and granular base as per OPSD 310.010 and 310.020.

.3 Frost slab assembly: 150 mm concrete over 300 mm depth of Granular A base, complete with key joints or dowels.

.6 Concrete Refuse Area:

~~.1~~ Concrete mix design and materials conforming to OPSS.MUNI 351 and 1350.

~~.2~~.1 Granular A base, a minimum of 150mm depth, compacted to 100% std. proctor density. ~~-~~

.2 Reinforcement: Welded wire mesh (150 mm x 150 mm weave, MW 18.7 x MW 18.7).

~~.3~~

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.7 Pavement Markings:

.1 Water-borne traffic paint conforming to OPSS.MUNI 1716, selected from the MTO Designated Sources for Materials (DSM) List number 5.85.40.

.8 Topsoil and Sod:

.1 Topsoil conforming to OPSS.MUNI 802.

.2 Sod conforming to OPSS.MUNI 803.

3 Execution

3.1 Examination

.1 Verify existing profiles, grades, dimensions, prior to the commencement of pavement milling, removals, or subgrade preparation.

.2 Verify moisture conditions of placed granular base before placing base course lifts.

.3 Verify base asphalt cleanliness prior to placing surface course lifts.

3.2 Preparation

.1 Perform pre-marking layout of all lanes and parking symbols for inspection by the Contract Administrator prior to applying permanent pavement markings.

.2 Thoroughly sweep and clean base asphalt free of debris and apply tack coat prior to surface paving.

3.3 Removals

.1 Remove all existing fencing within the property line.

.2 Complete all proposed curb cuts to the length and profile shown on the Contract Drawings.

.3 Excavate and remove concrete curb and gutter and concrete barrier curb as shown on the contract drawings.

.4 Excavate, sawcut, and remove concrete sidewalks, as shown on contract drawings.

.5 Remove asphalt and driveway pavements to the limits shown on the Contract Drawings.

3.4 Subgrade and Fine Grading

.1 Perform fine grading of walkway, sideways, and high residential driveway subgrades prior to placing Granular materials.

.2 Perform fine grading of all softscape and landscaping areas in preparation for topsoil placement.

.3 Supply, place, grade, and compact Granular A for road base, driveways, sidewalks, walkways, and refuse areas. Complete and submit Forms 1010-D and 1010-E for all QC testing performed.

.4 Supply, place, grade, and compact Granular B, Type I (Modified) for road base and driveways.

.5 Maintain subgrade and granular base in a smooth, well-drained, unsaturated condition prior to paving. Regrade and re-compact saturated areas at Contractor expense.

3.5 Hot Mix Asphalt and Tack Coat

.1 Asphalt depths shall be as per Contract Drawings.

.2 Supply, place, and grade HL8 base coat for Ontario Street Restoration. Construct a clean transverse joint at the end of each daily paving pass.

.3 Supply, place, and grade Heavy-duty HL8 base coat for drive aisles and Light duty HL8 base coat for parking spots in accordance with contract drawings.

.4 Construct 1 m long hand-placed temporary asphalt curbs behind new catch basins using HL 3 as per OPSD 601.010.

.5 Apply tack coat to milled pavement and previously paved surfaces prior to new paving in accordance with OPSS.MUNI 310 and OPSS.PROV 308.

.6 Adjust all maintenance holes, catch basin maintenance holes, and catch basins to final grade.

.7 Place surface hot mix asphalt (HL3) to the thicknesses and alignments shown on Contract Drawings. Construct a grinded lap joint for clean transverse matching on multi-day paving operations.

.8 Road restoration to be completed as per T.O.W. Standard Drawing 404.40, and 404.50.

.9 Treat all pavement joints according to RMDSD S-301.010. Utilize small cold planers as required.

.10 Where concrete curbs are installed after base asphalt placement, sawcut/trim base asphalt clean and place a 300 mm width HL8 asphalt infill to bridge and restore the base course gap.

3.6 Concrete Curbs, Sidewalks, walkways, and other concrete appurtenances

.1 Concrete Curb and gutter:

.1 Construct concrete curb and gutter according to RMDCS Section 600.040.

.2 Install concrete barrier curb as per OPSD 601.110.

.3 Install curb terminations in accordance with OPSD 608.010.

.4 Cut in and place monolithic curb returns at entrance curbs where new barrier curb transitions to existing curb and gutter, including all applicable surface and subgrade restoration as required.

.2 Concrete Sidewalks:

.1 Construct sidewalks according to OPSD 310.010 and 310.020.

.2 Construct contraction joints at maximum 2.0 m intervals and expansion joints at maximum 6.0 m intervals and where necessary.

.3 Isolate utility structures and poles within the sidewalk footprint according to OPSD 310.040.

.4 Increase the sidewalk width to 1.8 m where it is constructed adjacent to the curb and gutter.

.5 Place welded wire mesh reinforcement (150 mm x 150 mm weave, MW 18.7 x MW 18.7) for sidewalks abutting high-density residential driveways.

.6 Construct sidewalk ramps according to OPSD 310.033.

.7 Supply and install tactile walking surface indicator plates within the sidewalk ramps in accordance with Town of Whitby Standard Drawing 208.

.3 Concrete Walkways and Building Entrances:

.1 Construct concrete walkways and building entrances in accordance with RMDCS Section 02700 and OPSS.MUNI 351.

.2 Where walkways abut building entrances, construct a frost slab complete with key joint or dowels. Supply 150 mm concrete with a 300 mm depth of Granular A base graded to lead into the road makeup.

.3 Install Deck-O-Foam expansion joint filler along the building perimeter between the building foundation and abutting concrete walkways.

.4 Apply Formex Seal caulking sealant around the building perimeter to completely seal the concrete walkway interface against the building foundation.

.4 Concrete Refuse Area:

.1 Construct concrete refuse areas according to RMDCS Section 02700 and OPSS.MUNI 351.

.2 Shall be formed and poured flush to the abutting asphalt with a minimum 2% fall to drain towards the asphalt.

.3 Concrete mix design and materials conforming to OPSS.MUNI 351 and 1350.

.4 Concrete thickness of a minimum ~~15~~200mm

.5 Granular A base, a minimum of 150mm depth, compacted to 100% std. proctor density.

.6 Position the wire mesh at the mid-depth of the concrete slab, maintaining equal concrete cover above and below the reinforcement

.7 Place welded wire mesh reinforcement (150 mm x 150 mm weave, MW 18.7 x MW 18.7) within concrete refuse pad

.8 Support the wire mesh on approved concrete or plastic chairs, bolsters, or other suitable reinforcement supports at sufficient spacing to prevent sagging or displacement during concrete placement.

.9 Do not place the wire mesh directly on the granular base or pull it up through freshly placed concrete during pouring.

~~.1~~.10 Lap adjoining sheets by a minimum of one full mesh spacing plus 50 mm

~~.2 Place welded wire mesh reinforcement (150 mm x 150 mm weave, MW 18.7 x MW 18.7) below refuse area.~~

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3.7 Pavement Markings

.1 Remove all temporary tape line markings prior to permanent line application.

.2 Apply permanent water-borne traffic paint markings (solid white lanes, handicap symbols, and EV parking symbols) within two Working Days of completing the final lift of HMA.

3.8 Inspection and Testing

.1 Coordinate with the Contract Administrator for Quality Assurance grade checks to verify that base and subbase tolerances meet the requirements of OPSS.MUNI 314.

.2 Inspect prepared base for excess moisture prior to paving. Paving on soaked or wet granular base is strictly prohibited. Moisture levels and paving readiness must be approved by the Geotechnical Engineer.

.3 Inspect exposed base asphalt prior to surface course application to verify complete removal of dirt, sand, mud, and debris.

.4 Provide the Contract Administrator with access and opportunities to verify the gross mass and tare mass of delivery trucks as requested.

.5 When required by the Contract Administrator, provide samples for testing of diluted asphalt emulsion tack coats.

.6 Coordinate and facilitate any required asphalt coring, compaction testing, or referee laboratory testing programs to resolve disputes regarding Hot Mix Asphalt quality.

.7 Coordinate and facilitate any required concrete testing as required.

.8 Concrete curb and gutter shall meet the following expectations:

.1 Class of Concrete – 32 MPa at 28 days

.2 Maximum Slump – 60mm ± 20mm

.3 Air Content – 7.0 ± 1.5%

.9 Concrete for concrete sidewalk and refuse area shall meet the following expectations:

.1 Class of Concrete – 32 MPa at 28 days

.2 Maximum Slump – 90mm ± 20mm

.3 Air Content – 7.0 ± 1.5%

3.9 Clean up

.1 Removals

.1 Dispose of all removed materials off-site at a location arranged for by the Contractor.

.2 Topsoil and Sod Restoration:

.1 Place approved topsoil and sod over disturbed areas, maintaining moist conditions until root systems have successfully taken root in accordance with OPSS.MUNI 501, 802, 803, and 804.

.3 Site Restoration and Final Clean up:

.1 Restore all other areas disturbed by roadwork operations to match pre-construction conditions or specified finish grades.

.2 Dispose of all waste, debris, and excess materials off-site upon completion of paving, concrete, and restoration activities.

.3 All excavated spoils shall be removed offsite unless otherwise directed by the Contract Administrator.

End of section

1 General

1.1 Section includes

.1 Fixed stainless steel protective bollards at electric vehicle (EV) charging stations, including reflective marking, sleeves, anchorage, accessories and associated installation.

1.2 Related sections

.1 Section 03 30 00 - Cast-in-Place Concrete.

.2 Division 26 - Electrical: EV charging equipment, conduit, grounding and bonding, and related electrical work.

1.3 Reference standards

.1 Town of Whitby Engineering Services Standard Drawing 413.30 - EV Pad Installation Details, May 2025.

.2 ASTM A240/A240M - Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications, as applicable.

.3 ASTM A554 - Standard Specification for Welded Stainless Steel Mechanical Tubing, as applicable.

.4 ASTM D3350 - Standard Specification for Polyethylene Plastics Pipe and Fittings Materials, as applicable to HDPE sleeves.

1.4 Performance requirements

.1 Provide fixed bollards and associated components conforming to the configuration, dimensions and locations indicated on the Site Plan and Town of Whitby Standard Drawing 413.30.

.2 Coordinate bollard positioning with EV charging equipment, vehicle parking geometry, accessible routes, operating clearances, service doors, charging cables and electrical infrastructure. Do not obstruct required equipment access or accessibility clearances.

.3 Bollards are intended to provide physical protection of EV charging equipment.

.4 Where anchorage, footing or concrete pad modifications are proposed, obtain review by the electrical engineer or record and structural engineer of record on the project. Do not alter the indicated assembly without authorization.

1.5 Submittals

.1 See Section 01 33 00 - Submittal Procedures, for submittal procedures.

.2 Product Data: Submit material information for stainless steel bollards, finish, reflective markings, HDPE sleeves, fasteners and related accessories.

.3 Shop Drawings: Indicate bollard quantities, locations, diameters, heights, spacing, clearances, sleeve configuration, attachment details, and interface with concrete pad and EV charging equipment. Identify any proposed departures from Drawing 413.30.

.4 Provide samples or photographs identifying the proposed stainless steel finish and reflective tape for review, when requested.

1.6 Quality assurance

.1 Provide fabricated stainless steel products with consistent dimensions, workmanship and finish, free from sharp edges and exposed burrs.

.2 Installation to be carried out by personnel experienced in installing exterior metal bollards and associated anchors.

1.7 Delivery, storage, and handling

.1 Inspect materials upon delivery for damage, missing components and defects.

.2 Store components above ground and protect from contamination, mechanical damage, and contact with dissimilar metals and corrosive materials.

.3 Protect finished surfaces and reflective materials during transportation, storage and installation.

1.8 Sequencing

.1 Confirm bollard positions and anchorage requirements with the concrete and electrical trades before placement of concrete, drilling or installation.

.2 Arrange for installation to prevent conflict with embedded conduit, reinforcement, electrical grounding/bonding provisions and equipment mounting hardware.

1.9 Warranty

.1 Provide manufacturer’s standard written warranties, where available, for bollards, reflective materials and associated accessories.

2 Products

2.1 Bollard assembly

.1 Type: Fixed, stainless steel protective bollards.

.2 Bollard: Stainless steel tube, nominal 127 mm diameter. Submit shop drawings with dimensions and clearances prior to final fabrication for approval.

.3 Height: Approximately 900 mm above finished concrete pad. Coordinate overall tube length, sleeve and embedded/anchored portion with shop drawings.

.4 Provide smoothly finished rounded or domed top without exposed sharp edges. Provide secure, weather-resistant closure of hollow tubing. Bollard to include stainless steel bolts, anchorage and associated accessories.

.5 Sleeves: Provide HDPE plastic bollard sleeves where indicated, nominal 3.6 mm wall thickness. Coordinate sleeve size and interface with bollard.

.6 Reflective markings: Provide exterior-grade reflective tape bands on sleeve at locations indicated in Drawing 413.30. Securely adhere tape to cleaned HDPE surfaces.

2.2 Finish

.1 Stainless steel: Uniform mill or brushed finish, free of staining, dents, grinding marks, weld splatter and discoloration. Confirm finish appearance through product data or samples.

.2 Expose no unfinished sharp cut edges. Finish welded and cut surfaces to match adjacent visible material.

.3 Reflective tape: Durable, weather-resistant, suitable for outdoor vehicular environments; colour and placement to conform to the referenced detail and approved submittal.

2.3 Concrete pad and anchorage interface

.1 Concrete pad, reinforcement, base and associated electrical equipment foundations shall be provided under the applicable concrete and electrical sections, as assigned in the Contract Documents.

.2 Provide surface-mounted stainless steel bollards anchored to the reinforced concrete EV charging pad using stainless steel anchorage hardware, in accordance with Town of Whitby Standard Drawing 413.30 and reviewed shop drawings. Anchorage design and suitability of the supporting concrete pad, including pad thickness, reinforcement, anchor embedment, spacing, edge distances and anticipated vehicle-impact loads, shall be verified by the Structural Engineer. Provide separate concrete footings where requested by the Structural Engineer in review of shop drawings.

.3 Do not drill, cut or otherwise damage reinforcing steel, embedded conduit or equipment anchors. Refer conflicts to the Consultant before proceeding.

3 Execution

3.1 Examination

.1 Verify existing and proposed dimensions, finished grade, concrete pad dimensions, condition and suitability for the installation.

.2 Check equipment mounting layout, charger swing/operation, accessible route and electrical components for potential interference with bollards.

.3 Notify the Consultant of conflicts, unsuitable supporting conditions or dimensional discrepancies before proceeding.

3.2 Preparation

.1 Confirm bollard locations on site with the approved Contract Drawings and shop drawings before drilling or placing anchors.

.2 Protect adjacent concrete surfaces, EV charger components, cabling, wiring and other finished work.

3.3 Installation

.1 Install bollards plumb, true to line, at spacing and offsets shown on Contract Drawings.

.2 Surface-mount and securely anchor bollards to the reinforced concrete EV charging pad using stainless steel anchorage hardware, in accordance with Town of Whitby Standard Drawing 413.30, reviewed shop drawings and applicable anchor manufacturer instructions. Install separate concrete footings where directed by the Structural Engineer.

.3 Install HDPE sleeves with reflective bands neatly, level and consistent in position. Ensure sleeves are fitted without movement or exposed sharp edges.

.4 Maintain clearances to charging equipment, parking stalls and accessible circulation routes as required by the Contract Drawings and coordination with the Electrical Consultant.

.5 Do not modify electrical installations, determine electrical bonding requirements or certify electrical safety under this section. Refer such matters to the electrical contractor and Electrical Consultant.

3.4 Cleaning

.1 Remove debris, filings, adhesive residue and surface contaminants from installed bollards and surrounding concrete.

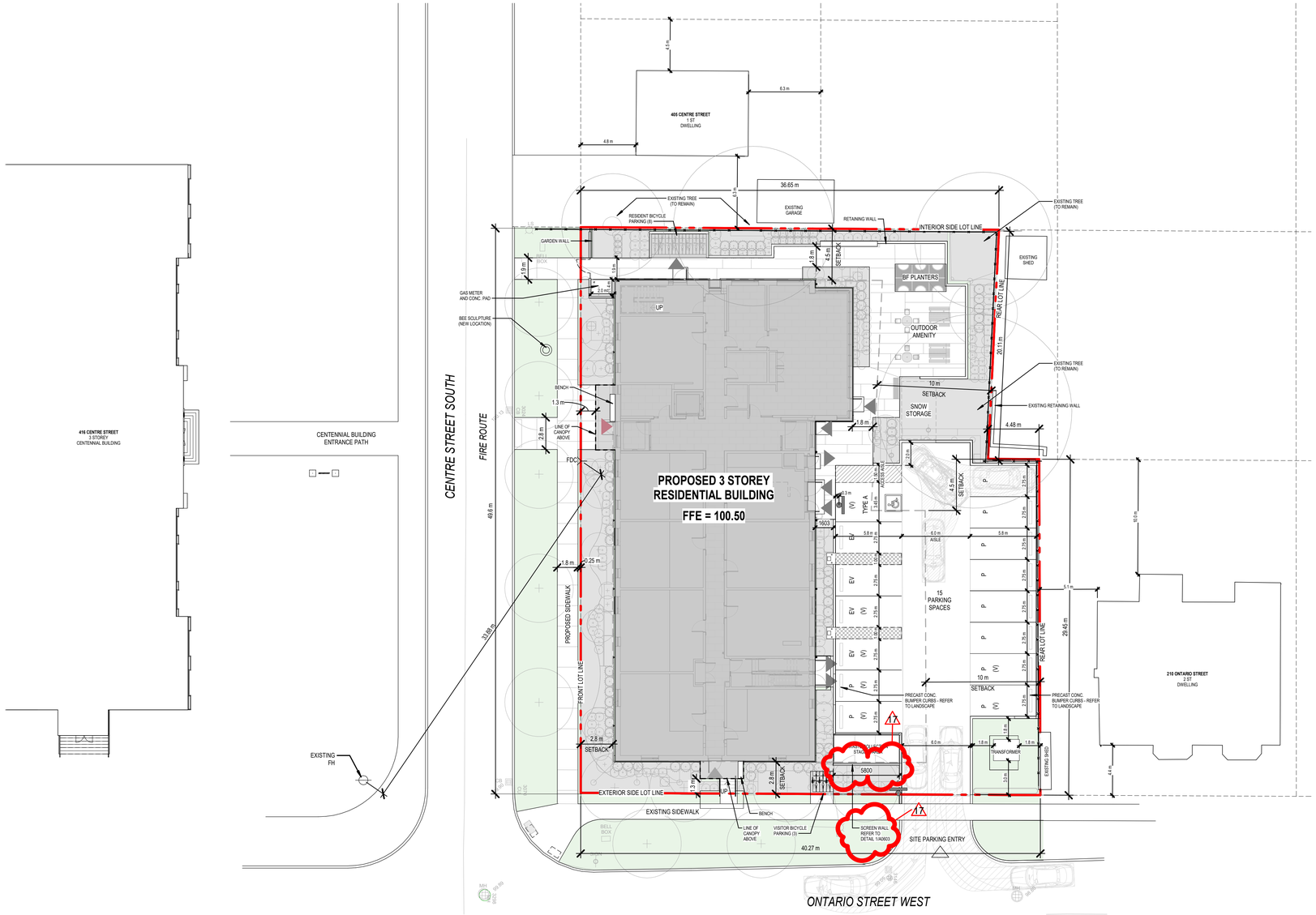
.2 Clean stainless steel using non-abrasive, manufacturer-recommended materials; do not use carbon steel brushes or tools that may contaminate the finish.

3.5 Protection

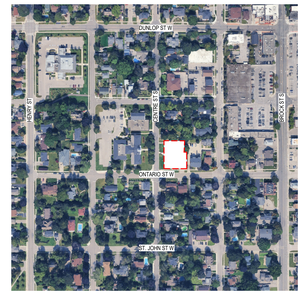
.1 Protect completed work from damage by vehicles, equipment and subsequent trades until Substantial Completion.

.2 Repair or replace damaged, loose, stained or misaligned bollards and defective reflective bands before Substantial Completion.

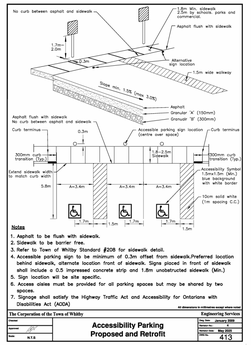
End of Section

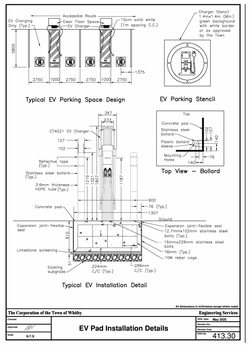


1 GROUND FLOOR SITE PLAN
A0601
1 : 200



Key Plan
1 : 5000





SITE PLAN LEGEND

- PROPERTY LINE
- SETBACK LINE
- ACCESSIBLE PARKING SPACE PER TOWN OF WHITBY ENGINEERING STANDARD 413
- TYPE A — ACCESSIBLE PARKING
- P — STANDARD PARKING SPACE
- EV — ELECTRIC VEHICLE PARKING SPACE PER TOWN OF WHITBY ENGINEERING STANDARD 413.30
- (V) - DESIGNATED VISITOR PARKING SPACE
- NO PARKING ZONE
- FIRE HYDRANT
- PRINCIPAL BUILDING ENTRANCE
- BUILDING ENTRANCE / EXIT
- PARKING ENTRANCE
- EXISTING TREE
- OPTIMAL TREE PROTECTION ZONE - REFER TO TREE PRESERVATION AND PROTECTION PLAN
- TACTILE SURFACE INDICATOR
- PROPOSED FENCE
- MH — EXISTING MANHOLE - REFER TO CIVIL
- CB — EXISTING CATCHBASIN - REFER TO CIVIL
- SOFT LANDSCAPING - REFER TO LANDSCAPE
- HARD LANDSCAPING - REFER TO LANDSCAPE
- PAVED AREA - REFER TO CIVIL
- A / B — TRAFFIC SIGN - REFER TO TRANSPORTATION AND CIVIL; POLE-MOUNTED ACCESSIBLE PARKING SIGN; POLE-MOUNTED STOP SIGN
- FDC — FIRE HOSE CONNECTION

The Regional Municipality of Durham
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MontgomerySisam

PROJECT NORTH / TRUE NORTH

#	DATE:	REVISION:	BY:
17	2026-10-30	ISSUED FOR TENDER - ADDENDUM 3	MSA
14	2026-09-14	ISSUED FOR SITE PLAN APPROVAL - REV.2	MSA
13	2026-09-16	ISSUED FOR TENDER	MSA
12	2026-05-15	ISSUED FOR SITE PLAN APPROVAL - REV.1	MSA
11	2026-03-27	ISSUED FOR 100% CONSTRUCTION DOCUMENTS	MSA
10	2026-02-20	ISSUED FOR 90% CD & CLASS A COSTING	MSA
9	2026-01-30	ISSUED FOR 75% CONSTRUCTION DOCUMENTS	MSA
8	2026-01-30	ISSUED FOR MINOR VARIANCE	MSA
7	2026-01-30	ISSUED FOR BUILDING PERMIT	MSA
6	2026-01-30	ISSUED FOR SITE SERVICES PERMIT	MSA
5	2025-12-17	ISSUED FOR 30% CONSTRUCTION DOCUMENTS	MSA
4	2025-11-21	ISSUED FOR SITE PLAN APPROVAL	MSA
3	2025-08-26	ISSUED FOR HERITAGE PERMIT	MSA
2	2025-07-14	ISSUED FOR STAGE 2 PRE-CONSULTATION	MSA
1	2025-06-20	ISSUED FOR 100% DD & CLASS B COSTING	MSA

REVISIONS

REDEVELOPMENT - 409 CENTRE ST S

25008

409 Centre St. South,
Whitby, ON L1N 4W3

Town of Whitby File Number: SP-26-25

SITE PLAN - GROUND PLAN

SCALE:	As indicated
DRAWN BY:	VN/MT
REVIEWED BY:	EM
JOB NUMBER:	25008
PLOT DATE:	11/10/25

DRAWING NUMBER:
A0601

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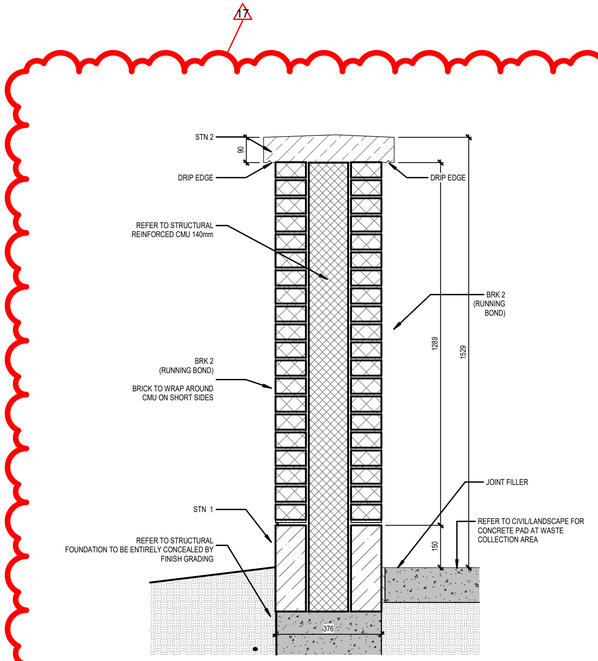
MontgomerySisam

PROJECT NORTH

TRUE NORTH



1 A0603 SCREEN WALL AT WAST COLLECTION AREA

1 : 10

#	DATE:	REVISION:	BY:
17	2026-10-30	ISSUED FOR TENDER - ADDENDUM 3	MSA

REVISIONS

REDEVELOPMENT - 409 CENTRE ST S

25008

409 Centre St. South,
Whitby, ON L1N 4W3

Town of Whitby File Number: SP-26-25

SITE PLAN DETAILS

SCALE:	1 : 10
DRAWN BY:	MT
REVIEWED BY:	EM
JOB NUMBER:	25008
PLOT DATE:	11/10/25

DRAWING NUMBER:

A0603

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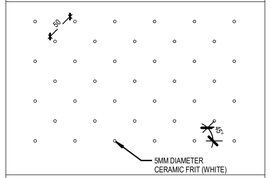
2 A2230 EAST ELEVATION

1 : 100

1 A2230 WEST (CENTRE ST) ELEVATION

1 : 100

BIRD FRIENDLY GLAZING PATTERN



APPLY BIRD FRIENDLY GLAZING PATTERN TO SURFACE 1 ON ALL EXTERIOR GLAZING.

MATERIAL SCHEDULE

BRK 1	BRICK MASONRY
BRK 2	BRICK MASONRY
MTL 1	METAL PANEL
MTL 2	STANDING SEAM METAL ROOF
STN 1	STONE BASE
STN 2	STONE SILL
STN 3	STONE CLADDING (LARGE FORMAT)
STN 4	STONE VENEER

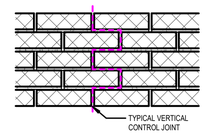
MATAL FINISHES

FINISH A	COLOUR MATCH ANODIZED ALUMINUM
FINISH B	COLOUR MATCH GALVALUM FINISH

ELEVATION LEGEND

RWL	RAINWATER LEADER
LF	EXTERIOR WALL-MOUNTED LIGHT FIXTURE - REFER TO ELEC

TYPICAL VERTICAL JOINT DETAIL



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REVISIONS

REDEVELOPMENT - 409 CENTRE ST S

25008

409 Centre St. South,
Whitby, ON L1N 4W3

Town of Whitby File Number: SP-26-25

SITE/BUILDING ELEVATIONS

SCALE:	As indicated
DRAWN BY:	MB
REVIEWED BY:	EM
JOB NUMBER:	25008
PLOT DATE:	2025-11-21

DRAWING NUMBER:

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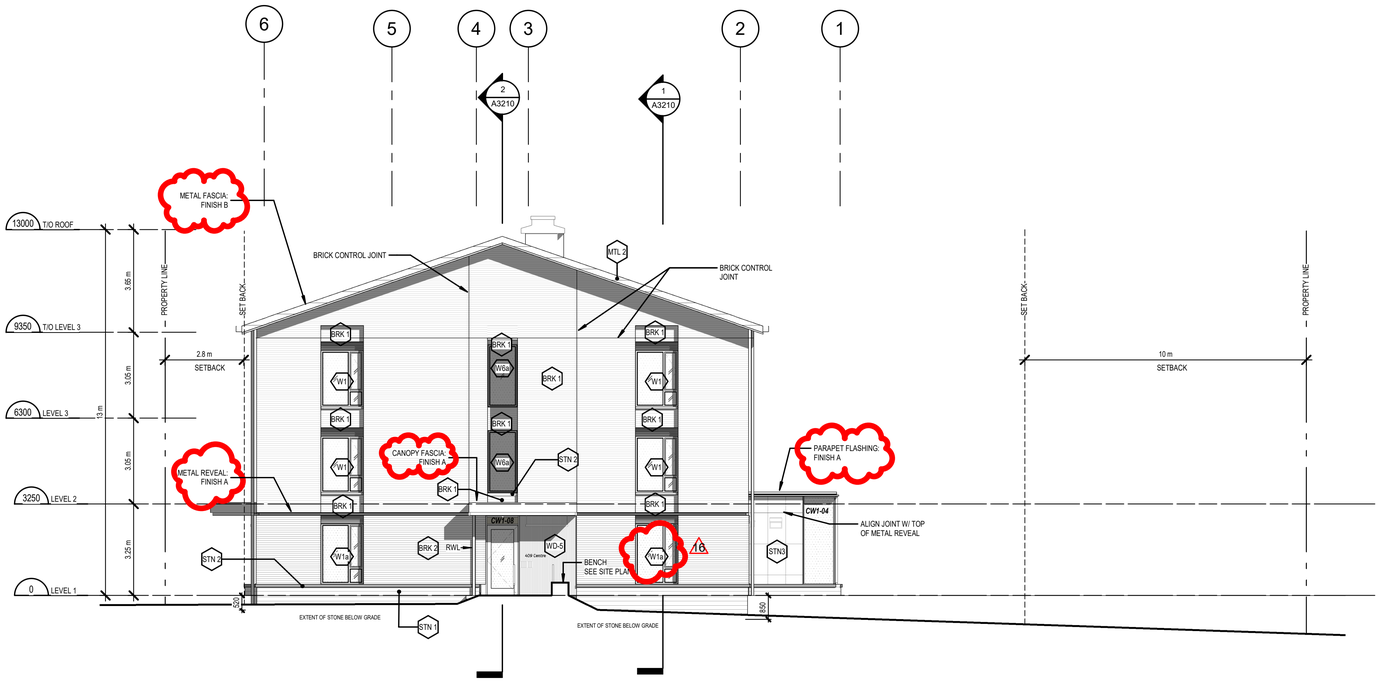

2 A2231 SOUTH ELEVATION (ONTARIO ST W)

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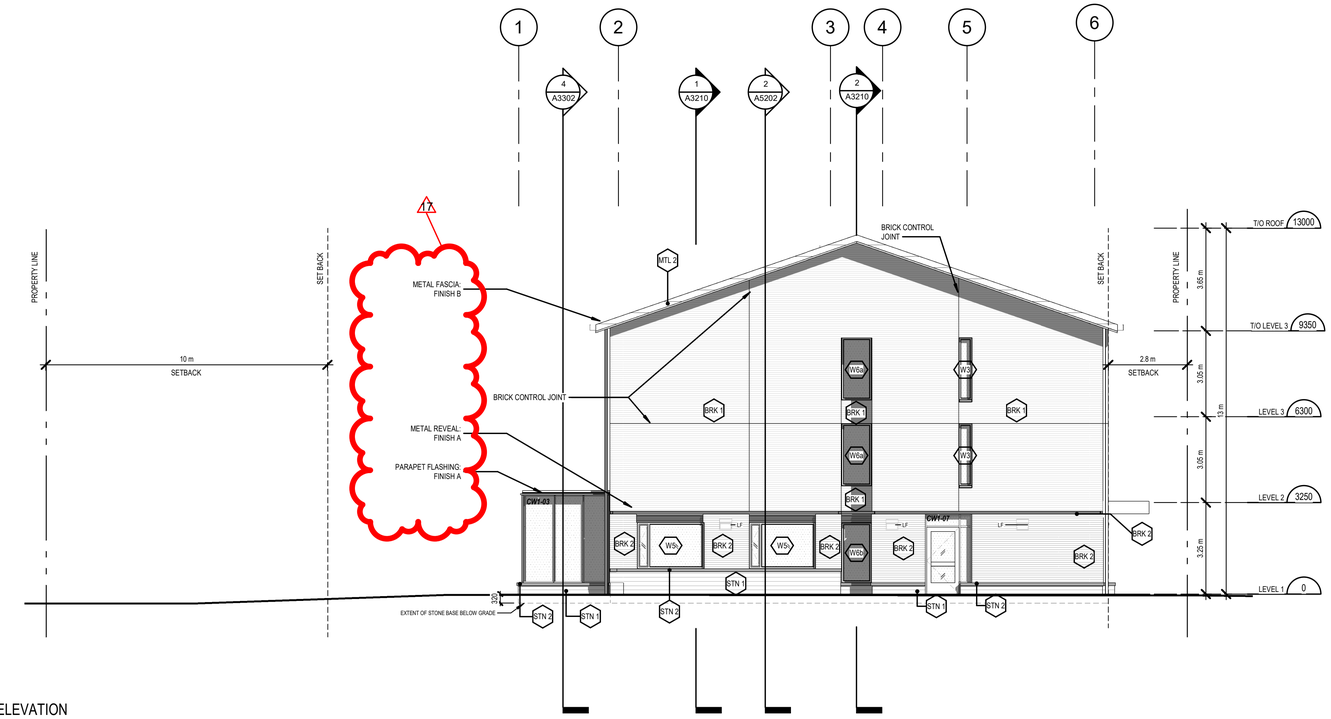
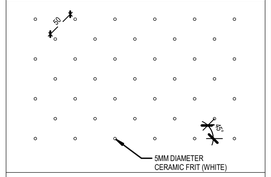


1 A2231 NORTH ELEVATION

1 : 100

BIRD FRIENDLY GLAZING PATTERN



APPLY BIRD FRIENDLY GLAZING PATTERN TO SURFACE 1 ON ALL EXTERIOR GLAZING.

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BRK 2	BRICK MASONRY
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MTL 2	STANDING SEAM METAL ROOF
STN 1	STONE BASE
STN 2	STONE SILL
WD	WOOD CLADDING
STN 3	STONE CLADDING (LARGE FORMAT)
STN 4	STONE VENEER

16

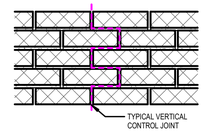
METAL FINISHES

FINISH A COLOUR MATCH ANODIZED ALUMINUM

ELEVATION LEGEND ALVALUM FINISH

RWL	RAINWATER LEADER
LF	EXTERIOR WALL-MOUNTED LIGHT FIXTURE - REFER TO ELEC

TYPICAL VERTICAL JOINT DETAIL



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REVISIONS

REDEVELOPMENT - 409 CENTRE ST S

25008

409 Centre St. South,
Whitby, ON L1N 4W3

Town of Whitby File Number: SP-26-25

SITE/BUILDING ELEVATIONS

SCALE:	As indicated
DRAWN BY:	MB
REVIEWED BY:	EM
JOB NUMBER:	25008
PLOT DATE:	10/17/22

DRAWING NUMBER:

A2231

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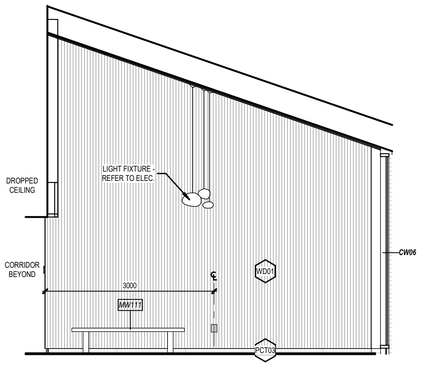
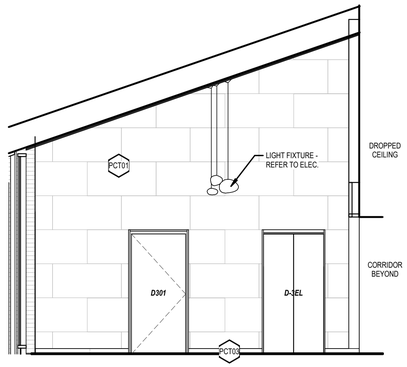


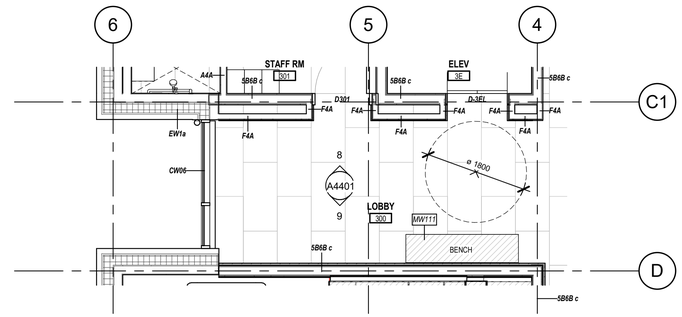
9 A4401 ELEVATOR LOBBY - LEVEL 3 - SOUTH WALL

1 : 50



8 A4401 ELEVATOR LOBBY - LEVEL 3 - NORTH WALL

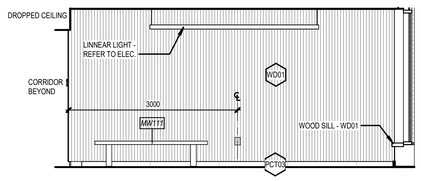
1 : 50



7 A4401 ENLARGED PLAN - L3 - LOBBY

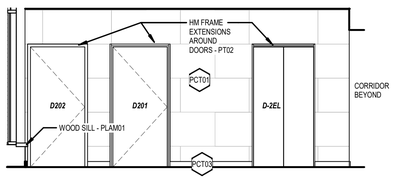
1 : 50

INTERIOR FINISH LEGEND	
PT01	PAINT - TYPICAL WHITE
PT02	PAINT - TYPICAL CHARCOAL
PLAM01	WOOD LOOK HIGH PRESSURE PLASTIC LAMINATE
PLAM02	SOLID CORE PLASTIC LAMINATE
BRK01	INTERIOR BRICK
RB01	RUBBER BASE
PCT01	PORCELAIN TILE (LOBBY/CORRIDORS) - RUNNING BOND
PCT02	PORCELAIN TILE (WC) - RUNNING BOND
PCT03	PORCELAIN TILE WALL BASE
WD01	WHITE OAK WOOD VENEER



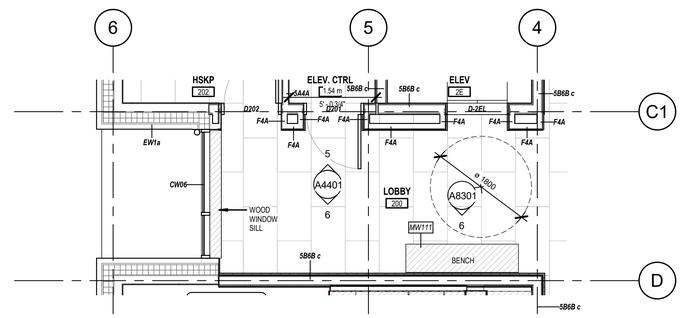
6 A4401 ELEVATOR LOBBY - LEVEL 2 - SOUTH WALL

1 : 50



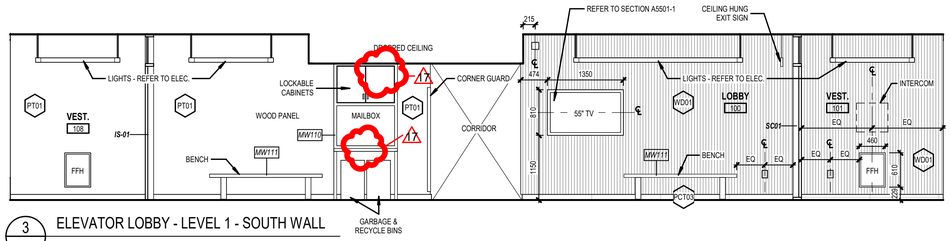
5 A4401 ELEVATOR LOBBY - LEVEL 2 - NORTH WALL

1 : 50



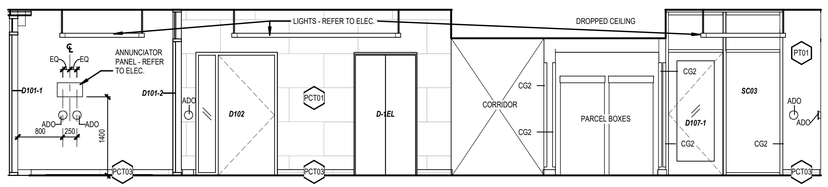
4 A4401 ENLARGED PLAN - L2 - LOBBY

1 : 50



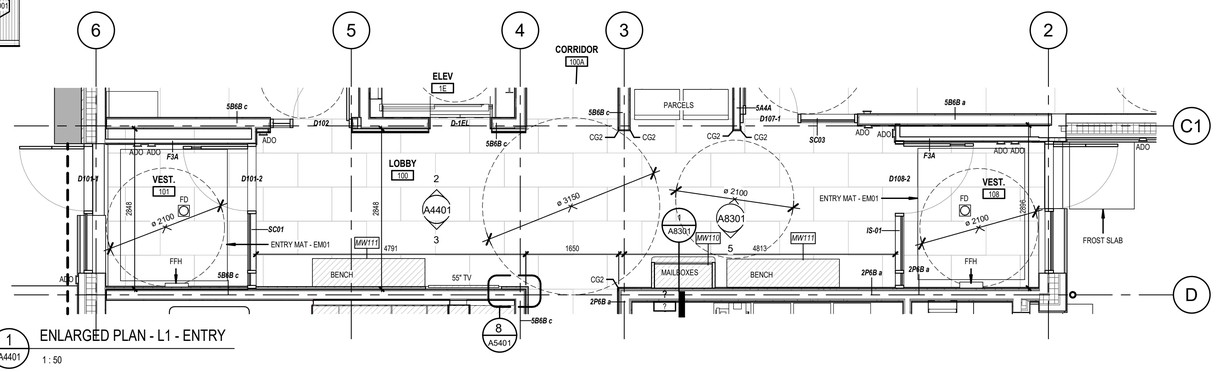
3 A4401 ELEVATOR LOBBY - LEVEL 1 - SOUTH WALL

1 : 50



2 A4401 ELEVATOR LOBBY - LEVEL 1 - NORTH WALL

1 : 50



1 A4401 ENLARGED PLAN - L1 - ENTRY

1 : 50

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17	2026-10-30	ISSUED FOR TENDER - ADDENDUM 3	MSA
13	2026-09-16	ISSUED FOR TENDER	MSA
11	2026-03-27	ISSUED FOR 100% CONSTRUCTION DOCUMENTS	MSA
10	2026-02-20	ISSUED FOR 90% CD & CLASS A COSTING	MSA
1	2025-06-20	ISSUED FOR 100% DD & CLASS B COSTING	MSA

REDEVELOPMENT - 409 CENTRE ST S

25008

409 Centre St. South,
Whitby, ON L1N 4W3

Town of Whitby File Number: SP-26-25

ENLARGED PLANS & INTERIOR ELEVATIONS - ENTRY & LOBBY

SCALE:	As indicated
DRAWN BY:	Author
REVIEWED BY:	Checker
JOB NUMBER:	25008
PLOT DATE:	05/22/25

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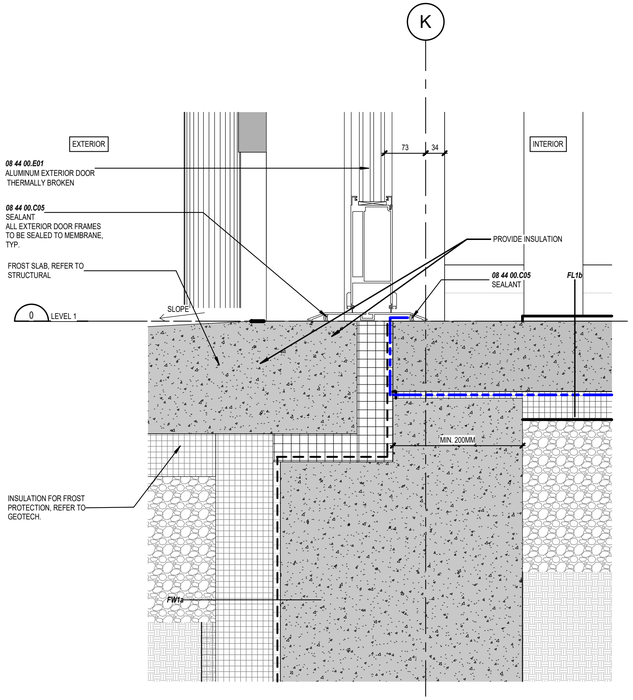
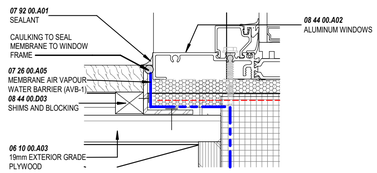


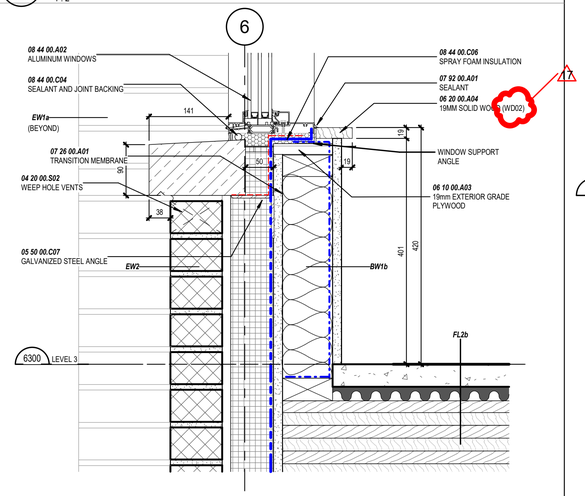
2 A5200 SECTION DETAIL - ALUM DOOR THRESHOLD

1 : 5



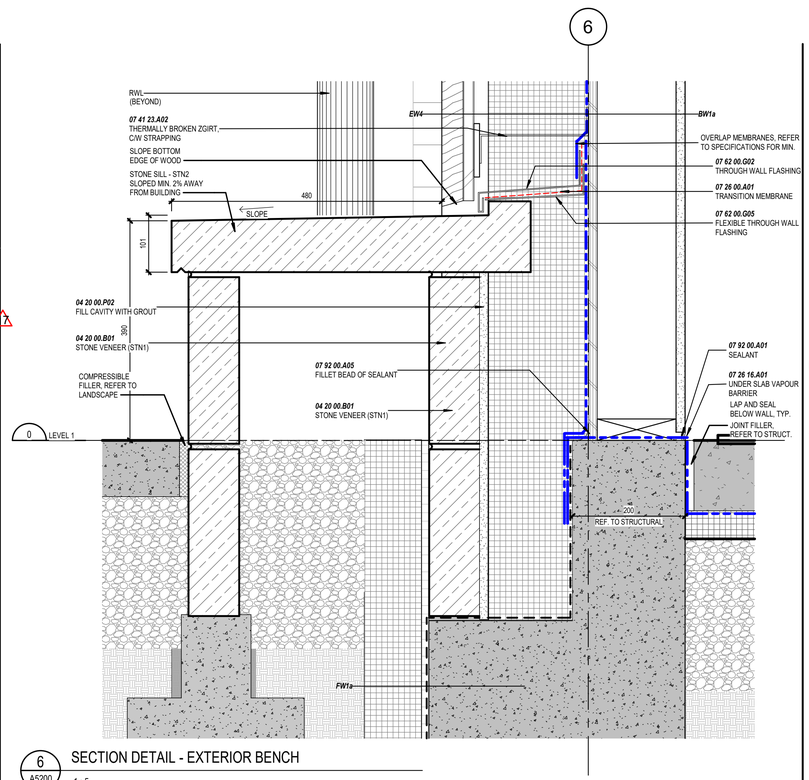
7 A5200 TYP. WINDOW SEAL/MEMBRANE DETAIL

1 : 2



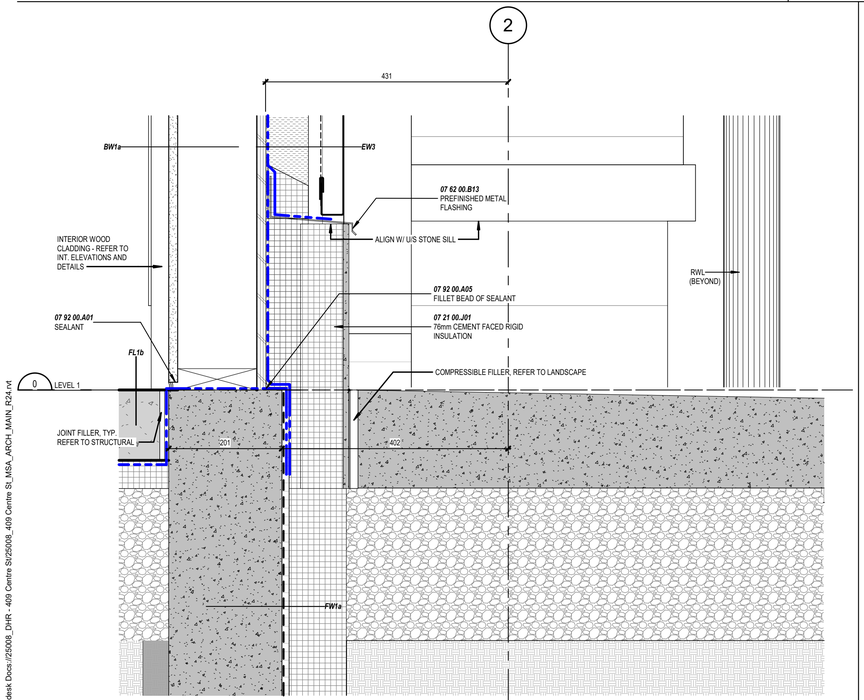
5 A5200 SECTION DETAIL @ WINDOW SILL AT EW2

1 : 5



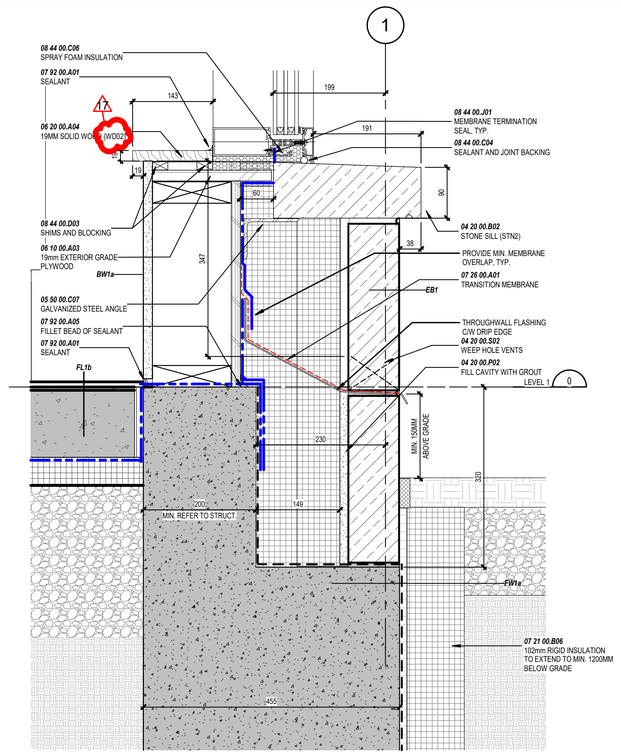
6 A5200 SECTION DETAIL - EXTERIOR BENCH

1 : 5



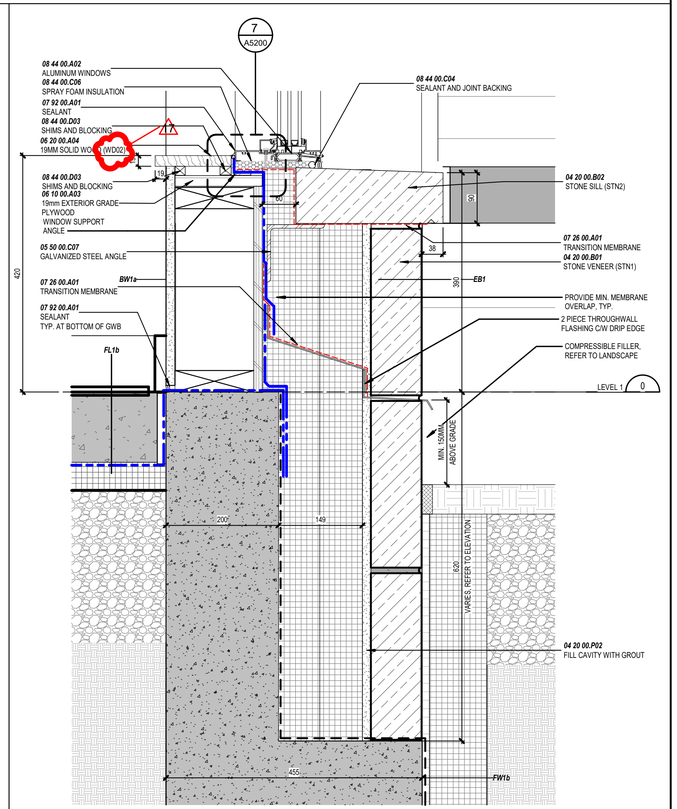
3 A5200 SECTION DETAIL - EW3 BASE AT SOG

1 : 5



4 A5200 SECTION DETAIL - CW BASE DETAIL AT EB1

1 : 5



1 A5200 SECTION DETAIL - EB1 BASE AT FOUNDATION

1 : 5

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#	DATE:	REVISION:	BY:
17	2026-10-30	ISSUED FOR TENDER - ADDENDUM 3	MSA
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9	2026-01-30	ISSUED FOR 75% CONSTRUCTION DOCUMENTS	MSA
7	2026-01-30	ISSUED FOR BUILDING PERMIT	MSA

REVISIONS

REDEVELOPMENT - 409 CENTRE ST S

25008

409 Centre St. South,
Whitby, ON L1N 4W3

Town of Whitby File Number: SP-26-25

ENVELOPE DETAILS - SECTIONS - BASE & SILL CONDITIONS

SCALE:	As indicated
DRAWN BY:	MB
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JOB NUMBER:	25008
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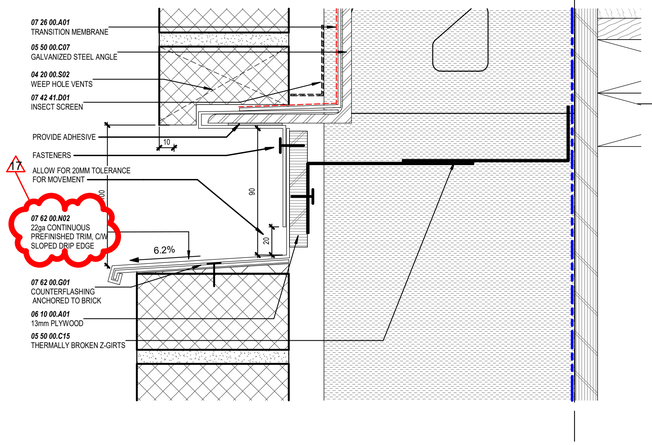
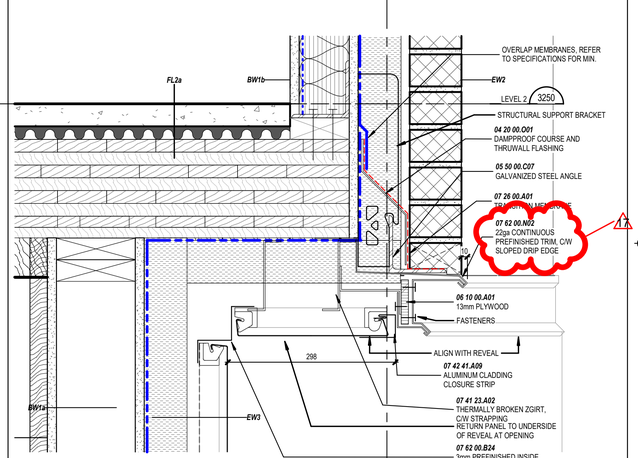
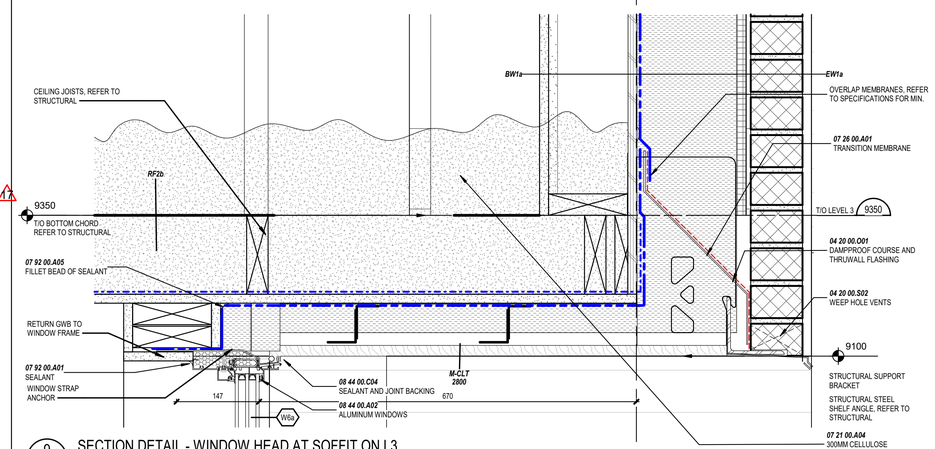
*ON SITE-MOCKUP REQUIRED



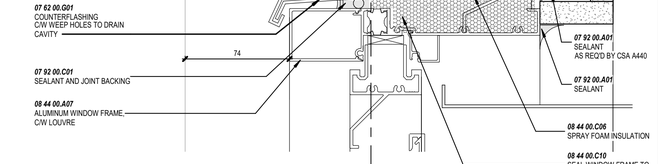
11 A5201 ENLARGED DETAIL - METAL REVEAL AT BRICK TRANSITION
1 : 2



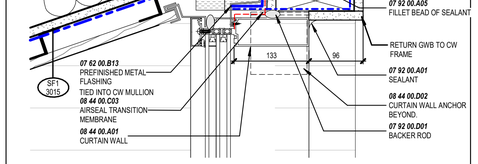
10 A5201 SECTION DETAIL - INSET DOOR HEAD AT REVEAL
1 : 5



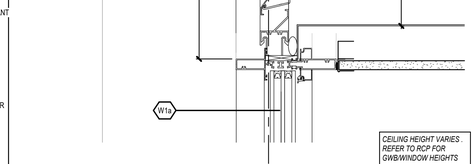
9 A5201 SECTION DETAIL - WINDOW HEAD AT SOFFIT ON L3
1 : 5



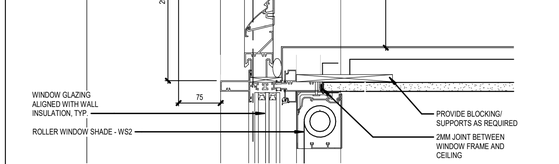
8 A5201 ENLARGED DETAIL - METAL REVEAL AT WINDOW/LOUVRE HEAD
1 : 2



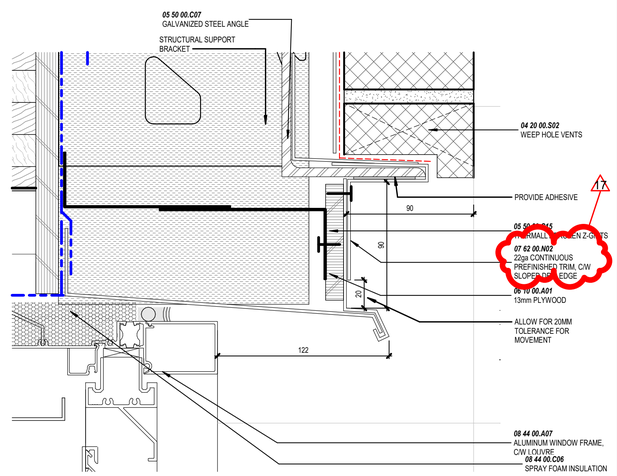
7 A5201 SECTION DETAIL - CW AT ROOF
1 : 5



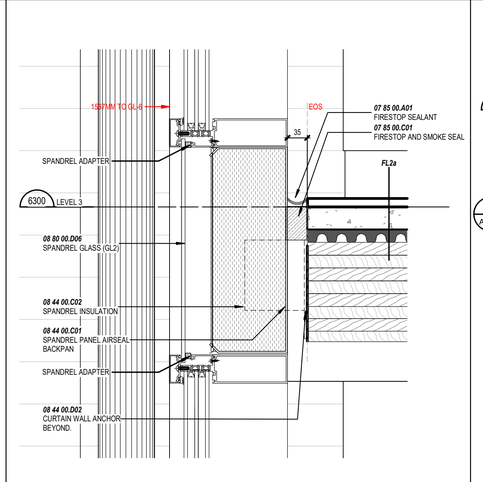
6 A5201 SECTION DETAIL - WINDOW/LOUVRE HEAD AT REVEAL AT EW2
1 : 5



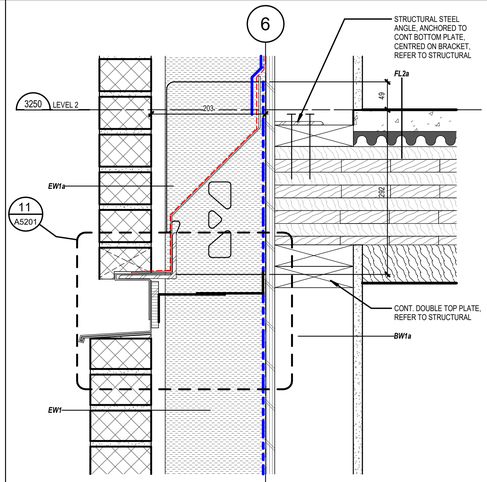
5 A5201 SECTION DETAIL - WINDOW/LOUVRE HEAD AT REVEAL AT EW2
1 : 5



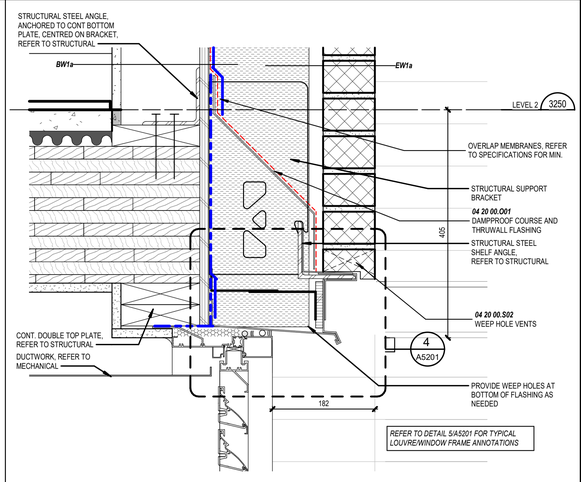
4 A5201 ENLARGED DETAIL - METAL REVEAL AT BRICK/WINDOW
1 : 2



3 A5201 SECTION DETAIL - CW AT FLOOR SLAB
1 : 5



2 A5201 SECTION DETAIL - TYPICAL REVEAL AT BRICK
1 : 5



1 A5201 SECTION DETAIL - WINDOW/LOUVRE HEAD AT REVEAL AT EW1a
1 : 5

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REVISIONS

REDEVELOPMENT - 409 CENTRE ST S

25008

409 Centre St. South,
Whitby, ON L1N 4W3

Town of Whitby File Number: SP-26-25

ENVELOPE DETAILS - SECTIONS - LOUVRE/WINDOWS, & REVEALS

SCALE:	As indicated
DRAWN BY:	MB
REVIEWED BY:	EM
JOB NUMBER:	25008
PLOT DATE:	11/08/22

DRAWING NUMBER:
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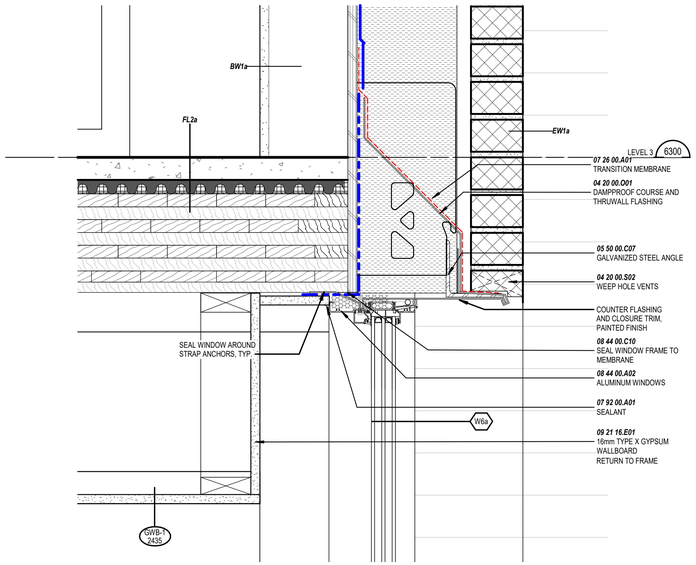
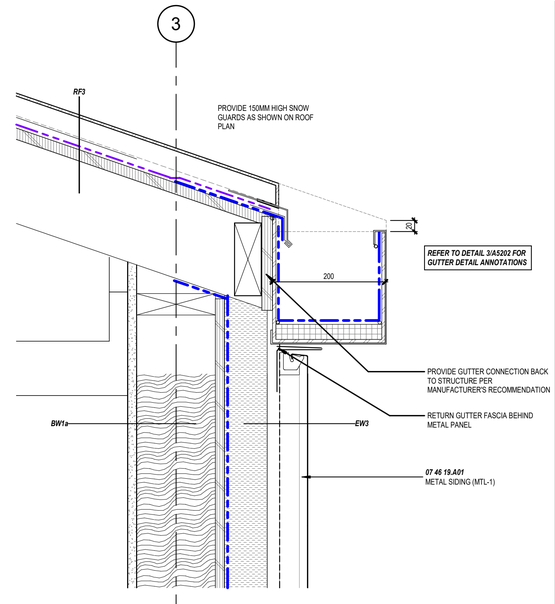


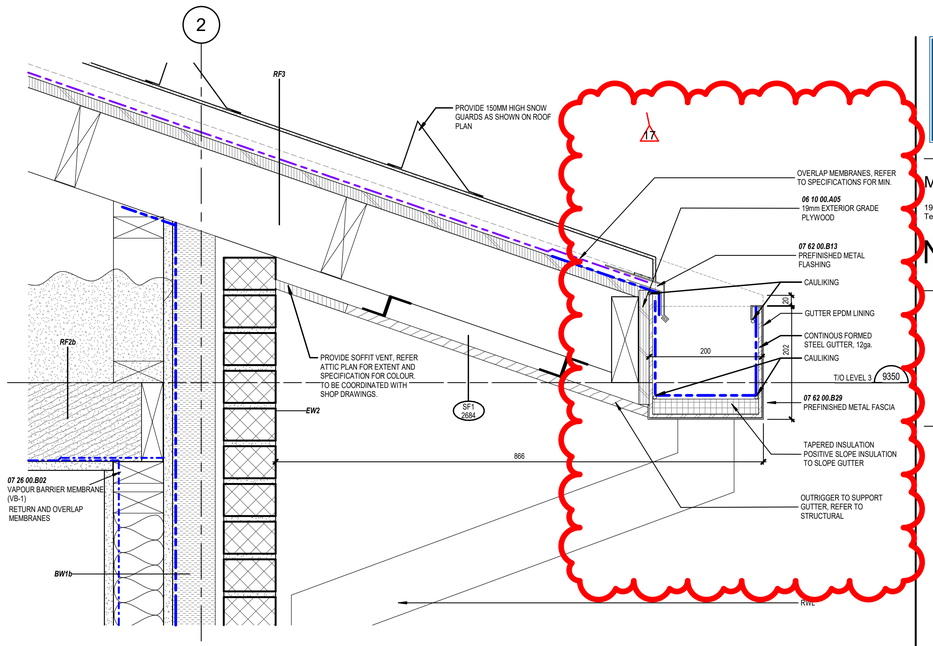
5 A5202 SECTION DETAIL - CORRIDOR WINDOW HEAD DETAIL

1 : 5



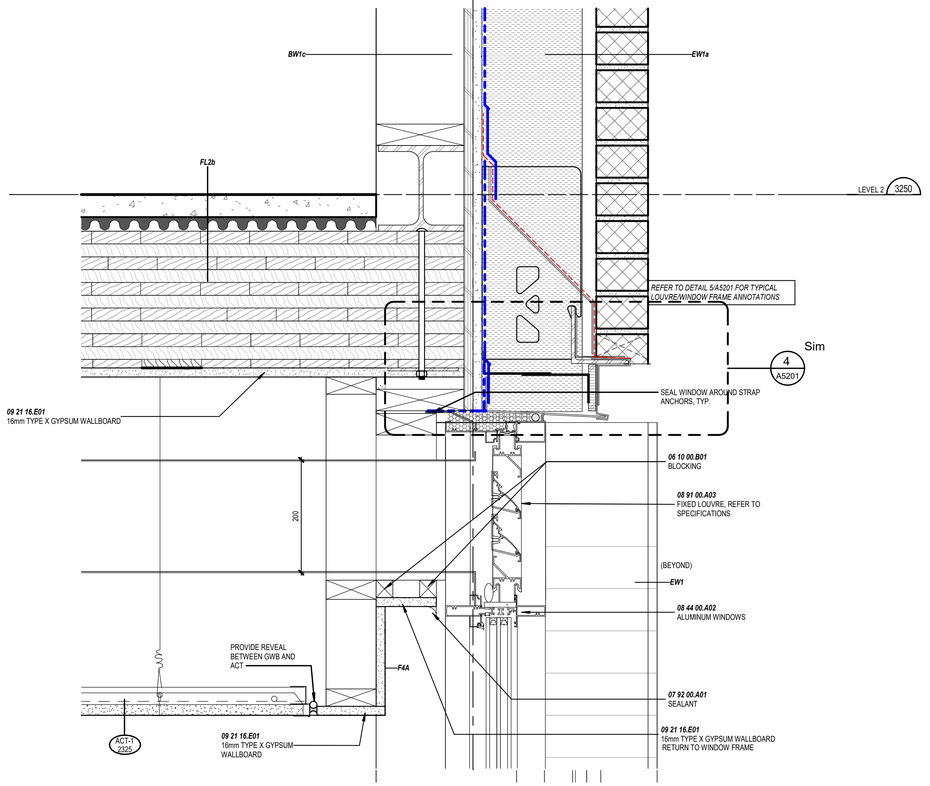
4 A5202 SECTION DETAIL - ROOF GUTTER AT EW3

1 : 5



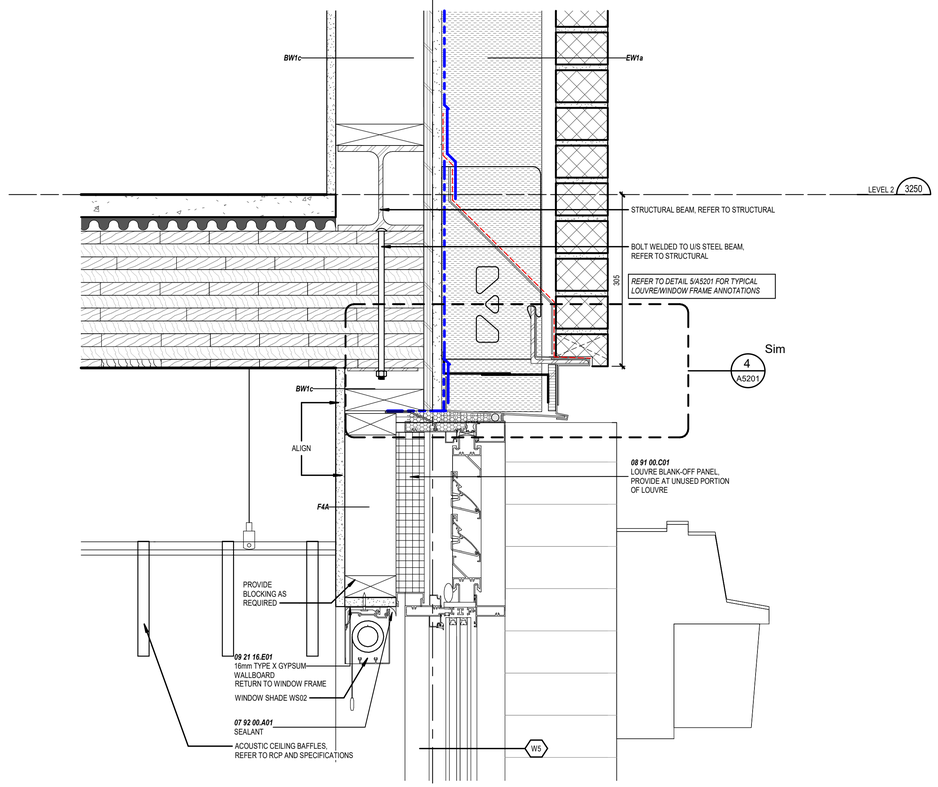
3 A5202 SECTION DETAIL - ROOF GUTTER

1 : 5



2 A5202 SECTION DETAIL - LAUNDRY WINDOW HEAD

1 : 5



1 A5202 SECTION DETAIL - AMENITY NORTH WINDOW - HEAD

1 : 5

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REVISIONS

REDEVELOPMENT - 409 CENTRE ST S

25008

409 Centre St. South,
Whitby, ON L1N 4W3

Town of Whitby File Number: SP-26-25

ENVELOPE DETAILS - SECTIONS - AMENITY & GUTTER DETAILS

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DRAWN BY:	Author
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JOB NUMBER:	25008
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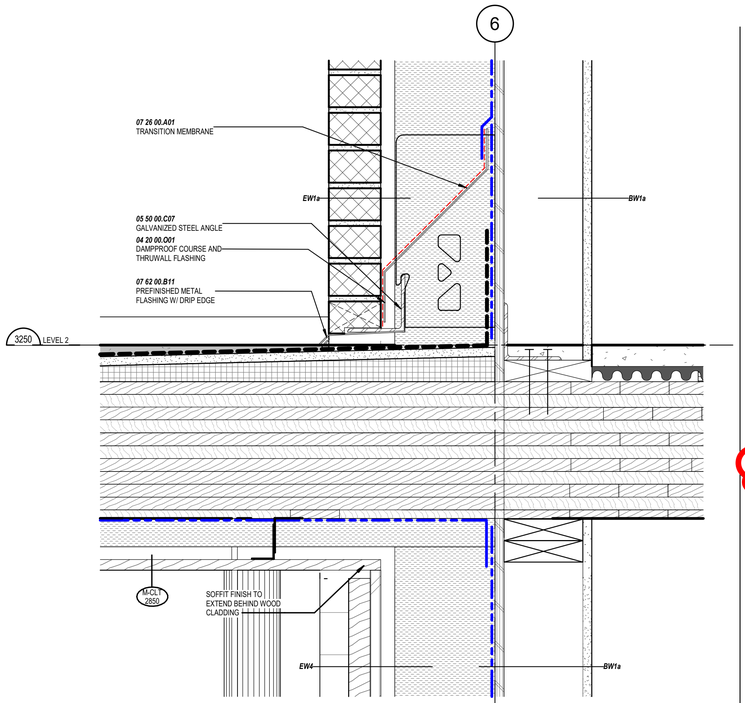

4 A5203 SECTION DETAIL - EW4 AT CANOPY

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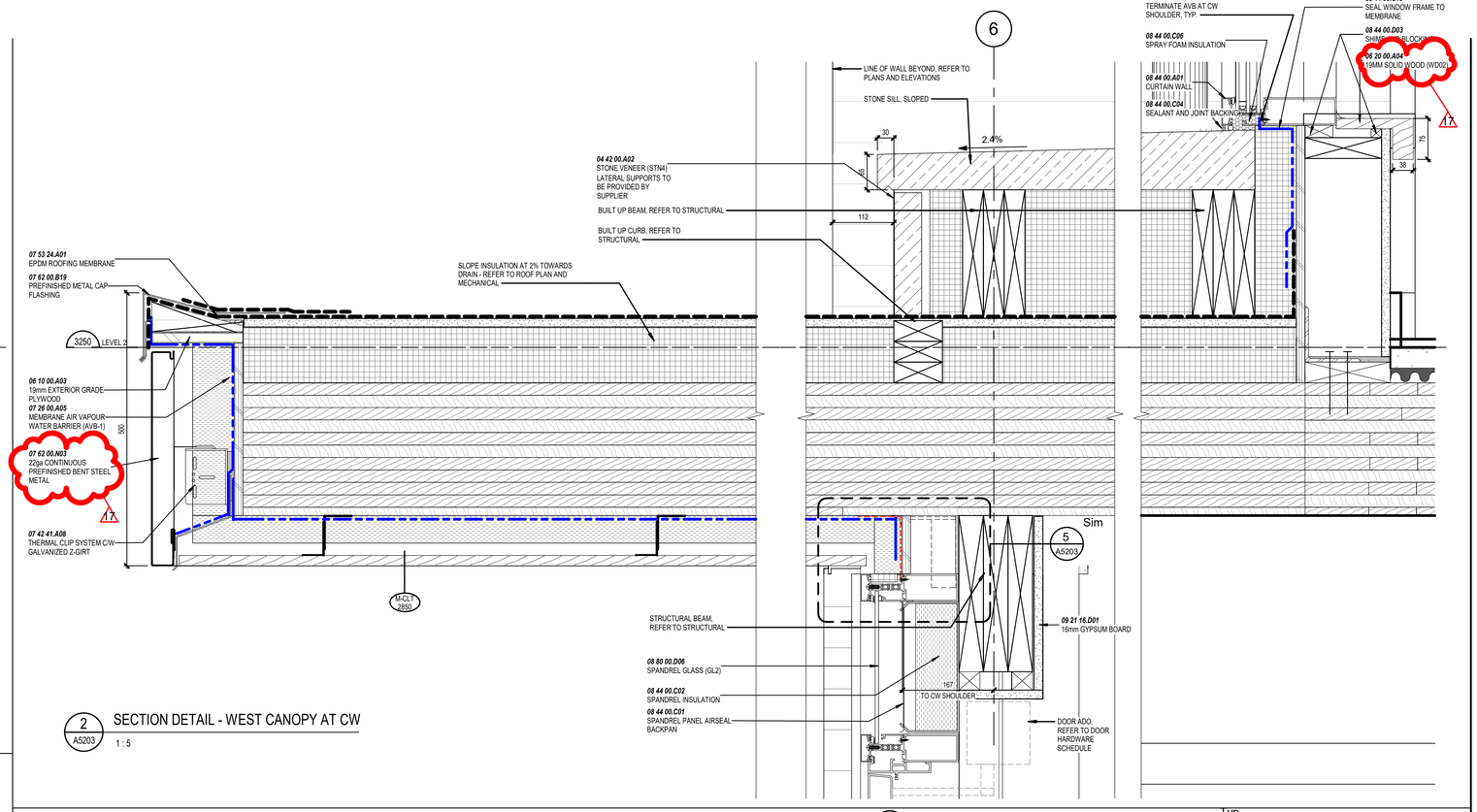

2 A5203 SECTION DETAIL - WEST CANOPY AT CW

1 : 5

5 A5203 ENLARGED DETAIL - CW HEAD AT CANOPY

1 : 2

1 A5203 SECTION DETAIL - CANOPY @ SOUTH ENTRANCE

1 : 5

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9	2026-01-30	ISSUED FOR 75% CONSTRUCTION DOCUMENTS	MSA

REVISIONS

REDEVELOPMENT - 409 CENTRE ST S

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409 Centre St. South,
Whitby, ON L1N 4W3

Town of Whitby File Number: SP-26-25

ENVELOPE DETAILS - SECTIONS - CANOPY

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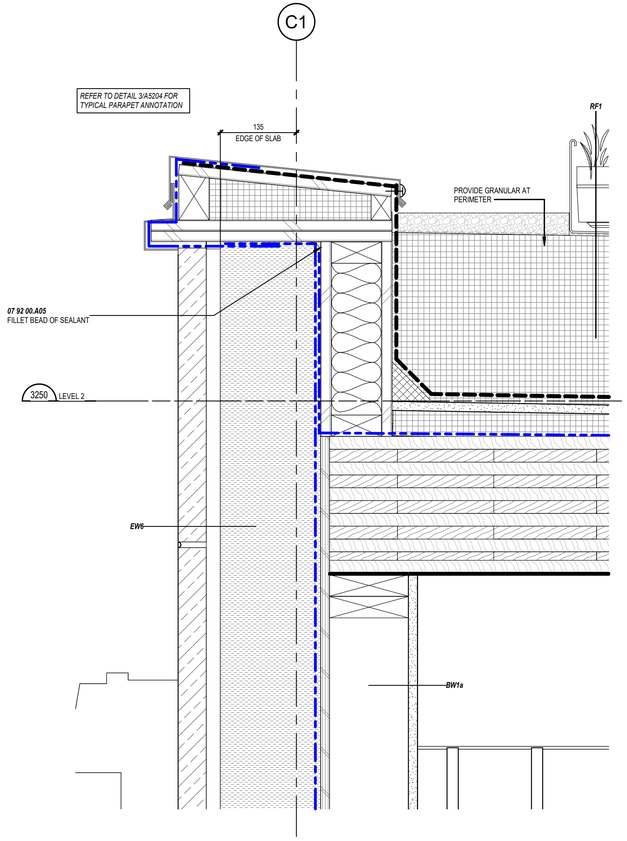
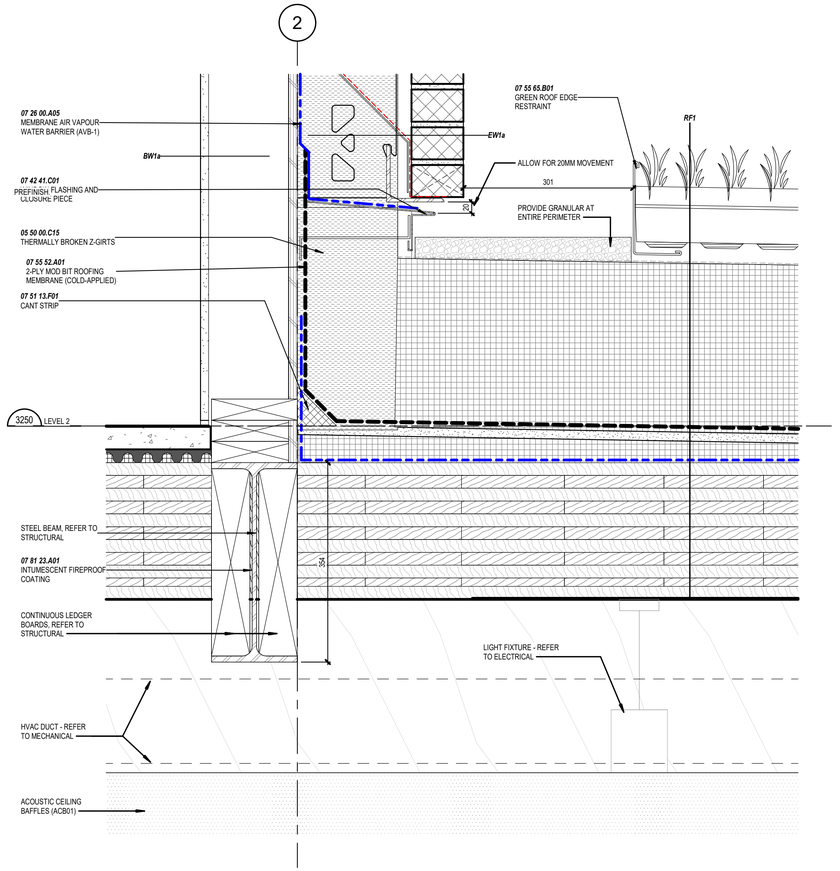


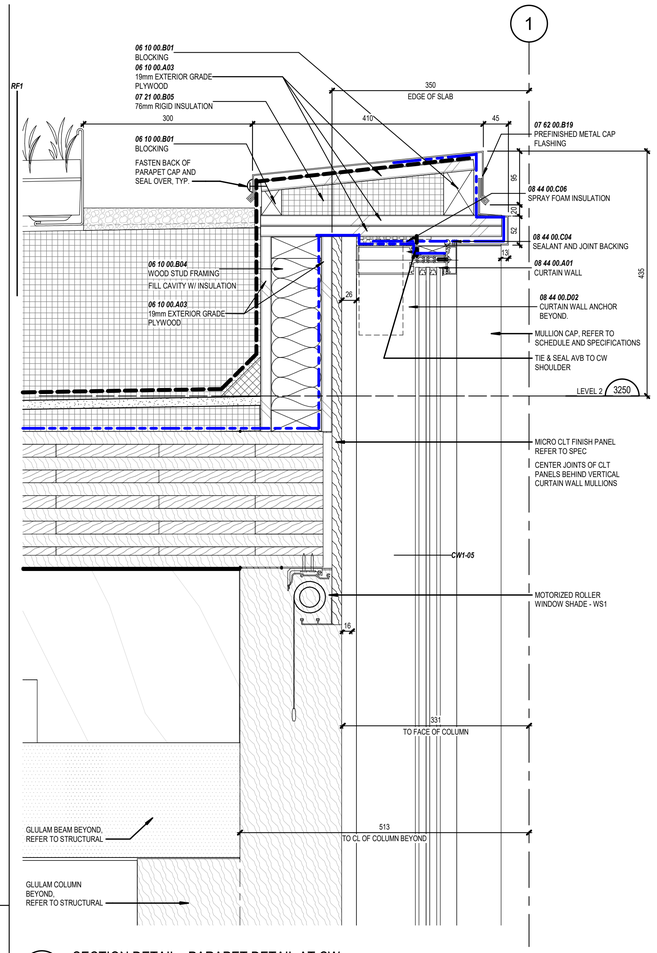
5 A5204 SECTION DETAIL - PARAPET AT EW6

1 : 5



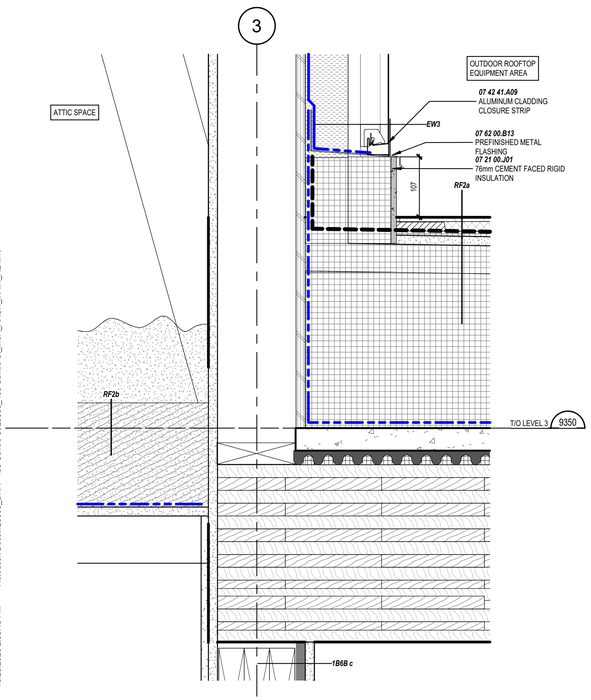
4 A5204 SECTION DETAIL - EW1a TO RF1 TRANSITION

1 : 5



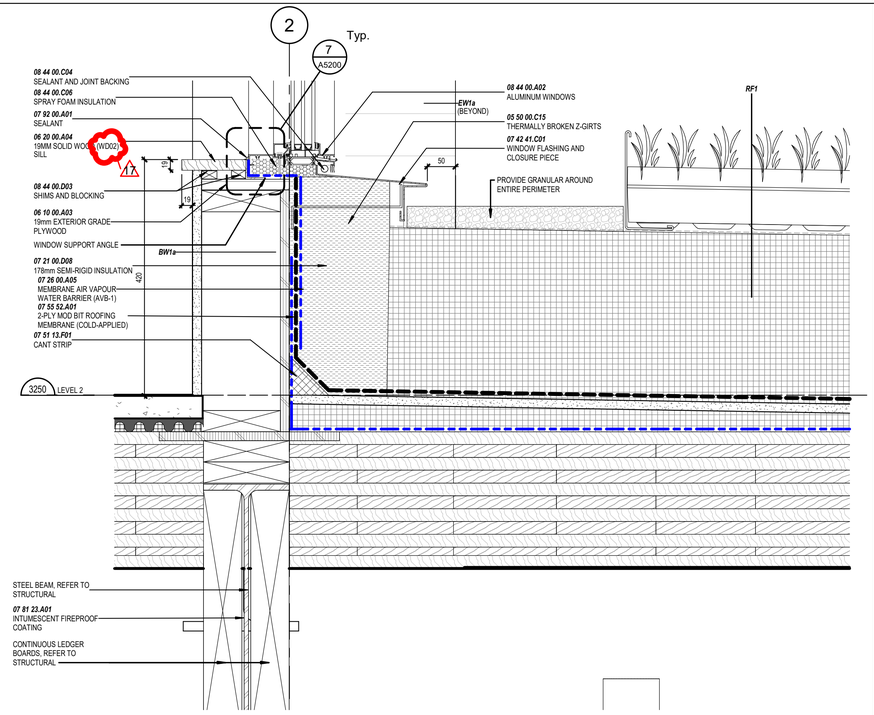
3 A5204 SECTION DETAIL - PARAPET DETAIL AT CW

1 : 5



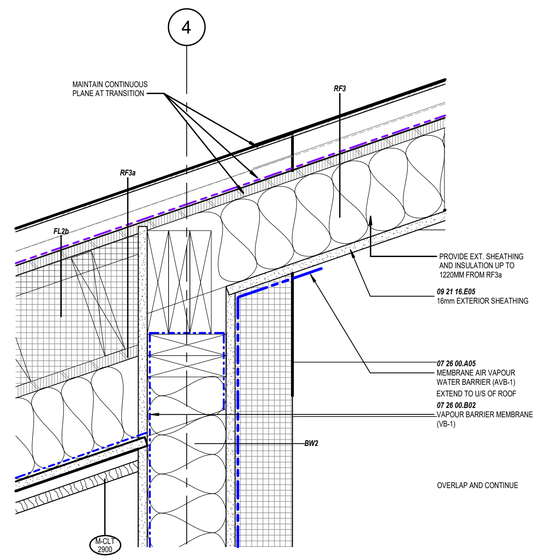
1 A5204 SECTION DETAIL - EW3 TRANSITION TO RF2a

1 : 5



2 A5204 SECTION DETAIL - WINDOW SILL AT GREEN ROOF

1 : 5



6 A5204 SECTION DETAIL - RF3 TO RF3a TRANSITION

1 : 5

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REVISIONS

REDEVELOPMENT - 409 CENTRE ST S

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Town of Whitby File Number: SP-26-25

ENVELOPE DETAILS - SECTIONS - ROOF & PARAPET

SCALE:	1 : 5
DRAWN BY:	MB
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PLOT DATE:	12/02/25

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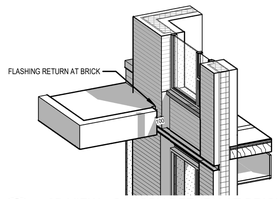
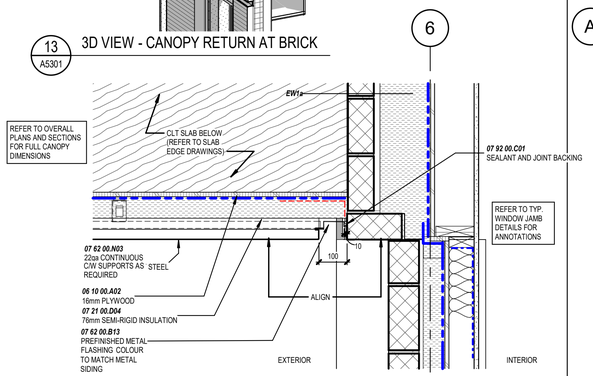
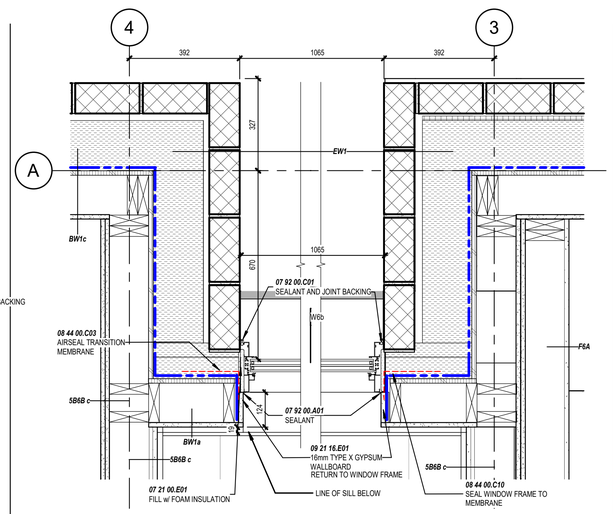


13 A5301 3D VIEW - CANOPY RETURN AT BRICK



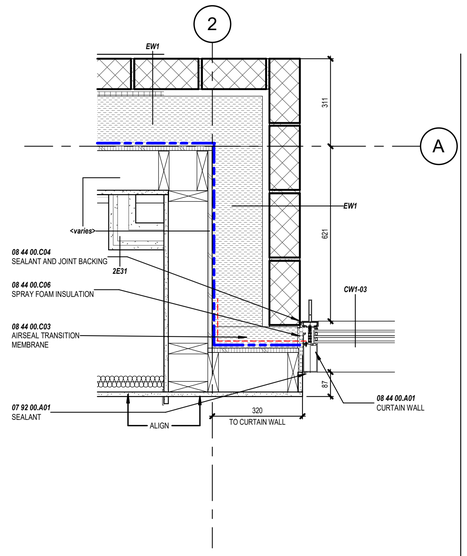
12 A5301 PLAN DETAIL - CANOPY FASCIA RETURN AT BRICK

1 : 10



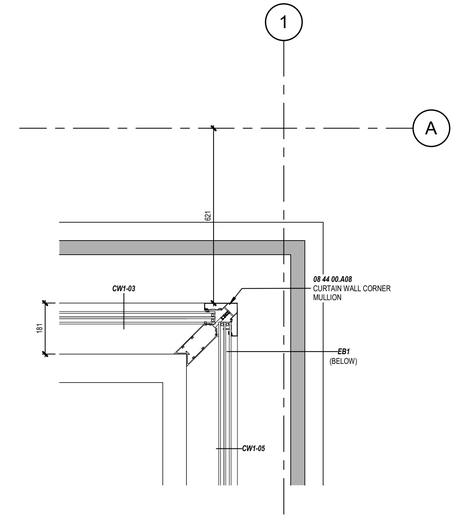
10 A5301 PLAN DETAIL - INSET WINDOW - CORRIDOR ENDS

1 : 10



9 A5301 PLAN DETAIL - CW AT EW1 RETURN

1 : 10



8 A5301 PLAN DETAIL - CORNER MULLION

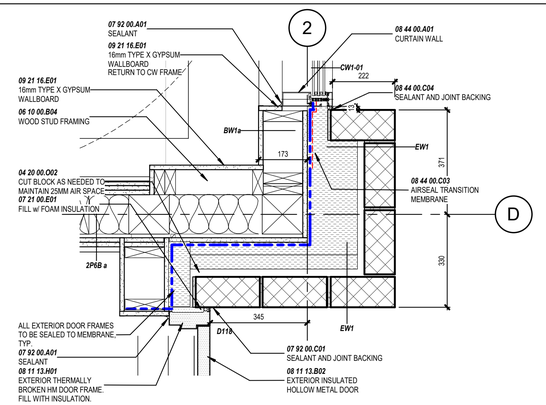
1 : 10

11 A5301 PLAN DETAIL - TYP. BRICK INSET DETAIL (BELOW WINDOW)

1 : 10

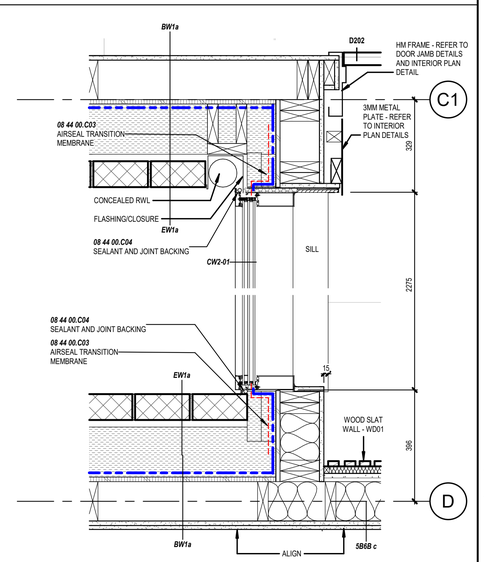
7 A5301 PLAN DETAIL - TYP. WINDOW JAMB IN EW1a

1 : 10



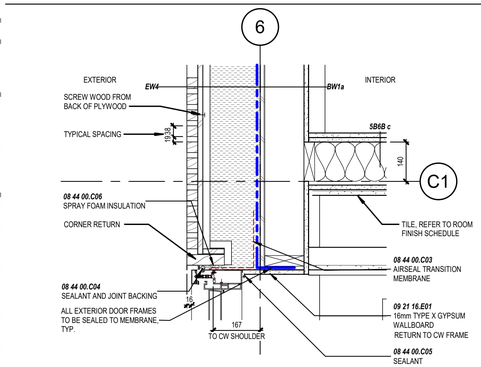
6 A5301 PLAN DETAIL - EAST VESTIBULE WALL RETURN

1 : 10



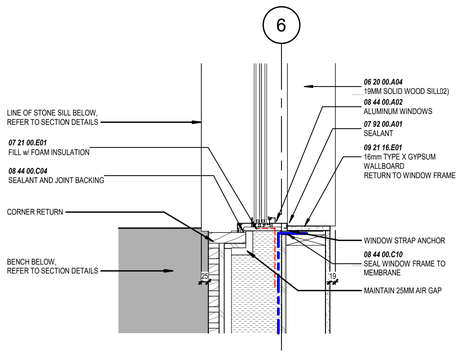
5 A5301 PLAN DETAIL - INSET CW IN EW1a

1 : 10



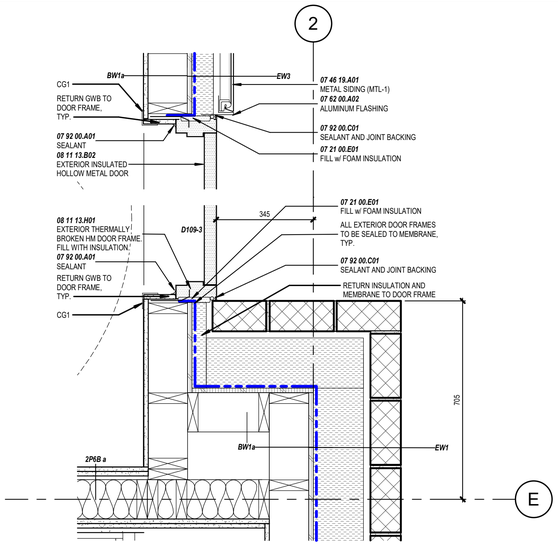
4 A5301 PLAN DETAIL - CW JAMB AT EW4

1 : 10



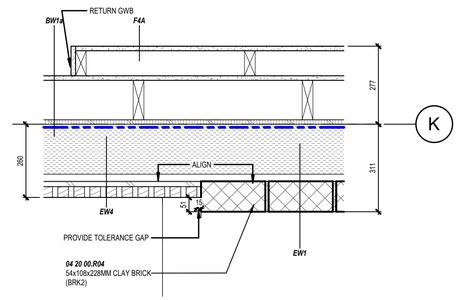
3 A5301 PLAN DETAIL - WINDOW JAMB AT EW4

1 : 10



2 A5301 PLAN DETAIL - DOOR JAMB AT EW1/EW3

1 : 10



1 A5301 PLAN DETAIL - EW1 TO EW4 TRANSITION

1 : 10

The Regional Municipality of Durham
605 Rossland Road East
Whitby, Ontario L1N 6A3
Tel 905.668.7711

Montgomery Sisam Architects Inc.
197 Spadina Avenue, Toronto, Ontario M5T 2C8 montgomerysisam.com
Tel 416.364.8079 Fax 416.364.7723

MontgomerySisam

PROJECT NORTH
TRUE NORTH

#	DATE:	REVISION:	BY:
17	2026-10-30	ISSUED FOR TENDER - ADDENDUM 3	MSA
13	2026-09-16	ISSUED FOR TENDER	MSA
11	2026-03-27	ISSUED FOR 100% CONSTRUCTION DOCUMENTS	MSA
10	2026-02-20	ISSUED FOR 50% CD & CLASS A COSTING	MSA
9	2026-01-30	ISSUED FOR 75% CONSTRUCTION DOCUMENTS	MSA
7	2026-01-30	ISSUED FOR BUILDING PERMIT	MSA

REVISIONS

REDEVELOPMENT - 409 CENTRE ST S

25008

409 Centre St. South,
Whitby, ON L1N 4W3

Town of Whitby File Number: SP-26-25

ENVELOPE DETAILS - PLANS

SCALE: 1 : 10
DRAWN BY: MB
REVIEWED BY: EM
JOB NUMBER: 25008
PLOT DATE: 11/08/22

DRAWING NUMBER:
A5301

10/29/2026 2:38:56 PM Autodesk Docs://25008_DHR_-_409 Centre St_MSA_ARCH_MAIN_R24.rvt

ROOM FINISH SCHEDULE

Level	Number	Name	Room Key	Floor Finish	Base Finish	Wall Finish	Remarks (Key)
Not Placed	114D	Room	(none)				
BASEMENT	B100	CORR	SERVICE CORRIDOR	CONS01	RB01	PT01	
BASEMENT	B101	ELEC & COMMS	ELEC & COMMS RM	CONS01	RB01	PT01	
BASEMENT	B102	INCOMING WATER RM	INCOMING WATER RM	WPC01	WPC01	PT01A	
BASEMENT	B103	CORR	SERVICE CORRIDOR	CONS01	RB01	PT01	
BASEMENT	BB	STAIR B	STAIRS	RBF01/RBST01	RB01	PT01	PROVIDE RUBBER BASE AT LANDINGS
LEVEL 1	1A	STAIR A	STAIRS	RBF01/RBST01	RB01	PT01	PROVIDE RUBBER BASE AT LANDINGS
LEVEL 1	1B	STAIR B	STAIRS	RBF01/RBST01	RB01	PT01	PROVIDE RUBBER BASE AT LANDINGS
LEVEL 1	1E	ELEV	ELEVATOR	PCT01	PCT03	PT01	
LEVEL 1	100	LOBBY	LOBBY (L1 MAIN ENTRY)	PCT01	PCT03	PT01	PCT01 AT ELEVATOR WALL, REFER TO INTERIOR ELEVATIONS
LEVEL 1	100A	CORRIDOR	CORRIDOR (L1)	PCT01	PCT03	PT01	DOOR SURROUND AT EACH SUITE ENTRY DOOR - REFER TO INTERIOR DETAIL & MILLWORK ELEVATION
LEVEL 1	100B	CORRIDOR	CORRIDOR (L1)	PCT01	PCT03	PT01	DOOR SURROUND AT EACH SUITE ENTRY DOOR - REFER TO INTERIOR DETAIL & MILLWORK ELEVATION
LEVEL 1	101	VEST.	VESTIBULE	EM01	PCT03	PT01	
LEVEL 1	102	OFFICE	OFFICE	PCT01	PCT03	PT01	
LEVEL 1	103	1-BD	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 1	103A	BEDROOM	SUITE - BEDROOM	LVP01	WB01	PT01	
LEVEL 1	103B	BATHROOM	SUITE - WC	PCT02	PCT02	PT01A/CT01-B/CT03	CT01-B AT VANITY BACKSPLASH, REFER TO ENLARGED SUITE PLAN FOR EXTENTS, CT03, REFER TO ENLARGED SUITE PLAN FOR EXTENTS
LEVEL 1	103C	DINING / KITCHEN	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 1	104	LAUNDRY	LAUNDRY	PCT01	PCT03	PT01A	
LEVEL 1	105	U WC	UNIVERSAL WC	PCT01	PCT03	PT01A/CT02	CT01 AT VANITY BACKSPLASH, REFER TO WASHROOM ELEVATION FOR EXTENTS, CT02, REFER TO WASHROOM ELEVATIONS FOR EXTENTS
LEVEL 1	106	STORAGE	STORAGE RM (PROGRAM)	PCT01	PCT03	PT01	
LEVEL 1	107	AMENITY	AMENITY	PCT01	PCT03	PT01/CT01	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 1	108	VEST.	VESTIBULE	EM01	PCT03	PT01	
LEVEL 1	109	WASTE	WASTE RM	EFS01	CB/EFS01	PT01A/FRP	PROVIDE RUBBER CRASH RAILS (RCR) AROUND THE ROOM AT ALL WALLS, PROVIDE FRP PANELS ON ALL WALLS TO THE UNDERSIDE OF CEILING
LEVEL 1	110	1-BD	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 1	110A	BEDROOM	SUITE - BEDROOM	LVP01	WB01	PT01	
LEVEL 1	110B	BATHROOM	SUITE - WC	PCT02	PCT02	PT01A/CT01-B/CT03	CT01-B AT VANITY BACKSPLASH, REFER TO ENLARGED SUITE PLAN FOR EXTENTS, CT03, REFER TO ENLARGED SUITE PLAN FOR EXTENTS
LEVEL 1	110C	DINING / KITCHEN	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 1	111	1-BD	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 1	111A	BEDROOM	SUITE - BEDROOM	LVP01	WB01	PT01	
LEVEL 1	111B	BATHROOM	SUITE - WC	PCT02	PCT02	PT01A/CT01-B/CT03	CT01-B AT VANITY BACKSPLASH, REFER TO ENLARGED SUITE PLAN FOR EXTENTS, CT03, REFER TO ENLARGED SUITE PLAN FOR EXTENTS
LEVEL 1	111C	DINING / KITCHEN	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 1	112	1-BD	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 1	112A	BEDROOM	SUITE - BEDROOM	LVP01	WB01	PT01	
LEVEL 1	112B	BATHROOM	SUITE - WC	PCT02	PCT02	PT01A/CT01-B/CT03	CT01-B AT VANITY BACKSPLASH, REFER TO ENLARGED SUITE PLAN FOR EXTENTS, CT03, REFER TO ENLARGED SUITE PLAN FOR EXTENTS
LEVEL 1	112C	DINING / KITCHEN	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 1	113	1-BD BF	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 1	113A	BEDROOM	SUITE - BEDROOM	LVP01	WB01	PT01	
LEVEL 1	113B	BF WC	SUITE - WC (BF)	PCT02	PCT02	PT01A/CT01-B/CT03	CT01-B AT VANITY BACKSPLASH, REFER TO ENGLARGED SUITE PLAN FOR EXTENTS, CT03, REFER TO ENLARGED SUITE PLAN FOR EXTENTS
LEVEL 1	113C	KITCHEN / DINING	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 1	114	1-BD BF	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 1	114A	BEDROOM	SUITE - BEDROOM	LVP01	WB01	PT01	
LEVEL 1	114B	BF WC	SUITE - WC (BF)	PCT02	PCT02	PT01A/CT01-B/CT03	CT01-B AT VANITY BACKSPLASH, REFER TO ENGLARGED SUITE PLAN FOR EXTENTS, CT03, REFER TO ENLARGED SUITE PLAN FOR EXTENTS
LEVEL 1	114C	KITCHEN / DINING	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 1	115	1-BD	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 1	115A	BEDROOM	SUITE - BEDROOM	LVP01	WB01	PT01	
LEVEL 1	115B	BATHROOM	SUITE - WC	PCT02	PCT02	PT01A/CT01-B/CT03	CT01-B AT VANITY BACKSPLASH, REFER TO ENLARGED SUITE PLAN FOR EXTENTS, CT03, REFER TO ENLARGED SUITE PLAN FOR EXTENTS
LEVEL 1	115C	DINING / KITCHEN	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 1	116	1-BD	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 1	116A	BEDROOM	SUITE - BEDROOM	LVP01	WB01	PT01	
LEVEL 1	116B	BATHROOM	SUITE - WC	PCT02	PCT02	PT01A/CT01-B/CT03	CT01-B AT VANITY BACKSPLASH, REFER TO ENLARGED SUITE PLAN FOR EXTENTS, CT03, REFER TO ENLARGED SUITE PLAN FOR EXTENTS
LEVEL 1	116C	DINING / KITCHEN	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 1	117	ELEC	ELEC CLOSET	PCT01	PCT03	PT01	
LEVEL 1	118	CORR	SERVICE CORRIDOR	CONS01	RB01	PT01	
LEVEL 2	2A	STAIR A	STAIRS	RBF01/RBST01	RB01	PT01	PROVIDE RUBBER BASE AT LANDINGS
LEVEL 2	2B	STAIR B	STAIRS	RBF01/RBST01	RB01	PT01	PROVIDE RUBBER BASE AT LANDINGS
LEVEL 2	2E	ELEVATOR	ELEVATOR	PCT01	PCT03	PT01	
LEVEL 2	200	LOBBY	LOBBY (L2-3)	PCT01	PCT03	PT01	PCT01 AT ELEVATOR WALL, REFER TO INTERIOR ELEVATIONS
LEVEL 2	200A	CORRIDOR	CORRIDOR (L2-3)	PCT01	PCT03	PT01	DOOR SURROUND AT EACH SUITE ENTRY DOOR - REFER TO INTERIOR DETAIL & MILLWORK ELEVATION
LEVEL 2	201	ELEV. CTRL	ELEV CONTROL RM	CONS01	RB01	PT01	
LEVEL 2	202	HSKP	HSKP	EFC01	CB/EFC01	PT01A/FRP	PROVIDE FRP PANELS ON ALL WALLS TO THE UNDERSIDE OF CEILING
LEVEL 2	203	1-BD	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 2	203A	BEDROOM	SUITE - BEDROOM	LVP01	WB01	PT01	
LEVEL 2	203B	BATHROOM	SUITE - WC	PCT02	PCT02	PT01A/CT01-B/CT03	CT01-B AT VANITY BACKSPLASH, REFER TO ENLARGED SUITE PLAN FOR EXTENTS, CT03, REFER TO ENLARGED SUITE PLAN FOR EXTENTS
LEVEL 2	203C	DINING / KITCHEN	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 2	205	MECH. RM	MECH RM	WPC01	WPC01	PT01A	
LEVEL 2	206	1-BD	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 2	206A	BEDROOM	SUITE - BEDROOM	LVP01	WB01	PT01	
LEVEL 2	206B	BATHROOM	SUITE - WC	PCT02	PCT02	PT01A/CT01-B/CT03	CT01-B AT VANITY BACKSPLASH, REFER TO ENLARGED SUITE PLAN FOR EXTENTS, CT03, REFER TO ENLARGED SUITE PLAN FOR EXTENTS
LEVEL 2	206C	DINING / KITCHEN	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 2	207	1-BD	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 2	207A	BEDROOM	SUITE - BEDROOM	LVP01	WB01	PT01	
LEVEL 2	207B	BATHROOM	SUITE - WC	PCT02	PCT02	PT01A/CT01-B/CT03	CT01-B AT VANITY BACKSPLASH, REFER TO ENLARGED SUITE PLAN FOR EXTENTS, CT03, REFER TO ENLARGED SUITE PLAN FOR EXTENTS
LEVEL 2	207C	DINING / KITCHEN	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 2	208	CHUTE	WASTE CHUTE VEST.	PCT01	PCT03	PT01A/FRP	PROVIDE FRP PANELS ON ALL WALLS TO THE UNDERSIDE OF CEILING
LEVEL 2	209	1-BD	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 2	209A	BEDROOM	SUITE - BEDROOM	LVP01	WB01	PT01	
LEVEL 2	209B	BATHROOM	SUITE - WC	PCT02	PCT02	PT01A/CT01-B/CT03	CT01-B AT VANITY BACKSPLASH, REFER TO ENLARGED SUITE PLAN FOR EXTENTS, CT03, REFER TO ENLARGED SUITE PLAN FOR EXTENTS
LEVEL 2	209C	DINING / KITCHEN	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 2	210	1-BD	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 2	210A	BEDROOM	SUITE - BEDROOM	LVP01	WB01	PT01	
LEVEL 2	210B	BATHROOM	SUITE - WC	PCT02	PCT02	PT01A/CT01-B/CT03	CT01-B AT VANITY BACKSPLASH, REFER TO ENLARGED SUITE PLAN FOR EXTENTS, CT03, REFER TO ENLARGED SUITE PLAN FOR EXTENTS
LEVEL 2	210C	DINING / KITCHEN	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 2	211	1-BD	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 2	211A	BEDROOM	SUITE - BEDROOM	LVP01	WB01	PT01	
LEVEL 2	211B	BATHROOM	SUITE - WC	PCT02	PCT02	PT01A/CT01-B/CT03	CT01-B AT VANITY BACKSPLASH, REFER TO ENLARGED SUITE PLAN FOR EXTENTS, CT03, REFER TO ENLARGED SUITE PLAN FOR EXTENTS
LEVEL 2	211C	DINING / KITCHEN	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 2	212	1-BD	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 2	212A	BEDROOM	SUITE - BEDROOM	LVP01	WB01	PT01	
LEVEL 2	212B	BATHROOM	SUITE - WC	PCT02	PCT02	PT01A/CT01-B/CT03	CT01-B AT VANITY BACKSPLASH, REFER TO ENLARGED SUITE PLAN FOR EXTENTS, CT03, REFER TO ENLARGED SUITE PLAN FOR EXTENTS
LEVEL 2	212C	DINING / KITCHEN	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 2	213	1-BD BF	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 2	213A	BEDROOM	SUITE - BEDROOM	LVP01	WB01	PT01	
LEVEL 2	213B	BF WC	SUITE - WC (BF)	PCT02	PCT02	PT01A/CT01-B/CT03	CT01-B AT VANITY BACKSPLASH, REFER TO ENGLARGED SUITE PLAN FOR EXTENTS, CT03, REFER TO ENLARGED SUITE PLAN FOR EXTENTS
LEVEL 2	213C	KITCHEN / DINING	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 2	214	1-BD BF	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 2	214A	BEDROOM	SUITE - BEDROOM	LVP01	WB01	PT01	
LEVEL 2	214B	BF WC	SUITE - WC (BF)	PCT02	PCT02	PT01A/CT01-B/CT03	CT01-B AT VANITY BACKSPLASH, REFER TO ENGLARGED SUITE PLAN FOR EXTENTS, CT03, REFER TO ENLARGED SUITE PLAN FOR EXTENTS
LEVEL 2	214C	KITCHEN / DINING	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 2	215	1-BD	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 2	215A	BEDROOM	SUITE - BEDROOM	LVP01	WB01	PT01	
LEVEL 2	215B	BATHROOM	SUITE - WC	PCT02	PCT02	PT01A/CT01-B/CT03	CT01-B AT VANITY BACKSPLASH, REFER TO ENLARGED SUITE PLAN FOR EXTENTS, CT03, REFER TO ENLARGED SUITE PLAN FOR EXTENTS
LEVEL 2	215C	DINING / KITCHEN	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 2	216	1-BD	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 2	216A	BEDROOM	SUITE - BEDROOM	LVP01	WB01	PT01	
LEVEL 2	216B	BATHROOM	SUITE - WC	PCT02	PCT02	PT01A/CT01-B/CT03	CT01-B AT VANITY BACKSPLASH, REFER TO ENLARGED SUITE PLAN FOR EXTENTS, CT03, REFER TO ENLARGED SUITE PLAN FOR EXTENTS
LEVEL 2	216C	DINING / KITCHEN	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 2	217	ELEC	ELEC CLOSET	PCT01	PCT03	PT01	
LEVEL 3	3A	STAIR A	STAIRS	RBF01/RBST01	RB01	PT01	PROVIDE RUBBER BASE AT LANDINGS
LEVEL 3	3B	STAIR B	STAIRS	RBF01/RBST01	RB01	PT01	PROVIDE RUBBER BASE AT LANDINGS
LEVEL 3	3E	ELEV	ELEVATOR	PCT01	PCT03	PT01	
LEVEL 3	300	LOBBY	LOBBY (L2-3)	PCT01	PCT03	PT01	PCT01 AT ELEVATOR WALL, REFER TO INTERIOR ELEVATIONS
LEVEL 3	300A	CORRIDOR	CORRIDOR (L2-3)	PCT01	PCT03	PT01	DOOR SURROUND AT EACH SUITE ENTRY DOOR - REFER TO INTERIOR DETAIL & MILLWORK ELEVATION
LEVEL 3	301	STAFF RM	STAFF RM	EFC01	CB/EFC01	PT01A	
LEVEL 3	303	1-BD	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 3	303A	BEDROOM	SUITE - BEDROOM	LVP01	WB01	PT01	
LEVEL 3	303B	BATHROOM	SUITE - WC	PCT02	PCT02	PT01A/CT01-B/CT03	CT01-B AT VANITY BACKSPLASH, REFER TO ENLARGED SUITE PLAN FOR EXTENTS, CT03, REFER TO ENLARGED SUITE PLAN FOR EXTENTS
LEVEL 3	303C	DINING / KITCHEN	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 3	305	MECH. RM	MECH RM	WPC01	WPC01	PT01A	
LEVEL 3	306	1-BD	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 3	306A	BEDROOM	SUITE - BEDROOM	LVP01	WB01	PT01	
LEVEL 3	306B	BATHROOM	SUITE - WC	PCT02	PCT02	PT01A/CT01-B/CT03	CT01-B AT VANITY BACKSPLASH, REFER TO ENLARGED SUITE PLAN FOR EXTENTS, CT03, REFER TO ENLARGED SUITE PLAN FOR EXTENTS
LEVEL 3	306C	DINING / KITCHEN	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 3	307	1-BD	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 3	307A	BEDROOM	SUITE - BEDROOM	LVP01	WB01	PT01	
LEVEL 3	307B	BATHROOM	SUITE - WC	PCT02	PCT02	PT01A/CT01-B/CT03	CT01-B AT VANITY BACKSPLASH, REFER TO ENLARGED SUITE PLAN FOR EXTENTS, CT03, REFER TO ENLARGED SUITE PLAN FOR EXTENTS
LEVEL 3	307C	DINING / KITCHEN	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 3	308	CHUTE	WASTE CHUTE VEST.	PCT01	PCT03	PT01A/FRP	PROVIDE FRP PANELS ON ALL WALLS TO THE UNDERSIDE OF CEILING
LEVEL 3	309	1-BD	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 3	309A	BEDROOM	SUITE - BEDROOM	LVP01	WB01	PT01	
LEVEL 3	309B	BATHROOM	SUITE - WC	PCT02	PCT02	PT01A/CT01-B/CT03	CT01-B AT VANITY BACKSPLASH, REFER TO ENLARGED SUITE PLAN FOR EXTENTS, CT03, REFER TO ENLARGED SUITE PLAN FOR EXTENTS
LEVEL 3	309C	DINING / KITCHEN	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 3	310	1-BD	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 3	310A	BEDROOM	SUITE - BEDROOM	LVP01	WB01	PT01	
LEVEL 3	310B	BATHROOM	SUITE - WC	PCT02	PCT02	PT01A/CT01-B/CT03	CT01-B AT VANITY BACKSPLASH, REFER TO ENLARGED SUITE PLAN FOR EXTENTS, CT03, REFER TO ENLARGED SUITE PLAN FOR EXTENTS
LEVEL 3	310C	DINING / KITCHEN	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 3	311	1-BD	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 3	311A	BEDROOM	SUITE - BEDROOM	LVP01	WB01	PT01	
LEVEL 3	311B	BATHROOM	SUITE - WC	PCT02	PCT02	PT01A/CT01-B/CT03	CT01-B AT VANITY BACKSPLASH, REFER TO ENLARGED SUITE PLAN FOR EXTENTS, CT03, REFER TO ENLARGED SUITE PLAN FOR EXTENTS
LEVEL 3	311C	DINING / KITCHEN	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 3	312	1-BD	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 3	312A	BEDROOM	SUITE - BEDROOM	LVP01	WB01	PT01	
LEVEL 3	312B	BATHROOM	SUITE - WC	PCT02	PCT02	PT01A/CT01-B/CT03	CT01-B AT VANITY BACKSPLASH, REFER TO ENLARGED SUITE PLAN FOR EXTENTS, CT03, REFER TO ENLARGED SUITE PLAN FOR EXTENTS
LEVEL 3	312C	DINING / KITCHEN	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 3	313	1-BD BF	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 3	313A	BEDROOM	SUITE - BEDROOM	LVP01	WB01	PT01	
LEVEL 3	313B	BF WC	SUITE - WC (BF)	PCT02	PCT02	PT01A/CT01-B/CT03	CT01-B AT VANITY BACKSPLASH, REFER TO ENGLARGED SUITE PLAN FOR EXTENTS, CT03, REFER TO ENLARGED SUITE PLAN FOR EXTENTS
LEVEL 3	313C	KITCHEN / DINING	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 3	314	1-BD BF	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 3	314A	BEDROOM	SUITE - BEDROOM	LVP01	WB01	PT01	
LEVEL 3	314B	BF WC	SUITE - WC (BF)	PCT02	PCT02	PT01A/CT01-B/CT03	CT01-B AT VANITY BACKSPLASH, REFER TO ENGLARGED SUITE PLAN FOR EXTENTS, CT03, REFER TO ENLARGED SUITE PLAN FOR EXTENTS
LEVEL 3	314C	KITCHEN / DINING	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 3	315	1-BD	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 3	315A	BEDROOM	SUITE - BEDROOM	LVP01	WB01	PT01	
LEVEL 3	315B	BATHROOM	SUITE - WC	PCT02	PCT02	PT01A/CT01-B/CT03	CT01-B AT VANITY BACKSPLASH, REFER TO ENLARGED SUITE PLAN FOR EXTENTS, CT03, REFER TO ENLARGED SUITE PLAN FOR EXTENTS
LEVEL 3	315C	DINING / KITCHEN	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 3	316	1-BD	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 3	316A	BEDROOM	SUITE - BEDROOM	LVP01	WB01	PT01	
LEVEL 3	316B	BATHROOM	SUITE - WC	PCT02	PCT02	PT01A/CT01-B/CT03	CT01-B AT VANITY BACKSPLASH, REFER TO ENLARGED SUITE PLAN FOR EXTENTS, CT03, REFER TO ENLARGED SUITE PLAN FOR EXTENTS
LEVEL 3	316C	DINING / KITCHEN	SUITE - LIVING / DINING	LVP01	WB01	PT01/CT01-A	CT01 AT SERVERY BACKSPLASH, REFER TO CASEWORK ELEVATION FOR EXTENTS
LEVEL 3	317	ELEC	ELEC CLOSET	PCT01	PCT03	PT01	

GENERAL NOTES:

1. DELETED

2. PROVIDE UNCOUPLING MEMBRANE UNDER PCT01 FLOOR APPLICATION IN ALL LOCATIONS

The Regional Municipality of Durham

605 Rossland Road East
Whitby, Ontario L1N 6A3

Tel 905.668.7711

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MontgomerySisam

PROJECT NORTH / TRUE NORTH

#	DATE:	REVISION:	BY:
REVISIONS			
17	2026-10-30	ISSUED FOR TENDER - ADDENDUM 3	MSA
16	2026-10-16	ISSUED FOR TENDER - ADDENDUM 2	MSA
13	2026-09-16	ISSUED FOR TENDER	MSA
11	2026-03-27	ISSUED FOR 100% CONSTRUCTION DOCUMENTS	MSA
10	2026-02-20	ISSUED FOR 90% CD & CLASS A COSTING	MSA
9	2026-01-30	ISSUED FOR 75% CONSTRUCTION DOCUMENTS	MSA
7	2026-01-30	ISSUED FOR BUILDING PERMIT	MSA

REDEVELOPMENT - 409 CENTRE ST S

25008

409 Centre St. South,
Whitby, ON L1N 4W3

Town of Whitby File Number: SP-26-25

SCHEDULE - ROOM FINISH

SCALE:

DRAWN BY: OE

REVIEWED BY: EM

JOB NUMBER: 25008

PLOT DATE: 10/17/22

DRAWING NUMBER:

A6110

10/29/2026 2:35:56 PM Autodesk Docs://25008_DHR - 409 Centre St/25008_409 Centre St_MSA_ARCH_MAIN_R24.rvt

DOOR SCHEDULE (SUITES)

DOOR NUMBER	FROM ROOM	TO ROOM	MRT_DR_TYPE	PANEL				FRAME			COMMENTS
				WIDTH	HEIGHT	MATERIAL	FINISH	TYPE	MATERIAL	FINISH	
LEVEL 1											
D103-A	BATHROOM	DINING / KITCHEN	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D103-B	BEDROOM	1-BD	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D103-C		BEDROOM	M	1829	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D103-D	DINING / KITCHEN	DINING / KITCHEN	M	800	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D110-A	BATHROOM	DINING / KITCHEN	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D110-B	BEDROOM	1-BD	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D110-C		BEDROOM	M	1829	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D110-D	DINING / KITCHEN	DINING / KITCHEN	M	800	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D111-A	DINING / KITCHEN	BATHROOM	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D111-B	1-BD	BEDROOM	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D111-C	BEDROOM		M	1829	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D111-D	DINING / KITCHEN	DINING / KITCHEN	M	800	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D112-A	BATHROOM	DINING / KITCHEN	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D112-B	BEDROOM	1-BD	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D112-C		BEDROOM	M	1829	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D112-D	DINING / KITCHEN	DINING / KITCHEN	M	800	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D113-A	KITCHEN / DINING	BF WC	G	965	2100	WD	PLAM	F0B	WD	PLAM	
D113-B	1-BD BF	BEDROOM	A	1000	2135	WD	PLAM	F1A	WD	PLAM	
D113-C		BEDROOM	M	1220	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D113-D		KITCHEN / DINING	M	800	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D114-A	BF WC	KITCHEN / DINING	G	965	2100	WD	PLAM	F0B	WD	PLAM	
D114-B	BEDROOM	1-BD BF	A	1000	2135	WD	PLAM	F1A	WD	PLAM	
D114-C	BEDROOM		M	1220	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D114-D	KITCHEN / DINING		M	800	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D115-A	BATHROOM	DINING / KITCHEN	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D115-B	BEDROOM	1-BD	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D115-C		BEDROOM	M	1829	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D115-D	DINING / KITCHEN	DINING / KITCHEN	M	800	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D116-A	BATHROOM	DINING / KITCHEN	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D116-B	BEDROOM	1-BD	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D116-C		BEDROOM	M	1829	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D116-D	DINING / KITCHEN	DINING / KITCHEN	M	800	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
LEVEL 2											
D203-A	BATHROOM	DINING / KITCHEN	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D203-B	BEDROOM	1-BD	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D203-C		BEDROOM	M	1829	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D203-D	DINING / KITCHEN	DINING / KITCHEN	M	800	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D206-A	BATHROOM	DINING / KITCHEN	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D206-B	BEDROOM	1-BD	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D206-C		BEDROOM	M	1829	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D206-D	DINING / KITCHEN	DINING / KITCHEN	M	800	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D207-A	BATHROOM	DINING / KITCHEN	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D207-B	BEDROOM	1-BD	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D207-C		BEDROOM	M	1829	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D207-D	DINING / KITCHEN	DINING / KITCHEN	M	800	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D209-A	BATHROOM	DINING / KITCHEN	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D209-B	BEDROOM	1-BD	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D209-C		BEDROOM	M	1524	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D209-D		DINING / KITCHEN	A	438	2135	WD	PLAM	F1A	WD	PLAM	
D210-A	BATHROOM	DINING / KITCHEN	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D210-B	BEDROOM	1-BD	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D210-C		BEDROOM	M	1829	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D210-D	DINING / KITCHEN	DINING / KITCHEN	M	800	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D211-A	DINING / KITCHEN	BATHROOM	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D211-B	1-BD	BEDROOM	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D211-C	BEDROOM		M	1829	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D211-D	DINING / KITCHEN	DINING / KITCHEN	M	800	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D212-A	BATHROOM	DINING / KITCHEN	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D212-B	BEDROOM	1-BD	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D212-C		BEDROOM	M	1829	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D212-D	DINING / KITCHEN	DINING / KITCHEN	M	800	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D213-A	KITCHEN / DINING	BF WC	G	965	2100	WD	PLAM	F0B	WD	PLAM	
D213-B	1-BD BF	BEDROOM	A	1000	2135	WD	PLAM	F1A	WD	PLAM	
D213-C		BEDROOM	M	1220	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D213-D		KITCHEN / DINING	M	800	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D214-A	BF WC	KITCHEN / DINING	G	965	2100	WD	PLAM	F0B	WD	PLAM	
D214-B	BEDROOM	1-BD BF	A	1000	2135	WD	PLAM	F1A	WD	PLAM	
D214-C	BEDROOM		M	1220	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D214-D	KITCHEN / DINING		M	800	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D215-A	BATHROOM	DINING / KITCHEN	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D215-B	BEDROOM	1-BD	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D215-C		BEDROOM	M	1829	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D215-D	DINING / KITCHEN	DINING / KITCHEN	M	800	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D216-A	BATHROOM	DINING / KITCHEN	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D216-B	BEDROOM	1-BD	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D216-C		BEDROOM	M	1829	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D216-D	DINING / KITCHEN	DINING / KITCHEN	M	800	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
LEVEL 3											
D303-A	BATHROOM	DINING / KITCHEN	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D303-B	BEDROOM	1-BD	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D303-C		BEDROOM	M	1829	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D303-D	DINING / KITCHEN	DINING / KITCHEN	M	800	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D306-A	BATHROOM	DINING / KITCHEN	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D306-B	BEDROOM	1-BD	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D306-C		BEDROOM	M	1829	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D306-D	DINING / KITCHEN	DINING / KITCHEN	M	800	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D307-A	BATHROOM	DINING / KITCHEN	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D307-B	BEDROOM	1-BD	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D307-C		BEDROOM	M	1829	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D307-D	DINING / KITCHEN	DINING / KITCHEN	M	800	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D309-A	BATHROOM	DINING / KITCHEN	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D309-B	BEDROOM	1-BD	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D309-C		BEDROOM	M	1524	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D309-D		DINING / KITCHEN	A	438	2135	WD	PLAM	F1A	WD	PLAM	
D310-A	BATHROOM	DINING / KITCHEN	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D310-B	BEDROOM	1-BD	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D310-C		BEDROOM	M	1829	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D310-D	DINING / KITCHEN	DINING / KITCHEN	M	800	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D311-A	DINING / KITCHEN	BATHROOM	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D311-B	1-BD	BEDROOM	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D311-C	BEDROOM		M	1829	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D311-D	DINING / KITCHEN	DINING / KITCHEN	M	800	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D312-A	BATHROOM	DINING / KITCHEN	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D312-B	BEDROOM	1-BD	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D312-C		BEDROOM	M	1829	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D312-D	DINING / KITCHEN	DINING / KITCHEN	M	800	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D313-A	KITCHEN / DINING	BF WC	G	965	2100	WD	PLAM	F0B	WD	PLAM	
D313-B	1-BD BF	BEDROOM	A	1000	2135	WD	PLAM	F1A	WD	PLAM	
D313-C		KITCHEN / DINING	M	800	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D313-E		BEDROOM	M	1220	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D314-A	BF WC	KITCHEN / DINING	G	965	2100	WD	PLAM	F0B	WD	PLAM	
D314-B	BEDROOM	1-BD BF	A	1000	2135	WD	PLAM	F1A	WD	PLAM	
D314-C	KITCHEN / DINING		M	800	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D314-E	BEDROOM		M	1220	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D315-A	BATHROOM	DINING / KITCHEN	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D315-B	BEDROOM	1-BD	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D315-C		BEDROOM	M	1829	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D315-D	DINING / KITCHEN	DINING / KITCHEN	M	800	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D316-A	BATHROOM	DINING / KITCHEN	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D316-B	BEDROOM	1-BD	A	915	2135	WD	PLAM	F1A	WD	PLAM	
D316-C		BEDROOM	M	1829	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR
D316-D	DINING / KITCHEN	DINING / KITCHEN	M	800	2032	WD	PLAM	F2A	WD	PLAM	CLOSET DR

DOOR SCHEDULE

DOOR NUMBER	FROM ROOM	TO ROOM	PANEL						FRAME				FRR	COMMENTS	ADD
			TYPE	GLAZING	WIDTH	HEIGHT	MATERIAL	FINISH	TYPE	GLAZING	MATERIAL	FINISH			
BASEMENT															
DB101	CORR	ELEC & COMMS	D	-	1410	2134	HM	PT	F2A	-	HM	PT		A-Symmetrical panels (W1=965mm & W2=445mm)	
DB102	INCOMING WATER RM	CORR	A	-	965	2135	HM	PT	F1A	-	HM	PT	45MIN		
DBB	STAIR B	CORR	A	-	965	2135	HM	PT	F1A	-	HM	PT	20MIN		
LEVEL 1															
D1A-1	CORRIDOR	STAIR A	A	-	1000	2134	HM	PT	F1D	IGL-F	HM	PT	20MIN		
D1A-2	STAIR A		D	GL3	1150	2375	ALUM	AB	F1D		ALUM	AB			
D1B-1	CORRIDOR	STAIR B	A	-	1000	2134	HM	PT	F1D	IGL-F	HM	PT	20MIN		
D1B-2	STAIR B		A	-	965	2135	ALUM	AB	F1A		ALUM	AB			
D1B-3	STAIR B		A	-	965	2135	ALUM	AB	F1A		ALUM	AB			
D100B		CORRIDOR	D	GL3	1120	2495	ALUM	AB	F1A		ALUM	AB			Yes
D101-1	VEST.		A	-	1145	2445	WD-5	WD-5	F1A		ALUM	AB			Yes
D101-2	LOBBY	VEST.	E	IGL-1	1145	2325	WD	WD01	F1A		ALUM	AB			Yes
D102	OFFICE	LOBBY	A	-	1000	2134	SCW	WD01	F1D	IGL-F	HM	PT	20MIN		
D103	CORRIDOR	DINING / KITCHEN	A	-	1000	2135	WD	PLAM01	F1A		WD	PT	20MIN		
D104-1	CORRIDOR	LAUNDRY	A	-	1000	2134	HM	PT	F1D	IGL-F	HM	PT	20MIN		Yes
D104-2	AMENITY	LAUNDRY	A	-	1000	2134	HM	PT	F1D	IGL-F	HM	PT	20MIN		Yes
D105	CORRIDOR	U WC	A	-	1000	2135	HM	PT	F1A		HM	PT	20MIN		Yes
D106	CORRIDOR	STORAGE	A	-	1000	2135	HM	PT	F1A		HM	PT	20MIN		
D107-1	LOBBY	AMENITY	D	IGL-F	965	2184	SCW	WD01	F1D		HM	PT	20MIN		Yes
D107-2	AMENITY		D		1043	2377	ALUM	PT	F1A		ALUM	PT			Yes
D107-3	AMENITY	STOR	AA	-	1202	2137	SCW	PLAM03	F0A		WD	PT		REFER TO MILLWORK SCHEDULE	
D107-4	AMENITY	STOR	AA	-	1202	2137	SCW	PLAM03	F0A		WD	PT		REFER TO MILLWORK SCHEDULE	
D108-1		VEST.	A		1145	2445	WD-5	WD-5	F1A		ALUM	AB			Yes
D108-2	VEST.	LOBBY	E	IGL-1	1239	2375	WD	WD01	F1A		ALUM	AB			Yes
D109-1	CORRIDOR	WASTE	A	-	1000	2135	HM	PT	F1A		HM	PT	1.5HR		
D109-2	WASTE		A		1219	2135	ALUM	AB	F1A		ALUM	PT			
D109-3	WASTE		A		965	2135	ALUM	AB	F1A		ALUM	PT			
D110	CORRIDOR	DINING / KITCHEN	A	-	1000	2135	WD	PLAM01	F1A	-	WD	PT	20MIN		
D111	DINING / KITCHEN	CORRIDOR	A	-	1000	2135	WD	PLAM01	F1A	-	WD	PT	20MIN		
D112	CORRIDOR	DINING / KITCHEN	A	-	1000	2135	WD	PLAM01	F1A	-	WD	PT	20MIN		
D113	KITCHEN / DINING	CORRIDOR	A	-	1000	2135	WD	PLAM01	F1A	-	WD	PT	20MIN		
D114	CORRIDOR	KITCHEN / DINING	A	-	1000	2135	WD	PLAM01	F1A	-	WD	PT	20MIN		
D115	CORRIDOR	DINING / KITCHEN	A	-	1000	2135	WD	PLAM01	F1A	-	WD	PT	20MIN		
D116	CORRIDOR	DINING / KITCHEN	A	-	1000	2135	WD	PLAM01	F1A	-	WD	PT	20MIN		
D117	ELEC	CORRIDOR	AA		1730	2135	HM		F2A		HM	PT	45MIN		
D118	CORR		A		965	2135	ALUM	AB	F1A		ALUM	PT			
D-1EL	ELEV	LOBBY			1067	2134		SS				SS		ELEVATOR DR	
LEVEL 2															
D2A	CORRIDOR	STAIR A	A	-	1000	2134	HM	PT	F1D	IGL-F	HM	PT	20MIN		
D2B	CORRIDOR	STAIR B	A	-	1000	2134	HM	PT	F1D	IGL-F	HM	PT	20MIN		
D201	ELEV. CTRL	LOBBY	A	-	965	2135	HM	PT	F1A	-	HM	PT	20MIN	COORDINATE GRILLE ON DOOR. REFER TO MECHANICAL	
D202	HSKP	LOBBY	A	-	965	2135	HM	PT	F1A	-	HM	PT	20MIN		
D203	CORRIDOR	DINING / KITCHEN	A	-	1000	2135	WD	PLAM01	F1A	-	WD	PT	20MIN		
D205	CORRIDOR	MECH. RM	A	-	965	2135	HM	PT	F1A	-	HM	PT	45MIN		
D206	CORRIDOR	DINING / KITCHEN	A	-	1000	2135	WD	PLAM01	F1A	-	WD	PT	20MIN		
D207	CORRIDOR	DINING / KITCHEN	A	-	1000	2135	WD	PLAM01	F1A	-	WD	PT	20MIN		
D208	CHUTE	CORRIDOR	A	-	1000	2135	HM	PT	F1A	-	HM	PT	45MIN		Yes
D209	CORRIDOR	DINING / KITCHEN	A	-	1000	2135	WD	PLAM01	F1A	-	WD	PT	20MIN		
D210	CORRIDOR	DINING / KITCHEN	A	-	1000	2135	WD	PLAM01	F1A	-	WD	PT	20MIN		
D211	DINING / KITCHEN	CORRIDOR	A	-	1000	2135	WD	PLAM01	F1A	-	WD	PT	20MIN		
D212	CORRIDOR	DINING / KITCHEN	A	-	1000	2135	WD	PLAM01	F1A	-	WD	PT	20MIN		
D213	KITCHEN / DINING	CORRIDOR	A	-	1000	2135	WD	PLAM01	F1A	-	WD	PT	20MIN		
D214	CORRIDOR	KITCHEN / DINING	A	-	1000	2135	WD	PLAM01	F1A	-	WD	PT	20MIN		
D215	CORRIDOR	DINING / KITCHEN	A	-	1000	2135	WD	PLAM01	F1A	-	WD	PT	20MIN		
D216	CORRIDOR	DINING / KITCHEN	A	-	1000	2135	WD	PLAM01	F1A	-	WD	PT	20MIN		
D217	ELEC	CORRIDOR	AA		1730	2135	HM	PT	F2A		HM	PT	45MIN		
D-2EL	ELEV	LOBBY			1067	2134		SS				SS		ELEVATOR DR	
LEVEL 3															
D3A	CORRIDOR	STAIR A	A	-	1000	2134	HM	PT	F1D	IGL-F	HM	PT	20MIN		
D3B	CORRIDOR	STAIR B	A	-	1000	2134	HM	PT	F1D	IGL-F	HM	PT	20MIN		
D301	LOBBY	STAFF RM	A	-	965	2135	HM	PT	F1A	-	HM	PT	20MIN		
D303	CORRIDOR	DINING / KITCHEN	A	-	1000	2135	WD	PLAM01	F1A	-	WD	PT	20MIN		
D305	CORRIDOR	MECH. RM	A	-	965	2135	HM	PT	F1A	-	HM	PT	45MIN		
D306	CORRIDOR	DINING / KITCHEN	A	-	1000	2135	WD	PLAM01	F1A	-	WD	PT	20MIN		
D307	CORRIDOR	DINING / KITCHEN	A	-	1000	2135	WD	PLAM01	F1A	-	WD	PT	20MIN		
D308	CHUTE	CORRIDOR	A	-	1000	2135	HM	PT	F1A	-	HM	PT	45MIN		Yes
D309	CORRIDOR	DINING / KITCHEN	A	-	1000	2135	WD	PLAM01	F1A	-	WD	PT	20MIN		
D310	CORRIDOR	DINING / KITCHEN	A	-	1000	2135	WD	PLAM01	F1A	-	WD	PT	20MIN		
D311	DINING / KITCHEN	CORRIDOR	A	-	1000	2135	WD	PLAM01	F1A	-	WD	PT	20MIN		
D312	CORRIDOR	DINING / KITCHEN	A	-	1000	2135	WD	PLAM01	F1A	-	WD	PT	20MIN		
D313	CORRIDOR	KITCHEN / DINING	A	-	1000	2135	WD	PLAM01	F1A	-	WD	PT	20MIN		
D314	CORRIDOR	KITCHEN / DINING	A	-	1000	2135	WD	PLAM01	F1A	-	WD	PT	20MIN		
D315	CORRIDOR	DINING / KITCHEN	A	-	1000	2135	WD	PLAM01	F1A	-	WD	PT	20MIN		
D316	CORRIDOR	DINING / KITCHEN	A	-	1000	2135	WD	PLAM01	F1A	-	WD	PT	20MIN		
D317	ELEC	CORRIDOR	AA		1730	2135	HM	PT	F2A		HM	PT	45MIN		
D-3EL	ELEV	LOBBY			1067	2134		SS				SS		ELEVATOR DR	
T/O LEVEL 3															
D1A-3		OUTDOOR ROOFTOP EQMT AREA	A		965	2032	HM	PT	F1A		HM	PT	-		

DOOR SCHEDULE GENERAL NOTES

1. DOOR OFFSET TO BE 100MM FROM INTERIOR PARTITION UNLESS OTHERWISE NOTED, WITH A MINIMUM CLEAR DOOR WIDTH OPENING OF 860MM

2. DOORS FOR ALL ROOMS WHERE NOISE MITIGATION IS REQUIRED (ALL SUITE ENTRY DOORS, OFFICE, AMENITY, UNIV. WC, STAFF WC, HSKP, ELEV. CONTROL ROOM MECHANICAL ROOMS, WASTE ROOM, CHUTE ROOM,) ACOUSTIC SILICONE SEALS SHOULD BE PROVIDED AT SIDES AND BOTTOM OF DOOR

3. OFFICE AND AMENITY SPACE ENTRANCE DOOR LEAFS TO HAVE A MINIMUM SURFACE MASS OF 25 kg/m².

4. ALL SUITE ENTRY , MECHANICAL, LAUNDRY AND WASTE ROOMS DOOR LEAFS TO HAVE A MINIMUM SURFACE MASS OF 35 kg/m².

5. COORDINATE WITH DOOR AND FRAME SCHEDULE FOR DOOR DIMENSIONS.

6. COORDINATE WITH SPECIFICATIONS FOR APPROVED MANUFACTURERS. DIMENSIONS MAY VARY TO SUIT FRAME MATERIAL AND PROFILES AVAILABLE.

7. COORDINATE WITH DOOR, FRAME & SCREEN SCHEDULE FOR TYPES OF GLAZING INSERTS.

8. REFER TO PLANS AND INTERIOR ELEVATIONS FOR SWING AND OPERATION OF DOOR.

9. COORDINATE WITH HARDWARE SCHEDULE FOR HARDWARE AND ACCESSORIES.

10. COORDINATE WITH ELECTRICAL FOR OPERATORS, HOLD-OPENS, CARD READER ACCESS, ETC.

11. REFER TO DOOR & FRAME SCHEDULE FOR DOOR DIMENSIONS.

12. ALL HM FRAMES TO BE PAINTED PT02, EXCEPT ELECTRICAL CLOSET AND MECHANICAL ROOM DOOR FRAMES TO BE PAINED PT01, & ALL SUITE ENTRY DOOR FRAMES TO BE PAINTED TO MATCH PLAM02.

13. ALL HM DOORS TO BE PAINTED PT03, EXCEPT ELECTRICAL CLOSET AND MECHANICAL ROOM DOORS TO BE PAINED PT01.

14. ALL DOORS INSIDE THE SUITES TO HAVE A WHITE PLAM03 FINISH

15. ALL EXTERIOR ALUMINUM DOORS & FRAMES TO BE INSULATED

16. PROVIDE JOINT SEALANT AROUND ALL EXTERIOR AND INTERIOR STEEL FRAMES (BOTH SIDES)

17. WHERE DOORS ARE TO BE FIRE RATED, DOOR FRAMES AND HARDWARE SHALL HAVE EQUAL FIRE RATING.

18. EVERY DOOR SHALL HAVE A DOOR STOP TO PROTECT THE INTEGRITY OF THE DOOR AND PROVIDE ADDITIONAL PROTECTION DAMAGE TO WALL FINISHES

19. GLASS FILM TO BE PROVIDED ON ALL GLAZED PORTIONS OF DOORS/SIDELIGHTS AT 1350MM ABOVE FFE.

ABBREVIATIONS

DOOR

ALUM	ALUMINUM
HM	HOLLOW METAL
PLAM	PLASTIC LAMINATE VENEER
PT	PAINT
WD	SOLID CORE WOOD DOOR
U/C	UNDER CUT

SECURITY

ADO	AUTOMATIC DOOR OPERATOR
ADO-DBL	AUTOMATIC DOOR OPERATOR-DOUBLE
CR	CARD READER
CR-DBL	CARD READER-DOUBLE
DA	DOOR ALARM
DB	DOOR BUZZER
DC	DOOR CONTACT
DLC	DELAYED CLOSING
DO	DOOR OPERATOR
EPH	ELECTRIFIED PANIC HARDWARE
ES	ELECTRIC STRIKE
HO	HOLD OPEN DEVICE
HO-CLR	HOLD OPEN DEVICE - CLOSURE
HO-MAG	HOLD OPEN DEVICE - MAGNETIC
KP	KEY PAD LOCK
KP-DBL	KEY PAD LOCK-DOUBLE
KS	KEY SWITCH
ML	MAGNETIC LOCK
OL	OCCUPIED INDICATOR
PB	PUSH BUTTON
PHL	PHYSICAL LOCK
PL	PUSH TO LOCK
REX	REQUEST TO EXIT
RWS	RESIDENT WANDERING SYSTEM
TID	TAG ID SYSTEM
TRK	ALUMINUM TRACK

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PROJECT NORTH / TRUE NORTH

#	DATE:	REVISION:	BY:
17	2026-10-30	ISSUED FOR TENDER - ADDENDUM 3	MSA
16	2026-10-16	ISSUED FOR TENDER - ADDENDUM 2	MSA
13	2026-09-16	ISSUED FOR TENDER	MSA
11	2026-03-27	ISSUED FOR 100% CONSTRUCTION DOCUMENTS	MSA
10	2026-02-20	ISSUED FOR 90% CD & CLASS A COSTING	MSA
9	2026-01-30	ISSUED FOR 75% CONSTRUCTION DOCUMENTS	MSA
7	2026-01-30	ISSUED FOR BUILDING PERMIT	MSA

REVISIONS

REDEVELOPMENT - 409 CENTRE ST S

25008

409 Centre St. South,
Whitby, ON L1N 4W3

Town of Whitby File Number: SP-26-25

SCHEDULE - DOOR

SCALE: 1 : 50
DRAWN BY: OE
REVIEWED BY: EM
JOB NUMBER: 25008
PLOT DATE: 10/17/22

DRAWING NUMBER:
A6120

10/30/2026 12:21:22 PM Autodesk Docs://25008_DIR - 409 Centre St/25008_409 Centre St_MSA_ARCH_MAIN_R24.rvt

CASEWORK / MILLWORK ASSEMBLY SCHEDULE

MILLWORK #	DESCRIPTION	MATERIAL	COUNTERTOP MATERIAL	BASE HEIGHT	HARDWARE FINISH	NOTES
CW100	SERVERY - L1 AMENITY	PLAM03	SS01	230mm	BLACK	PROVDE KEYED LOCKING HARDWARE AT CABINET ABOVE FRIDGE
CW101	SERVERY - SUITE	PLAM02	SS01	TO MATCH SUITE BASE	NICKEL	
CW102	SERVERY - SUITE BF	PLAM02	SS01	TO MATCH SUITE BASE	NICKEL	
CW103	STORAGE - SUITE WC	PLAM02	-	TO MATCH SUITE BASE	NICKEL	
CW104	CLOSET COATS - AMENITY	PLAM03	-	230mm	BLACK	
MW101	VANITY - SUITE	-	SS02	-	-	
MW102	VANITY - SUITE CHUTE	PLAM02	SS02	TO MATCH SUITE BASE	NICKEL	
MW103	VANITY - SUITE BF	-	SS02	-	-	
MW104	VANITY - STAFF RM	-	SS02	-	-	
MW105	FOLDING - L1 LAUNDRY	PLAM03	SS01	TO MATCH ROOM BASE	-	
MW106	CLOSET STORAGE - AMENITY	PLAM03	-	-	BLACK	PROVIDE KEYED LOCKING HARDWARE, CONCEALED HINGES
MW110	MAILBOX SURROUND	WD01	-	TO MATCH ROOM BASE	BLACK	PROVIDE KEYED LOCKING HARDWARE AT CABINET ABOVE MAILBOXES
MW111	BENCH 2	WD01	-	-	-	REFER TO PLANS FOR LENGTH

17

17

17

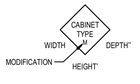
CASEWORK / MILLWORK PLAN TAG LEGEND

CW### — CASEWORK ASSEMBLY REFERENCE NUMBER

MW### — MILLWORK ASSEMBLY REFERENCE NUMBER

m — m - SUFFIX DENOTES MIRRORED FROM ELEVATION SHOWN

CASEWORK ELEVATION TAG LEGEND



ELEVATION TAG

ELEVATION TAG NOTES:
- HEIGHT OF BASE CABINETS INCLUDES COUNTERTOP THICKNESS.
- HEIGHT OF WALL-HUNG CABINETS DOES NOT INCLUDE LIGHT VALANCE.
- DEPTH OF CABINETS DOES NOT INCLUDE CABINET DOOR THICKNESS.
- DEPTH OF OPEN CABINETS INDICATES DEPTH OF CABINET ONLY.

GENERAL CABINET TYPE NOTES **[#TEAMS TO EDIT]**
1. ALL CABINETS TO BE FITTED WITH KEY LOCKS UNLESS OTHERWISE NOTED.

MODIFICATIONS LEGEND **[#TEAMS TO EDIT]**
A ANTI-LIGATURE HINGES AND PULL
L KEY LOCK
S PROVIDE EXTRA SHELF

COMPONENT DETAIL REFERENCE. REFER TO A8111, A8112.

B BASE OF CABINET DETAIL
BS BACKSPLASH DETAIL
C COUNTERTOP AT FRONT OF CABINET DETAIL
F FILLER DETAIL
G GABLE DETAIL
S SHELF DETAIL
T TOP OF CABINET DETAIL
U UNDERSIDE OF CABINET DETAIL

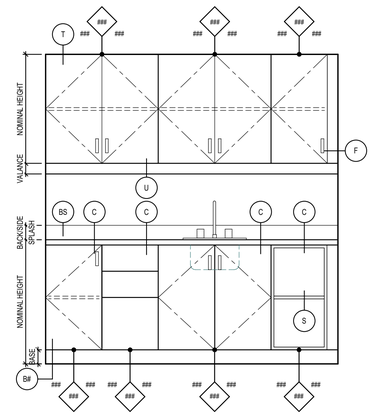
T1

##

UNIQUE CASEWORK/MILLWORK FINISH TAG. REFER TO MATERIAL & FINISH SCHEDULE IN SPECIFICATOPMS

EXAMPLE CASEWORK COMPONENT REFERENCE ELEVATION

ELEVATION HAS BEEN PROVIDED AS A GENERIC REFERENCE AND DOES NOT INCLUDE ALL COMPONENTS OR CABINET TYPES.



CASEWORK AND MILLWORK GENERAL NOTES

1. REFER TO 06 20 00 FINISH CARPENTRY SPECIFICATIONS FOR ADDITIONAL INFORMATION.
2. TYPICAL CABINET SERIES TYPES ARE LABELED WITH A CABINET DESIGN SERIES (CDS) NUMBER IN ACCORDANCE WITH THE MOST CURRENT NORTH AMERICAN ARCHITECTURAL WOODWORK STANDARDS (NAAWS).
3. REFER TO CASEWORK AND MILLWORK ELEVATIONS FOR CABINET SIZES, MOUNTING HEIGHTS AND ADDITIONAL DESIGN DETAILS.
4. REFER TO THE CASEWORK AND MILLWORK SCHEDULE FOR CABINET AND COUNTER FINISHES.
   - FASCIA FINISHES SHALL MATCH FRONT OF CABINETS UNLESS OTHERWISE NOTED
   - INTERIOR CABINET FINISHES SHALL BE ML01 UNLESS OTHERWISE NOTED
   - WHERE APPLICABLE, FOR BOOKCASES, MAIL SLOTS AND OPEN CABINETS, ALL EXPOSED SURFACES SHALL MATCH CABINET FINISH AS PER CASEWORK / MILLWORK FINISH SCHEDULE.
   - WHERE END OF CABINET DOES NOT MEET A PARTITION, FINISH END OF CABINET TO MATCH ADJACENT FINISH.
5. REFER TO THE ROOM FINISHES SCHEDULE FOR APPLIED BASE FINISHES AND WALL FINISHES, INCLUDING APPLIED BACKSPLASHES.
6. BACKSPLASHES AND SIDE SPLASHES TO BE FROM TOP OF COUNTER TO UNDERSIDE OF FINISHED CABINET UNLESS NOTED OTHERWISE.
7. CASEWORK AND MILLWORK TYPES AND ELEVATIONS ARE TYPICAL. REFER TO FLOOR PLANS FOR LOCATIONS, SPACE CONSTRAINTS, QUANTITIES AND ORIENTATION.
8. ELEVATIONS HAVE BEEN PROVIDED FOR DESIGN INTENT. FILLERS HAVE BEEN SHOWN ON TYPICALS WHERE THEY ARE ANTICIPATED, HOWEVER SITE CONDITIONS FROM ROOM TO ROOM MAY VARY. ALL FILLER PANELS TO BE NOT LESS THAN 38MM FACE WIDTH. PROVIDE FILLER PANELS, AS REQUIRED, BETWEEN MILLWORK AND ADJACENT WALL AND CEILING SURFACES. FILLER PANELS TO MATCH FINISH OF ADJACENT MILLWORK UNLESS NOTED OTHERWISE. FILLER PANELS TO BE INSTALLED FLUSH WITH DOOR / DRAWER FRONTS.
9. VERIFY IN FIELD ALL FASCIA PANEL HEIGHTS RELATIVE TO CEILING.
10. WHERE APPLICABLE, COORDINATE MILLWORK DIMENSIONS AND SHELF LOCATIONS WITH APPLIANCE PRODUCT SPECIFICATIONS. WHERE EQUIPMENT HAS BEEN NOTED ON ELEVATIONS, REFER TO FFE PLANS FOR ADDITIONAL INFORMATION.
11. CONTRACTOR SHALL PROVIDE ALL INFRASTRUCTURE MATERIALS, SERVICES AND CONNECTIONS FOR FURNITURE, APPLIANCES AND EQUIPMENT AS DESCRIBED IN ENGINEERS' DOCUMENTS, INCLUDING BUT NOT LIMITED TO POWER, DATA AND PLUMBING.
12. ALL HARDWARE TO BE TYPICAL, UNLESS NOTED OTHERWISE. (REFER TO 06 20 00 FINISH CARPENTRY SPECIFICATIONS)
13. ALL CABINETS ARE NON-LOCKABLE UNLESS NOTED OTHERWISE.
14. ALL SHELVES TO BE ADJUSTABLE UNLESS NOTED OTHERWISE.
15. MOUNT CABINET HARDWARE AS PER TYPICAL NOTES ON A8101 CASEWORK / MILLWORK LEGEND AND NOTES, UNLESS NOTED OTHERWISE.
16. UNLESS NOTED OTHERWISE, ALL CASEWORK / MILLWORK TO BE CONSTRUCTED OF VENEER CORE PLYWOOD, MINIMUM THICKNESS 3/4" (19MM).
   - BACK PANELS MAY HAVE A MINIMUM THICKNESS OF 1/2" (13MM).
   - CABINET DOORS AND DRAWER FRONTS SHALL BE CONSTRUCTED OF MDF, MINIMUM THICKNESS 3/4" (19MM).
   - PROVIDE EXTERIOR-GRADE VENEER CORE PLYWOOD TO BOTTOMS OF SINK BOXES AND AREAS THAT COME INTO CONTACT WITH WATER.
   - PROVIDE 1" (25MM) SUBSTRATES FOR SHELVES THAT HAVE LARGER SPANS, AS REQUIRED TO MEET NAAWS STANDARDS.
17. PROVIDE CLEAR SILICONE SEALANT AT ALL INTERFACES BETWEEN DISSIMILAR MATERIALS, VERTICAL & HORIZONTAL (COUNTERTOPS, WALLS, ETC.)
18. PROVIDE GROMMETS WHERE APPLICABLE, EXACT LOCATIONS COORDINATED ON SITE.
19. ENSURE MILLWORK / CASEWORK IS MOUNTED TO BLOCKING AS REQUIRED TO SUPPORT CASEWORK / MILLWORK FOR ITS INTENDED FUNCTION. REFER TO MOUNTING HEIGHT SHEETS FOR ADDITIONAL INFORMATION.
20. ALL INTERIOR WINDOW SILLS TO BE SOLID WOOD UNLESS NOTED OTHERWISE.
21. WHERE PIANO HINGES ARE REQUIRED, PROVIDE A 50MM FILLER BETWEEN CABINETS TO ACCOMODATE CABINET DOOR SWINGS. OVERALL LENGTH OF ASSEMBLIES SHALL NOT CHANGE. CABINET SIZES TO BE ADJUSTED TO MEET THE DESIGN INTENT WITHIN THE OVERALL LENGTH THAT HAS CURRENTLY BEEN PLANNED. ELEVATIONS TO BE FURTHER DEVELOPED DURING SHOP DRAWING SUBMITTAL REVIEWS.
22. APPLY INSECT REPELLANT POWDER (DIATOMACEOUS EARTH) UNDER ALL LOWER CABINETS BEHIND KICKPLATES, AT ALL CABINETS AT SINKS. SEAL ALL CORNERS BETWEEN CABINET BOTTOM, SIDES, AND BACKS.
23. PROVIDE ALL SUPPORTS FOR WALL- OR CABINET-MOUNTED APPLIANCES (E.G. MICROWAVES, WALL OVENS, HOODS) AS REQUIRED, INCLUDING BUT NOT LIMITED TO METAL SUPPORT BRACKETS, BLOCKING, AND TOP PLATES.
24. CONTRACTOR SHALL SUPPLY ALL APPLIANCES AS PER APPLIANCE SCHEDULE

15

HOW TO READ CASEWORK AND MILLWORK DRAWINGS

THE DOCUMENTS IDENTIFY ASSEMBLIES THAT ARE INTENDED TO HAVE THE SAME BASIS OF DESIGN. CONTRACTOR SHALL SITE VERIFY SPACES PRIOR TO CONSTRUCTION TO ENSURE SITE CONDITIONS SUPPORT THE DESIGN INTENT AND IDENTIFY ADJUSTMENTS, AS REQUIRED IN SHOP DRAWINGS.

CW# - REFERS TO A CASEWORK CONSTRUCTION ASSEMBLY AND MAY INCLUDE A GROUP OF UPPER AND LOWER CABINETS, COUNTERS, FILLERS AND SUPPORTS

MW# - REFERS TO A MILLWORK CONSTRUCTION ASSEMBLY DESIGNED TO SUIT A SPECIFIC CONDITION AND MAY INCLUDE A GROUP OF UPPER AND LOWER CABINETS, COUNTERS, FILLERS AND SUPPORTS.

COMPONENT DETAIL - REFERS TO A SELECTION OF PREDETERMINED DETAILS SUCH AS TOP OF CABINET, UNDERSIDE OF CABINET, BACKSPLASH, BASE AND SHELF DETAILS

CASEWORK AND MILLWORK ASSEMBLIES ARE TAGGED ON FLOOR PLANS. THE TAGS IDENTIFY THE CONSTRUCTION ASSEMBLY TYPE AND LOCATION OF THE ASSEMBLY TYPE WITHIN THE BUILDING.

ELEVATIONS OF CONSTRUCTION ASSEMBLY TYPES ARE TAGGED TO IDENTIFY MILLWORK TYPES AND COMPONENT DETAILS NECESSARY TO COMPLETE THE SCOPE OF WORK.

CASEWORK TAGS REFLECT TYPICAL CABINET SERIES TYPES AS PER NORTH AMERICAN ARCHITECTURAL WOODWORK STANDARDS, UNLESS NOTED OTHERWISE.

CASEWORK VARIANTS

THE FOLLOWING ADDITIONAL CASEWORK TYPES HAVE BEEN CREATED AND HAVE BEEN IDENTIFIED WITH A LETTER MODIFICATION. REFER TO CASEWORK LEGEND FOR ADDITIONAL INFORMATION:

A PROVIDE PULL OUT PANTRY DRAWER HARDWARE SYSTEM - REFER TO SPECIFICATIONS
P PROVIDE PULL DOWN CABINET HARDWARE SYSTEM - REFER TO SPECIFICATIONS
S PROVIDE PULL OUT COUNTERTOP EXTENSION SYSTEM WITHIN TOP CABINET DRAWER. COUNTERTOP EXTENSION PANEL TO MATCH COUNTERTOP AS NOTED IN CASEWORK/MILLWORK SCHEDULE. REFER TO SPECIFICATIONS

The Regional Municipality of Durham
605 Rossland Road East
Whitby, Ontario L1N 6A3
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197 Spadina Avenue, Toronto, Ontario M5T 2C8 montgomerysisam.com
Tel 416.364.8079 Fax 416.364.7723

MontgomerySisam

#	DATE:	REVISION:	BY:
17	2026-10-30	ISSUED FOR TENDER - ADDENDUM 3	MSA
15	2026-10-13	ISSUED FOR TENDER - ADDENDUM 1	MSA
13	2026-09-16	ISSUED FOR TENDER	MSA
11	2026-03-27	ISSUED FOR 100% CONSTRUCTION DOCUMENTS	MSA
10	2026-02-20	ISSUED FOR 90% CD & CLASS A COSTING	MSA
9	2026-01-30	ISSUED FOR 75% CONSTRUCTION DOCUMENTS	MSA
5	2025-12-17	ISSUED FOR 30% CONSTRUCTION DOCUMENTS	MSA
1	2025-06-20	ISSUED FOR 100% DD & CLASS B COSTING	MSA

REVISIONS

REDEVELOPMENT - 409 CENTRE ST S

25008

409 Centre St. South,
Whitby, ON L1N 4W3

Town of Whitby File Number: SP-26-25

CASEWORK / MILLWORK NOTES & LEGEND, SCHEDULE

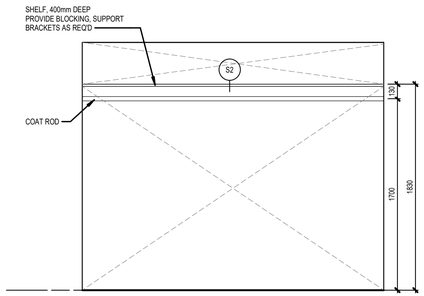
SCALE:	1 : 20
DRAWN BY:	Author
REVIEWED BY:	Checker
JOB NUMBER:	25008
PLOT DATE:	06/06/25

DRAWING NUMBER:
A8100

10/29/2026 2:38:56 PM Autodesk Docs://25008_DHR - 409 Centre St/25008_409 Centre St_MSA_ARCH_MAIN_R24.rvt

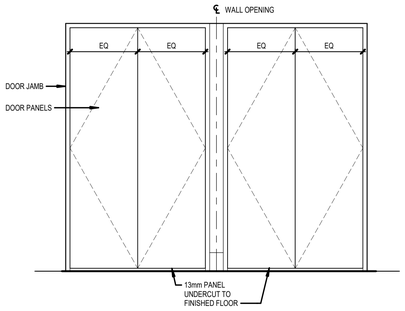
AMENITY MILLWORK

MW106



8 / A8301 MW106 - AMENITY STORAGE CLOSET - ELEVATION - INTERIOR

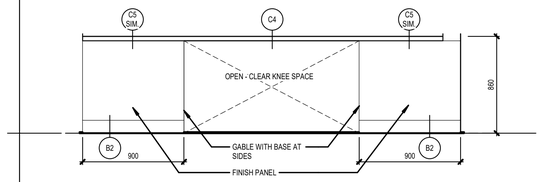
1 : 25



12 / A8301 MW106 - CLOSET STORAGE CLOSET - ELEVATION - CLOSET DOORS

1 : 25

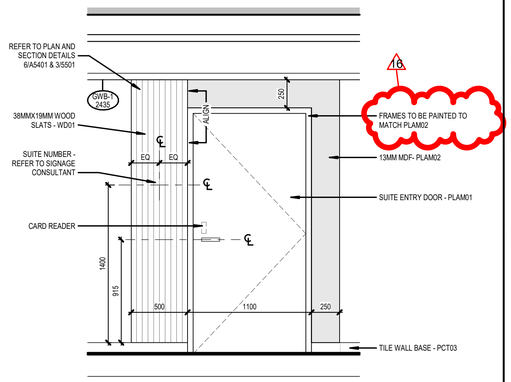
MW105



13 / A8301 MW105 - LAUNDRY FOLDING COUNTER ELEVATION

1 : 25

SUITE DOOR SURROUND (TYP.)

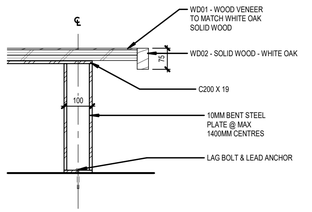


10 / A8301 ENLARGED ELEVATION - SUITE ENTRY DOOR SURROUND

1 : 25

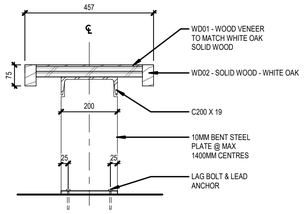
LOBBY MILLWORK

MW111



11 / A8301 MW111 - ELEVATOR LOBBY FLOOR BENCH SECTION DETAIL 2

1 : 10



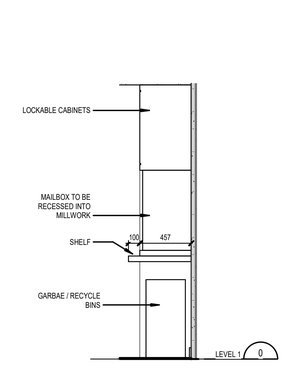
9 / A8301 MW111 - ELEVATOR LOBBY FLOOR BENCH SECTION DETAIL 1

1 : 10

6 / A8301 MW111 - ELEVATOR LOBBY FLOOR BENCH

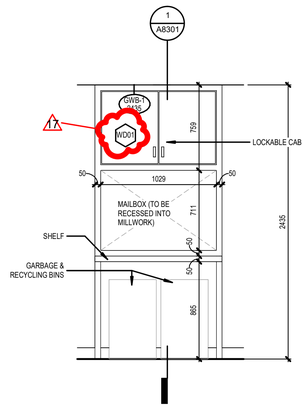
1 : 25

MW110



1 / A8301 MW110 - MAILBOX - SECTION

1 : 25

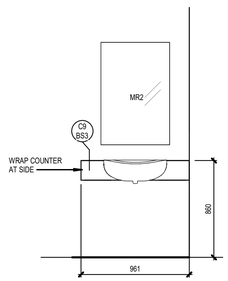


5 / A8301 MW110 - MAILBOX - ELEVATION

1 : 25

VANITIES

MW104

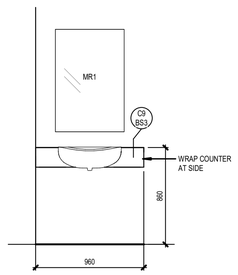


7 / A8301 MW104 - STAFF WC VANITY

1 : 25

MILLWORK MAY BE MIRRORED PER SUITE ORIENTATION.
REFER TO FLOOR PLANS.

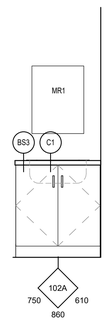
MW103



4 / A8301 MW103 - SUITE ACCESSIBLE WC VANITY

1 : 25

MW102

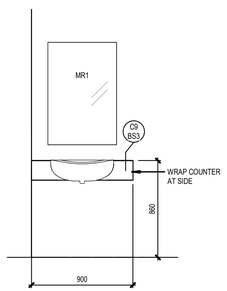


3 / A8301 MW102 - SUITE CHUTE WC VANITY

1 : 25

MILLWORK MAY BE MIRRORED PER SUITE ORIENTATION.
REFER TO FLOOR PLANS.

MW101



2 / A8301 MW101 - SUITE TYP. WC VANITY

1 : 25

The Regional Municipality of Durham

605 Rossland Road East
Whitby, Ontario L1N 6A3

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MontgomerySisam

PROJECT NORTH
TRUE NORTH

#	DATE:	REVISION:	BY:
17	2026-10-30	ISSUED FOR TENDER - ADDENDUM 3	MSA
16	2026-10-16	ISSUED FOR TENDER - ADDENDUM 2	MSA
13	2026-09-16	ISSUED FOR TENDER	MSA
11	2026-03-27	ISSUED FOR 100% CONSTRUCTION DOCUMENTS	MSA
10	2026-02-20	ISSUED FOR 90% CD & CLASS A COSTING	MSA

REVISIONS

REDEVELOPMENT - 409 CENTRE ST S

25008

409 Centre St. South,
Whitby, ON L1N 4W3

Town of Whitby File Number: SP-26-25

MILLWORK - ELEVATIONS, SECTIONS & SECTION DETAILS

SCALE:	As indicated
DRAWN BY:	Author
REVIEWED BY:	Checker
JOB NUMBER:	25008
PLOT DATE:	02/18/26

DRAWING NUMBER:

A8301

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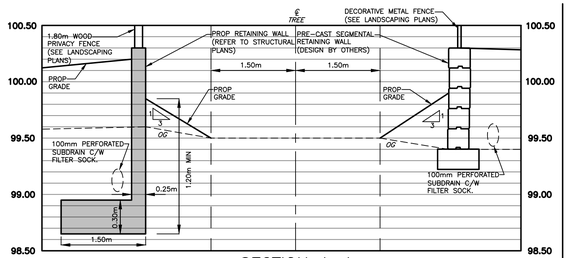

SECTION A–A
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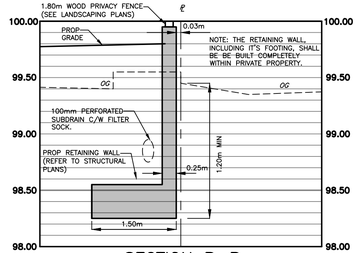
V SCALE: 1:25



SECTION B–B
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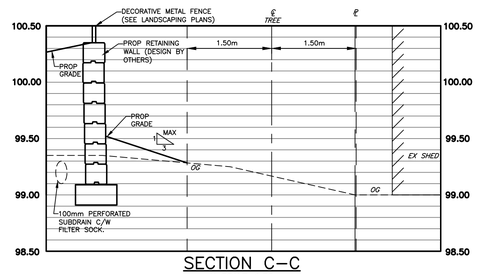
V SCALE: 1:25



SECTION C–C
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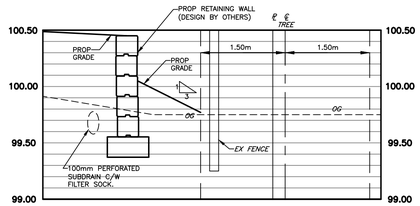
V SCALE: 1:25



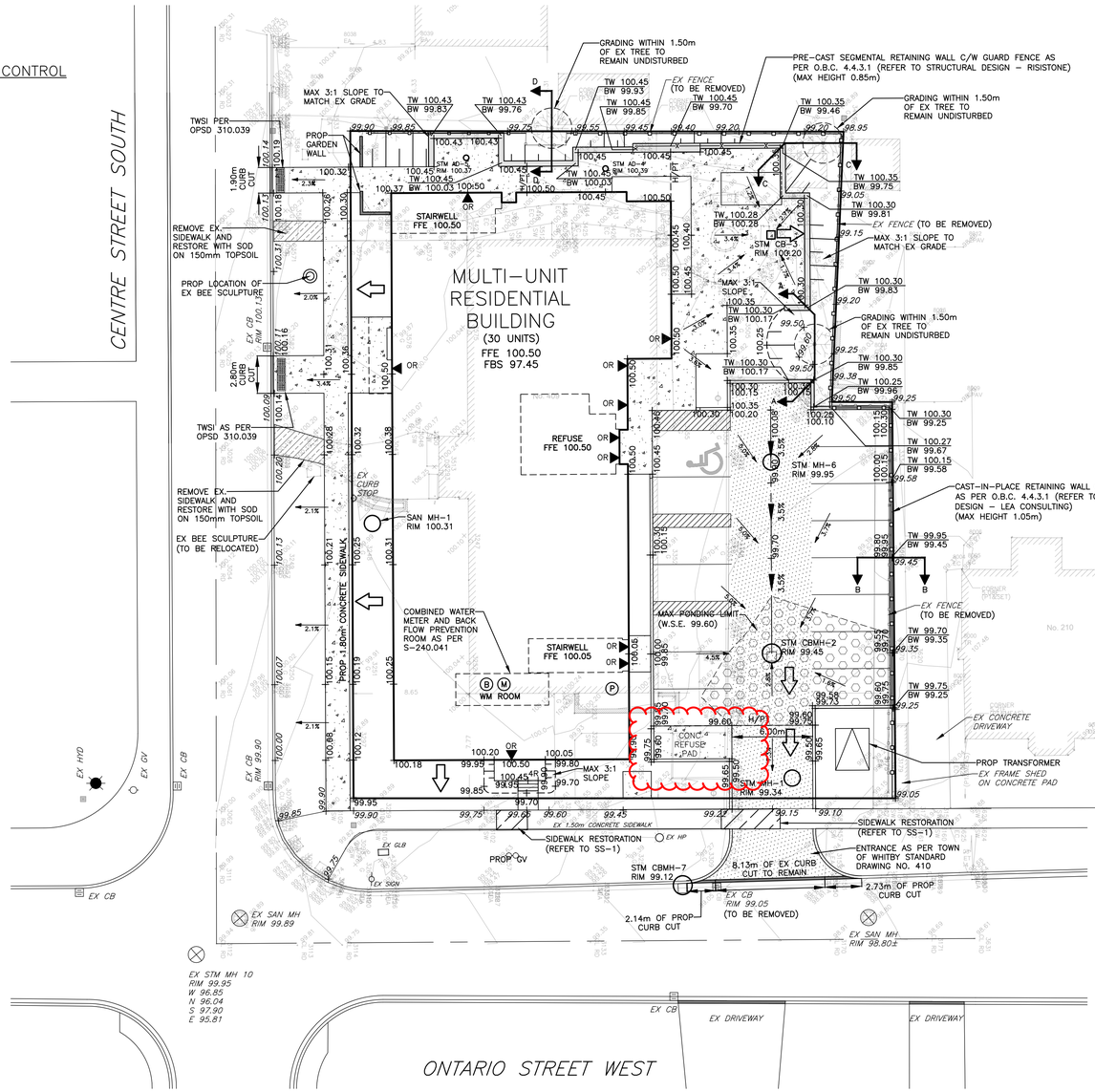
SECTION D–D
H SCALE: 1:50
V SCALE: 1:25

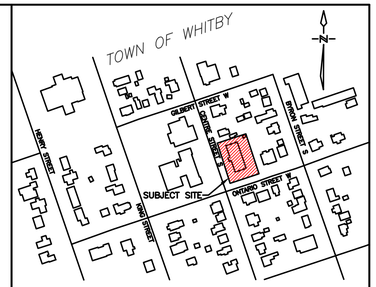
RETAINING WALL NOTES:

1. RETAINING WALL SECTIONS OVER 0.60m SHALL BE INSTALLED WITH GUARDS AS PER O.B.C. 4.4.3.1. REFER TO ENGINEERED DESIGN DRAWINGS (LEA CONSULTING, RISISTONE) FOR DETAILS.
2. SUBDRAINS SHALL BE INSTALLED AT AN ELEVATION LOWER THAN THE FINISHED GRADE ON THE LOW SIDE OF THE RETAINING WALL AND SHALL DISCHARGE BY GRAVITY TO THE PROPOSED STORM SEWER SYSTEM. REFER TO SS–1 FOR STORM SEWER CONNECTION DETAILS.

HOUSE KEEPING MEASURES – EROSION AND SEDIMENT CONTROL

1. THE CONTRACTOR SHALL BE RESPONSIBLE FOR THE PROPER INSTALLATION, MAINTENANCE AND REMOVAL OF ALL TEMPORARY EROSION AND SEDIMENT CONTROL MEASURES DURING CONSTRUCTION, AS DIRECTED BY THE ENGINEER.
2. ADDITIONAL EROSION AND SEDIMENT CONTROL MEASURES MAY BE REQUIRED AND SHALL BE DETERMINED BY THE ENGINEER.
3. SILT CONTROL FENCING SHALL BE INSTALLED ACCORDING TO THE ES–1 DRAWINGS AND MAINTAINED UNTIL COMPLETION OF THE LANDSCAPING AND SITE STABILIZATION.
4. NO CONSTRUCTION ACTIVITY OR MACHINERY SHALL INTRUDE BEYOND THE SILT/SNOW FENCE OR LIMIT OF DEVELOPMENT. ALL CONSTRUCTION VEHICLES SHALL LEAVE THE SITE AT DESIGNATED LOCATIONS AS SHOWN ON THE PLANS. ALL MATERIAL AND EQUIPMENT SHALL BE STORED ON SITE IN A DESIGNATED AREA. NO MATERIAL OR EQUIPMENT SHALL BE STORED ON THE MUNICIPAL RIGHT OF WAY. NO CONSTRUCTION VEHICLES WILL PARK ON THE MUNICIPAL ROADS.
5. ALL EXPOSED SOILS SHALL BE IMMEDIATELY STABILIZED WITH A SEED AND MULCH APPLICATION AS DIRECTED BY THE ENGINEER.
6. STOCKPILES SHALL BE SET BACK FROM ANY WATERCOURSE AND STABILIZED AGAINST EROSION AS SOON AS POSSIBLE. A SETBACK OF AT LEAST 15m FROM ANY TOP OF BANK OR WATERCOURSE IS REQUIRED.
7. CLEANING AND REPAIRS OF CHECK DAMS, MUD MATS AND ANY OTHER TEMPORARY SEDIMENT CONTROL MEASURES SHALL BE DONE AS NECESSARY THROUGH REGULAR INSPECTION OR AS DIRECTED BY THE ENGINEER.
8. SEDIMENT/SILT SHALL BE REMOVED FROM THE SEDIMENT CONTROL DEVICES AND THE CATCHBASIN BUFFERS AFTER STORM EVENTS AND DISPOSED OF IN AREAS AS APPROVED BY THE ENGINEER.
9. ALL RE–GRADED AREAS WITHIN THE SITE WHICH ARE NOT OCCUPIED BY BUILDINGS, ROADWAYS, SIDEWALKS OR DRIVEWAYS SHALL BE TOPSOILED AND SODDED/SEEDED IMMEDIATELY AFTER COMPLETION OF FINAL GRADING OPERATIONS OR AS DIRECTED BY THE ENGINEER.
10. INSPECTION OF THE PROPOSED EROSION AND SEDIMENT CONTROL MEASURES WILL OCCUR ON A WEEKLY BASIS, AFTER SIGNIFICANT RAINFALL OR SNOW MELT EVENTS AND DAILY DURING EXTENDED RAIN OR SNOW MELT PERIODS.
11. MATERIALS TO REPAIR DAMAGED ESC MEASURES MUST BE KEPT ON–SITE AT ALL TIMES.
12. THE ESC STRATEGIES ON THESE PLANS ARE NOT STATIC AND MAY NEED TO BE UPGRADED / AMENDED AS SITE CONDITIONS CHANGE TO PREVENT SEDIMENT RELEASE TO THE NATURAL ENVIRONMENT. FAILED ESC MEASURES MUST BE REPAIRED IMMEDIATELY.
13. SERVICING OF CONSTRUCTION EQUIPMENT ON SITE IS PROHIBITED.
14. CLEANING OF EXISTING ROAD(S) AT SITE ACCESS POINTS SHALL BE DONE DAILY DURING CONSTRUCTION OR AS NECESSARY THROUGH REGULAR INSPECTION OR AS DIRECTED BY THE ENGINEER.
15. DUST CONTROL TO BE REVIEWED DAILY. WATER TRUCK TO BE PROVIDED ON–SITE AND ALL HAUL ROAD /WORKING AREAS TO BE SPRAYED WITH WATER AS REQUIRED TO ENSURE DUST IS CONTROLLED ON–SITE.
16. ROCK CHECK DAMS ARE TO BE CLEANED OF ALL ACCUMULATED SEDIMENT AS SOON AS SEDIMENT HAS ACCUMULATED TO A DEPTH GREATER THAN 50% OF THE UPSTREAM CHECK DAM.
17. ALL LITTER AND DEBRIS SHALL BE MONITORED AND DISPOSED OF DAILY OR AS NECESSARY THROUGH REGULAR INSPECTION.
18. THE SILT FENCE MUST BE INSPECTED WEEKLY AND IMMEDIATELY AFTER RAINFALL OR SIGNIFICANT SNOW MELT EVENTS FOR RIPS AND TEARS, BROKEN STAKES, BLOW OUTS (STRUCTURAL FAILURE) AND ACCUMULATION OF SEDIMENT. THE SILT FENCE MUST BE FIXED AND/OR REPLACED IMMEDIATELY WHEN DAMAGED. ACCUMULATED SEDIMENT MUST BE REMOVED FROM THE SILT FENCE WHEN ACCUMULATION REACHES 50% OF THE HEIGHT OF THE FENCE.
19. SILT SACKS (OR APPROVED EQUIVALENT) ARE TO BE INSTALLED AT ALL CATCHBASINS AND CATCHBASIN MANHOLES LOCATIONS UPON COMPLETION OF SERVICING.
20. SITE CONTACT FOR EROSION AND SEDIMENT MEASURES:
    REGION OF DURHAM HOUSING SERVICES ON–CALL
    RYAN VOSCEK, FACILITIES MAINTENANCE TECHNICIAN
    905–261–0335
21. STREET TO BE LEFT IN A BROOM SWEPT CONDITION AT THE END OF EACH DAY AND SWEPT THROUGHOUT AS REQUIRED AND/OR DIRECTED BY THE TOWN.
22. THE CONTRACTOR SHALL KEEP ON SITE, AT ALL TIMES, SUFFICIENT ADDITIONAL MATERIALS AND EROSION AND SEDIMENT CONTROL DEVICES REQUIRED TO COMPLETE EMERGENCY REPAIRS TO EROSION AND SEDIMENT CONTROL MEASURES.





KEY PLAN
N.T.S.

BENCHMARK

ELEVATIONS SHOWN ON THIS PLAN ARE RELATED TO CANADIAN GEODETIC VERTICAL DATUM (1978 ADJUSTMENT) AND ARE DERIVED FROM BENCHMARK No. 00819668302 HAVING A PUBLISHED ELEVATION OF 101.580 METERS.

REFER TO SURVEY DONE BY ROUSE SURVEYORS INC. DATED: FEBRUARY 25, 2025. REFERENCE No. 23–431–06

SITE GRADING NOTES

1. A ROAD OCCUPANCY PERMIT WILL BE REQUIRED FOR ANY WORKS WITHIN THE MUNICIPAL RIGHT OF WAY.
2. ALL WORKS OR RESTORATION WITHIN THE RIGHT OF WAY SHALL BE COMPLETED AS PER MUNICIPAL FIELD STAFF DIRECTION.
3. THE LOCATION AND ELEVATION OF ALL EXISTING UTILITIES AND SERVICES TO BE VERIFIED BY THE CONTRACTOR PRIOR TO COMMENCING CONSTRUCTION. CONCERNED UTILITIES TO BE GIVEN ADVANCED NOTICE FOR STAKE OUT. THE CONSULTANT ASSUMES NO RESPONSIBILITY FOR THE ACCURACY OF THE UTILITIES SHOWN ON THE DRAWINGS.
4. QUANTITIES, DIMENSIONS AND ELEVATIONS TO BE VERIFIED BY THE CONTRACTOR PRIOR TO ORDERING MATERIALS OR COMMENCING CONSTRUCTION.
5. ALL SLOPES SHALL BE CONSTRUCTED AT 3:1 (MAXIMUM) UNLESS OTHERWISE NOTED ON THE DRAWINGS.
6. ALL SWALES SHALL BE CONSTRUCTED WITH 2% MINIMUM GRADE UNLESS OTHERWISE NOTED ON THE DRAWINGS.
7. ALL SLOPES, SWALES AND OPEN SPACE AREAS SHALL BE SODDED ON A MINIMUM OF 150mm OF TOPSOIL.
8. ALL DISTURBED AREAS WITHIN ABUTTING THE REGIONAL RIGHT OF WAY SHALL BE SODDED ON A MINIMUM OF 150mm OF TOPSOIL.
9. RETAINING WALLS SHALL BE CONSTRUCTED IN ACCORDANCE WITH SHOP DRAWINGS PROVIDED BY RISISTONE AND LEA CONSULTING LTD. THE RETAINING WALL, INCLUDING IT'S FOOTING, MUST BE CONSTRUCTED COMPLETELY WITHIN THE PROPERTY LIMIT. GUARD, WHERE REQUIRED, SHALL BE CONSTRUCTED AS PER T.O.W. STD DWG NO. 730.
10. PAVEMENT STRUCTURE FOR LIGHT–DUTY CAR PARKING SHALL CONSIST OF THE FOLLOWING UNLESS OTHERWISE DIRECTED BY THE GEOTECHNICAL ENGINEER:
    –200mm GRANULAR 'B' SUBBASE (AS PER OPSS 1010, 100% SPMDD)
    –150mm GRANULAR 'A' BASE (AS PER OPSS 1010, 100% SPMDD)
    –50mm HL8 ASPHALT (MINIMUM 92% MRD)
    –40mm HL3 ASPHALT
    –CONCRETE BARRIER CURB PER OPSD 600.110
11. PAVEMENT STRUCTURE FOR HEAVY–DUTY DRIVE AISLE SHALL CONSIST OF THE FOLLOWING UNLESS OTHERWISE DIRECTED BY THE GEOTECHNICAL ENGINEER:
    –300mm GRANULAR 'B' SUBBASE (AS PER OPSS 1010, 100% SPMDD)
    –150mm GRANULAR 'A' BASE (AS PER OPSS 1010, 100% SPMDD)
    –75mm HL8 ASPHALT (MINIMUM 92% MRD)
    –40mm HL3 ASPHALT
    –CONCRETE BARRIER CURB PER OPSD 600.110
12. CONCRETE REFUSE PAD SHALL BE 200mm THICK REINFORCED CONCRETE SLAB REINFORCED WITH WELDED WIRE MESH (150mm x 150mm WEAVE, MW 18.7x MW 18.7), 32MPa (C–2) CONCRETE, BEDDED WITH MINIMUM 150mm GRANULAR 'A'.

LEGEND

ORIGINAL SURVEY ELEVATION	103.15
ORIGINAL ELEVATION TO REMAIN	101.35
PROPOSED ELEVATIONS	103.15
EXISTING CONTOURS	125.00
PROPOSED SWALE	
DIRECTION OF DRAINAGE	
DIRECTION OF OVERLAND FLOW	
100–YEAR PONDING LIMIT ELEV = 99.60	
HEAVY–DUTY ASPHALT	
EXISTING BOARD FENCE (TO BE REMOVED)	
DECORATIVE METAL FENCE (RET. WALL GUARD – REFER TO LANDSCAPE PLANS)	
1.80m WOOD PRIVACY FENCE (REFER TO LANDSCAPE PLANS)	

NO.	SUBMISSION	MONTH YEAR
12.	ISSUED FOR ADDENDUM 3	OCT 30/2026
11.	ISSUED FOR TENDER	SEP 16/2026
10.	ISSUED FOR SITE PLAN APPROVAL – REV.1	MAY 15/2026
9.	ISSUED FOR 100% CD	MAR 25/2026
8.	ISSUED FOR INTERIM T.O.W. REVIEW #2	MAR 23/2026
7.	ISSUED FOR INTERIM T.O.W. REVIEW	MAR 17/2026
6.	ISSUED FOR 90% CD & CLASS A COSTING	FEB 20/2026
5.	ISSUED FOR SITE SERVICES PERMIT	JAN 30/2026
5.	ISSUED FOR 75% CD & BUILDING PERMIT SUBMISSION	JAN 30/2026
4.	ISSUED FOR 30% CD SUBMISSION	DEC 17/2025
3.	ISSUED FOR SITE PLAN APPLICATION	NOV 17/2025
2.	ISSUED FOR 100% DD & CLASS B COSTING	JUNE 20/2025
1.	ISSUED FOR STAGE 2 PRE–CONSULTATION	JUNE 13/2025

DURHAM REGIONAL HOUSING
409 CENTRE STREET, WHITBY ON (SP–26–25)

SITE GRADING PLAN

D.G. BIDDLE & ASSOCIATES
CONSULTING ENGINEERS & PLANNERS
96 King Street East
Oshawa, Ontario, L1H 1B6
Phone: 905-576-8500
info@dgbiddle.com
dgbiddle.com

SCALE:	1:200
DESIGNED BY:	B.K.Y.
CHECKED BY:	C.J.F.
PROJECT NO.:	125012
DRAWING NO.:	SG–1

29/10/2026
X:\DWG\JOB FILES\125000\125012 409 CENTER STREET – DURHAM HOUSING PROJECT\125012 DRAWINGS\125012 DRAWINGS CIVIL\125012 ENGINEERING DRAWINGS\125012–20261026–SG–SITE PLAN

SOIL VOLUME CHART

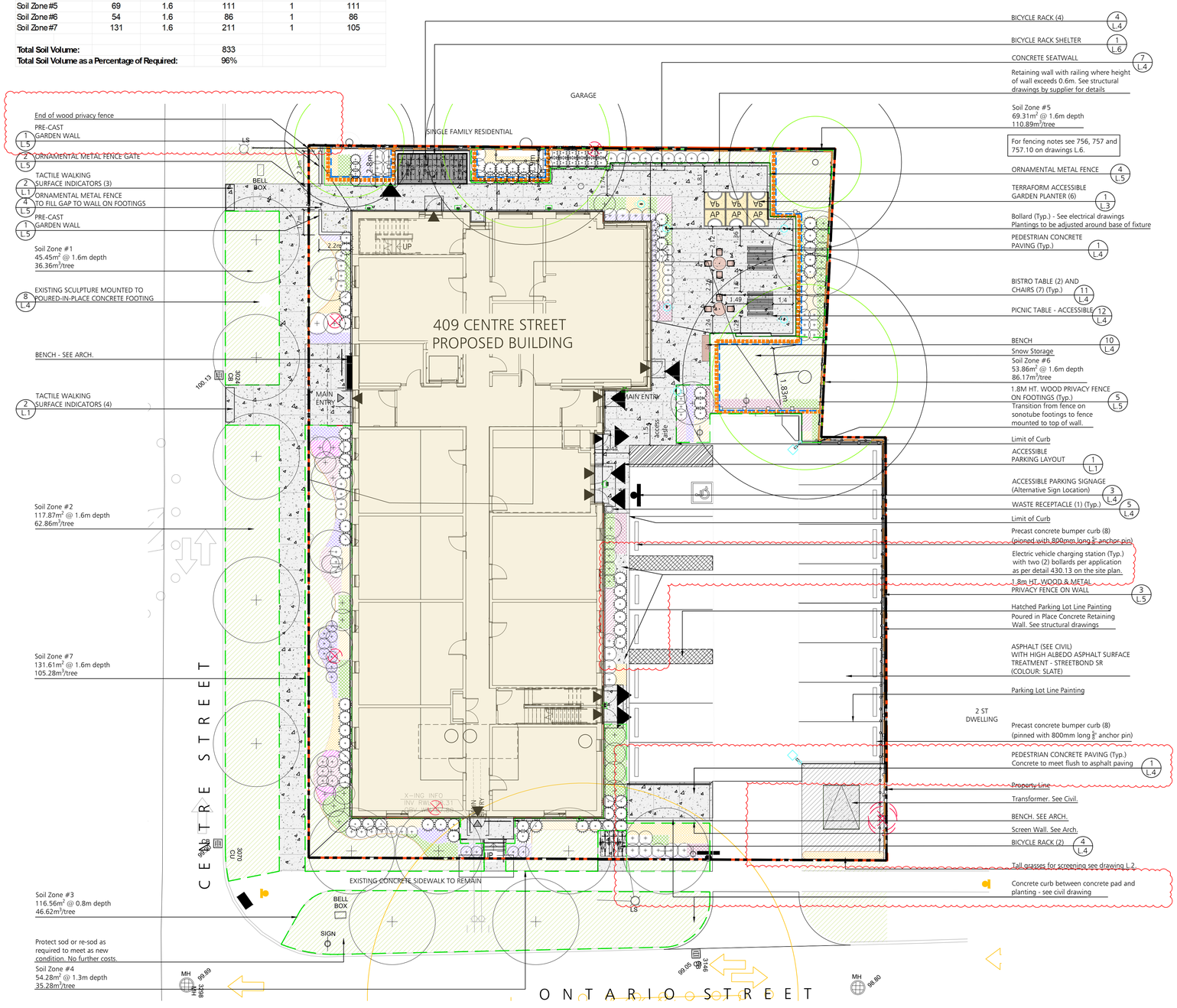
Total Site Area: 1,912 m^2

Total Soil Volume Required: 869 m^3

	Area (m^2)	Soil Depth (m)	Soil Volume (m^3)	Planted Trees	Volume/Tree (m^3)
Soil Zone #1	45	1.6	72	2	36
Soil Zone #2	118	1.6	189	5	63
Soil Zone #3	117	0.8	93	2	47
Soil Zone #4	54	1.3	70	2	35
Soil Zone #5	69	1.6	111	1	111
Soil Zone #6	54	1.6	86	1	86
Soil Zone #7	131	1.6	211	1	105
Total Soil Volume:			833		
Total Soil Volume as a Percentage of Required:			96%		

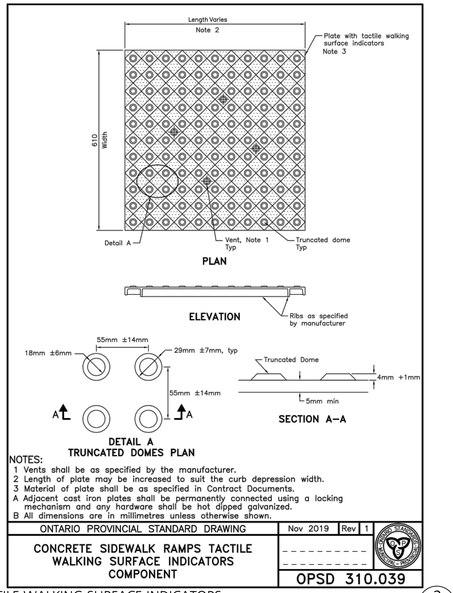
* Soil volume number for areas with existing trees to be protected is not certain. The depth of proposed soil to be used in built garden beds has been extended to these areas.

* See topsoil notes 753; topsoil placement and fine grading notes; structural soil standards; and sodding notes on drawing L.7



LANDSCAPE LAYOUT PLAN
1:150

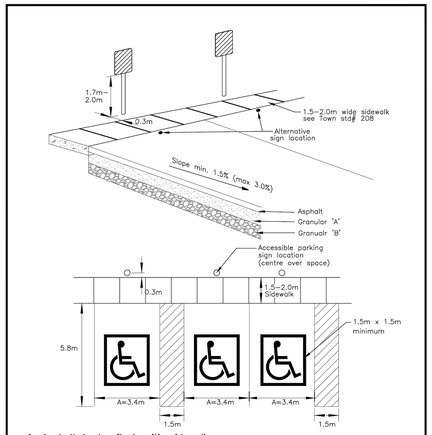
KEY MAP



NOTES:
1 Vents shall be as specified by the manufacturer.
2 Length of plate may be increased to suit the curb depression width.
3 Material of plate shall be as specified in Contract Documents.
A Adjacent cast iron plates shall be permanently connected using a locking mechanism and any hardware shall be hot dipped galvanized.
B All dimensions are in millimetres unless otherwise shown.

ONTARIO PROVINCIAL STANDARD DRAWING	Nov 2019	Rev 1
CONCRETE SIDEWALK RAMPS TACTILE WALKING SURFACE INDICATORS COMPONENT		OPSD 310.039

TACTILE WALKING SURFACE INDICATORS
N.T.S. — 2 L.1



1. Asphalt to be flush with sidewalk.
2. Sidewalk to be barrier free.
3. Refer to Town of Whitby Standard #208 for sidewalk detail.
4. Accessible parking sign to be minimum of 0.3m offset from sidewalk. Preferred location behind sidewalk, alternate location front of sidewalk.
5. Sign location will be site specific.
6. Access aisles must be provided for all parking spaces and may be shared by two spaces.

All dimensions in millimetres except where noted

The Corporation of the Town of Whitby		Engineering Services
Checked	Accessibility Parking Proposed and Retrofit	Dwg. Date: January 2009
Approved		Revision No.: 3
Scale: N.T.S		Revision Date: October 2019
		DWG No. 413

ACCESSIBILITY PARKING LAYOUT
N.T.S. — 1 L.1

LEGEND

- Property Line
- 1.8m Solid Wood Fence
- Ornamental Metal Railing
- Tree Protection hoarding. See TPP
- Root sensitive excavation line. See TPP
- Soil Zone Calculation Area
- Proposed Deciduous Tree
- Existing tree to be preserved. See TPP
- Optimal Tree Protection Zone (TPZ). See TPP
- Proposed Evergreen Shrub
- Proposed Deciduous Shrub
- Fine Grade and Sod (see soil volume chart for topsoil depth)
- Grasses
- Perennial
- Concrete Paving

Note: All Drawings by Baker Turner inc. to be Printed in Colour.

REVISIONS

DATE	DESCRIPTION
2026.10.29	Issued for Addendum 3
2026.10.21	Issued for Addendum 2
2026.09.16	Issued for Tender
2026.06.30	Issued for Tender
2026.05.15	Issued for Site Plan Approval - Rev. 1
2026.03.24	Issued for 100% Construction Documents
2026.02.18	Issued for 90% CD and Class A Costing
2026.01.30	Issued for 75% CD, Bldg. Permit Sub, and Site Services Permit
2025.12.17	Issued for 30% Construction Documents
2025.11.28	Progress Set to 30% Construction Documents

NOTE: Contractor is to check and verify all dimensions and conditions on the project, and is to immediately report any discrepancies to the landscape architect before proceeding with the work.

Applicant name & Contact Information:
Durham Regional Local Housing Corporation
605 Rossland Road E, Level 5
PO Box 623, Whitby ON
L1N 6A3

Baker Turner inc.
Landscape Architecture | Site Design
Suite 200, 2nd floor
2010 Winston Park Drive
Oakville Ontario L6H 5R7
tel: (289) 291-7620
web: www.bakerturner.com
email: tba@bakerturner.com

Project Title
Durham Region Housing

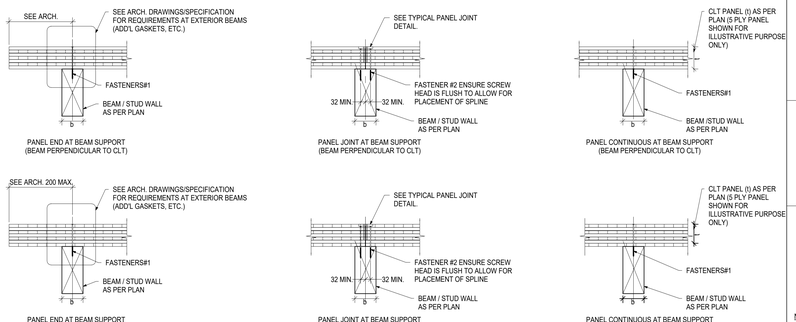
Centre Street Housing
409 Centre St S.
Whitby, ON

Landscape Plan

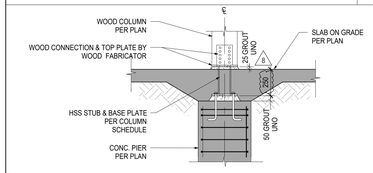
Date	Issued
November 2024	
Job Number	Drawn By
BTI-1775	JW
Scale	Checked By
As Shown	MT
Sheet Number	File Number
L.1	#SP-26-25

S:\BTI_CURRENT\BTI-Projects_Current\1775 – Centre Street Housing\Drawings\Working Drawings\1775-CentreStreetHsng_LP_Whitby-Addendum3.dwg
Sep 30, 2026 - 3:59pm

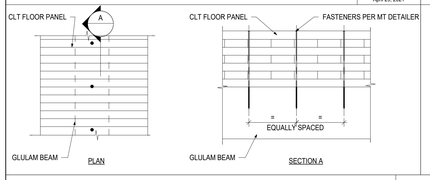
CLT PANELS SUPPORTED ON WOOD BEAMS — WD-012
April 29, 2021



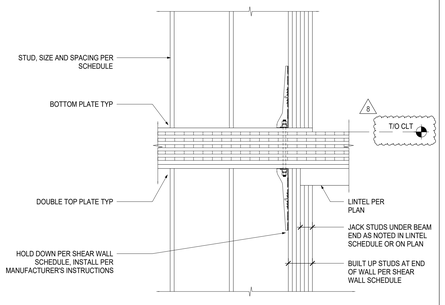
WOOD COLUMN BASE DETAIL — WD-013
April 29, 2021



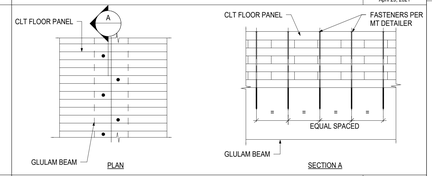
CONTINUOUS CLT FLOOR PANEL ON GLULAM BEAM CONNECTION — WD-014
April 29, 2021



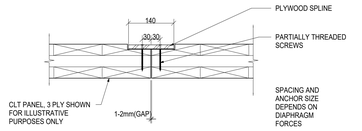
SHEAR WALL HOLD DOWN — WD-015



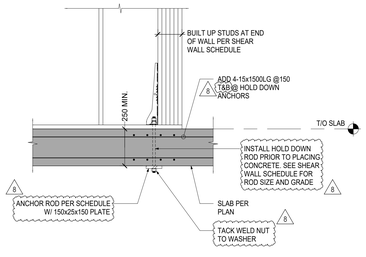
DISCONTINUOUS CLT FLOOR PANEL ON GLULAM BEAM CONNECTION — WD-016
April 29, 2021



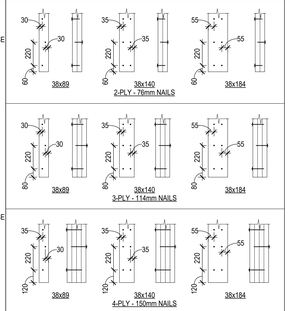
CLT PANEL-PANEL BUTT JOINT/SPLINE CONNECTION — WD-017
September 9, 2025



SHEAR WALL HOLD DOWN AT SLAB — WD-018

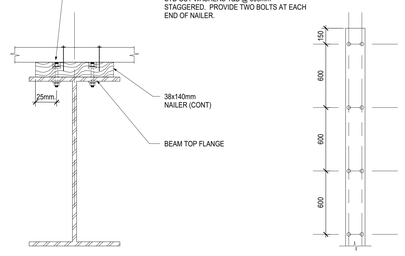


TIMBER BUILT UP COLUMN NAILING — WD-019

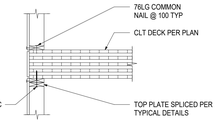


NOTE:
1. NAILING PATTERN SHOW FOR D. FIR-L AND HEM-FIR. REDUCE END DISTANCES FOR SPF OR NORTHERN TO 80% OF THOSE SHOWN

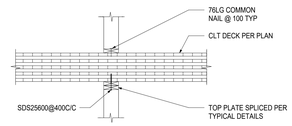
WOOD NAILERS ON STEEL BEAMS — WD-021



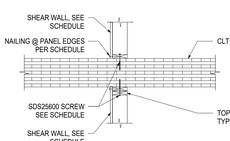
TYPICAL EXTERIOR LOAD BEARING WALL, NON SHEAR WALL — WD-023



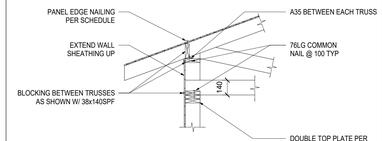
TYPICAL INTERIOR LOAD BEARING WALL, NON SHEAR WALL — WD-025



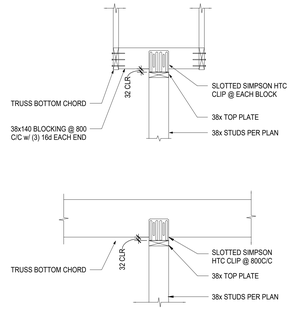
TYPICAL INTERIOR LOAD BEARING SHEAR WALL — WD-024



TOP OF EXTERIOR WALL AT ROOF TRUSSES — WD-026

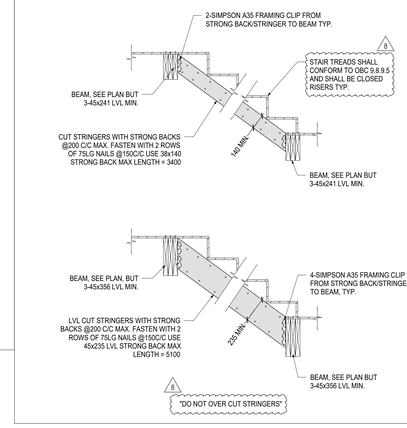


NON-LOAD BEARING PARTITION WALL - TOP — WD-020

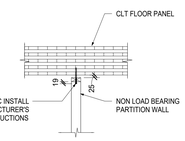


NOTE: INSTALL PER MANUFACTURER'S INSTRUCTIONS

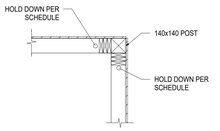
TYPICAL STAIR STRINGER — WD-022



NON LOAD BEARING PARTITION WALL - CLT — WD-027



SHEAR WALL INTERSECTIONS — WD-028



NOTES:
1. WHERE ENDS OF SHEAR WALL MEET AT A CORNER, PROVIDE ONE HOLD DOWN FOR EACH WALL.
2. GRADE TO MATCH SHEAR WALL SCHEDULE FOR ALL MEMBERS.

REV.	DATE	DESCRIPTION
8	2026-10-30	ISSUED FOR STRUCTURAL TENDER ADDENDUM NO.3
7	2026-09-16	ISSUED FOR TENDER
6	2026-03-20	ISSUED FOR 100% CD
5	2026-02-18	ISSUED FOR 90% CD AND CLASS A COSTING
4	2026-01-30	ISSUED FOR BUILDING PERMIT
3	2026-01-30	ISSUED FOR 75% CD
2	2025-12-17	Issued for 30 % Construction Documents
1	2025-06-20	Issued for 100% DD & Class B Costing

REDEVELOPMENT - 409 CENTRE ST S - 25008

409 Centre Street, Whitby, ON L1N 4W3

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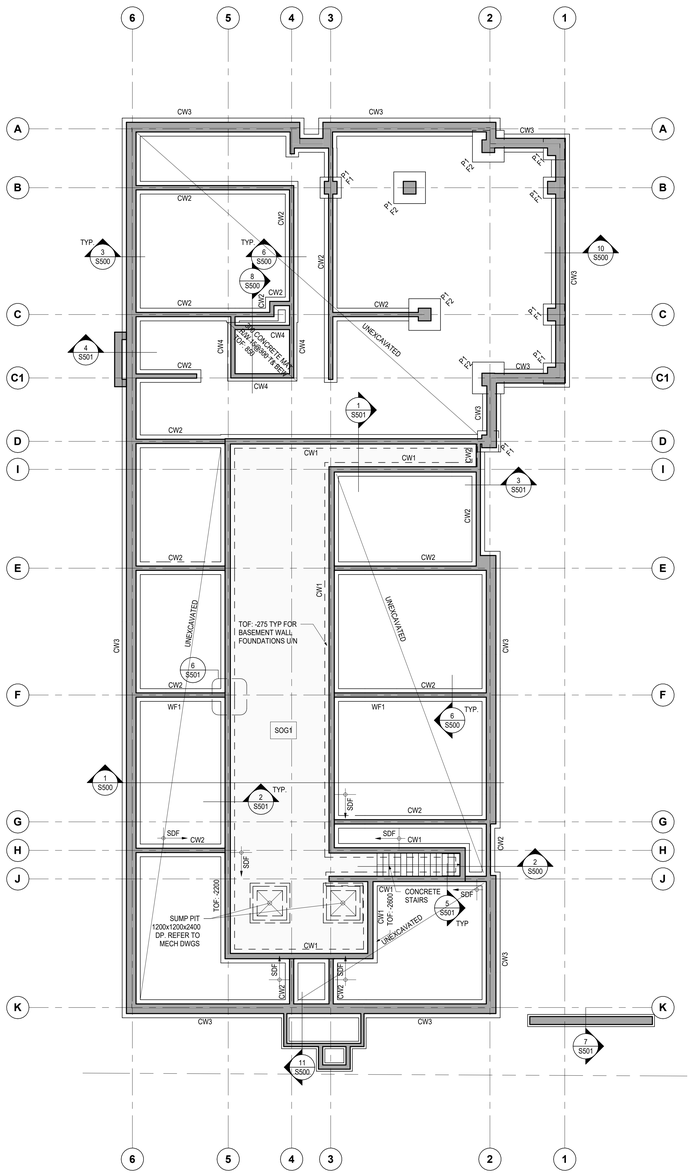
LICENSED PROFESSIONAL ENGINEER
J.M. KACZMAREK
100148194
2026-10-30
PROVINCE OF ONTARIO

TYPICAL DETAILS

Scale	Date
1 : 20	2026-06-12
Drawn: Author	Checked By: Checker
Project No.: 25356	Drawing No.: S104

2026-10-30 2:23:3 PM Autodesk Docs://25008_DWG - 409 Centre St/25008_409 Centre St_LEA_STRU_MAIN_R24.rvt



1 S200 BASEMENT & FOUNDATION PLAN

1 : 100

10 NOTES

1. SEE GENERAL REQUIREMENTS AND TYPICAL DETAILS (ON DRAWINGS S100 SERIES) AND READ IN CONJUNCTION WITH ALL FOOTING SCHEDULES.
2. ELEVATIONS AND GEOMETRY
   1. BASEMENT FLOOR DATUM ELEVATION IS -3050 (GEODETIC EL. 97450)
   2. "TOP OF SLAB IS 0 FROM DATUM ELEVATION EXCEPT AS CROSSED AND NOTED +/-X ON PLAN.
   3. "TOF" DENOTES TOP OF FOOTING ELEVATION. UNLESS NOTED SET TOF AT ELEVATION -275 FROM TOP OF BASEMENT SLAB.
3. DESIGN LOADING
   1. DESIGN LIVE LOAD IS 3.6 kPa.
   2. SUPERIMPOSED DEAD LOAD IS 1.5 kPa. THIS INCLUDES PARTITION ALLOWANCE OF 1.0 kPa.
4. FRAMING
   1. SLAB ON GRADE (SOG1 -)125 THICK CONCRETE SLAB-ON-GRADE (WITH 152x152 MW 18.7/MW 18.7 W.W.F. PLACED 50 BELOW TOP OF SLAB)
   2. COORDINATE WITH NEW (AND EXISTING SERVICES AND FOUNDATION PRIOR TO EXCAVATION FOR FOOTING. REFER TO ARCHITECTURAL, MECHANICAL, ELECTRICAL, AND SITE SERVICING DRAWINGS.

REV.	DATE	DESCRIPTION
10	2026-10-30	ISSUED FOR STRUCTURAL TENDER ADDENDUM NO.3
9	2026-10-29	ISSUED FOR STRUCTURAL TENDER ADDENDUM NO.2
8	2026-10-13	ISSUED FOR STRUCTURAL TENDER ADDENDUM NO.1
7	2026-09-16	ISSUED FOR TENDER
6	2026-03-20	ISSUED FOR 100% CD
5	2026-02-18	ISSUED FOR 90% CD AND CLASS A COSTING
4	2026-01-30	ISSUED FOR BUILDING PERMIT
3	2026-01-30	ISSUED FOR 75% CD
2	2025-12-17	Issued for 30 % Construction Documents

REDEVELOPMENT - 409 CENTRE ST S - 25008

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LICENSED PROFESSIONAL ENGINEER
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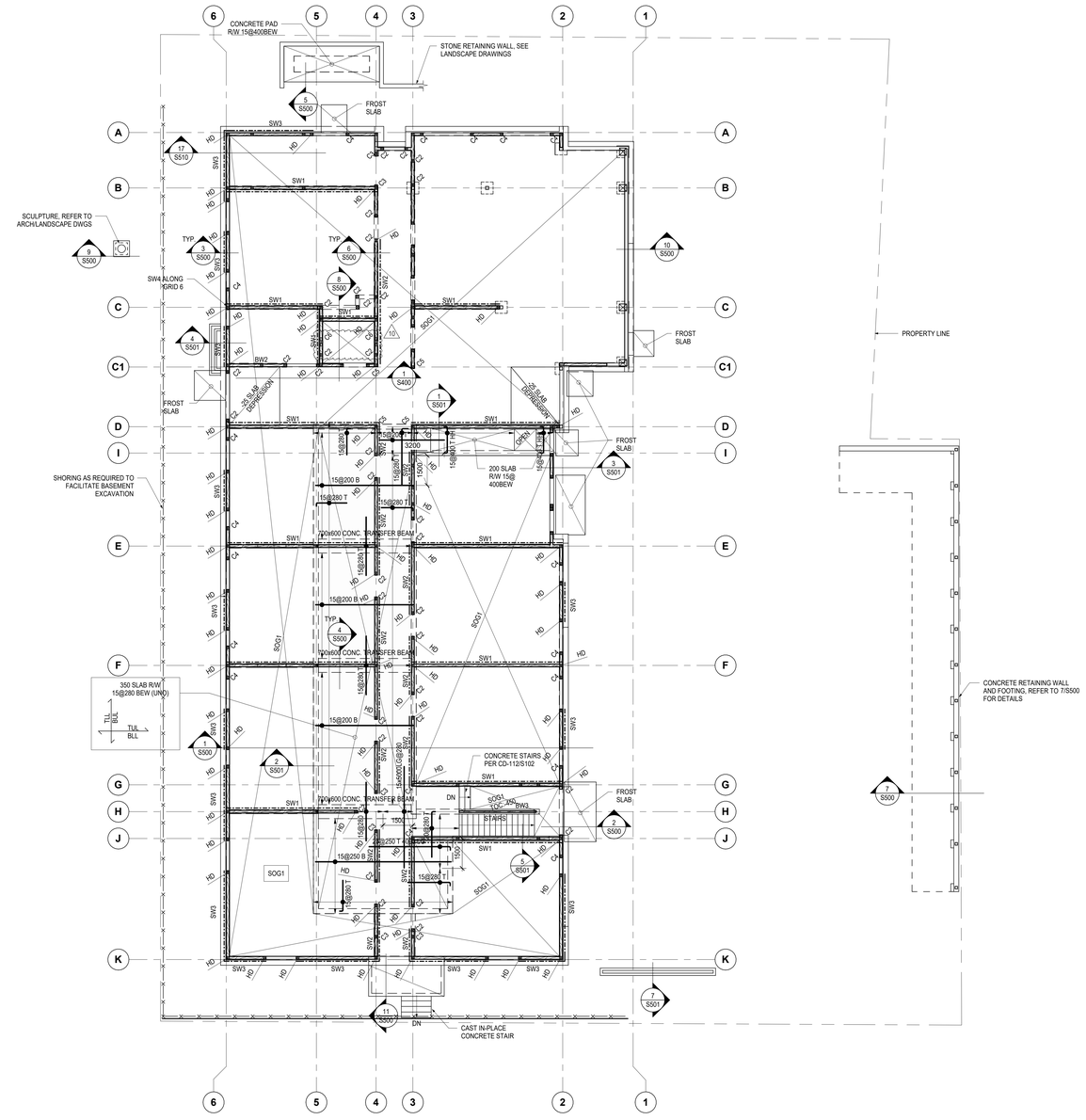
100146134
2026-10-30
PROVINCE OF ONTARIO

BASEMENT & FOUNDATION PLAN

Scale	Date
As indicated	2026-06-12
Drawn: K.N.	Checked By: J.K.
Project No.: 25356	Drawing No.: S200

2026-10-30 2:23:35 PM Autodesk Docs://25008_DWG - 409 Centre St/25008_409 Centre St_LEA_STRU_MAIN_R24.rvt



1 S201 LEVEL 1
1 : 100

NOTES

1. SEE GENERAL REQUIREMENTS AND TYPICAL DETAILS ON (DRAWING S100 SERIES).
2. ELEVATIONS AND GEOMETRY
   1. LEVEL 1 FLOOR DATUM ELEVATION IS 0.0 (GEODETIC EL. 100500).
   2. TOP OF SLAB IS 0 FROM DATUM ELEVATION EXCEPT AS CROSSED AND NOTED ±X ON PLAN.
3. DESIGN LOADING
   1. DESIGN LIVE LOAD IS 4.8kPa.
   2. SUPERIMPOSED DEAD LOAD IS 2.0kPa. THIS INCLUDES PARTITION ALLOWANCE OF 1.0kPa.
4. FRAMING
   1. COORDINATE WITH OPENINGS BY OTHER TRADES.
   2. TIMBER COLUMNS NOTED ON PLAN EXTEND FROM THIS LEVEL TO THE U/S OF THE LEVEL ABOVE.

SHEAR WALLS ARE IDENTIFIED ON PLAN AS:

⊠ INDICATES COLUMN IN WALL. MINIMUM 2 PLYS OF STUDS TO MATCH WALL CONSTRUCTION U/N ON PLAN, SHEAR WALL SCHEDULE OR LINTEL SCHEDULE. EVEN IF COLUMNS ARE NOT CALLED UP ON PLAN DRAWINGS OF THE FLOORS BELOW, ALL POSTS TO BE CONTINUOUS TO MAIN FLOOR SLAB, BOTH JACK STUDS AND KING POSTS.

REV.	DATE	DESCRIPTION
10	2026-10-30	ISSUED FOR STRUCTURAL TENDER ADDENDUM NO.3
9	2026-10-29	ISSUED FOR STRUCTURAL TENDER ADDENDUM NO.2
8	2026-10-13	ISSUED FOR STRUCTURAL TENDER ADDENDUM NO.1
7	2026-09-16	ISSUED FOR TENDER
6	2026-03-20	ISSUED FOR 100% CD
5	2026-02-18	ISSUED FOR 90% CD AND CLASS A COSTING
4	2026-01-30	ISSUED FOR BUILDING PERMIT
3	2026-01-30	ISSUED FOR 75% CD
2	2025-12-17	Issued for 30 % Construction Documents

REDEVELOPMENT - 409 CENTRE ST S - 25008

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LICENSED PROFESSIONAL ENGINEER
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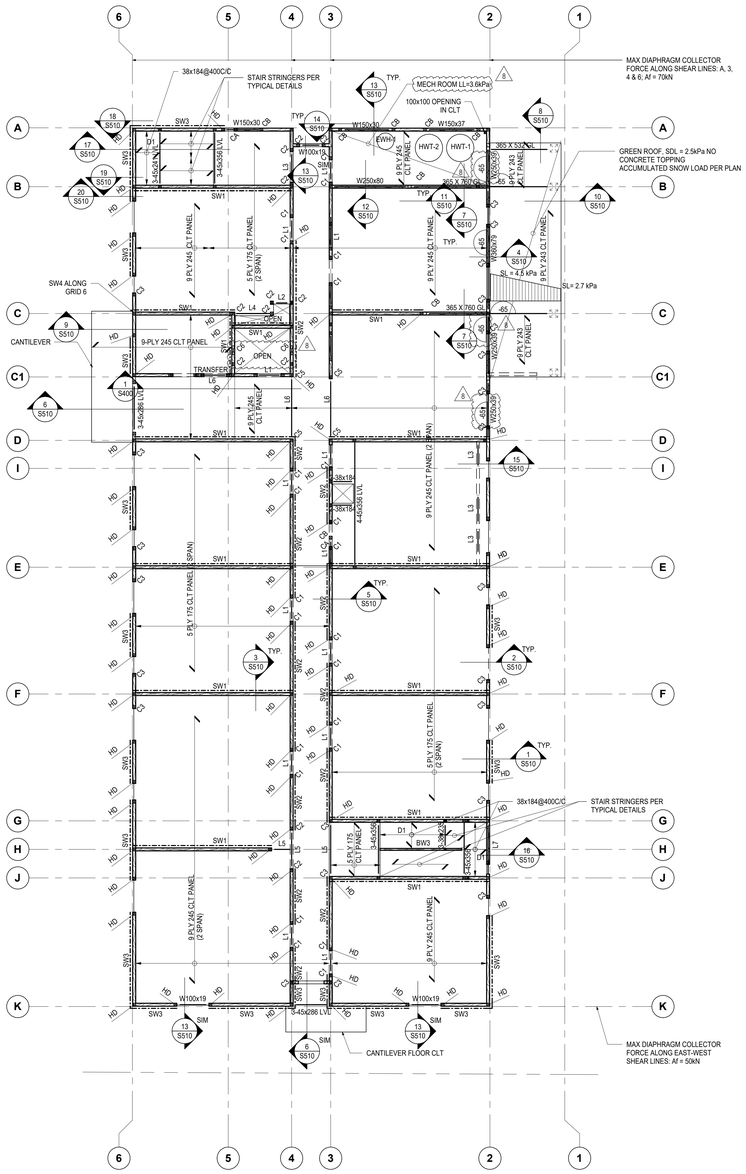
100146194
2026-10-30
PROVINCE OF ONTARIO

LEVEL 1 FRAMING PLAN

Scale	Date
As indicated	2026-06-12
Drawn: K.N.	Checked By: J.K.
Project No.: 25356	Drawing No.: S201

2026-10-30 2:23:38 PM Autodesk Docs://25008_409 Centre St/25008_409 Centre St_LEA_STRU_MAIN_R24.rvt



1 S202 LEVEL 2

1 : 100

NOTES

1. SEE GENERAL REQUIREMENTS AND TYPICAL DETAILS ON (DRAWING S100 SERIES).
2. ELEVATIONS AND GEOMETRY
   1. LEVEL 2 FLOOR DATUM ELEVATION IS 3250 (GEODETIC EL. 103750).
   2. TOP OF CLT DECK IS -65 FROM DATUM ELEVATION EXCEPT AS CROSSED AND NOTED ±X ON PLAN.
   3. TOP OF WOOD BEAMS ARE AT UNDERSIDE OF CLT DECK.
   4. WHERE BEAMS NOTED THUS (±X), X IS DISTANCE TO TOP OF BEAM FROM FLOOR DATUM.
3. DESIGN LOADING
   1. DESIGN LIVE LOAD FOR CORRIDORS AND STAIRS IS 4.8 kPa AND FOR OTHER AREAS IS 1.9 kPa.
   2. SUPERIMPOSED DEAD LOAD IS 3.2 kPa. THIS INCLUDES PARTITION ALLOWANCE OF 1.0kPa.
4. FRAMING
   1. PROVIDE 40 CONCRETE TOPPING OVER ACOUSTIC MAT ON CLT DECK. REINFORCE TOPPING WITH FIBRILLATED POLYPROPYLENE FIBRES AT LEAST 19mm IN LENGTH, DOSAGE 0.9kg/m³.
   2. TIMBER COLUMNS NOTED ON PLAN EXTEND FROM THIS LEVEL TO THE U/S OF THE LEVEL ABOVE.
   3. CONNECT CLT PANELS FOR DIAPHRAGM SHEAR FORCE OF Vf = 7.5kN/m

REV.	DATE	DESCRIPTION
8	2026-10-30	ISSUED FOR STRUCTURAL TENDER ADDENDUM NO.3
7	2026-09-16	ISSUED FOR TENDER
6	2026-03-20	ISSUED FOR 100% CD
5	2026-02-18	ISSUED FOR 90% CD AND CLASS A COSTING
4	2026-01-30	ISSUED FOR BUILDING PERMIT
3	2026-01-30	ISSUED FOR 75% CD
2	2025-12-17	Issued for 30 % Construction Documents
1	2025-06-20	Issued for 100% DD & Class B Costing

REDEVELOPMENT - 409 CENTRE ST S - 25008

409 Centre Street, Whitby, ON L1N 4W3

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LICENSED PROFESSIONAL ENGINEER
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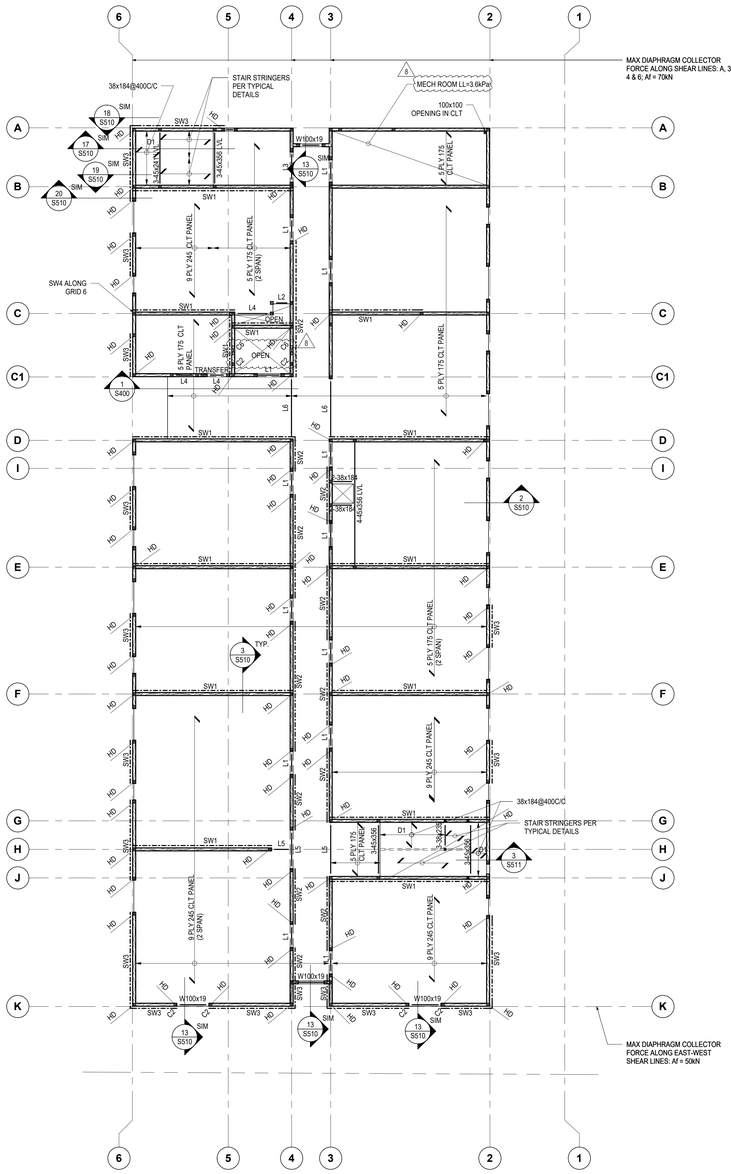
100146194
2026-10-30
PROVINCE OF ONTARIO

LEVEL 2 FRAMING PLAN

Scale: As indicated	Date: 2026-06-12
Drawn: K.N.	Checked By: J.K.
Project No.: 25356	Drawing No.: S202

2026-10-30 2:23:18 PM Autodesk Docs://25008_409 Centre St/25008_409 Centre St_LEA_STRU_MAIN_R24.rvt



1 S203 LEVEL 3
1 : 100

NOTES

1. SEE GENERAL REQUIREMENTS AND TYPICAL DETAILS ON (DRAWING S100 SERIES).
2. ELEVATIONS AND GEOMETRY
   1. LEVEL 3 FLOOR DATUM ELEVATION IS 6300 (GEODETIC EL. 106800).
   2. TOP OF CLT DECK IS -65 FROM DATUM ELEVATION EXCEPT AS CROSSED AND NOTED ±X ON PLAN.
   3. TOP OF WOOD BEAMS ARE AT UNDERSIDE OF CLT DECK.
   4. WHERE BEAMS NOTED THUS (±X) , X IS DISTANCE TO TOP OF BEAM FROM FLOOR DATUM.
3. DESIGN LOADING
   1. DESIGN LIVE LOAD FOR CORRIDORS AND STAIRS IS 4.8 kPa AND FOR OTHER AREAS IS 1.9 kPa.
   2. SUPERIMPOSED DEAD LOAD IS 3.2 kPa. THIS INCLUDES PARTITION ALLOWANCE OF 1.0kPa.
4. FRAMING
   1. PROVIDE 40 CONCRETE TOPPING OVER ACOUSTIC MAT ON CLT DECK. REINFORCE TOPPING WITH FIBRILLATED POLYPROPYLENE FIBRES AT LEAST 19mm IN LENGTH. DOSAGE 0.9kg/mt.
   2. TIMBER COLUMNS NOTED ON PLAN EXTEND FROM THIS LEVEL TO THE U/S OF THE LEVEL ABOVE.
   3. CONNECT CLT PANEL FOR DIAPHRAGM SHEAR FORCE OF Vf = 7.5kN/m

REV.	DATE	DESCRIPTION
8	2026-10-30	ISSUED FOR STRUCTURAL TENDER ADDENDUM NO.3
7	2026-09-16	ISSUED FOR TENDER
6	2026-03-20	ISSUED FOR 100% CD
5	2026-02-18	ISSUED FOR 90% CD AND CLASS A COSTING
4	2026-01-30	ISSUED FOR BUILDING PERMIT
3	2026-01-30	ISSUED FOR 75% CD
2	2025-12-17	Issued for 30 % Construction Documents
1	2025-06-20	Issued for 100% DD & Class B Costing

REDEVELOPMENT - 409 CENTRE ST S - 25008

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LICENSED PROFESSIONAL ENGINEER
J.M.KACZMAREK
100146134
2026-10-30
PROVINCE OF ONTARIO

LEVEL 3 FRAMING PLAN

Scale	Date
As indicated	2026-06-12
Drawn: K.N.	Checked By: J.K.
Project No.: 25356	Drawing No.: S203

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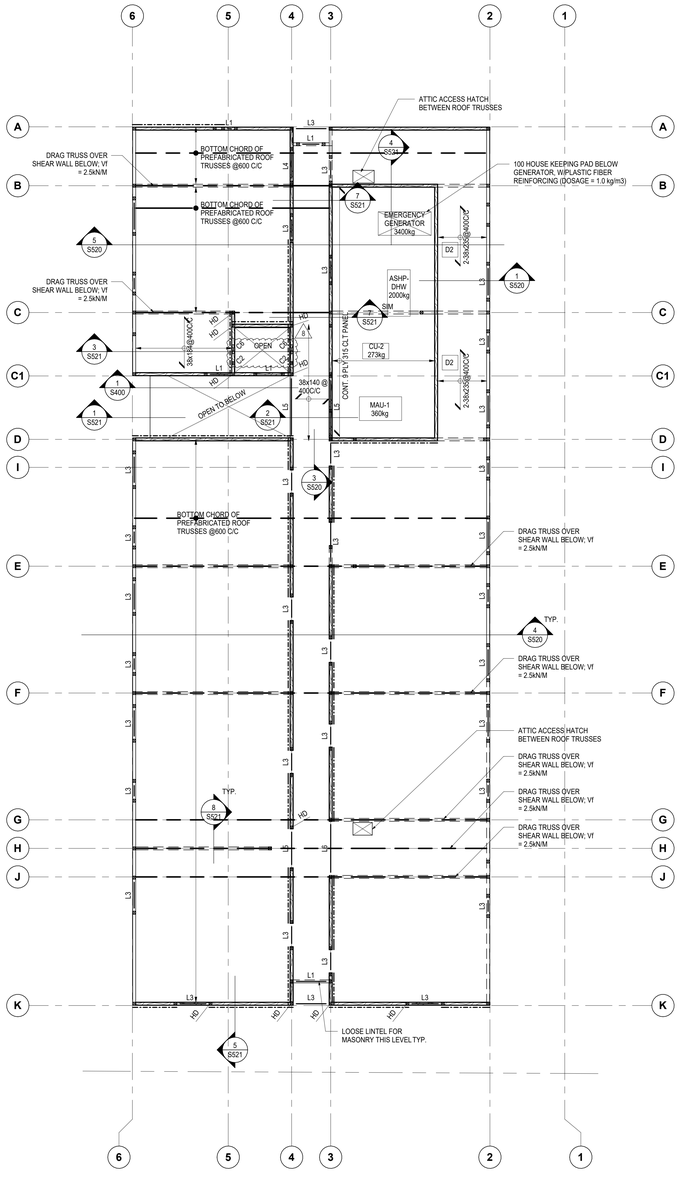

1 / S204 T/O LEVEL 3

1 : 100

NOTES

1. SEE GENERAL REQUIREMENTS AND TYPICAL DETAILS ON (DRAWING S100 SERIES).
2. ELEVATIONS AND GEOMETRY
   1. ATTIC DATUM ELEVATION IS 9350 (GEODETIC EL. 109850).
   2. TOP OF TRUSS BOTTOM CHORDS SET AT DATUM ELEVATIONS.
   3. TOP OF CLT DECK IS -65 FROM DATUM ELEVATION EXCEPT AS CROSSED AND NOTED ±X ON PLAN.
   4. TOP OF WOOD BEAMS ARE SET AT THE UNDERSIDE OF THE TRUSS BOTTOM CHORD.
   5. WHERE BEAMS NOTED THUS (±X) , X IS DISTANCE TO TOP OF BEAM FROM FLOOR DATUM.
3. DESIGN LOADING
   1. DESIGN LIVE LOAD FOR THE ATTIC IS 1.0kPa
   2. DESIGN LIVE LOAD FOR THE ROOFTOP EQUIPMENT AREA IS 3.6kPa.
   3. SUPERIMPOSED DEAD FOR THE ROOFTOP EQUIPMENT AREA LOAD IS 3.2 kPa.
4. FRAMING
   1. PROVIDE 40 CONCRETE TOPPING OVER ACOUSTIC MAT ON CLT DECK. REINFORCE TOPPING WITH FIBRILLATED POLYPROPYLENE FIBRES AT LEAST 19mm IN LENGTH, DOSAGE 0.9kg/m³.

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2026-10-30
PROVINCE OF ONTARIO

ATTIC FRAMING PLAN

Scale	Date
As indicated	2026-06-12
Drawn: K.N.	Checked By: J.K.
Project No.: 25356	Drawing No.: S204

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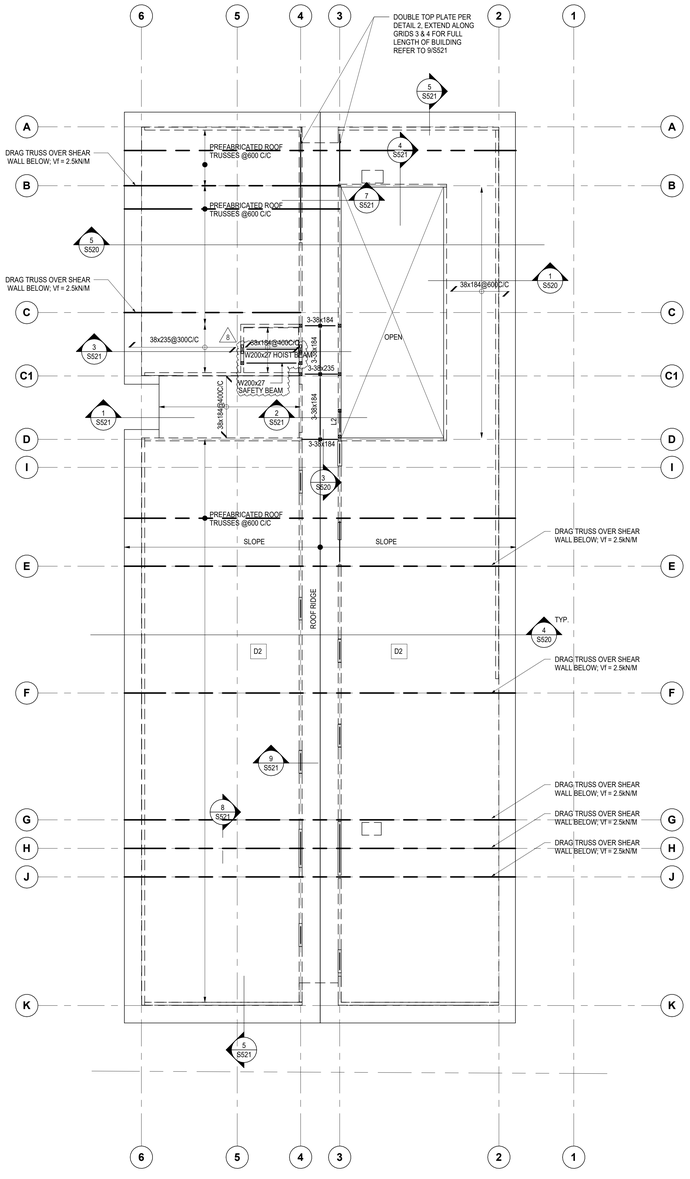

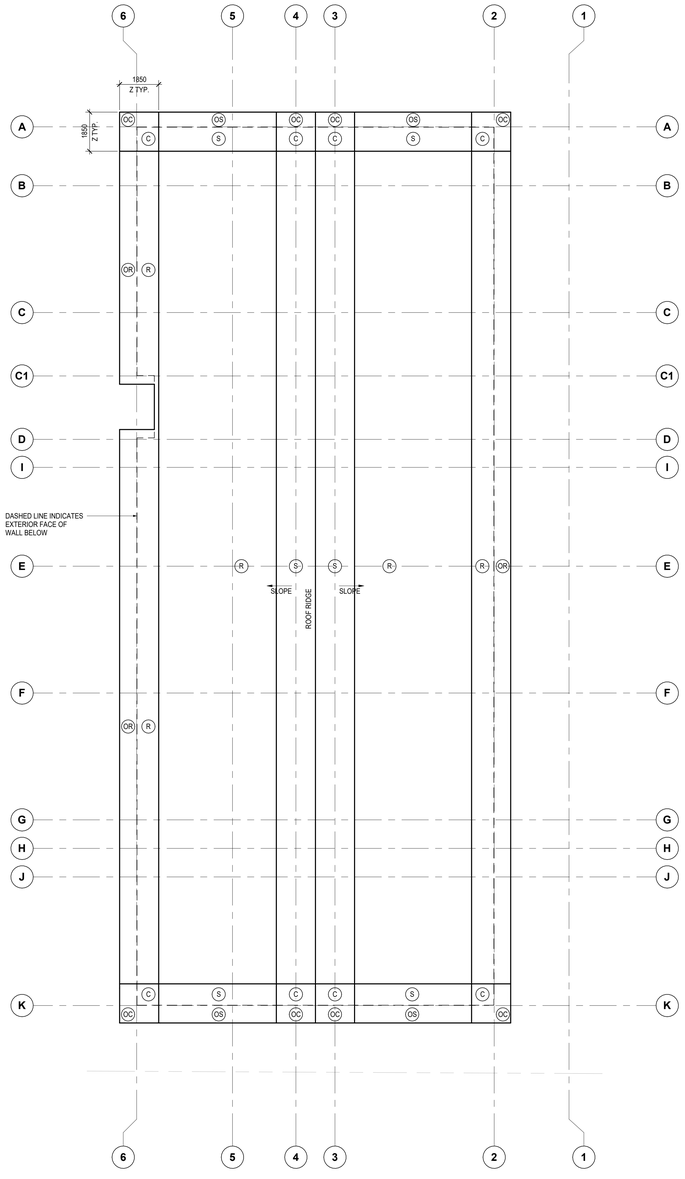

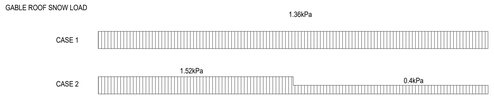

1 S205 T/O ROOF
1 : 100

2 S205 ROOF UPLIFT DIAGRAM
1 : 100

REGION		GROSS PRESSURE (kPa)
R	MAX	+0.47
	MIN	-0.90
S	MAX	+0.47
	MIN	-1.09
C	MAX	+0.47
	MIN	-1.56
OR	MAX	-0.57
	MIN	-0.57
OS	MAX	-1.41
	MIN	-1.41
OC	MAX	-1.58
	MIN	-1.58

NOTES:

1. POSITIVE WIND PRESSURE INDICATES DOWNWARD LOAD. NEGATIVE VALUES INDICATE UPLIFT ON THE ROOF.
2. "O" PREFIX FOR (OR, OC, OS) REGIONS REFERS TO OVERHANGING ROOF REGIONS

NOTES

1. SEE GENERAL REQUIREMENTS AND TYPICAL DETAILS ON DRAWING S100 SERIES.
2. ELEVATIONS AND GEOMETRY
   1. ROOF SLOPED, SEE ARCHITECTURAL DRAWINGS FOR ELEVATIONS AND GEOMETRY.
   2. TOP OF PLYWOOD SHEATHING IS THE TOP OF THE STRUCTURE AND SLOPES FROM ROOF RIDGE AS NOTED ON PLAN
   3. TOP OF TRUSS TOP CHORDS ARE AT UNDERSIDE OF PLYWOOD SHEATHING.
   4. TRIM ALL SIDES OF FRAMED OPENINGS IN ROOF WITH 2-38x184 UNO. COORDINATE WITH OPENINGS BY OTHER TRADES. ALL OPENINGS REQUIRE APPROVAL FROM STRUCTURAL CONSULTANT.
3. DESIGN LOADING
   1. SNOW LOADING IS SHOWN ON PLAN
   2. REFER TO 2/S205 FOR WIND UPLIFT ON THE ROOF.
   3. SUPERIMPOSED DEAD LOAD IS 1.5 kN/m².
   4. ALLOW FOR 0.5kPa FOR FUTURE PHOTOVOLTAIC PANEL INSTALLATION
4. FRAMING
   1. DESIGN ROOF TRUSSES, DESIGN LOADS UNLESS OTHERWISE NOTED
   2. TRUSSES TO BEAR ON EXTERIOR AND CORRIDOR WALLS UNLESS NOTED OTHERWISE ON PLAN.
   3. PROVIDE HOLD DOWNS AT ALL BEARING POINTS FOR UPLIFT FORCES.

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LICENSED PROFESSIONAL ENGINEER
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100148194
2026-10-30
PROVINCE OF ONTARIO

ROOF FRAMING PLAN

Scale	Date
As indicated	2026-06-12
Drawn: K.N.	Checked By: J.K.
Project No.: 25356	Drawing No.: S205

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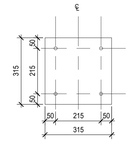
GLULAM AND STEEL COLUMN SCHEDULE

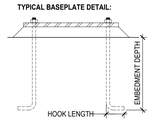
T/O ROOF												T/O ROOF
13000 T/O LEVEL 3												13000 T/O LEVEL 3
9350 LEVEL 3												9350 LEVEL 3
6300 LEVEL 2												6300 LEVEL 2
3250 LEVEL 1	360X342 GL 200 TYP. 4	HSS152x152x6.4	360X342 GL	HSS152x152x7.9	HSS152x152x9.5	360X342 GL	HSS152x152x5.5	360X342 GL	HSS152x152x6.4	HSS152x152x6.4		3250 LEVEL 1
0 BASEMENT	BP1	BP1	BP1	BP1	BP1	BP1	BP1	BP1	BP1	BP1		0 BASEMENT
-3050												-3050
Column Locations	A(-965)-1(-514)	A(-823)-2(-77)	B-1(-513)	B-3	B-3(3734)	C-1(-514)	C-2(-3089)	C1(208)-1(-514)	C1-2(-86)	D-2(-86)	D-2(-232)	

STEEL/GLULAM COLUMN SCHEDULE NOTES
1. SEE GENERAL REQUIREMENTS AND TYPICAL DETAILS ON THE S100 DRAWING SERIES.
2. ALL LOADS ARE ULTIMATE LIMIT STATE (ULS) UNLESS NOTED OTHERWISE.
3. PROVIDE 4-19Ø ANCHOR RODS (400 EMBEDMENT PLUS 75 HOOK) FOR EACH BASE PLATE, UNLESS OTHERWISE NOTED IN SCHEDULE.
4. COLUMN BASES TO BE GROUTED WITH 50mm OF GROUT. SEE SPECIFICATIONS.
5. PROVIDE LEVELLING SCREWS FOR BASE PLATES.
6. WELD BASE PLATES TO COLUMNS TO DEVELOP ANCHOR RODS.
7. PROVIDE CAP PLATES ON ALL HSS COLUMNS. PROVIDE CAP PLATES ON ALL OTHER COLUMNS WHERE REQUIRED FOR SUPPORT OF JOISTS, DECK, WINDOW WASHING ANCHORS ETC.
8. PROVIDE HSS152X152X13 STUB AT ALL GLULAM COLUMNS
4 9. SET U/S OF BASEPLATE -250 FROM FLOOR DATUM, UNO

BASEPLATE SCHEDULE

MARK	SIZE	ANCHORS	DETAILS
BP1	315x19x315	4-19Ø ANCHOR RODS x 400 LG. EMBED + 75 HOOK	

TYPICAL BASEPLATE DETAIL:



STEEL / TIMBER COLUMN SCHEDULE LEGEND:

COLUMN TAG(S):

W360x60	COLUMN SECTION
COMMENTS	COLUMN-SPECIFIC COMMENTS (IF APPLICABLE)

BASE TAG:

BP1	BASE PLATE (REFER TO BASE PLATE SCHEDULE)
SLS P=1000 (600)	SLS (UNFACTORED) AXIAL REACTION (UPLIFT IN BRACKETS, IF APPLICABLE)
ULS P=1800 (800)	ULS (FACTORED) AXIAL REACTION (UPLIFT IN BRACKETS, IF APPLICABLE)
V=250 / 150	ULS (FACTORED) SHEAR REACTION(S), IF APPLICABLE (REFER TO NOTES)
M=150 / 120	ULS (FACTORED) BENDING MOMENT REACTION(S), IF APPLICABLE (REFER TO NOTES)
TM=60	ULS (FACTORED) TORSIONAL MOMENT REACTION, IF APPLICABLE

STEEL COLUMN SYMBOL LEGEND

BASE CONNECTIONS:
- BASE PLATE
- STEEL BEAM BEARING
- CONCRETE BEAM BEARING

TOP CONNECTIONS:
- COLUMN SPLICE

BUILT UP COLUMN SCHEDULE	
MARK	SIZE
C1	3-38x140
C2	4-38x140
C3	5-38x140
C4	6-38x140
C5	5-38x140 LSL
C6	133x406 PSL 4

NOTES:
1. AT MINIMUM, COLUMNS SHALL HAVE THE SAME NUMBER OF PLY AS THE BEAM THEY ARE SUPPORTING. IF BEAM PLYS ARE WIDER THAN STUDS, ENSURE COLUMN IS WIDER THAN THE BEAM.
2. CARRY COLUMN DOWN TO CONCRETE EVEN IF NOT NOTED ON PLANS.
3. BUILT UP POSTS TO BE FASTENED TOGETHER PER TYPICAL DETAILS.

CONCRETE PIER SCHEDULE

MARK	SIZE (mm)	REINFORCING		REMARKS
		VERTICAL	TIES	
P1	600x600	4-25V	10@400	

NOTE: CONCRETE PIERS CENTERED BELOW COLUMNS TYP.

SPREAD FOOTING SCHEDULE

MARK	SIZE (mm)	REINFORCING	BEARING CAPACITY (kN)		REMARKS
			SLS	ULS	
F1	1000x1000x300	4-15BEW	360	540	
F2	1500x1500x350	6-15BEW	675	1000	

TIMBER LINTEL SCHEDULE

LINTEL MARK	LINTEL	MIN JACK STUDS	KING STUDS	COMMENTS
L1	3-38x140	2	1	
L2	3-38x184	3	1	
L3	3-38x235	2	1	
L4	3-38x286	2	1	
L5	3-45x241 LVL	3	1	
L6	3-45x302 LVL	3	1	
L7	3-45x457 LVL	3	1	

NOTES: 1. JACK STUDS ARE A MINIMUM, WHERE COLUMN IS NOTED ON PLAN AT THE END OF A LINTEL, USE THE COLUMN AS THE JACK STUDS.
2. NAIL BUILT UP JACK STUDS AND KING STUDS TOGETHER PER TYPICAL DETAILS.
3. CARRY BUILT UP JACK STUD AND KING STUD POSTS DOWN TO THE FOUNDATION, TYP.

SHEATING SCHEDULE

DECK MARK	SHEATING	NAILS	NAIL SPACING (mm)			BLOCK AT PANEL EDGES	COMMENTS
			OVER SHEARLINES	PANEL EDGES	INTERMEDIATE FRAMING		
D1	15.5 CSP	75 NAILS	150	150	300	REQUIRED	NAILED AND GLUED TO JOISTS
D2	12.5 CSP	75 NAILS	100	100	300	REQUIRED	NAILED AND GLUED TO JOISTS

SLAB-ON-GRADE SCHEDULE

MARK	THICKNESS (mm)	REINFORCEMENT	REMARKS
SOG1	125	152x152 MW 18.7 / MW 18.7 W.W.F.	TYPICAL AT GROUND FLOOR

SHEAR WALL SCHEDULE

SW1

LEVEL	STUD SIZE, TYPE & GRADE	ONE-SIDED SHEATING THICKNESS AND TYPE	NAILING@PANEL EDGES	PILES PER POSTS AT ENDS OF WALL & SIDES OF OPENING	HOLD DOWN ANCHORS AT ENDS OF WALL PANEL (ASTM F1554 GRADE 36) 4	TOP AND BOTTOM PLATES	SIMPSON SDS25600 SCREW TOP AND BOTTOM PLATE CONNECTION	BOTTOM PLATE ANCHOR BOLT
ROOF								
L3	38x140@400 C/C	12.5 CSP	75 NAIL @ 75	6	HDUE13-SDS3.5 + 25mm ANCHOR BOLTS	38 THK 1.3E LSL	SCREW @ 400	N/A
L2	38x140@400 C/C	12.5 CSP	75 NAIL @ 75	6	HDUE13-SDS3.5 + 25mm ANCHOR BOLTS	38 THK 1.3E LSL	SCREW @ 250	N/A
L1	2-38x140@400 C/C SPF SS	12.5 CSP	75 NAIL @ 50	6	HDUE13-SDS3.5 + 25mm ANCHOR BOLTS	38 THK 1.3E LSL	SCREW @ 150	16Ø@300C/C, w/ 200 EMBED + 50mm HOOK

SW2

LEVEL	STUD SIZE, TYPE & GRADE	ONE-SIDED SHEATING THICKNESS AND TYPE	NAILING@PANEL EDGES	PILES PER POSTS AT ENDS OF WALL & SIDES OF OPENING	HOLD DOWN ANCHORS AT ENDS OF WALL PANEL (ASTM F1554 GRADE 36) 4	TOP AND BOTTOM PLATES	SIMPSON SDS25600 SCREW TOP AND BOTTOM PLATE CONNECTION	BOTTOM PLATE ANCHOR BOLT
ROOF								
L3	38x140@400 C/C	12.5 CSP	75 NAIL @ 75	6	HDUE13-SDS3.5 + 25mm ANCHOR BOLTS	38 THK 1.3E LSL	SCREW @ 400	N/A
L2	38x140@400 C/C	12.5 CSP	75 NAIL @ 75	6	HDUE13-SDS3.5 + 25mm ANCHOR BOLTS	38 THK 1.3E LSL	SCREW @ 250	N/A
L1	2-38x140@400 C/C	12.5 CSP	75 NAIL @ 50	6	HDUE13-SDS3.5 + 25mm ANCHOR BOLTS	38 THK 1.3E LSL	SCREW @ 150	16Ø@300C/C, w/ 200 EMBED + 50mm HOOK

SW3

LEVEL	STUD SIZE, TYPE & GRADE	ONE-SIDED SHEATING THICKNESS AND TYPE	NAILING@PANEL EDGES	PILES PER POSTS AT ENDS OF WALL & SIDES OF OPENING	HOLD DOWN ANCHORS AT ENDS OF WALL PANEL (ASTM F1554 GRADE 36) 4	TOP AND BOTTOM PLATES	SIMPSON SDS25600 SCREW TOP AND BOTTOM PLATE CONNECTION	BOTTOM PLATE ANCHOR BOLT
ROOF								
L3	2-38x140@400 C/C	12.5 CSP	75 NAIL @ 75	6	HDUE13-SDS3.5 + 25mm ANCHOR BOLTS	38 THK 1.3E LSL	SCREW @ 400	N/A
L2	2-38x140@400 C/C	12.5 CSP	75 NAIL @ 75	6	HDUE13-SDS3.5 + 25mm ANCHOR BOLTS	38 THK 1.3E LSL	SCREW @ 250	N/A
L1	2-38x140@400 C/C	12.5 CSP	75 NAIL @ 50	6	HDUE13-SDS3.5 + 25mm ANCHOR BOLTS	38 THK 1.3E LSL	SCREW @ 150	16Ø@300C/C, w/ 200 EMBED + 50mm HOOK

NOTE: AT LEVEL 1, ALL HOLD DOWN ANCHORS TO INCLUDE WELDABLE DOUBLE NUT W/ 100x100x12 PLATE, EMBED 650 INTO FOUNDATION WALLS OR CONNECT TO SLAB PER TYPICAL DETAILS.

LOAD BEARING WALLS (NON SHEAR)		
MARK	STUDS	NOTES
BW1	38x140@400C/C	ALL INTERIOR WALLS U/N
BW2	2-38x140@400C/C	ALL EXTERIOR WALLS U/N
BW3	38x89@400C/C	

NOTES:
1. SHEATHING PER ARCHITECTURAL
2. SEE TYPICAL DETAILS

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2026-10-30
PROVINCE OF ONTARIO

SCHEDULES

Scale: As indicated	Date: 2026-06-12
Drawn: K.N.	Checked By: J.K.
Project No.: 25356	Drawing No.: S300

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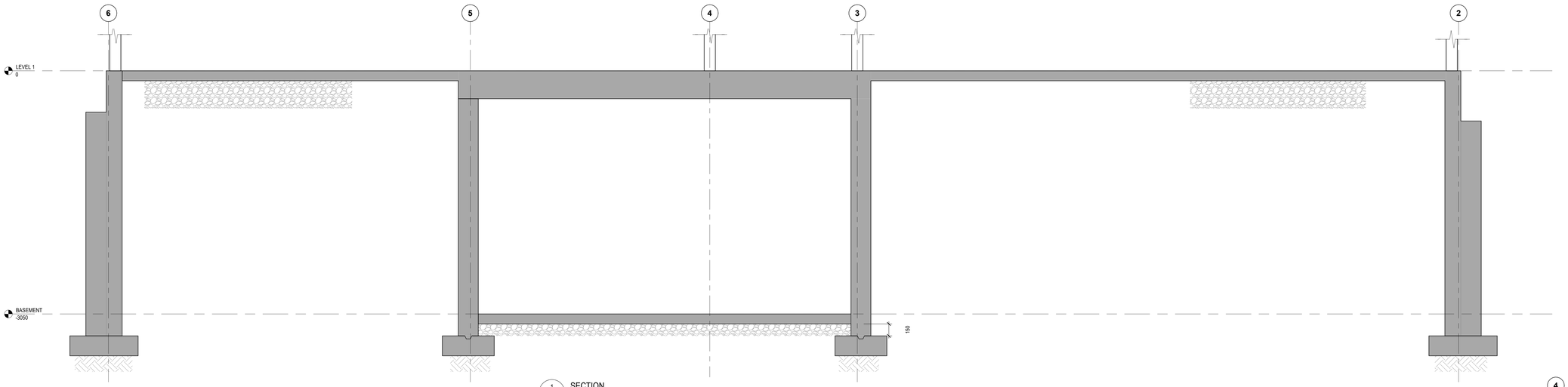
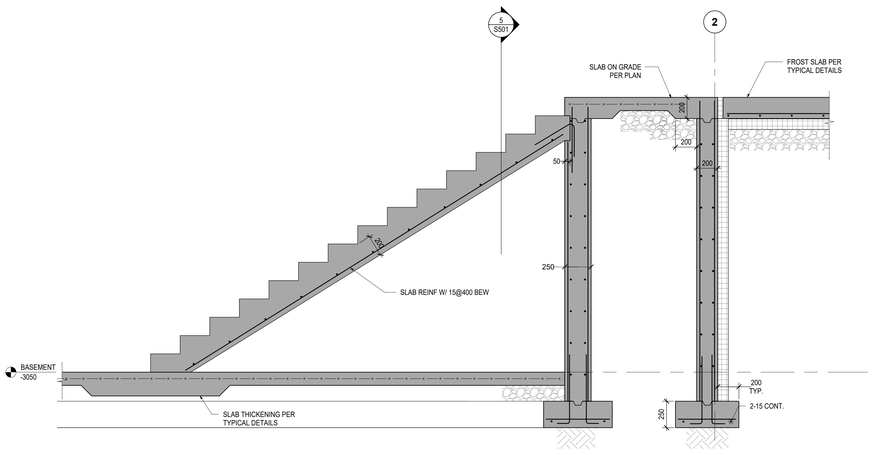


1 S500 SECTION 1 : 20



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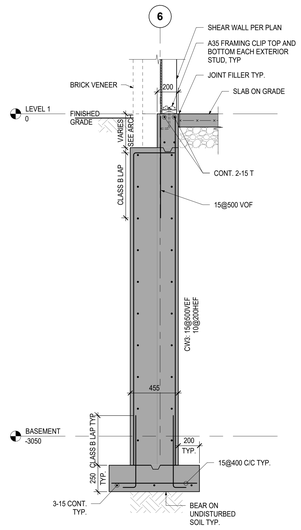

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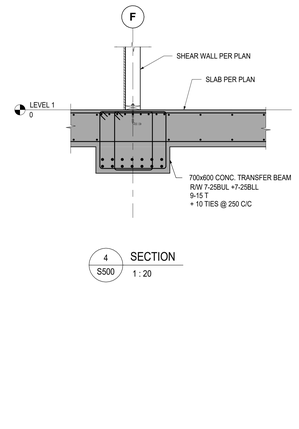

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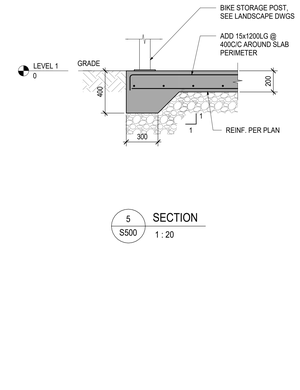

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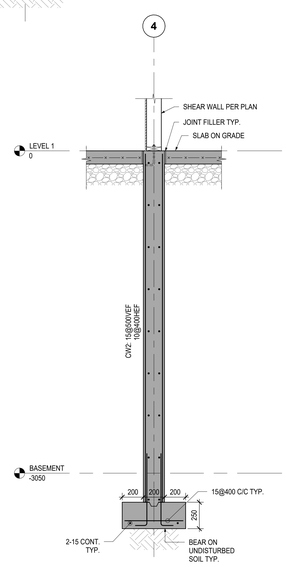
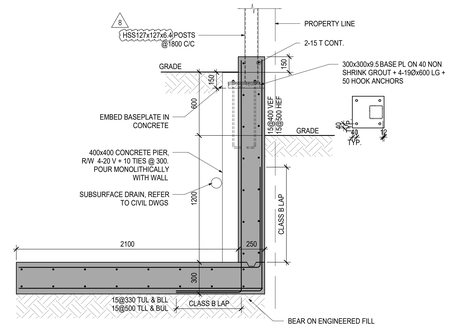
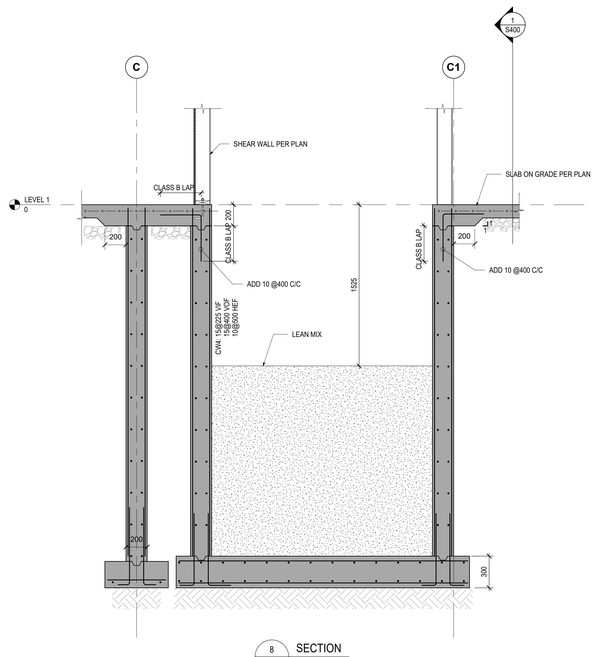


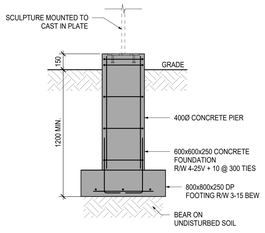
6 S500 SECTION 1 : 20



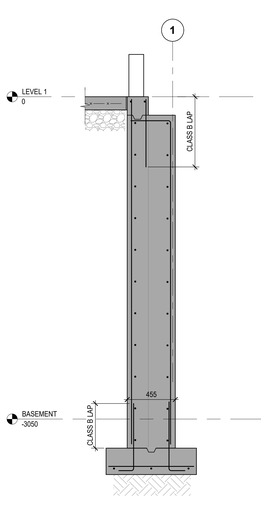
7 S500 TYPICAL RETAINING WALL DETAIL 1 : 20



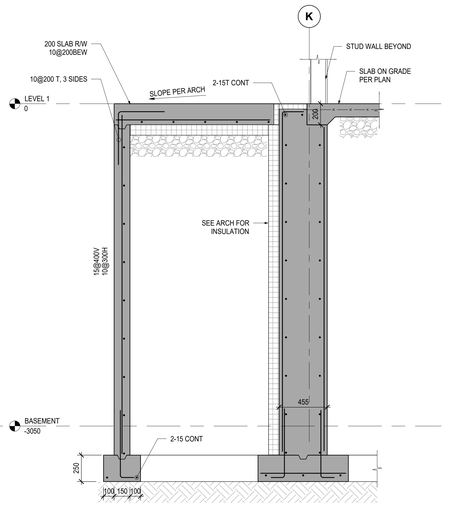
8 S500 SECTION 1 : 20



9 S500 SCULPTURE FOUNDATION 1 : 20



10 S500 SECTION 1 : 20



11 S500 SECTION 1 : 20

REV.	DATE	DESCRIPTION
8	2026-10-29	ISSUED FOR STRUCTURAL TENDER ADDENDUM NO.2
7	2026-09-16	ISSUED FOR TENDER
6	2026-03-20	ISSUED FOR 100% CD
5	2026-02-18	ISSUED FOR 90% CD AND CLASS A COSTING
4	2026-01-30	ISSUED FOR BUILDING PERMIT
3	2026-01-30	ISSUED FOR 75% CD
2	2025-12-17	Issued for 30 % Construction Documents
1	2025-06-20	Issued for 100% DD & Class B Costing

REDEVELOPMENT - 409 CENTRE ST S - 25008

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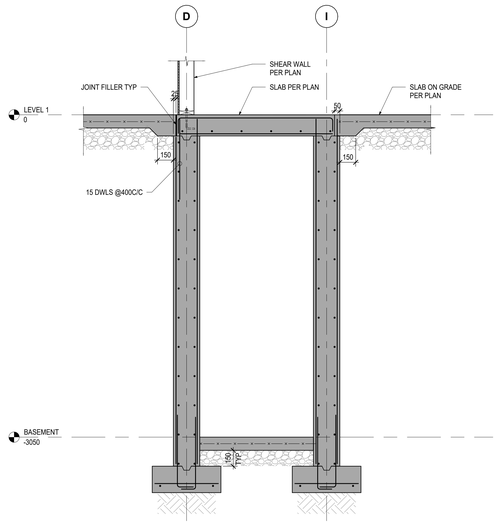
LICENSED PROFESSIONAL ENGINEER
J.M. KACZMAREK
100146134
2026-10-29
PROVINCE OF ONTARIO

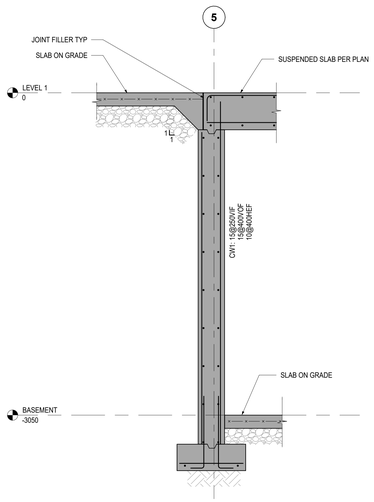
FOUNDATION SECTIONS

Scale	Date
1 : 20	2026-06-12
Drawn: K.N.	Checked By: J.K.
Project No.: 25356	Drawing No.: S500

2026-10-29 5:14:59 PM Autodesk Docs://25008_DHR - 409 Centre St/25008_409 Centre St_LEA_STRU_MAIN_R24.rvt



1 / S501 SECTION
1 : 20



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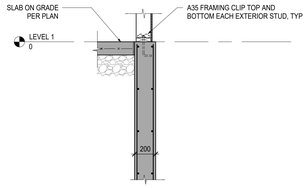
1 : 20



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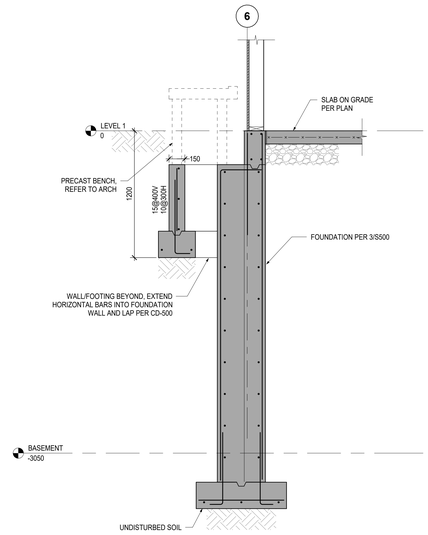
1 : 20



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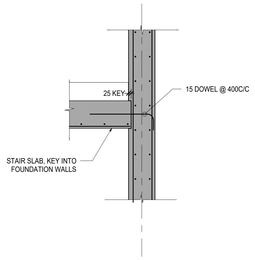
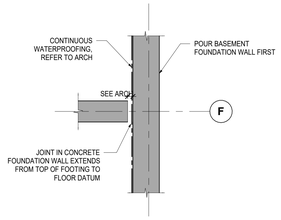
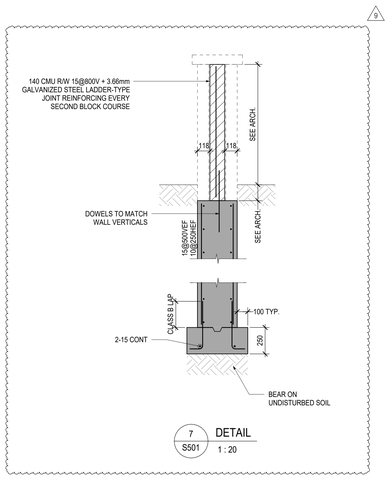
1 : 20



5 / S501 SECTION
1 : 20



6 / S501 DETAIL
1 : 20



7 / S501 DETAIL
1 : 20

REV.	DATE	DESCRIPTION
9	2026-10-29	ISSUED FOR STRUCTURAL TENDER ADDENDUM NO.2
8	2026-10-13	ISSUED FOR STRUCTURAL TENDER ADDENDUM NO.1
7	2026-09-16	ISSUED FOR TENDER
6	2026-03-20	ISSUED FOR 100% CD
5	2026-02-18	ISSUED FOR 90% CD AND CLASS A COSTING
4	2026-01-30	ISSUED FOR BUILDING PERMIT
3	2026-01-30	ISSUED FOR 75% CD
2	2025-12-17	Issued for 30 % Construction Documents
1	2025-06-20	Issued for 100% DD & Class B Costing

REDEVELOPMENT - 409 CENTRE ST S - 25008

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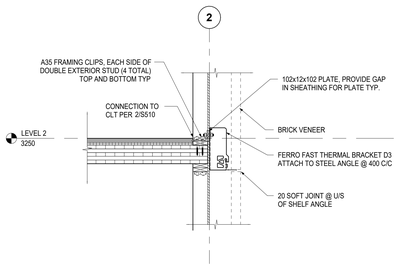
LICENSED PROFESSIONAL ENGINEER
J.M. KACZMAREK
100146134
2026-10-29
PROVINCE OF ONTARIO

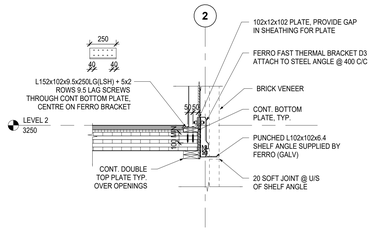
FOUNDATION SECTIONS

Scale: 1 : 20	Date: 2026-06-12
Drawn: K.N.	Checked By: J.K.
Project No.: 25356	Drawing No.: S501

2026-10-29 5:14:19 PM Autodesk Docs://25008_DHS - 409 Centre St/25008_409 Centre St_LEA_STRU_MAIN_R24.rvt



1 S510 SECTION 1 : 20



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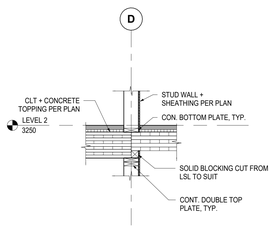

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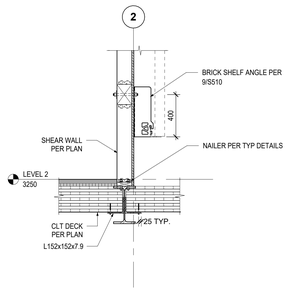

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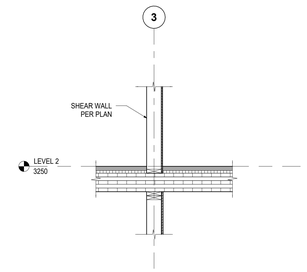

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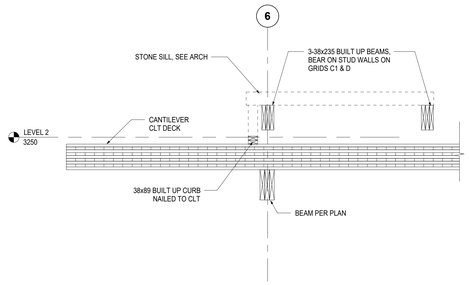

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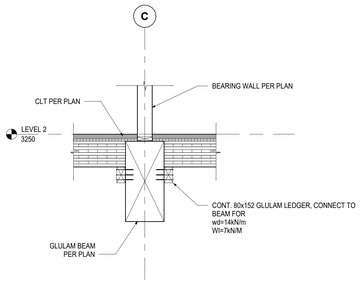

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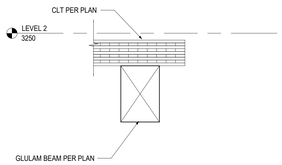

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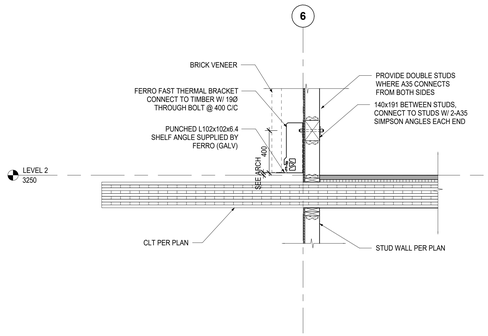

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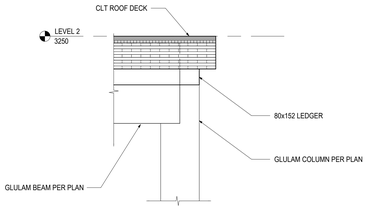

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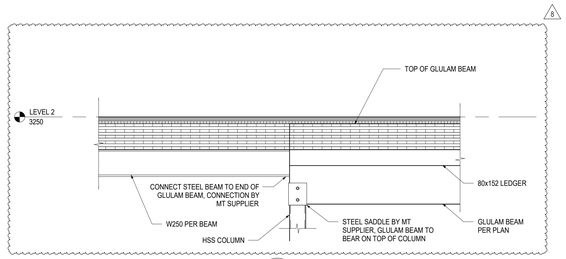

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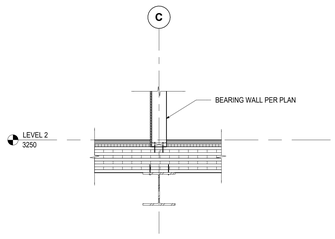

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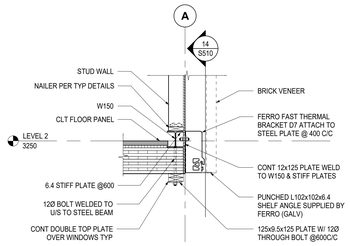

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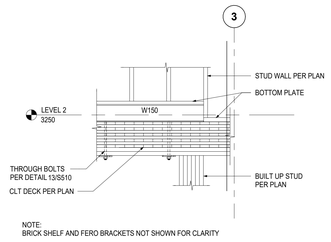

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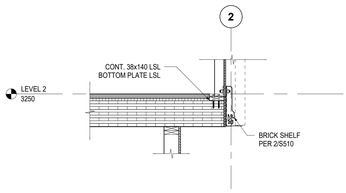
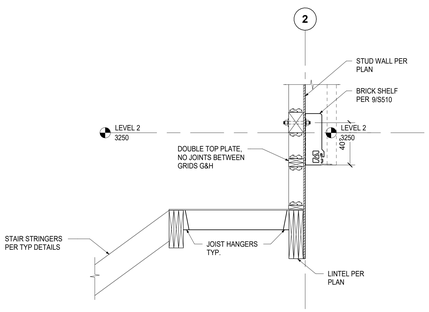


15 S510 SECTION 1 : 20



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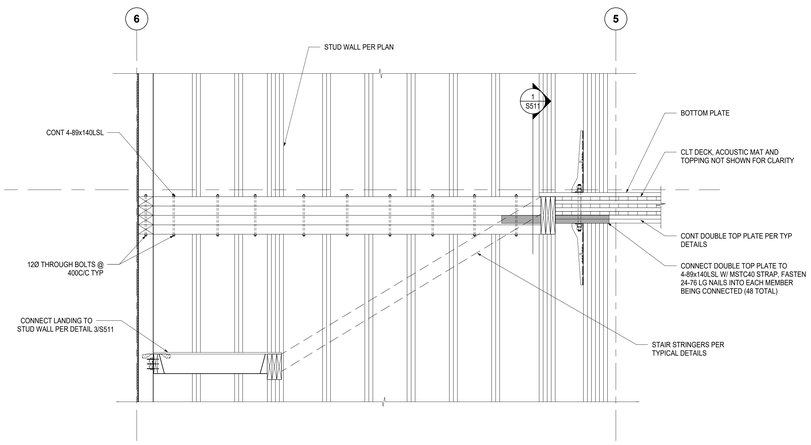
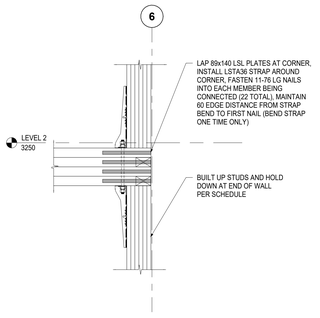


17 S510 SECTION 1 : 20



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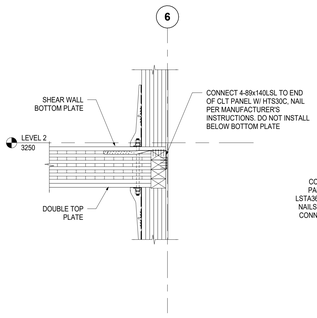

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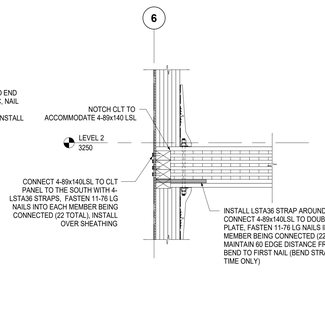

20 S510 SECTION 1 : 20

REV.	DATE	DESCRIPTION
8	2026-10-30	ISSUED FOR STRUCTURAL TENDER ADDENDUM NO.3
7	2026-09-16	ISSUED FOR TENDER
6	2026-03-20	ISSUED FOR 100% CD
5	2026-02-18	ISSUED FOR 90% CD AND CLASS A COSTING
4	2026-01-30	ISSUED FOR BUILDING PERMIT
3	2026-01-30	ISSUED FOR 75% CD
2	2025-12-17	Issued for 30 % Construction Documents
1	2025-06-20	Issued for 100% DD & Class B Costing

REDEVELOPMENT - 409 CENTRE ST S - 25008

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LICENSED PROFESSIONAL ENGINEER
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100146194
2026-10-30
PROVINCE OF ONTARIO

LEVEL 2 SECTIONS

Scale: 1 : 20	Date: 2026-06-12
Drawn: K.N.	Checked By: J.K.
Project No.: 25356	Drawing No.: S510

2026-10-30 3:23:44 PM Autodesk Docs://25008_BHR - 409 Centre St/25008_409 Centre St_LEA_STRU_MAIN_R24.rvt

MAKE-UP AIR UNIT - AIR-SOURCE HEAT PUMP

NOTES:
1. PROVIDE WITH SINGLE POINT POWER CONNECTION.

TAG	MANUFACTURER	MODEL	AIRFLOW (L/s)	E.S.P (Pa)	PRE-FILTER	FINAL FILTER	REFRIGERANT		COOLING - AIR SOURCE HEAT PUMP (°C)								HEATING						ELECTRICAL						SOUND POWER (dB)									DIMENSIONS (mm)			WEIGHT (kg)	NOTES
							TYPE	CHARGE (g)	TOTAL CAPACITY (kW)	SENSIBLE CAPACITY (kW)	HOT GAS REHEAT (kW)	OUTDOOR AIR TEMPERATURE		AIR TEMPERATURE BEFORE REHEAT		SUPPLY AIR TEMPERATURE		AIR SOURCE HEAT PUMP					ELECTRIC HEATER CAPACITY (kW)	V/Ph/Hz	FLA (A)	MCA (A)	MOCP (A)	VFD	EMERGENCY POWER	63 Hz	125 Hz	250 Hz	500 Hz	1000 Hz	2000 Hz	4000 Hz	8000 Hz	LENGTH	WIDTH	HEIGHT		
												DB	WB	DB	WB	DB	WB	CAPACITY (kW)	OUTDOOR AIR TEMPERATURE (°C)		SUPPLY AIR TEMPERATURE (°C)																									
																			DB	WB	DB	WB																				
MAU-1	AAON	RQA-005-A-A-8-GJB0A-A03NA	360	187	MERV 7	MERV 13	R-454B	28.3 g	16.94	8.94	2.14 kW	30.0	23.0	9.6	9.2	14.0	11.2	11.69	-2.8	-4.3	19.2	8.0	22.50	208/3/60	70	87	90	YES	YES	75	76	77	64	59	57	52	46	2096	1124	1283	357.0	1

AIR-SOURCE HEAT PUMPS - OUTDOOR

TAG	MANUFACTURER	MODEL	LOCATION	SERVICE	FLUID	REFRIGERANT	HEATING								ELECTRICAL					SOUND POWER DATA									DIMENSIONS (mm)			WEIGHT (kg)	REMARKS
						TYPE	CAPACITY (W)	COP	OUTDOOR AIR TEMPERATURE (°C)		FLUID				V/Ph/Hz	FLA (A)	MCA (A)	MOCP (A)	EMERGENCY POWER	OCTAVE BANDS (dB)									LENGTH	WIDTH	HEIGHT		
									DB	WB	EFT (°C)	LFT (°C)	FLOW RATE (L/s)	PRESSURE DROP (kPa)						63 Hz	125 Hz	250 Hz	500 Hz	1000 Hz	2000 Hz	4000 Hz	8000 Hz					
ASHP-DHW	ADVANCE INDUSTRIAL MECHANICAL	ASB-25	ROOF	DOMESTIC HOT WATER	WATER	HIGH TEMP CIRCUIT: R515B, LOW TEMP CIRCUIT R448A	53398	1.75	-22.2		54.4	65.6	1.17	36.5	208/3/60	175	195	220	NO		73	75.0	77	80.0	78	76	70	2200	1080	2375	1905	REMOTE CONDENSER HEAT EXCHANGER

AIR-SOURCE HEAT PUMPS - CEILING CONCEALED

NOTES:
1. ALL CEILING CONCEALED AIR-SOURCE HEAT PUMPS LOCATED IN RESIDENTIAL SUITES: SHALL HAVE ERV FANS AND SUPPLY FAN ENABLED BY SWITCH IN BATHROOM TO OPERATE AT MEDIUM SPEED. IF FANS ARE ALREADY OPERATING ON HIGH SPEED, FAN SPEED SHALL NOT BE OVERRIDDEN.
2. ALL CEILING CONCEALED AIR-SOURCE HEAT PUMPS LOCATED IN PUBLIC SPACES: SHALL HAVE ERV FANS AND SUPPLY FAN OPERATE CONTINUOUSLY AT LOW SPEED. FAN SPEED SHALL BE INCREASED OR DECREASED AS REQUIRED TO AID IN MEETING THERMOSTAT SETPOINTS BUT SHALL NOT BE DISABLED..

TAG	MANUFACTURER	MODEL	SERVICE	FANS						REFRIGERANT		COOLING						HEATING					ELECTRICAL						SOUND POWER DATA										DIMENSIONS (mm)			WEIGHT (kg)	REMARKS
				SUPPLY		ERV EXHAUST INTAKE		OUTDOOR AIR INTAKE & EXHAUST TO OUTSIDE		TYPE	CHARGE (kg)	CAPACITY (W)	COP	OUTDOOR AIR TEMPERATURE (°C)		INDOOR AIR TEMPERATURE (°C)		ERV RECOVERY (W)		CAPACITY (W)	COP	OUTDOOR AIR TEMPERATURE (°C)		INDOOR AIR TEMPERATURE(°C)		ERV TOTAL RECOVERY (W)	V/Ph/Hz	FLA (A)	MCA (A)	MOCP (A)	ELECTRIC PRE-HEATER (kW)	ELECTRIC POST-HEATER (W)	EMERGENCY POWER	OCTAVE BANDS (dB)									WEIGHTED SOUND POWER (dBA)	LENGTH	WIDTH	HEIGHT		
				AIRFLOW (L/s)	E.S.P. (Pa)	AIRFLOW (L/s)	E.S.P. (Pa)	TOTAL AIRFLOW (L/s)	OUTDOOR AIR DIVERTED TO ERV (L/s)					DB	WB	DB	WB	SENSIBLE	LATENT			DB	WB	DB	WB								63 Hz	125 Hz	250 Hz	500 Hz	1000 Hz	2000 Hz	4000 Hz	8000 Hz					
ASHP-101	INSPIRONAIR	IAIRHP-35-CC-ERV-B	RESIDENTIAL SUITE	189	124	38	75	227	38	124	R32	0.55 kg	4396	3.49	35.0	23.9	26.7	19.4	293	193	2550	2.42	-10.6	-12.8	21.1	15.6	1260	208/1/60	15.3	26	35	0.35	2500	YES	23.2	38.8	48.2	50.5	50.0	46.1	41.8	32.4	60.7	1152	1033	280	106	
ASHP-102	INSPIRONAIR	IAIRHP-35-CC-ERV-B	RESIDENTIAL SUITE	189	124	38	75	227	38	124	R32	0.55 kg	4396	3.49	35.0	23.9	26.7	19.4	293	193	2550	2.42	-10.6	-12.8	21.1	15.6	1260	208/1/60	15.3	26	35	0.35	2500	YES	23.2	38.8	48.2	50.5	50.0	46.1	41.8	32.4	60.7	1152	1033	280	106	
ASHP-103	INSPIRONAIR	IAIRHP-35-CC-ERV-B	RESIDENTIAL SUITE	189	124	38	75	227	38	124	R32	0.55 kg	4396	3.49	35.0	23.9	26.7	19.4	293	193	2550	2.42	-10.6	-12.8	21.1	15.6	1260	208/1/60	15.3	26	35	0.35	2500	NO	23.2	38.8	48.2	50.5	50.0	46.1	41.8	32.4	60.7	1152	1033	280	106	
ASHP-110	INSPIRONAIR	IAIRHP-35-CC-ERV-B	RESIDENTIAL SUITE	189	124	38	75	227	38	124	R32	0.55 kg	4396	3.49	35.0	23.9	26.7	19.4	293	193	2550	2.42	-10.6	-12.8	21.1	15.6	1260	208/1/60	15.3	26	35	0.35	2500	NO	23.2	38.8	48.2	50.5	50.0	46.1	41.8	32.4	60.7	1152	1033	280	106	
ASHP-111	INSPIRONAIR	IAIRHP-35-CC-ERV-B	RESIDENTIAL SUITE	189	124	38	75	227	38	124	R32	0.55 kg	4396	3.49	35.0	23.9	26.7	19.4	293	193	2550	2.42	-10.6	-12.8	21.1	15.6	1260	208/1/60	15.3	26	35	0.35	2500	NO	23.2	38.8	48.2	50.5	50.0	46.1	41.8	32.4	60.7	1152	1033	280	106	
ASHP-112	INSPIRONAIR	IAIRHP-35-CC-ERV-B	RESIDENTIAL SUITE	189	124	38	75	227	38	124	R32	0.55 kg	4396	3.49	35.0	23.9	26.7	19.4	293	193	2550	2.42	-10.6	-12.8	21.1	15.6	1260	208/1/60	15.3	26	35	0.35	2500	NO	23.2	38.8	48.2	50.5	50.0	46.1	41.8	32.4	60.7	1152	1033	280	106	
ASHP-113	INSPIRONAIR	IAIRHP-35-CC-ERV-B	RESIDENTIAL SUITE	189	124	38	75	227	38	124	R32	0.55 kg	4396	3.49	35.0	23.9	26.7	19.4	293	193	2550	2.42	-10.6	-12.8	21.1	15.6	1260	208/1/60	15.3	26	35	0.35	2500	NO	23.2	38.8	48.2	50.5	50.0	46.1	41.8	32.4	60.7	1152	1033	280	106	
ASHP-114	INSPIRONAIR	IAIRHP-35-CC-ERV-B	RESIDENTIAL SUITE	189	124	38	75	227	38	124	R32	0.55 kg	4396	3.49	35.0	23.9	26.7	19.4	293	193	2550	2.42	-10.6	-12.8	21.1	15.6	1260	208/1/60	15.3	26	35	0.35	2500	NO	23.2	38.8	48.2	50.5	50.0	46.1	41.8	32.4	60.7	1152	1033	280	106	
ASHP-115	INSPIRONAIR	IAIRHP-35-CC-ERV-B	RESIDENTIAL SUITE	189	124	38	75	227	38	124	R32	0.55 kg	4396	3.49	35.0	23.9	26.7	19.4	293	193	2550	2.42	-10.6	-12.8	21.1	15.6	1260	208/1/60	15.3	26	35	0.35	2500	NO	23.2	38.8	48.2	50.5	50.0	46.1	41.8	32.4	60.7	1152	1033	280	106	
ASHP-116	INSPIRONAIR	IAIRHP-35-CC-ERV-B	RESIDENTIAL SUITE	189	124	38	75	227	38	124	R32	0.55 kg	4396	3.49	35.0	23.9	26.7	19.4	293	193	2550	2.42	-10.6	-12.8	21.1	15.6	1260	208/1/60	15.3	26	35	0.35	2500	NO	23.2	38.8	48.2	50.5	50.0	46.1	41.8	32.4	60.7	1152	1033	280	106	
ASHP-203	INSPIRONAIR	IAIRHP-35-CC-ERV-B	RESIDENTIAL SUITE	189	124	38	75	227	38	124	R32	0.55 kg	4396	3.49	35.0	23.9	26.7	19.4	293	193	2550	2.42	-10.6	-12.8	21.1	15.6	1260	208/1/60	15.3	26	35	0.35	2500	NO	23.2	38.8	48.2	50.5	50.0	46.1	41.8	32.4	60.7	1152	1033	280	106	
ASHP-206	INSPIRONAIR	IAIRHP-35-CC-ERV-B	RESIDENTIAL SUITE	189	124	38	75	227	38	124	R32	0.55 kg	4396	3.49	35.0	23.9	26.7	19.4	293	193	2550	2.42	-10.6	-12.8	21.1	15.6	1260	208/1/60	15.3	26	35	0.35	2500	NO	23.2	38.8	48.2	50.5	50.0	46.1	41.8	32.4	60.7	1152	1033	280	106	
ASHP-207	INSPIRONAIR	IAIRHP-35-CC-ERV-B	RESIDENTIAL SUITE	189	124	38	75	227	38	124	R32	0.55 kg	4396	3.49	35.0	23.9	26.7	19.4	293	193	2550	2.42	-10.6	-12.8	21.1	15.6	1260	208/1/60	15.3	26	35	0.35	2500	NO	23.2	38.8	48.2	50.5	50.0	46.1	41.8	32.4	60.7	1152	1033	280	106	
ASHP-209	INSPIRONAIR	IAIRHP-35-CC-ERV-B	RESIDENTIAL SUITE	189	124	38	75	227	38	124	R32	0.55 kg	4396	3.49	35.0	23.9	26.7	19.4	293	193	2550	2.42	-10.6	-12.8	21.1	15.6	1260	208/1/60	15.3	26	35	0.35	2500	NO	23.2	38.8	48.2	50.5	50.0	46.1	41.8	32.4	60.7	1152	1033	280	106	
ASHP-210	INSPIRONAIR	IAIRHP-35-CC-ERV-B	RESIDENTIAL SUITE	189	124	38	75	227	38	124	R32	0.55 kg	4396	3.49	35.0	23.9	26.7	19.4	293	193	2550	2.42	-10.6	-12.8	21.1	15.6	1260	208/1/60	15.3	26	35	0.35	2500	NO	23.2	38.8	48.2	50.5	50.0	46.1	41.8	32.4	60.7	1152	1033	280	106	
ASHP-211	INSPIRONAIR	IAIRHP-35-CC-ERV-B	RESIDENTIAL SUITE	189	124	38	75	227	38	124	R32	0.55 kg	4396	3.49	35.0	23.9	26.7	19.4	293	193	2550	2.42	-10.6	-12.8	21.1	15.6	1260	208/1/60	15.3	26	35	0.35	2500	NO	23.2	38.8	48.2	50.5	50.0	46.1	41.8	32.4	60.7	1152	1033	280	106	
ASHP-212	INSPIRONAIR	IAIRHP-35-CC-ERV-B	RESIDENTIAL SUITE	189	124	38	75	227	38	124	R32	0.55 kg	4396	3.49	35.0	23.9	26.7	19.4	293	193	2550	2.42	-10.6	-12.8	21.1	15.6	1260	208/1/60	15.3	26	35	0.35	2500	NO	23.2	38.8	48.2	50.5	50.0	46.1	41.8	32.4	60.7	1152	1033	280	106	
ASHP-213	INSPIRONAIR	IAIRHP-35-CC-ERV-B	RESIDENTIAL SUITE	189	124	38	75	227	38	124	R32	0.55 kg	4396	3.49	35.0	23.9	26.7	19.4	293	193	2550	2.42	-10.6	-12.8	21.1	15.6	1260	208/1/60	15.3	26	35	0.35	2500	NO	23.2	38.8	48.2	50.5	50.0	46.1	41.8	32.4	60.7	1152	1033	280	106	
ASHP-214	INSPIRONAIR	IAIRHP-35-CC-ERV-B	RESIDENTIAL SUITE	189	124	38	75	227	38	124	R32	0.55 kg	4396	3.49	35.0	23.9	26.7	19.4	293	193	2550	2.42	-10.6	-12.8	21.1	15.6	1260	208/1/60	15.3	26	35	0.35	2500	NO	23.2	38.8	48.2	50.5	50.0	46.1	41.8	32.4	60.7	1152	1033	280	106	
ASHP-215	INSPIRONAIR	IAIRHP-35-CC-ERV-B	RESIDENTIAL SUITE	189	124	38	75	227	38	124	R32	0.55 kg	4396	3.49	35.0	23.9	26.7	19.4	293	193	2550	2.42	-10.6	-12.8	21.1	15.6	1260	208/1/60	15.3	26	35	0.35	2500	NO	23.2	38.8	48.2	50.5	50.0	46.1	41.8	32.4	60.7	1152	1033	280	106	
ASHP-216	INSPIRONAIR	IAIRHP-35-CC-ERV-B	RESIDENTIAL SUITE	189	124	38	75	227	38	124	R32	0.55 kg	4396	3.49	35.0	23.9	26.7	19.4	293	193	2550	2.42	-10.6	-12.8	21.1	15.6	1260	208/1/60	15.3	26	35	0.35	2500	NO	23.2	38.8	48.2	50.5	50.0	46.1	41.8	32.4	60.7	1152	1033	280	106	
ASHP-303	INSPIRONAIR	IAIRHP-35-CC-ERV-B	RESIDENTIAL SUITE	189	124	38	75	227	38	124	R32	0.55 kg	4396	3.49	35.0	23.9	26.7	19.4	293	193	2550	2.42	-10.6	-12.8	21.1	15.6	1260	208/1/60	15.3	26	35	0.35	2500	NO	23.2	38.8	48.2	50.5	50.0	46.1	41.8	32.4	60.7	1152	1033	280	106	
ASHP-306	INSPIRONAIR	IAIRHP-35-CC-ERV-B	RESIDENTIAL SUITE	189	124	38	75	227	38	124	R32	0.55 kg	4396	3.49	35.0	23.9	26.7	19.4	293	193	2550	2.42	-10.6	-12.8	21.1	15.6	1260	208/1/60	15.3	26	35	0.35	2500	NO	23.2	38.8	48.2	50.5	50.0	46.1	41.8	32.4	60.7	1152	1033	280	106	MIRRORED CONFIGURATION
ASHP-307	INSPIRONAIR	IAIRHP-35-CC-ERV-B	RESIDENTIAL SUITE	189	124	38	75	227	38	124	R32	0.55 kg	4396	3.49	35.0	23.9	26.7	19.4	293	193	2550	2.42	-10.6	-12.8	21.1	15.6	1260	208/1/60	15.3	26	35	0.35	2500	NO	23.2	38.8	48.2	50.5	50.0	46.1	41.8	32.4	60.7	1152	1033	280	106	
ASHP-309	INSPIRONAIR	IAIRHP-35-CC-ERV-B	RESIDENTIAL SUITE	189	124	38	75	227	38	124	R32	0.55 kg	4396	3.49	35.0	23.9	26.7	19.4	293	193	2550	2.42	-10.6	-12.8	21.1	15.6	1260	208/1/60	15.3	26	35	0.35	2500	NO	23.2	38.8	48.2	50.5	50.0	46.1	41.8	32.4	60.7	1152	1033	280	106	
ASHP-310	INSPIRONAIR	IAIRHP-35-CC-ERV-B	RESIDENTIAL SUITE	189	124	38	75	227	38	124	R32	0.55 kg	4396	3.49	35.0	23.9	26.7	19.4	293	193	2550	2.42	-10.6	-12.8	21.1	15.6	1260	208/1/60	15.3	26	35	0.35	2500	NO	23.2	38.8	48.2	50.5	50.0	46.1	41.8	32.4	60.7	1152	1033	280	106	
ASHP-311	INSPIRONAIR	IAIRHP-35-CC-ERV-B	RESIDENTIAL SUITE	189	124	38	75	227	38	124	R32	0.55 kg	4396	3.49	35.0	23.9	26.7	19.4	293	193	2550	2.42	-10.6	-12.8	21.1	15.6	1260	208/1/60	15.3	26	35	0.35	2500	NO	23.2	38.8	48.2	50.5	50.0	46.1	41.8	32.4	60.7	1152	1033	280	106	
ASHP-312	INSPIRONAIR	IAIRHP-35-CC-ERV-B	RESIDENTIAL SUITE	189	124	38	75	227	38	124	R32	0.55 kg	4396	3.49	35.0	23.9	26.7	19.4	293	193	2550	2.42	-10.6	-12.8	21.1	15.6	1260	208/1/60	15.3	26	35	0.35	2500	NO	23.2	38.8	48.2	50.5	50.0	46.1	41.8	32.4	60.7	1152	1033	280	106	
ASHP-313	INSPIRONAIR	IAIRHP-35-CC-ERV-B	RESIDENTIAL SUITE	189	124	38	75	227	38	124	R32	0.55 kg	4396	3.49	35.0	23.9	26.7	19.4	293	193	2550	2.42	-10.6	-12.8	21.1	15.6	1260	208/1/60	15.3	26	35	0.35	2500	NO	23.2	38.8	48.2	50.5	50.0	46.1	41.8	32.4	60.7	1152	1033	280	106	
ASHP-314	INSPIRONAIR	IAIRHP-35-CC-ERV-B	RESIDENTIAL SUITE	189	124	38	75	227	38	124	R32	0.55 kg	4396	3.49	35.0	23.9	26.7	19.4	293	193	2550	2.42	-10.6	-12.8	21.1	15.6	1260	208/1/60	15.3	26	35	0.35	2500	NO	23.2	38.8	48.2	50.5	50.0	46.1	41.8	32.4	60.7	1152	1033	280	106	
ASHP-315	INSPIRONAIR	IAIRHP-35-CC-ERV-B	RESIDENTIAL SUITE	189	124	38	75	227	38	124	R32	0.55 kg	4396	3.49	35.0	23.9	26.7	19.4	293	193	2550	2.42	-10.6	-12.8	21.1	15.6	1260	208/1/60	15.3	26	35	0.35	2500	NO	23.2	38.8	48.2	50.5	50.0	46.1	41.8	32.4	60.7	1152	1033	280	106	MIRRORED CONFIGURATION
ASHP-316	INSPIRONAIR	IAIRHP-35-CC-ERV-B	RESIDENTIAL SUITE	189	124	38	75	227	38	124	R32	0.55 kg	4396	3.49	35.0	23.9	26.7	19.4	293	193	2550	2.42	-10.6	-12.8	21.1	15.6	1260	208/1/60	15.3	26	35	0.35	2500	NO	23.2	38.8	48.2	50.5	50.0	46.1	41.8	32.4	60.7	1152	1033	280	106	

FANS

NOTES:
1. ALL FANS TO BE PROVIDED WITH DISCONNECT
2. ALL FANS TO BE PROVIDED WITH STARTER IF VFD IS NOT REQUIRED

TAG	MANUFACTURER	MODEL	TYPE	SERVICE	LOCATION	AIRFLOW (L/s)	E.S.P. (Pa)	FAN RPM	MOTOR RPM	SOUND POWER AT OUTLET (dB)										ELECTRICAL									WEIGHT (kg)	REMARKS
										63 HZ	125 HZ	250 HZ	500 HZ	1000 HZ	2000 HZ	4000 HZ	8000 HZ	LwA	dBA	MOTOR POWER (HP)	FLA	MCA	MOCP	VOLT. (V)	PH. (ϕ)	FREQ. (Hz)	VFD	EMERGENCY POWER		
EF-1	GREENHECK	CSP-A390-VG	DIRECT DRIVE CABINET FAN	ELEC & WATER RM EXHAUST	BASEMENT - ELEC RM	75	150	1335	1800												0.9	1.1	15	208	1	60	NO	NO	10.9	SOUND: 2.5 SONES AT INLET
EF-2	GREENHECK	SQ-9-M1-VG	DIRECT DRIVE INLINE FAN	WASTE ROOM EXHAUST	L1 - WASTE RM	400	150	1598	1800	81	69	63	65	67	61	50	44	70	58	0.25	2.1	2.6	15	208	1	60	NO	NO	23.6	
EF-3	GREENHECK	SP-A390-VG	DIRECT DRIVE CEILING EXHAUST	HOUSEKEEPING EXHAUST	202 - HOUSEKEEPING	40	150	1295	1295												0.9	1.1	15	208	1	60	NO	NO	11	SOUND: 4.0 SONES AT INLET, LIGHT SWITCH ON/OFF CONTROL
EF-4	GREENHECK	SP-A390-VG	DIRECT DRIVE CEILING EXHAUST	STAFF ROOM EXHAUST	301-STAFF RM	40	150	1295	1295												0.9	1.1	15	208	1	60	NO	NO	11	SOUND: 4.0 SONES AT INLET, LIGHT SWITCH ON/OFF CONTROL
EF-5	GREENHECK	CSP-A390-VG	DIRECT DRIVE CABINET FAN	MECHANICAL ROOMS EXHAUST	ATTIC	150	150	1470	1800												0.9	1.1	15	208	1	60	NO	NO	10.9	SOUND: 3.2 SONES AT INLET
EF-6	GREENHECK	CSP-A390-VG	DIRECT DRIVE CABINET FAN	ATTIC EXHAUST	ATTIC	150	150	1470	1800												0.9	1.1	15	208	1	60	NO	NO	10.9	SOUND:3.2 SONES AT INLET
EF-7	GREENHECK	SP-A390-VG	DIRECT DRIVE CEILING EXHAUST	ELEV CTRL EXHAUST	201 - ELEV CTRL	40	150	1295	1295												0.9	1.1	15	208	1	60	NO	NO	11	SOUND: 4.0 SONES AT INLET
SF-1	GREENHECK	CSP-A390-VG	DIRECT DRIVE CABINET FAN	ELEC & WATER RM FRESH AIR	BASEMENT - ELEC RM	60	150	1333	1800												0.9	1.1	15	208	1	60	NO	NO	10.9	SOUND:2.4 SONES AT INLET
SF-2	GREENHECK	CSP-A700-VG	DIRECT DRIVE CABINET FAN	LAUNDRY MAKE UP AIR	ATTIC	240	150	1232	1800												2.4	3.0	15	208	1	60	YES	NO	17.7	SOUND: 2.0 SONES

GRILLES, DIFFUSERS, AND LOUVERS

TAG	MANUFACTURER	MODEL	DESCRIPTION	VOLUME CONTROL	SIZE (mm x mm)	FINISH	MATERIAL	REMARKS
A	EH PRICE	ASCD	SQUARE CONE DIFFUSER	YES	600x600	-	ALUMINUM	
A1	EH PRICE	ASCD	SQUARE CONE DIFFUSER	YES	300x300	-	ALUMINUM	
B	EH PRICE	80	EGG CRATE RETURN GRILLE	YES	AS SHOWN	-	ALUMINUM	
B1	EH PRICE	80	EGG CRATE RETURN GRILLE	YES	150x150	-	ALUMINUM	
B2	EH PRICE	80	EGG CRATE RETURN GRILLE	YES	300x300	-	ALUMINUM	
BF	EH PRICE	80-FR	EGG CRATE RETURN GRILLE C/W FIRE DAMPER	YES	AS SHOWN	-	ALUMINUM	C/W INTEGRATED FIRE DAMPER
C	EH PRICE	ARCD	ROUND CONE DIFFUSER	YES	AS SHOWN	-	ALUMINUM	
D	EH PRICE	630	LOUVERED RETURN/EXHAUST GRILLE	YES	AS SHOWN	-	ALUMINUM	
DR	EH PRICE	630FR	FIRE RATED LOUVERED RETURN/EXHAUST GRILLE	NO	AS SHOWN	-	ALUMINUM	
E	EH PRICE	820	LOUVERED DOUBLE DEFLECTION SUPPLY DIFFUSER	YES	AS SHOWN	-	ALUMINUM	
E1	EH PRICE	640	LOUVERED DOUBLE DEFLECTION RESIDENTIAL SUPPLY	YES	AS SHOWN	-	ALUMINUM	
M	EH PRICE	DE635	DRAINABLE FIXED BLADE EXTERNAL LOUVER	NO	AS SHOWN	-	ALUMINUM	

NOTES:
1. ALL DIFFUSER AND GRILLE FRAMES SHALL SUIT CEILING CONSTRUCTION.
2. ALL DAMPERS SHALL BE OF THE SAME MATERIAL AS THE DIFFUSER OR GRILLE.
3. REFER TO ARCHITECTURAL DRAWINGS FOR CEILING CONSTRUCTION.
4. DIFFUSERS IN DRYWALL CEILING TO BE C/W ADAPTOR FRAME TO ACCOMODATE INSTALLATION. DIFFUSER ADAPTOR FRAME SHALL BE OF THE SAME MATERIAL AS THE DIFFUSER.
5. GRILLE AND DIFFUSER COLOUR TO BE B12 WHITE UNLESS OTHERWISE SPECIFIED BY THE ARCHITECT
6. NECK SIZE AS SHOWN ON PLANS.

No.	DATE	DESCRIPTION
10	2026-10-30	MECHANICAL ADDENDUM 01
9	2026-09-16	ISSUED FOR TENDER
8	2026-03-24	ISSUED FOR 100% CD
7	2026-02-20	ISSUED FOR 90% CD AND CLASS A COSTING
6	2026-01-30	ISSUED FOR 75%CD
5	2026-01-30	ISSUED FOR SITE SERVICES PERMIT
4	2026-01-30	ISSUED FOR BUILDING PERMIT
3	2025-12-17	ISSUED FOR 30%CD
2	2025-12-01	RE-ISSUED FOR SPA
1	2025-06-27	ISSUED FOR SPA

REVISIONS

THE SPECIFICATIONS ARE TO BE CONSIDERED AS AN INTEGRAL PART OF THESE DRAWINGS AND NEITHER THE DRAWINGS NOR THE SPECIFICATIONS SHALL BE USED ALONE. REFER TO ARCHITECTURAL DRAWINGS FOR DIMENSIONS. DO NOT SCALE.

PROJECT NORTH

QUASAR

250 ROWNTREE DAIRY RD, WOODBRIDGE, ON
TEL: 905-507-0800
WEB: WWW.QUASARCG.COM

PROJECT

DURHAM REGION
409 CENTRE ST S SENIORS HOUSING
409 CENTRE ST. SOUTH, WHITBY, ON L1N 4W3
TOWN FILE NUMBER: SP-26-25

TITLE

MECHANICAL SCHEDULES I

CHECKED	PROJECT No.
SA	RE-24-162
SCALE	DRAWING No.
N.T.S.	M-003

VRF & SPLIT FAN COIL UNITS

NOTES:
1. COOLING CAPACITIES AT NOMINAL CONDITIONS (INDOOR // OUTDOOR) 26.7°C DB, 19.4°C WB // 35°C DB.
2. HEATING CAPACITIES AT NOMINAL CONDITIONS (INDOOR // OUTDOOR) 21.1°C DB // 8.3°C DB, 6.1°C WB.
3. SOUND PRESSURE MEASURED IN ANECHOIC ROOM.
4. PROVIDE WITH THERMOSTAT MODEL PAR-42MAACAB.

TAG	ASSOCIATED CONDENSING UNIT	MANUFACTURER	MODEL	TYPE	SERVICE	REFRIGERANT	AIRFLOW (L/s)			TOTAL COOLING CAPACITY (kW)	HEATING CAPACITY (kW)	SOUND (DBA)	ELECTRICAL					WEIGHT (kg)	NOTES
							HIGH	MEDIUM	LOW			PRESSURE (H/M/L)	FLA	VOLT. (V)	PH. (ϕ)	FREQ. (Hz)	EMERGENCY POWER		
IDU-01	CU-1	MITSUBISHI	PKFY-M24NKMU	WALL-MOUNTED DUCTLESS	B - ELECTRICAL & COMMS	R454B	433		267	7.00	7.90	49/39	0.50	208	1	60	NO	21	1, 2, 3
IDU-02	CU-1	MITSUBISHI	PKFY-M24NKMU	WALL-MOUNTED DUCTLESS	B - ELECTRICAL & COMMS	R454B	433		267	7.00	7.90	49/39	0.50	208	1	60	NO	21	1, 2, 3
IDU-03	CU-2	MITSUBISHI	PKFY-M04NLMU	WALL-MOUNTED DUCTLESS	B - INCOMING WATER RM	R32	70	61	55	1.10	1.30	28/25/22	0.20	208	1	60	NO	11	1, 2, 3
IDU-04	CU-2	MITSUBISHI	PKFY-M24NKMU	WALL-MOUNTED DUCTLESS	L1 - WASTE	R32	433		267	7.00	7.90	49/39	0.50	208	1	60	NO	21	1, 2, 3
IDU-05	CU-2	MITSUBISHI	PKFY-M24NKMU	WALL-MOUNTED DUCTLESS	L1 - WASTE	R32	433		267	7.00	7.90	49/39	0.50	208	1	60	NO	21	1, 2, 3
IDU-06	CU-2	MITSUBISHI	PLFY-EM06NEMU	4-WAY CASSETTE	L1 - OFFICE	R32	233	208	142	1.80	2.00	27/24/19	0.19	208	1	60	NO	5	1, 2, 3
IDU-07	CU-2	MITSUBISHI	PKFY-M04NLMU	WALL-MOUNTED DUCTLESS	L2 - HOUSEKEEPING	R32	70	61	55	1.10	1.30	28/25/22	0.20	208	1	60	NO	11	1, 2, 3
IDU-08	CU-3	MITSUBISHI	PKFY-M08NLMU	WALL-MOUNTED DUCTLESS	L2 - ELEV. CTRL	R454B	112	83	67	2.30	2.60	35/29/22	0.30	208	1	60	NO	11	1, 2, 3
IDU-09	CU-4	MITSUBISHI	PKFY-M12NLMU	WALL-MOUNTED DUCTLESS	L2 - MECH RM	R454B	140	103	72	3.50	4.00	41/34/24	0.35	208	1	60	NO	13	1, 2, 3
IDU-10	CU-2	MITSUBISHI	PKFY-M04NLMU	WALL-MOUNTED DUCTLESS	L3 - STAFF RM	R32	70	61	55	1.10	1.30	28/25/22	0.20	208	1	60	NO	11	1, 2, 3
IDU-11	CU-4	MITSUBISHI	PKFY-M12NLMU	WALL-MOUNTED DUCTLESS	L3 - MECH RM	R454B	140	103	72	3.50	4.00	41/34/24	0.35	208	1	60	NO	13	1, 2, 3

VRF & SPLIT CONDENSING UNITS

NOTES:
1. AIR-SOURCE COOLING CAPACITIES AT NOMINAL CONDITIONS (INDOOR // OUTDOOR) 26.7°C DB, 19.4°C WB // 35°C DB.
2. AIR-SOURCE HEATING CAPACITIES AT NOMINAL CONDITIONS (INDOOR // OUTDOOR) 21.1°C DB // 8.3°C DB, 6.1°C WB.
3. AIR-SOURCE CONDENSING UNIT WITH HEATING CAPABILITY DOWN TO -30°C.
4. AIR-SOURCE COOLING CAPABILITY CAPABILITY AT LOW AMBIENT TEMPERATURES LESS THAN -30°C.
5. SOUND PRESSURE MEASURED IN ANECHOIC ROOM.
6. SOUND POWER MEASURED IN ANECHOIC ROOM.
7. C/W BRANCH SELECTOR BOX MODEL CMB-M108NU-MA1-SV, WITH 208V/1PH/60Hz, 0.13 MCA, 15 MOCP POWER CONNECTION. MECHANICAL CONTRACTOR TO ENSURE POWER IS PROVIDED TO BRANCH SELECTOR BOX.

TAG	MANUFACTURER	MODEL	LOCATION	REFRIGERANT TYPE	COOLING NOMINAL CAPACITY (kW) TOTAL	HEATING NOMINAL CAPACITY (kW)	SOUND PRESSURE/POWER (dBA)	ELECTRICAL MCA	MOCP	VOLT. (V)	PH. (ϕ)	FREQ. (Hz)	EMERGENCY POWER	WEIGHT (kg)	REMARKS
CU-1	MITSUBISHI	PUMY-L48NKMU	ROOF	R454B	14.07	14.65	54	36	40	208	1	60	NO	123	1, 2, 4, 5
CU-2	MITSUBISHI	PURY-HM72TXU-A	ROOF	R-32	21.10	23.45	77	55	90	208	3	60	NO	273	1, 2, 3, 6, 7
CU-3	MITSUBISHI	PUY-AK12NL-U1	ROOF	R454B	3.52		44	16	20	208	1	60	NO	50	1, 2, 4, 5
CU-4	MITSUBISHI	PUMY-L36NKMU	ROOF	R454B	10.55	12.02	53	36	40	208	1	60	NO	123	1, 2, 4, 5

(ELECTRIC HEATERS)

TAG	MANUFACTURER	MODEL	LOCATION	MOUNTING CONFIGURATION	AIRFLOW (L/s)	HEATING CAPACITY (kW)	# OF FANS	ELECTRICAL LOAD (W)	ELECTRICAL VOLT. (V)	ELECTRICAL PH. (ϕ)	ELECTRICAL FREQ. (Hz)	NOTES
EBH-1	OUELLET	OFM0508	MULTIPLE INSTANCES	ELEC. BASEBORD HEATER	0	0.50	0	500	208	1	60	
EBH-2	OUELLET	OFM1008	<varies>	ELEC. BASEBORD HEATER	0	1.00	0	1000	208	1	60	
EDH-1	GREENHECK	EDHE	B1 - ELEC RM	ELEC. DUCT HEATER	60	2.00	0	2000	208	3	60	SHALL MAINTAIN 18°C SUPPLY AIR SETPOINT
EDH-2	GREENHECK	EDHE	L1 - LAUNDRY	ELEC. DUCT HEATER	240	8.00	0	8000	208	3	60	SHALL MAINTAIN 18°C SUPPLY AIR SETPOINT
EFH-1	OUELLET	OAWH02008	MULTIPLE INSTANCES	ELEC. FORCED FLOW HEATER	0	2.00	1	2000	208	1	60	
FFH-1	OUELLET	OAC02008-T	MULTIPLE INSTANCES	ELEC. FORCED FLOW HEATER	76	2.00	1	2000	208	1	60	
FFH-2	OUELLET	OACP2008	L1 - CORRIDOR	ELEC. FORCED FLOW HEATER	76	2.00	1	2000	208	1	60	

PLUMBING DRAINS

UNIT TAG	USE	NOTES
FD-1	FLOOR DRAIN, FINISHED AREAS	NICKEL BRONZE STRAINER (SHAPE TO MATCH FLOOR), VANDAL RESISTANT, PRIMER TAPPING
FD-2	FLOOR DRAINS, UNFINISHED AREAS	HEAVY DUTY DUCTILE IRON STRAINER, PRIMER TAPPING
FD-3	FLOOR DRAIN, PEDIMAT	ENTRANCE MAT, 4-1/4" DIAMETER STAINLESS STEEL STRAINER, CAST BRASS DRAIN BODY, THREADED OUTLET
FFD-2	FUNNEL FLOOR DRAIN, UNFINISHED AREAS	CAST IRON FUNNEL (STRAINER SHAPE TO MATCH FLOOR), PRIMER TAPPING
FFD-3	FUNNEL FLOOR DRAIN, WATER ROOMS	6"DIA. OUTLET, OPEN THROAT CAST IRON FUNNEL, PRIMER TAPPING
RD-1	ROOF DRAIN, STEEL DECK	SUMP RECEIVER, UNDERLOCK CLAMP, BODY EXTENSION
RD-2	ROOF DRAIN, CAST DECK	UNDERDECK CLAMP, BODY EXTENSION
RD-3	ROOF DRAIN, MENTAL HEALTH	SUMP RECEIVER, UNDERDECK CLAMP, BODY EXTENSION
RD-4	ROOF DRAIN, EXTERIOR GRADE PLYWOOD	8-3/8" LATERAL ROOF DRAIN, UNDERDECK CLAMP
RD-5	ROOF DRAIN, GREEN ROOF	
SD-1	SCUPPER DRAIN, ELEVATOR PIT	REFER TO PLUMBING FIXTURES SPECIFICATIONS SECTION 22 42 00

PLUMBING PUMPS

NOTES:
1. ALL PUMPS WITHOUT A VFD OR ECM MOTOR MUST BE PROVIDED WITH A STARTER.
2. PROVIDE PUMP WITH ECM MOTOR FOR VARIABLE SPEED OPERATION.
3. CONSTANT SPEED PUMP
4. SET OF SUMP PUMPS TO BE PROVIDED WITH RAIL SYSTEMS AND FLOATS WITH NEMA 4X CONTROL PANEL.
5. PACKAGED DUPLEX SYSTEM.
6. PUMPS SELECTED AS DUTY-DUTY WITH 100% REDUNDANCY TO OPERATE BOTH PUMPS AT BEST EFFICIENCY POINT DURING DESIGN FLOW. IN THE EVENT OF ONE PUMP FAILING, A SINGLE PUMP CAN MEET DESIGN FLOW.

TAG	MANUFACTURER	MODEL	LOCATION	SERVICE	TYPE	PERFORMANCE CHARACTERISTICS FLOW (L/s)	PERFORMANCE CHARACTERISTICS HEAD (kPa)	MOTOR CHARACTERISTICS VFD	MOTOR CHARACTERISTICS MOTOR POWER (HP)	MOTOR CHARACTERISTICS MOTOR RPM	ELECTRICAL VOLT. (V)	ELECTRICAL PH. (ϕ)	ELECTRICAL FREQ. (Hz)	ELECTRICAL FLA (A)	REDUNDANCY	REMARKS
DHWP-1&2	ARMSTRONG	H ECM-HS3 ECM-0.5 HP	L3 - MECH RM	DOMESTIC HOT WATER HEAT PUMP LOAD SIDE	VERTICAL INLINE	1.20	70.0	NO	0.50	1750	208	1	60		DUTY-DUTY	2, 6
DRCP-1	ARMSTRONG	E12.2B AB	L3 - MECH RM	DOMESTIC HOT WATER RECIRCULATION	VERTICAL INLINE	0.45	120.0	NO	0.40	3400	208	1	60		N/A	1, 3
SP-1 & SP-2	LIBERTY	LE102M3	B - INCOMING WATER RM	BASEMENT FLOOR DRAINS AND ELEVATOR PIT	SUMP	4.30	90.0	NO	1.00	1725	208	1	60	8.0	DUTY-STANDBY	1, 4, 5
SP-3 & SP-4	LIBERTY	LE52M	B - INCOMING WATER RM	GROUNDWATER DEWATERING	SUMP	0.80	70.0	NO	0.50	1725	208	1	60	6.8	DUTY-STANDBY	1, 4, 5

DOMESTIC HOT WATER TANKS - ELECTRIC

NOTES:
1. PROVIDE A SINGLE TEMPERATURE SENSOR ON ALL TANKS INSTALLED AT PORT CLOSEST TO BOTTOM OF TANK.

TAG	MANUFACTURER	MODEL	LOCATION	VOLUME (L)	RECOVERY RATE @ 50°C RISE (LPH)	INPUT CAPACITY (kW)	ELECTRICAL FLA	ELECTRICAL VOLT. (V)	ELECTRICAL PH. (ϕ)	ELECTRICAL FREQ. (Hz)	NOTES
EWH-1	AO SMITH	DRE-80-13.5	L2 - MECH RM	302	233	13.5	38	208	3	60	ELECTRIC HEATER SERVING DHWR
HWT-1	AO SMITH	TJV-600A	L2 - MECH RM	2271							STORAGE TANK ONLY, NO HEATING ELEMENT
HWT-2	AO SMITH	TJV-600A	L2 - MECH RM	2271							STORAGE TANK ONLY, NO HEATING ELEMENT

EXPANSION TANKS

TAG	MANUFACTURER	MODEL	LOCATION	SERVICE	TOTAL VOLUME (L)	ACCEPTANCE VOLUME (L)	WEIGHT (kg)	NOTES
ET-DCW	WATTS	DETA 42	L3 - MECH RM	DOMESTIC COLD WATER	83	55	26	
ET-DHW	WATTS	DETA 160	L3 - MECH RM	DOMESTIC HOT WATER	265	178	80	

No.	DATE	DESCRIPTION
10	2026-10-30	MECHANICAL ADDENDUM 01
9	2026-09-16	ISSUED FOR TENDER
8	2026-03-24	ISSUED FOR 100% CD
7	2026-02-20	ISSUED FOR 90% CD AND CLASS A COSTING
6	2026-01-30	ISSUED FOR 75%CD
5	2026-01-30	ISSUED FOR SITE SERVICES PERMIT
4	2026-01-30	ISSUED FOR BUILDING PERMIT
3	2025-12-17	ISSUED FOR 30%CD
2	2025-12-01	RE-ISSUED FOR SPA
1	2025-06-27	ISSUED FOR SPA

REVISIONS

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PROJECT NORTH

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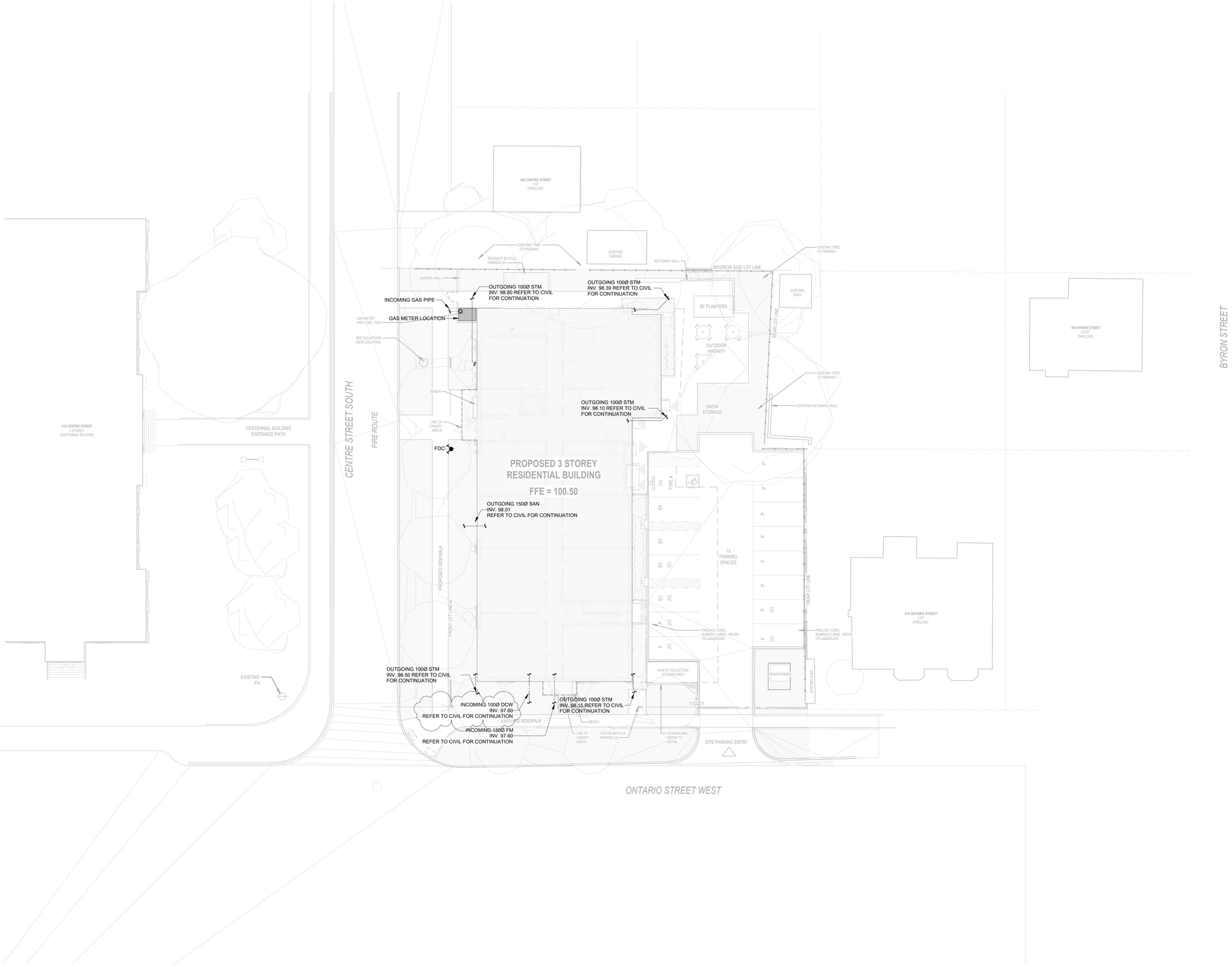
PROJECT

DURHAM REGION
409 CENTRE ST S SENIORS HOUSING
409 CENTRE ST. SOUTH, WHITBY, ON L1N 4W3
TOWN FILE NUMBER: SP-26-25

TITLE

MECHANICAL SCHEDULES II

CHECKED	PROJECT No.
SA	RE-24-162
SCALE	DRAWING No.
N.T.S.	M-004



No.	DATE	DESCRIPTION
9	2026-10-30	MECHANICAL ADDENDUM 01
8	2026-09-16	ISSUED FOR TENDER
7	2026-03-24	ISSUED FOR 100% CD
6	2026-02-20	ISSUED FOR 90% CD AND CLASS A COSTING
5	2026-01-30	ISSUED FOR 75%CD
4	2026-01-30	ISSUED FOR SITE SERVICES PERMIT
3	2026-01-30	ISSUED FOR BUILDING PERMIT
2	2025-12-17	ISSUED FOR 30%CD
1	2025-12-01	RE-ISSUED FOR SPA

REVISIONS

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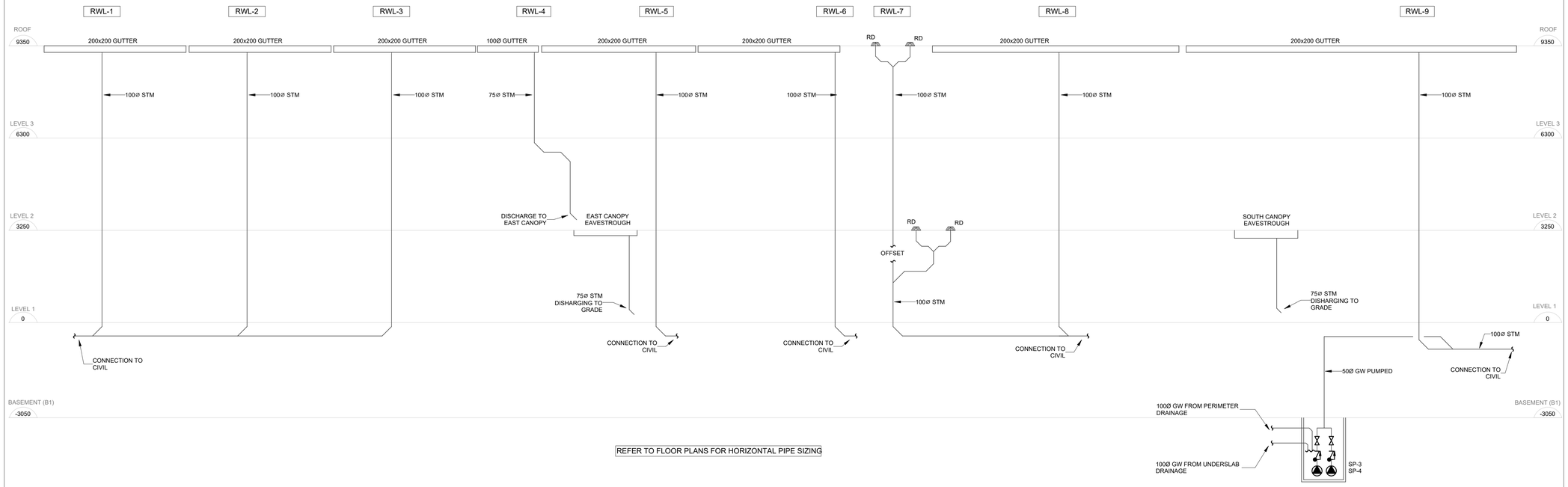
PROJECT

DURHAM REGION
409 CENTRE ST S SENIORS HOUSING
409 CENTRE ST. SOUTH, WHITBY, ON L1N 4W3
TOWN FILE NUMBER: SP-26-25

TITLE

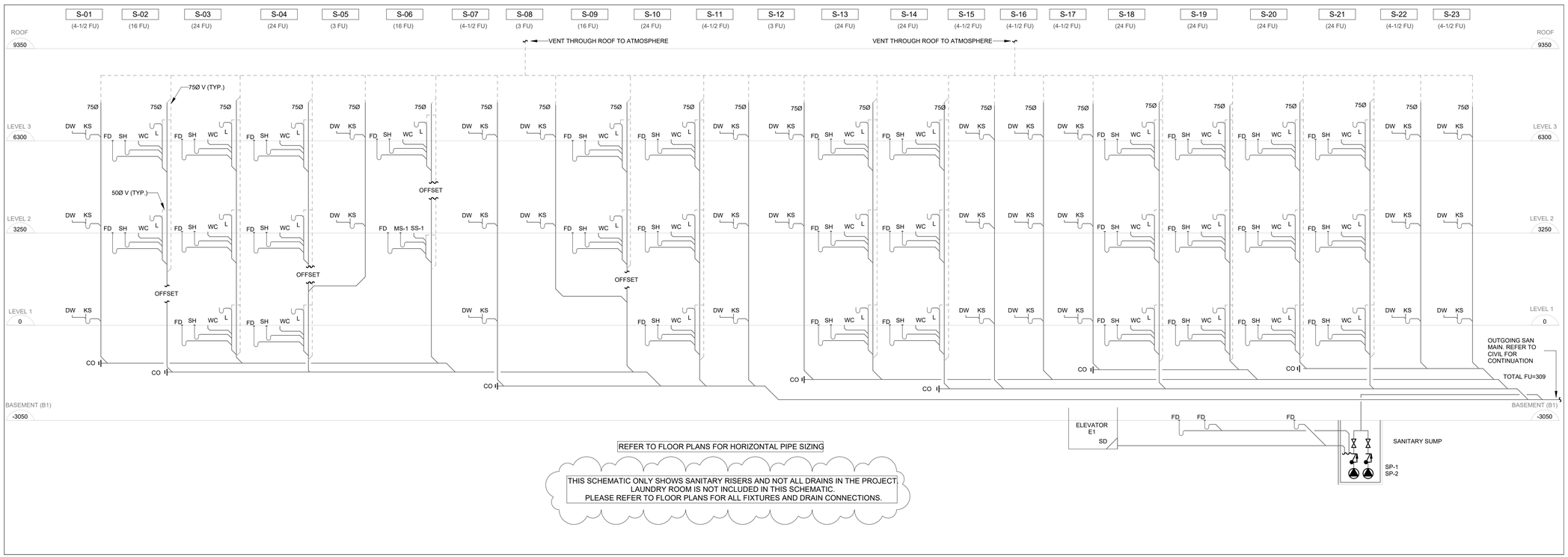
MECHANICAL SITE PLAN

CHECKED	PROJECT No.
SA	RE-24-162
SCALE	DRAWING No.
1 : 200	M-100



2 STORM SCHEMATIC

SCALE:N.T.S.



1 SANITARY SCHEMATIC

SCALE:N.T.S.

No.	DATE	DESCRIPTION
9	2026-10-30	MECHANICAL ADDENDUM 01
8	2026-09-16	ISSUED FOR TENDER
7	2026-03-24	ISSUED FOR 100% CD
6	2026-02-20	ISSUED FOR 90% CD AND CLASS A COSTING
5	2026-01-30	ISSUED FOR 75%CD
4	2026-01-30	ISSUED FOR SITE SERVICES PERMIT
3	2026-01-30	ISSUED FOR BUILDING PERMIT
2	2025-12-17	ISSUED FOR 30%CD
1	2025-12-01	RE-ISSUED FOR SPA

REVISIONS

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PROJECT NORTH

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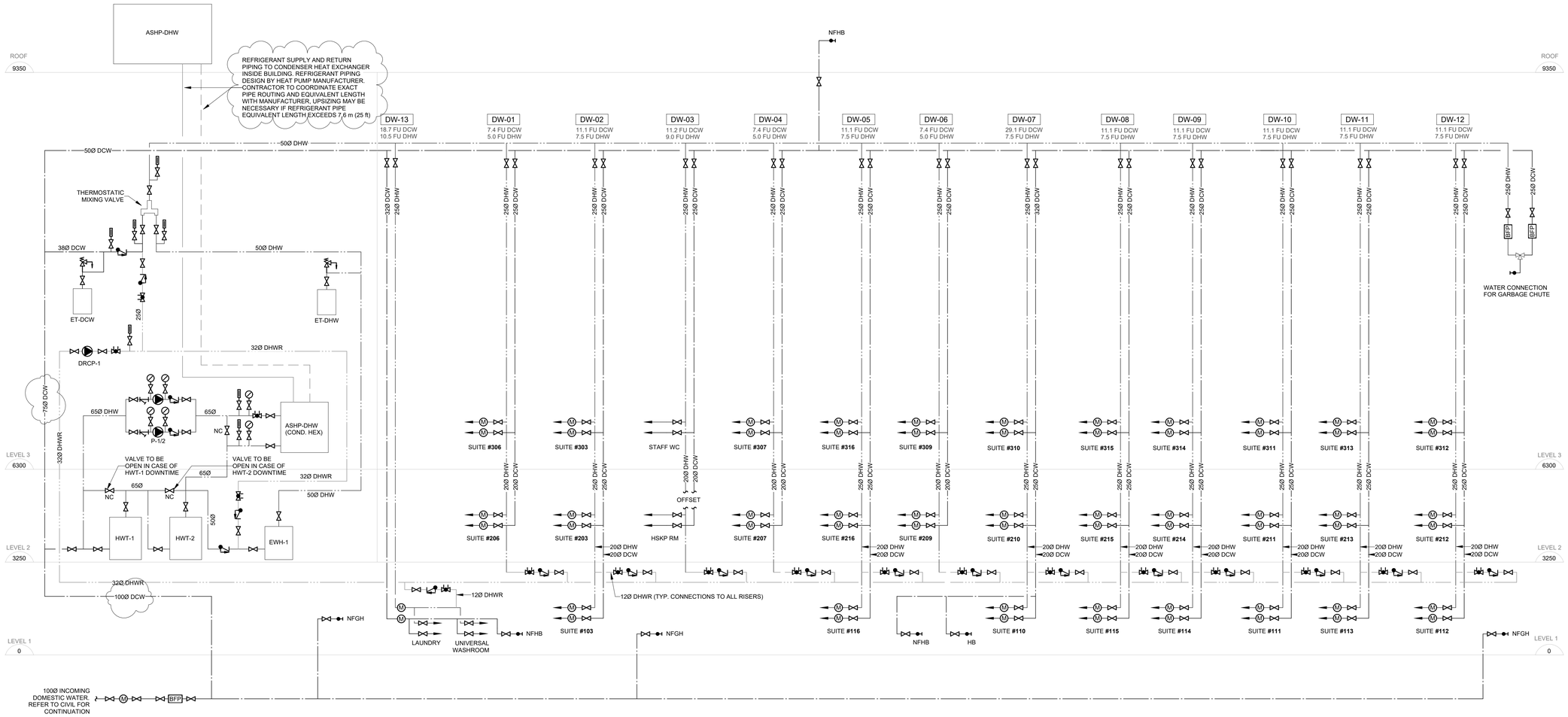
PROJECT

DURHAM REGION
409 CENTRE ST S SENIORS HOUSING
409 CENTRE ST. SOUTH, WHITBY, ON L1N 4W3
TOWN FILE NUMBER: SP-26-25

TITLE

SANITARY & STORM SCHEMATICS

CHECKED	PROJECT No.
SA	RE-24-162
SCALE	DRAWING No.
N.T.S.	M-120



No.	DATE	DESCRIPTION
8	2026-10-30	MECHANICAL ADDENDUM 01
7	2026-09-16	ISSUED FOR TENDER
6	2026-03-24	ISSUED FOR 100% CD
5	2026-02-20	ISSUED FOR 90% CD AND CLASS A COSTING
4	2026-01-30	ISSUED FOR 75%CD
3	2026-01-30	ISSUED FOR SITE SERVICES PERMIT
2	2026-01-30	ISSUED FOR BUILDING PERMIT
1	2025-12-01	RE-ISSUED FOR SPA

REVISIONS

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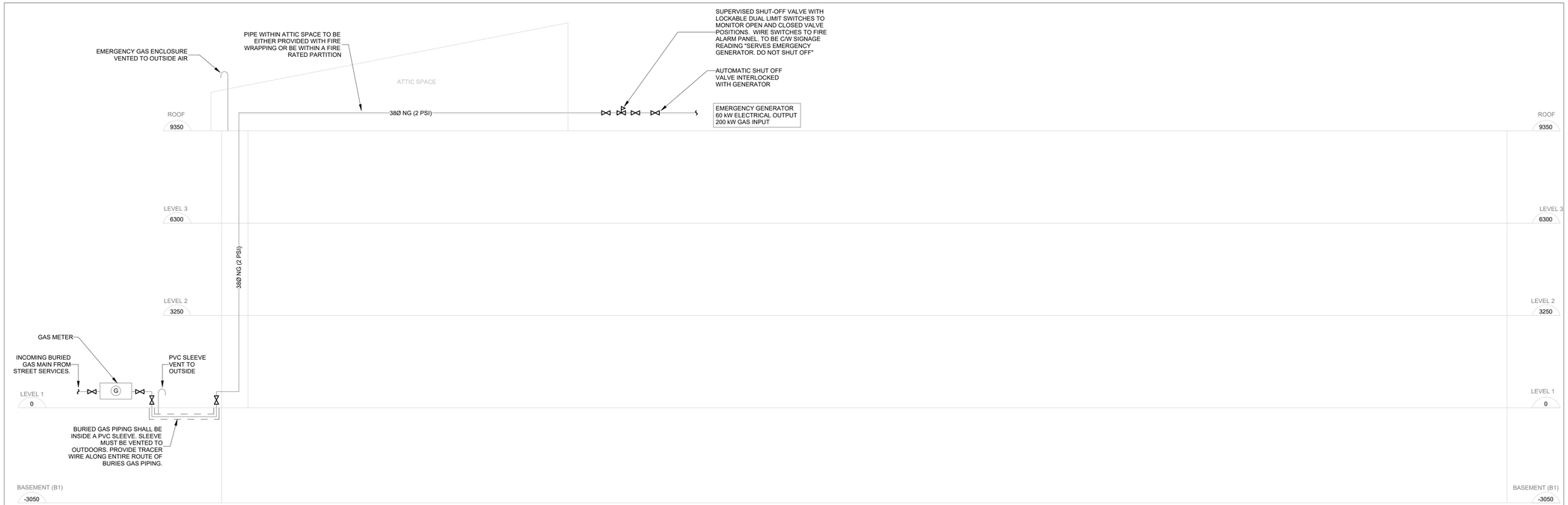
PROJECT

DURHAM REGION
409 CENTRE ST S SENIORS HOUSING
409 CENTRE ST. SOUTH, WHITBY, ON L1N 4W3
TOWN FILE NUMBER: SP-26-25

TITLE

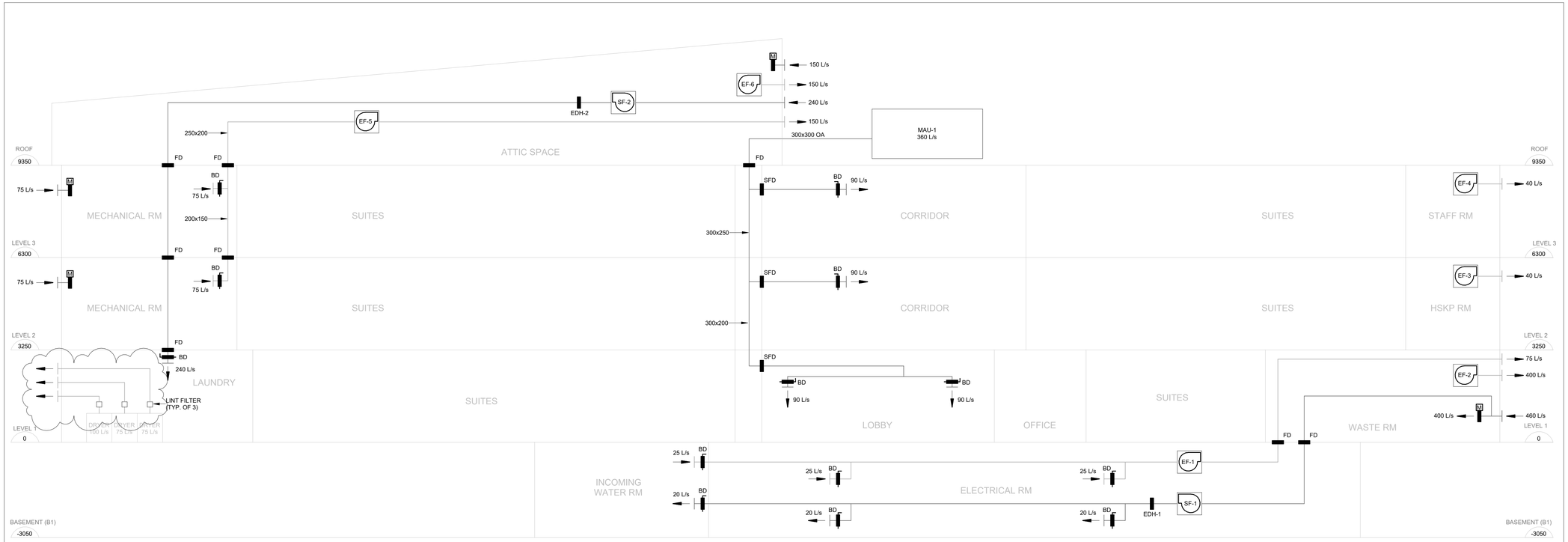
DOMESTIC WATER SCHEMATIC

CHECKED	PROJECT No.
SA	RE-24-162
SCALE	DRAWING No.
N.T.S.	M-121



2 NATURAL GAS SCHEMATIC

SCALE:N.T.S.



1 VENTILATION SCHEMATIC

SCALE:N.T.S.

No.	DATE	DESCRIPTION
8	2026-10-30	MECHANICAL ADDENDUM 01
7	2026-09-16	ISSUED FOR TENDER
6	2026-03-24	ISSUED FOR 100% CD
5	2026-02-20	ISSUED FOR 90% CD AND CLASS A COSTING
4	2026-01-30	ISSUED FOR 75%CD
3	2026-01-30	ISSUED FOR SITE SERVICES PERMIT
2	2026-01-30	ISSUED FOR BUILDING PERMIT
1	2025-12-01	RE-ISSUED FOR SPA

REVISIONS

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PROJECT

DURHAM REGION
409 CENTRE ST S SENIORS HOUSING
409 CENTRE ST. SOUTH, WHITBY, ON L1N 4W3
TOWN FILE NUMBER: SP-26-25

TITLE

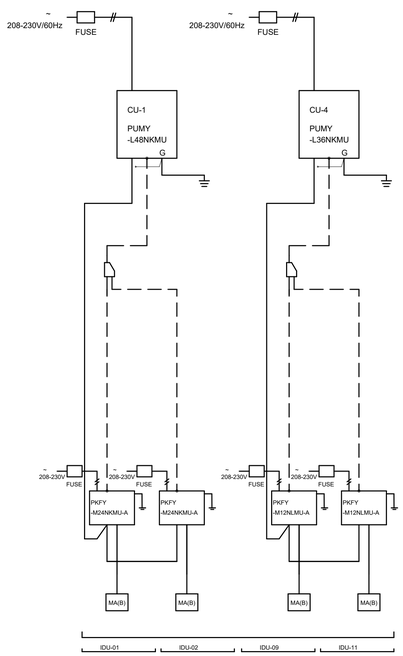
VENTILATION & NATURAL GAS SCHEMATIC

CHECKED	PROJECT No.
SA	RE-24-162
SCALE	DRAWING No.
N.T.S.	M-130

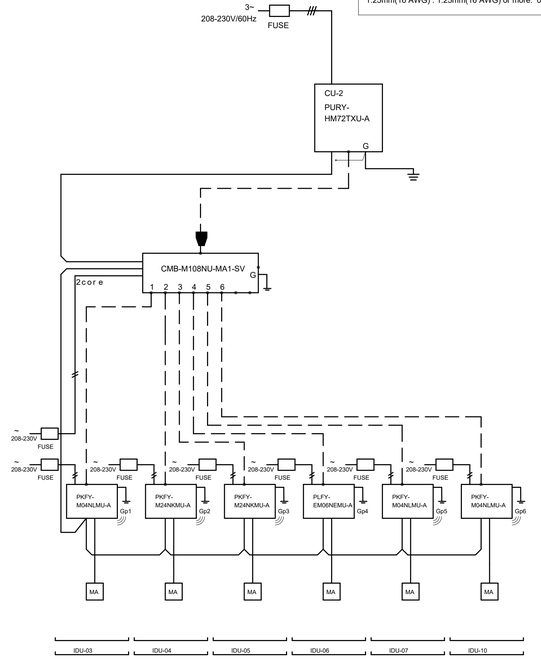
3

DIAGRAM SYMBOL LEGEND	
DISPLAY	DESCRIPTION
⫽ ⫽⫽	POWER WIRE
———	CONTROL WIRE
- - - - -	REF. PIPE / WATER PIPE
⫽⫽	POWER SIGNAL WIRE

Appropriate Circuit Protection Device in accordance with local government regulations are mandatory required such as GFI(Inverter type) and WB etc.
Please refer the amount of pre-charge and the formula of calculation which is mentioned on the data book.
1.25mm^2(16 AWG) : 1.25mm^2(16 AWG) or more. 1.25mm^2(16 AWG) : 1.25mm^2(16 AWG) or more.
HVRF pipe size is dependent on pipe length, please confirm before implementation.



This drawing is schematic in nature. Final routing of piping & wiring shall be determined by the installing contractor and/or designer of record
Additional refrigerant charge is needed depending on the size and length of extended piping.
Please refer the amount of pre-charge and the formula of calculation which is mentioned on the data book.
1.25mm^2(16 AWG) : 1.25mm^2(16 AWG) or more. 0.75mm^2(20 AWG) : between 0.5mm^2(24 AWG) and 0.75mm^2(20 AWG).



No.	DATE	DESCRIPTION
3	2026-10-30	MECHANICAL ADDENDUM 01
2	2026-09-16	ISSUED FOR TENDER
1	2026-03-24	ISSUED FOR 100% CD

REVISIONS

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PROJECT

DURHAM REGION
409 CENTRE ST S SENIORS HOUSING
409 CENTRE ST. SOUTH, WHITBY, ON L1N 4W3
TOWN FILE NUMBER: SP-26-25

TITLE

VRF SCHEMATICS

CHECKED	PROJECT No.
SA	RE-24-162
SCALE	DRAWING No.
N.T.S.	M-131

SUPERVISED VALVE SCHEDULE

TAG	SERVICE	LOCATION	COMMENTS
SV-1	INCOMING WATER LINE	INCOMING WATER ROOM	
SV-2	FIRE PROTECTION BACKFLOW INLET	INCOMING WATER ROOM	
SV-3	FIRE PROTECTION BACKFLOW OUTLET	INCOMING WATER ROOM	
SV-4	PRE-ACTION SYSTEM PA-B1-01	INCOMING WATER ROOM	
SV-5	WET SYSTEM RISER MANIFOLD	INCOMING WATER ROOM	
SV-6	BASEMENT LEVEL WET SPRINKLER WSZ-B1-01	INCOMING WATER ROOM	
SV-7	LEVEL 1 WET SPRINKLER WSZ-L1-01	STAIRWELL B	
SV-8	GARBAGE CHUTE WET SPRINKLER WSZ-L1-02 & WSZ-L3-01	WASTE ROOM	
SV-9	LEVEL 2 WET SPRINKLER WSZ-L2-01	STAIRWELL B	
SV-10	LEVEL 3 WET SPRINKLER WSZ-L3-01	STAIRWELL B	
SV-11	ATTIC DRY SPRINKLER DSZ-ATTIC-01	LEVEL 3 MECHANICAL ROOM	

NOTES:

FLOW SWITCH SCHEDULE

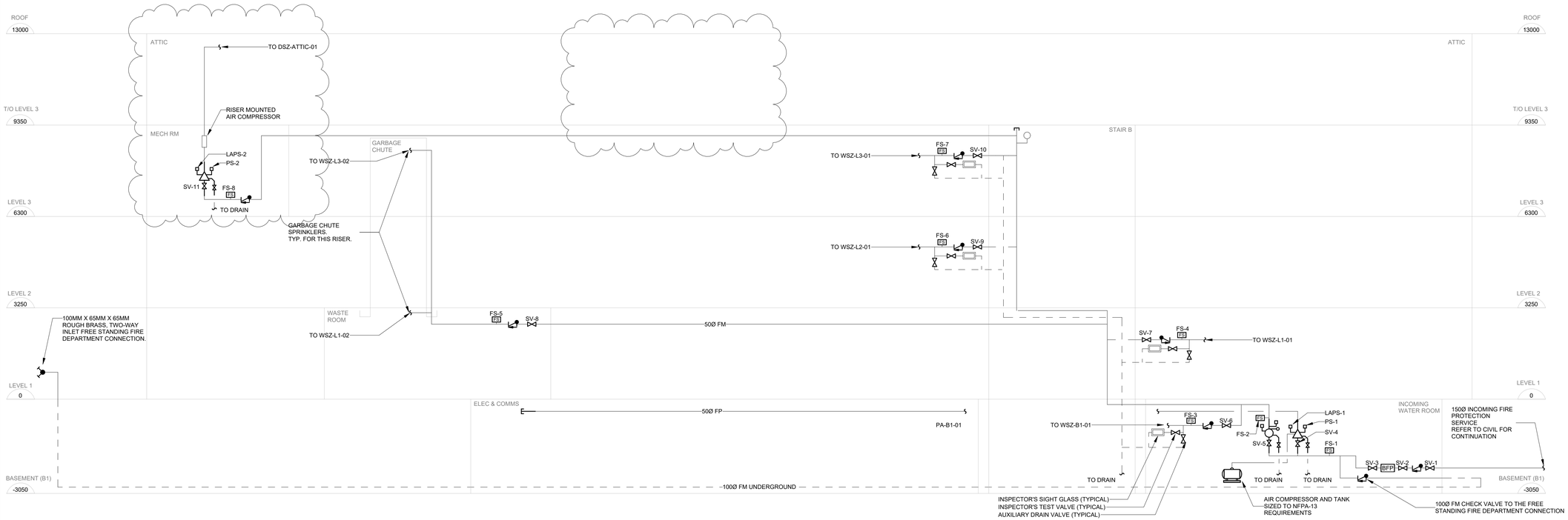
TAG	SERVICE	LOCATION	COMMENTS
FS-1	MAIN HEADER LOW FLOW SWITCH	INCOMING WATER ROOM	
FS-2	WET SYSTEM RISER MANIFOLD	INCOMING WATER ROOM	
FS-3	BASEMENT LEVEL WET SPRINKLER WSZ-B1-01	INCOMING WATER ROOM	
FS-4	LEVEL 1 WET SPRINKLER WSZ-L1-01	STAIRWELL B	
FS-5	GARBAGE CHUTE WET SPRINKLER WSZ-L1-02 & WSZ-L3-01	WASTE ROOM	
FS-6	LEVEL 2 WET SPRINKLER WSZ-L2-01	STAIRWELL B	
FS-7	LEVEL 3 WET SPRINKLER WSZ-L3-01	STAIRWELL B	
FS-8	ATTIC DRY SPRINKLER DSZ-ATTIC-01	LEVEL 3 MECHANICAL ROOM	

NOTES:

PRESSURE SWITCH SCHEDULE

TAG	SERVICE	LOCATION	COMMENTS
LAPS-1	PRE-ACTION SYSTEM PA-B1-01	INCOMING WATER ROOM	LOW AIR PRESSURE SWITCH
PS-1	PRE-ACTION SYSTEM PA-B1-01	INCOMING WATER ROOM	WATERFLOW INDICATING PRESSURE SWITCH
LAPS-2	ATTIC DRY SPRINKLER DSZ-ATTIC-01	LEVEL 3 MECHANICAL ROOM	LOW AIR PRESSURE SWITCH
PS-2	ATTIC DRY SPRINKLER DSZ-ATTIC-01	LEVEL 3 MECHANICAL ROOM	WATERFLOW INDICATING PRESSURE SWITCH

NOTES:



1 FIRE PROTECTION SCHEMATIC

SCALE:N.T.S.

No.	DATE	DESCRIPTION
9	2026-10-30	MECHANICAL ADDENDUM 01
8	2026-09-16	ISSUED FOR TENDER
7	2026-03-24	ISSUED FOR 100% CD
6	2026-02-20	ISSUED FOR 90% CD AND CLASS A COSTING
5	2026-01-30	ISSUED FOR 75%CD
4	2026-01-30	ISSUED FOR SITE SERVICES PERMIT
3	2026-01-30	ISSUED FOR BUILDING PERMIT
2	2025-12-17	ISSUED FOR 30%CD
1	2025-12-01	RE-ISSUED FOR SPA

REVISIONS

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PROJECT NORTH

QUASAR

250 ROWNTREE DAIRY RD, WOODBRIDGE, ON
TEL: 905-507-0800
WEB: WWW.QUASARCG.COM

PROJECT

DURHAM REGION
409 CENTRE ST S SENIORS HOUSING
409 CENTRE ST. SOUTH, WHITBY, ON L1N 4W3
TOWN FILE NUMBER: SP-26-25

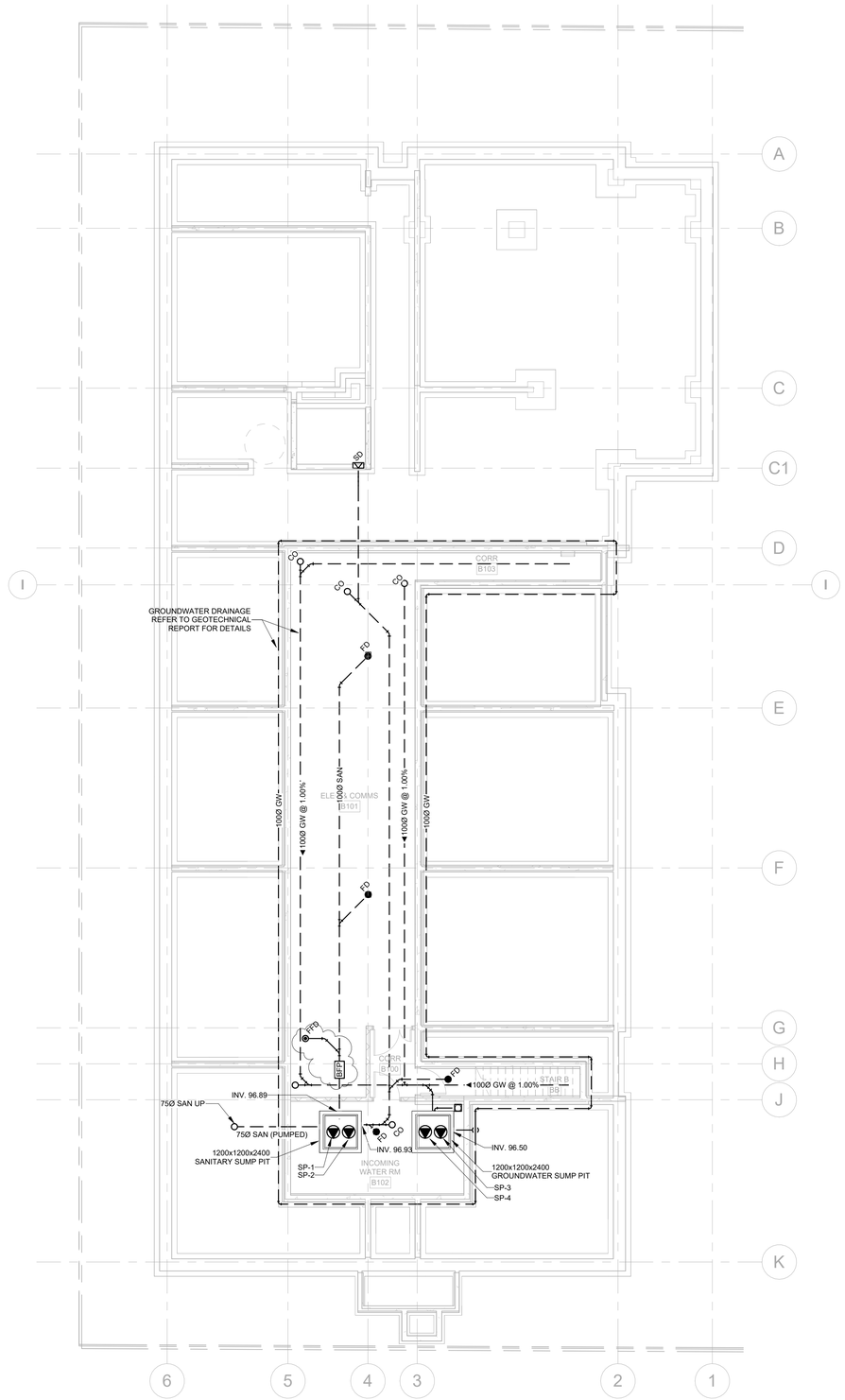
TITLE

FIRE PROTECTION SCHEMATIC

CHECKED	PROJECT No.
SA	RE-24-162
SCALE	DRAWING No.
N.T.S.	M-140

GENERAL NOTES - PLUMBING

1. MECHANICAL CONTRACTOR SHALL REVIEW STRUCTURAL DRAWINGS REGARDING SIZE AND LOCATIONS OF BEAMS AND EXPANSION JOINTS.
2. MECHANICAL CONTRACTOR SHALL COORDINATE ALL PIPING WITH STRUCTURAL BEAMS AND PROVIDE SLEEVING AS NECESSARY TO MAINTAIN MINIMUM HEADROOM AS INDICATED ON ARCHITECTURAL DRAWINGS.
3. MECHANICAL CONTRACTOR SHALL REVIEW ALL ARCHITECTURAL AND INTERIOR DESIGN DRAWINGS AND MAINTAIN MINIMUM HEADROOM AS INDICATED.
4. ALL EQUIPMENT SHALL BE INSTALLED IN STRICT ACCORDANCE WITH MANUFACTURER'S RECOMMENDATIONS.
5. PROVIDE ISOLATION VALVES FOR ALL RISERS AND AT EACH FIXTURE
6. ALL DRAINAGE PIPING SHALL BE 100Ø AND SLOPED AT 1% UNLESS NOTED OTHERWISE.
7. ANY DRAINAGE PIPING 75Ø OR LESS SHALL BE SLOPED AT 2% UNLESS NOTED OTHERWISE.
8. ALL TRAPPED SANITARY DRAINS LOCATED IN UNHEATED SPACE SHALL BE ELECTRICALLY HEAT TRACED AND INSULATED OVER ENTIRE LENGTH.
9. PROVIDE P-TRAPS, VENT TO OUTDOORS AND PRIMING TO ALL FLOOR DRAINS
10. PROVIDE FULL VENTING SYSTEM IN ACCORDANCE WITH OBC PART 7. COORDINATE ALL VENTS AND TERMINATION POINTS.
11. ALLOW FOR PIPING AND HEAT TRACING FOR ALL TRAP PRIMERS.
12. REFER TO SCHEMATICS FOR GAS, DOMESTIC WATER, SANITARY, VENTING AND STORM RISER DETAILS AND SIZING.
13. PROVIDE PROVISIONS FOR ABILITY TO ADD WATER METERS IN EACH RESIDENTIAL UNIT.



No.	DATE	DESCRIPTION
9	2026-10-30	MECHANICAL ADDENDUM 01
8	2026-09-16	ISSUED FOR TENDER
7	2026-03-24	ISSUED FOR 100% CD
6	2026-02-20	ISSUED FOR 90% CD AND CLASS A COSTING
5	2026-01-30	ISSUED FOR 75%CD
4	2026-01-30	ISSUED FOR SITE SERVICES PERMIT
3	2026-01-30	ISSUED FOR BUILDING PERMIT
2	2025-12-17	ISSUED FOR 30%CD
1	2025-12-01	RE-ISSUED FOR SPA

REVISIONS

THE SPECIFICATIONS ARE TO BE CONSIDERED AS AN INTEGRAL PART OF THESE DRAWINGS AND NEITHER THE DRAWINGS NOR THE SPECIFICATIONS SHALL BE USED ALONE. REFER TO ARCHITECTURAL DRAWINGS FOR DIMENSIONS. DO NOT SCALE.

PROJECT NORTH

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PROJECT

DURHAM REGION
409 CENTRE ST S SENIORS HOUSING
409 CENTRE ST. SOUTH, WHITBY, ON L1N 4W3
TOWN FILE NUMBER: SP-26-25

TITLE

PLUMBING - FOUNDATION

CHECKED	PROJECT No.
SA	RE-24-162
SCALE	DRAWING No.
1:100	M-200

GENERAL NOTES - PLUMBING

1. MECHANICAL CONTRACTOR SHALL REVIEW STRUCTURAL DRAWINGS REGARDING SIZE AND LOCATIONS OF BEAMS AND EXPANSION JOINTS.
2. MECHANICAL CONTRACTOR SHALL COORDINATE ALL PIPING WITH STRUCTURAL BEAMS AND PROVIDE SLEEVING AS NECESSARY TO MAINTAIN MINIMUM HEADROOM AS INDICATED ON ARCHITECTURAL DRAWINGS.
3. MECHANICAL CONTRACTOR SHALL REVIEW ALL ARCHITECTURAL AND INTERIOR DESIGN DRAWINGS AND MAINTAIN MINIMUM HEADROOM AS INDICATED.
4. ALL EQUIPMENT SHALL BE INSTALLED IN STRICT ACCORDANCE WITH MANUFACTURER'S RECOMMENDATIONS.
5. PROVIDE ISOLATION VALVES FOR ALL RISERS AND AT EACH FIXTURE
6. ALL DRAINAGE PIPING SHALL BE 100Ø AND SLOPED AT 1% UNLESS NOTED OTHERWISE.
7. ANY DRAINAGE PIPING 75Ø OR LESS SHALL BE SLOPED AT 2% UNLESS NOTED OTHERWISE.
8. ALL TRAPPED SANITARY DRAINS LOCATED IN UNHEATED SPACE SHALL BE ELECTRICALLY HEAT TRACED AND INSULATED OVER ENTIRE LENGTH.
9. PROVIDE P-TRAPS, VENT TO OUTDOORS AND PRIMING TO ALL FLOOR DRAINS
10. PROVIDE FULL VENTING SYSTEM IN ACCORDANCE WITH OBC PART 7. COORDINATE ALL VENTS AND TERMINATION POINTS.
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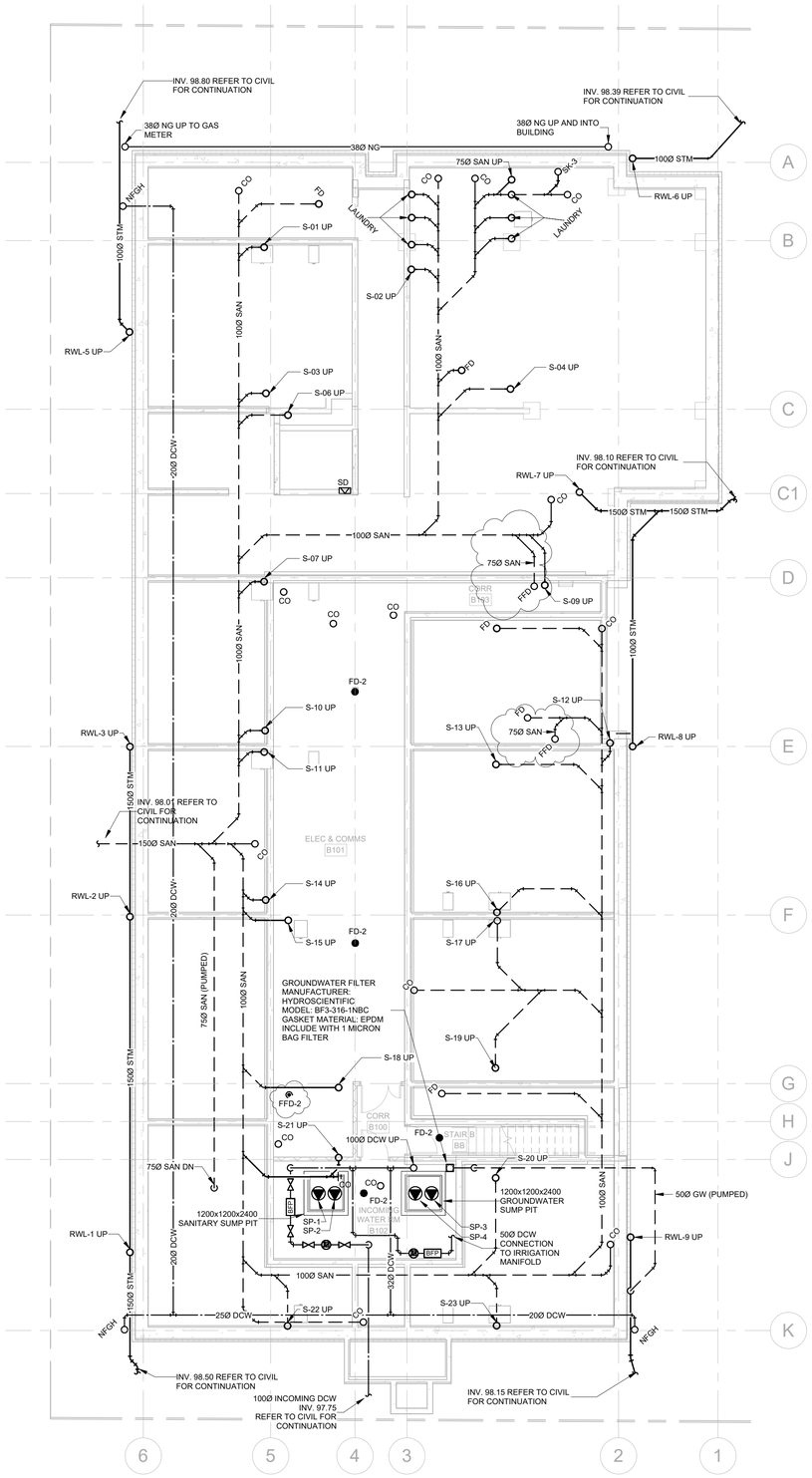

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1	2025-12-01	RE-ISSUED FOR SPA

REVISIONS

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PROJECT NORTH

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PROJECT

DURHAM REGION
409 CENTRE ST S SENIORS HOUSING
409 CENTRE ST. SOUTH, WHITBY, ON L1N 4W3
TOWN FILE NUMBER: SP-26-25

TITLE

PLUMBING - BASEMENT

CHECKED	PROJECT No.
SA	RE-24-162
SCALE	DRAWING No.
1:100	M-201

GENERAL NOTES - PLUMBING

1. MECHANICAL CONTRACTOR SHALL REVIEW STRUCTURAL DRAWINGS REGARDING SIZE AND LOCATIONS OF BEAMS AND EXPANSION JOINTS.
2. MECHANICAL CONTRACTOR SHALL COORDINATE ALL PIPING WITH STRUCTURAL BEAMS AND PROVIDE SLEEVING AS NECESSARY TO MAINTAIN MINIMUM HEADROOM AS INDICATED ON ARCHITECTURAL DRAWINGS.
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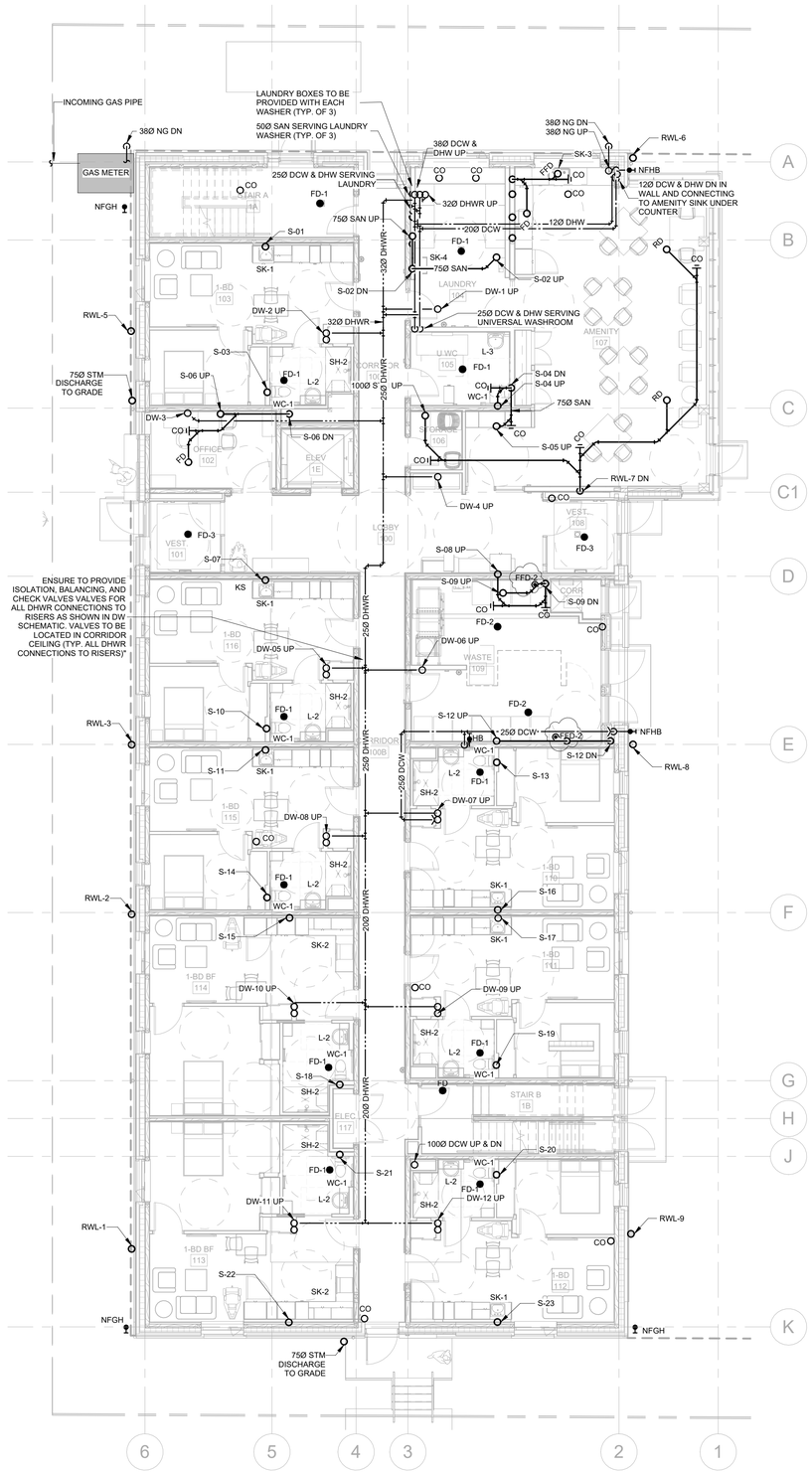

No.	DATE	DESCRIPTION
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4	2026-01-30	ISSUED FOR SITE SERVICES PERMIT
3	2026-01-30	ISSUED FOR BUILDING PERMIT
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1	2025-12-01	RE-ISSUED FOR SPA

REVISIONS

THE SPECIFICATIONS ARE TO BE CONSIDERED AS AN INTEGRAL PART OF THESE DRAWINGS AND NEITHER THE DRAWINGS NOR THE SPECIFICATIONS SHALL BE USED ALONE. REFER TO ARCHITECTURAL DRAWINGS FOR DIMENSIONS. DO NOT SCALE.

PROJECT NORTH

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PROJECT

DURHAM REGION
409 CENTRE ST S SENIORS HOUSING
409 CENTRE ST. SOUTH, WHITBY, ON L1N 4W3
TOWN FILE NUMBER: SP-26-25

TITLE

PLUMBING - LEVEL 1

CHECKED	PROJECT No.
SA	RE-24-162
SCALE	DRAWING No.
1:100	M-202

GENERAL NOTES - HVAC

1. DRAWINGS SHOW GENERAL ARRANGEMENT AND ARE NOT MEANT TO BE SCALED. REFER TO ARCHITECTURAL REFLECTED CEILING PLANS FOR EXACT LOCATIONS OF DIFFUSERS AND GRILLES.
2. MECHANICAL CONTRACTOR SHALL REVIEW STRUCTURAL DRAWINGS REGARDING SIZE AND LOCATIONS OF BEAMS AND EXPANSION JOINTS.
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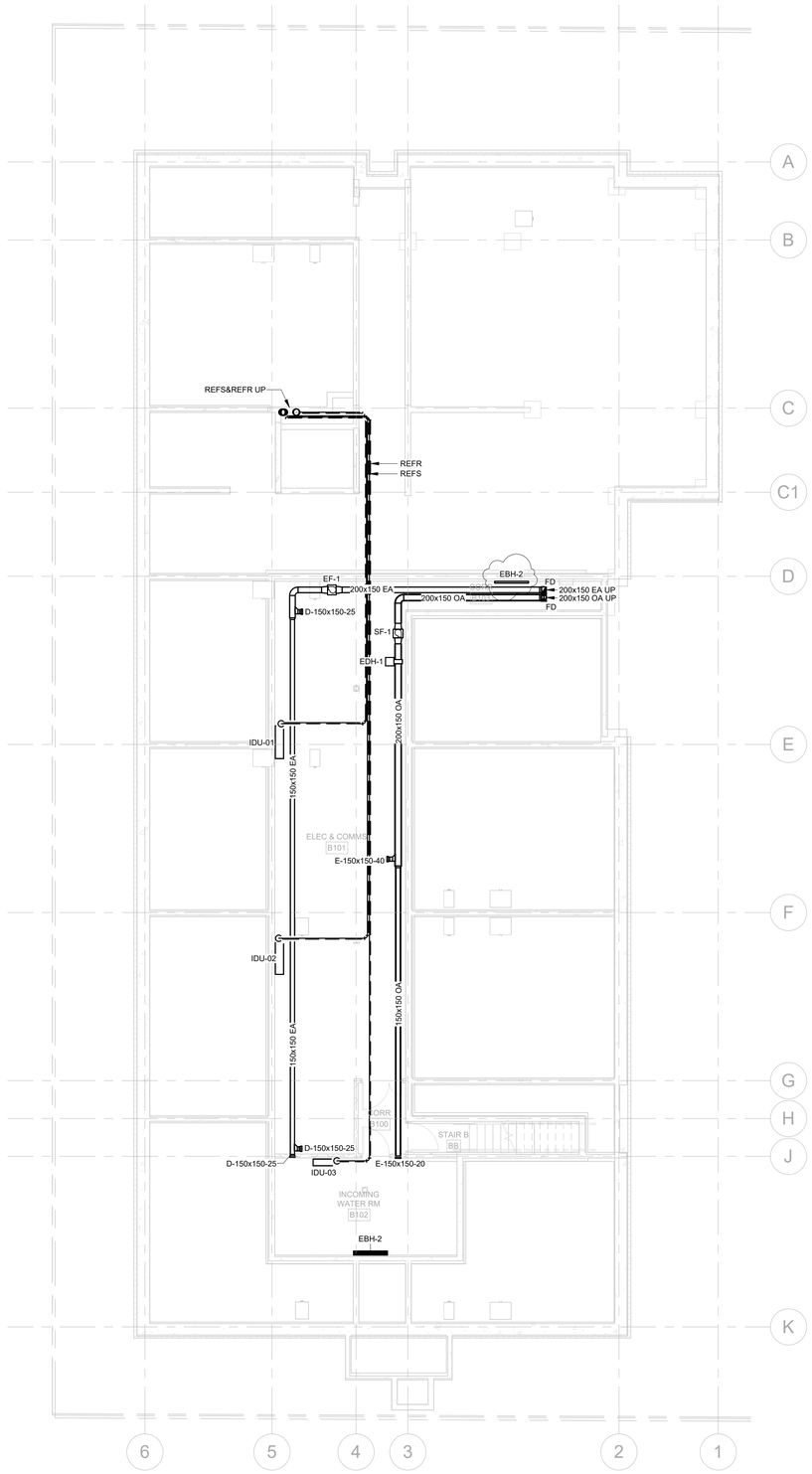
6. DUCTWORK SHALL BE INSULATED OR LINED PER SPECIFICATIONS AND/OR AS NOTED ON DRAWINGS. ALL DUCT JOINTS AND SEAMS SHALL BE PER SPECIFICATIONS.
7. ALL DUCTWORK INSTALLATION TO BE IN ACCORDANCE WITH OBC PART 6 AS WELL AS SMACNA GUIDELINES.
8. MANUAL BALANCING DAMPERS SHALL BE PROVIDED IN ALL DUCT BRANCHES AND IN ALL BRANCHES TO INDIVIDUAL GRILLES AND DIFFUSERS.
9. SUITE DUCT SIZES (UNLESS NOTED OTHERWISE):
   A. KITCHEN EXHAUST.....150Ø
   B. WASHROOM EXHAUST.....150Ø
   C. ERV INTAKE.....150Ø
   D. ERV EXHAUST.....150Ø
10. SUITE INTAKES AND EXHAUST TERMINATIONS TO BE AT EXTERIOR LOUVERS ABOVE WINDOWS. EXTERIOR LOUVERS PROVIDED BY ARCHITECT.
11. ALL EXTERIOR LOUVERS TO BE PROVIDED WITH BIRD SCREEN.
12. ALL OPEN-ENDED DUCTS (OED) TO BE C/W WIRE MESH
13. ALL CEILING UNITS HAVE AN ELEVATION OF AT LEAST 2100 FROM LEVEL.
14. PROVIDE PROVISIONS FOR ABILITY TO ADD WATER METERS IN EACH RESIDENTIAL UNIT.
15. PROVIDE ACCESS PANELS TO HEAT PUMPS EACH RESIDENTIAL SUITE TO ALLOW ACCESS TO THE ENTIRE UNIDERSIDE OF THE EQUIPMENT.
16. PROVIDE 300x300 ACCESS PANELS FOR ALL MOTORIZED DAMPERS.
17. PROVIDE 25 mm THICK INTERNAL DUCT LINING FOR FIRST 1500 mm ON ALL SUPPLY AND FIRST 1000 mm ON RETURN DUCTWORK ON ALL AIR-SOURCE HEAT PUMPS.
18. PROVIDE CONDENSATE PIPING FOR ALL CEILING CONCEALED HEAT PUMPS AND VRF FAN COIL UNITS. DRAIN TO NEAREST FLOOR DRAIN (SHEET M-802 DETAIL #1) OR LAVATORY P-TRAP (SHEET M-804 DETAIL #2).



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REVISIONS

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PROJECT

DURHAM REGION
409 CENTRE ST S SENIORS HOUSING
409 CENTRE ST. SOUTH, WHITBY, ON L1N 4W3
TOWN FILE NUMBER: SP-26-25

TITLE

HVAC - BASEMENT

CHECKED	PROJECT No.
SA	RE-24-162
SCALE	DRAWING No.
1:100	M-300

GENERAL NOTES - HVAC

1. DRAWINGS SHOW GENERAL ARRANGEMENT AND ARE NOT MEANT TO BE SCALED. REFER TO ARCHITECTURAL REFLECTED CEILING PLANS FOR EXACT LOCATIONS OF DIFFUSERS AND GRILLES.
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8. MANUAL BALANCING DAMPERS SHALL BE PROVIDED IN ALL DUCT BRANCHES AND IN ALL BRANCHES TO INDIVIDUAL GRILLES AND DIFFUSERS.
9. SUITE DUCT SIZES (UNLESS NOTED OTHERWISE):
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   - B. WASHROOM EXHAUST.....150Ø
   - C. ERV INTAKE.....150Ø
   - D. ERV EXHAUST.....150Ø
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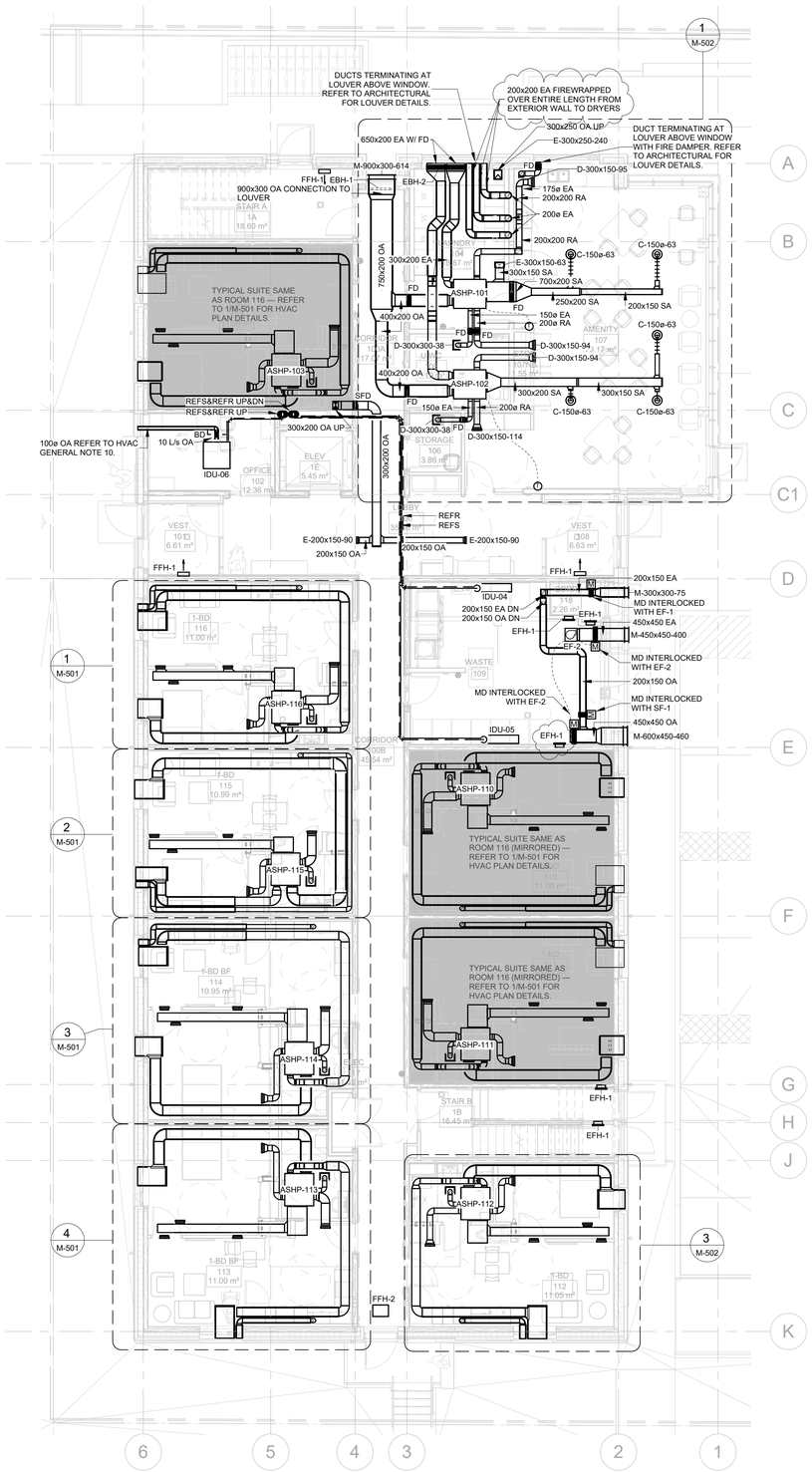
11. ALL EXTERIOR LOUVERS TO BE PROVIDED WITH BIRD SCREEN.
12. ALL OPEN-ENDED DUCTS (OED) TO BE C/W WIRE MESH
13. ALL CEILING UNITS HAVE AN ELEVATION OF AT LEAST 2100 FROM LEVEL.
14. PROVIDE PROVISIONS FOR ABILITY TO ADD WATER METERS IN EACH RESIDENTIAL UNIT.
15. PROVIDE ACCESS PANELS TO HEAT PUMPS EACH RESIDENTIAL SUITE TO ALLOW ACCESS TO THE ENTIRE UNIDERSIDE OF THE EQUIPMENT.
16. PROVIDE 300x300 ACCESS PANELS FOR ALL MOTORIZED DAMPERS.
17. PROVIDE 25 mm THICK INTERNAL DUCT LINING FOR FIRST 1500 mm ON ALL SUPPLY AND FIRST 1000 mm ON RETURN DUCTWORK ON ALL AIR-SOURCE HEAT PUMPS.
18. PROVIDE CONDENSATE PIPING FOR ALL CEILING CONCEALED HEAT PUMPS AND VRF FAN COIL UNITS. DRAIN TO NEAREST FLOOR DRAIN (SHEET M-802 DETAIL #1) OR LAVATORY P-TRAP (SHEET M-804 DETAIL #2).



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1	2025-12-01	RE-ISSUED FOR SPA

REVISIONS

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PROJECT NORTH

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PROJECT

DURHAM REGION
409 CENTRE ST S SENIORS HOUSING
409 CENTRE ST. SOUTH, WHITBY, ON L1N 4W3
TOWN FILE NUMBER: SP-26-25

TITLE

HVAC - LEVEL 1

CHECKED	PROJECT No.
SA	RE-24-162
SCALE	DRAWING No.
1:100	M-301

GENERAL NOTES - HVAC

1. DRAWINGS SHOW GENERAL ARRANGEMENT AND ARE NOT MEANT TO BE SCALED. REFER TO ARCHITECTURAL REFLECTED CEILING PLANS FOR EXACT LOCATIONS OF DIFFUSERS AND GRILLES.
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11. ALL EXTERIOR LOUVERS TO BE PROVIDED WITH BIRD SCREEN.
12. ALL OPEN-ENDED DUCTS (OED) TO BE C/W WIRE MESH
13. ALL CEILING UNITS HAVE AN ELEVATION OF AT LEAST 2100 FROM LEVEL.
14. PROVIDE PROVISIONS FOR ABILITY TO ADD WATER METERS IN EACH RESIDENTIAL UNIT.
15. PROVIDE ACCESS PANELS TO HEAT PUMPS EACH RESIDENTIAL SUITE TO ALLOW ACCESS TO THE ENTIRE UNIDERSIDE OF THE EQUIPMENT.
16. PROVIDE 300x300 ACCESS PANELS FOR ALL MOTORIZED DAMPERS.
17. PROVIDE 25 mm THICK INTERNAL DUCT LINING FOR FIRST 1500 mm ON ALL SUPPLY AND FIRST 1000 mm ON RETURN DUCTWORK ON ALL AIR-SOURCE HEAT PUMPS.
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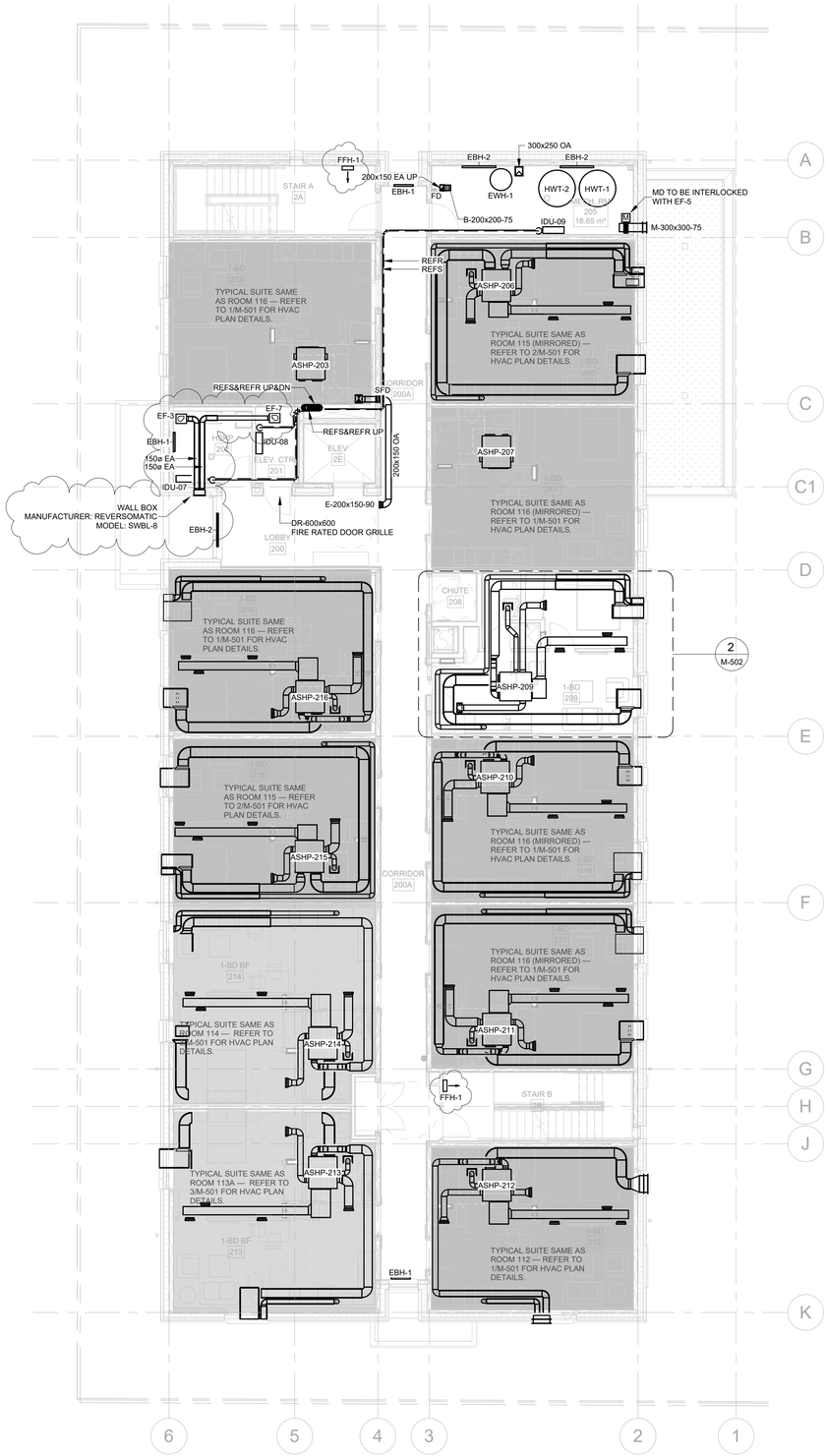

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PROJECT

DURHAM REGION
409 CENTRE ST S SENIORS HOUSING
409 CENTRE ST. SOUTH, WHITBY, ON L1N 4W3
TOWN FILE NUMBER: SP-26-25

TITLE

HVAC - LEVEL 2

CHECKED	PROJECT No.
SA	RE-24-162
SCALE	DRAWING No.
1:100	M-302

GENERAL NOTES - HVAC

1. DRAWINGS SHOW GENERAL ARRANGEMENT AND ARE NOT MEANT TO BE SCALED. REFER TO ARCHITECTURAL REFLECTED CEILING PLANS FOR EXACT LOCATIONS OF DIFFUSERS AND GRILLES.
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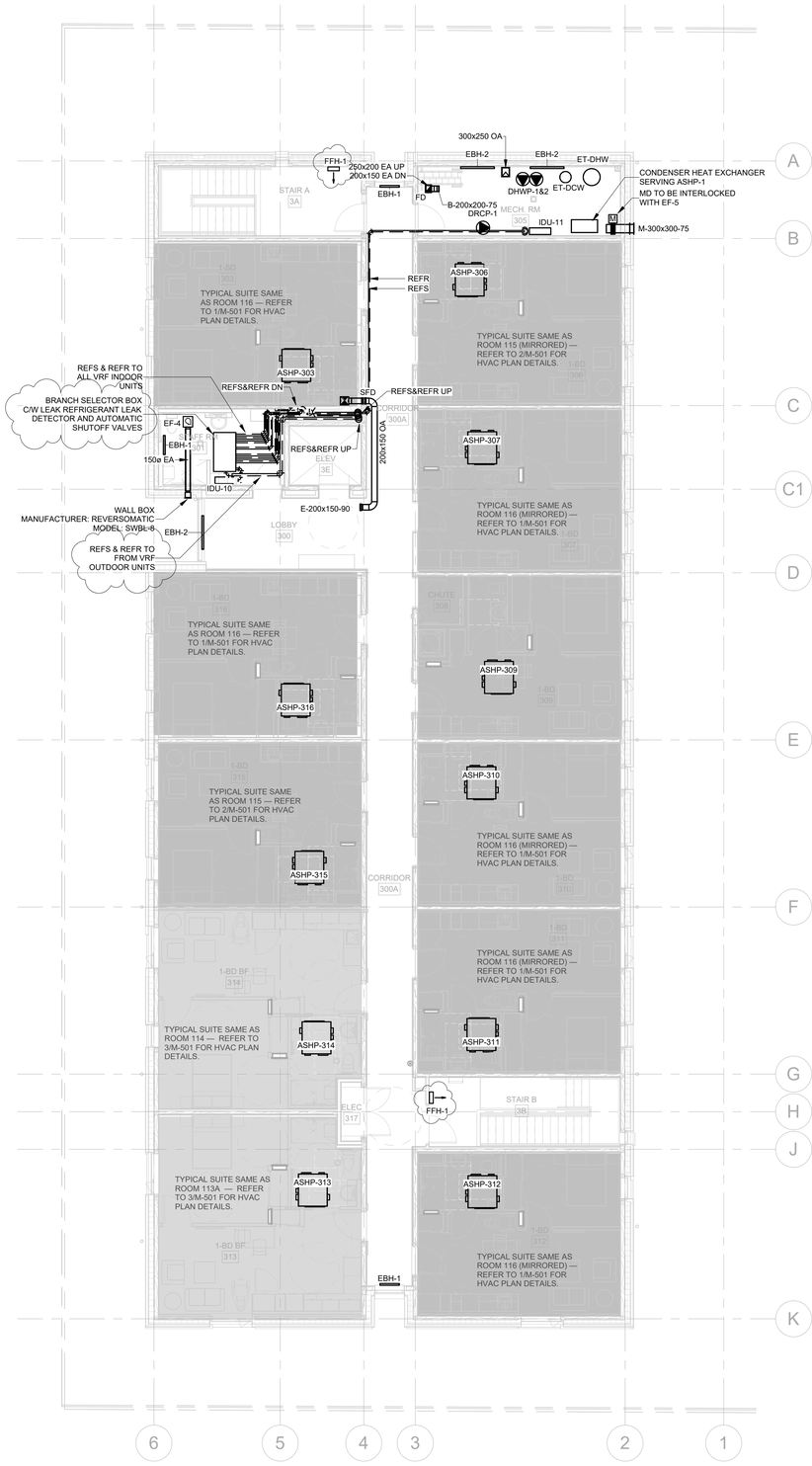
13. ALL CEILING UNITS HAVE AN ELEVATION OF AT LEAST 2100 FROM LEVEL.
14. PROVIDE PROVISIONS FOR ABILITY TO ADD WATER METERS IN EACH RESIDENTIAL UNIT.
15. PROVIDE ACCESS PANELS TO HEAT PUMPS EACH RESIDENTIAL SUITE TO ALLOW ACCESS TO THE ENTIRE UNDERSIDE OF THE EQUIPMENT.
16. PROVIDE 300x300 ACCESS PANELS FOR ALL MOTORIZED DAMPERS.
17. PROVIDE 25 mm THICK INTERNAL DUCT LINING FOR FIRST 1500 mm ON ALL SUPPLY AND FIRST 1000 mm ON RETURN DUCTWORK ON ALL AIR-SOURCE HEAT PUMPS.
18. PROVIDE CONDENSATE PIPING FOR ALL CEILING CONCEALED HEAT PUMPS AND VRF FAN COIL UNITS. DRAIN TO NEAREST FLOOR DRAIN (SHEET M-802 DETAIL #1) OR LAVATORY P-TRAP (SHEET M-804 DETAIL #2).



No.	DATE	DESCRIPTION
9	2026-10-30	MECHANICAL ADDENDUM 01
8	2026-09-16	ISSUED FOR TENDER
7	2026-03-24	ISSUED FOR 100% CD
6	2026-02-20	ISSUED FOR 90% CD AND CLASS A COSTING
5	2026-01-30	ISSUED FOR 75%CD
4	2026-01-30	ISSUED FOR SITE SERVICES PERMIT
3	2026-01-30	ISSUED FOR BUILDING PERMIT
2	2025-12-17	ISSUED FOR 30%CD
1	2025-12-01	RE-ISSUED FOR SPA

REVISIONS

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PROJECT NORTH

QUASAR
250 ROWNTREE DAIRY RD, WOODBRIDGE, ON
TEL: 905-507-0800
WEB: WWW.QUASARCG.COM

PROJECT

DURHAM REGION
409 CENTRE ST S SENIORS HOUSING
409 CENTRE ST. SOUTH, WHITBY, ON L1N 4W3
TOWN FILE NUMBER: SP-26-25

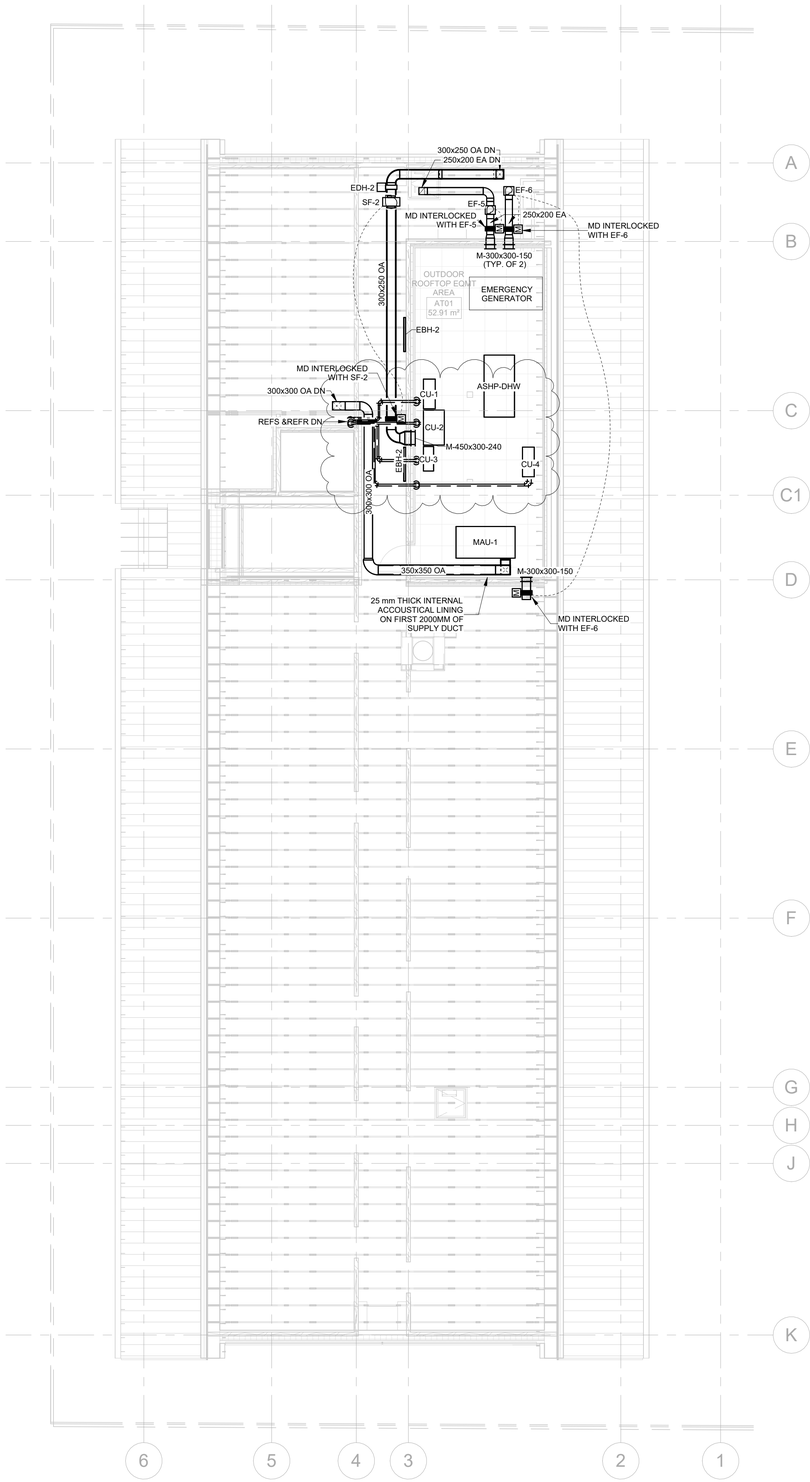
TITLE

HVAC - LEVEL 3

CHECKED	PROJECT No.
SA	RE-24-162
SCALE	DRAWING No.
1:100	M-303

GENERAL NOTES - HVAC

1. DRAWINGS SHOW GENERAL ARRANGEMENT AND ARE NOT MEANT TO BE SCALED. REFER TO ARCHITECTURAL REFLECTED CEILING PLANS FOR EXACT LOCATIONS OF DIFFUSERS AND GRILLES.
2. MECHANICAL CONTRACTOR SHALL REVIEW STRUCTURAL DRAWINGS REGARDING SIZE AND LOCATIONS OF BEAMS AND EXPANSION JOINTS.
3. MECHANICAL CONTRACTOR SHALL COORDINATE ALL PIPING AND DUCTWORK WITH STRUCTURAL BEAMS AND PROVIDE SLEEVING AS NECESSARY TO MAINTAIN MINIMUM HEADROOM AS INDICATED ON ARCHITECTURAL DRAWINGS.
4. MECHANICAL CONTRACTOR SHALL REVIEW ALL ARCHITECTURAL AND INTERIOR DESIGN DRAWINGS AND MAINTAIN MINIMUM HEADROOM AS INDICATED.
5. ALL EQUIPMENT SHALL BE INSTALLED IN STRICT ACCORDANCE WITH MANUFACTURER'S RECOMMENDATIONS.
6. DUCTWORK SHALL BE INSULATED OR LINED PER SPECIFICATIONS AND/OR AS NOTED ON DRAWINGS. ALL DUCT JOINTS AND SEAMS SHALL BE PER SPECIFICATIONS.
7. ALL DUCTWORK INSTALLATION TO BE IN ACCORDANCE WITH OBC PART 6 AS WELL AS SMACNA GUIDELINES.
8. MANUAL BALANCING DAMPERS SHALL BE PROVIDED IN ALL DUCT BRANCHES AND IN ALL BRANCHES TO INDIVIDUAL GRILLES AND DIFFUSERS.
9. SUITE DUCT SIZES (UNLESS NOTED OTHERWISE):
   A. KITCHEN EXHAUST.....150Ø
   B. WASHROOM EXHAUST.....150Ø
   C. ERV INTAKE.....150Ø
   D. ERV EXHAUST.....150Ø
10. SUITE INTAKES AND EXHAUST TERMINATIONS TO BE AT EXTERIOR LOUVERS ABOVE WINDOWS. EXTERIOR LOUVERS PROVIDED BY ARCHITECT.
11. ALL EXTERIOR LOUVERS TO BE PROVIDED WITH BIRD SCREEN.
12. ALL OPEN-ENDED DUCTS (OED) TO BE C/W WIRE MESH
13. ALL CEILING UNITS HAVE AN ELEVATION OF AT LEAST 2100 FROM LEVEL.
14. PROVIDE PROVISIONS FOR ABILITY TO ADD WATER METERS IN EACH RESIDENTIAL UNIT.
15. PROVIDE ACCESS PANELS TO HEAT PUMPS EACH RESIDENTIAL SUITE TO ALLOW ACCESS TO THE ENTIRE UNDERSIDE OF THE EQUIPMENT.
16. PROVIDE 300x300 ACCESS PANELS FOR ALL MOTORIZED DAMPERS.
17. PROVIDE 25 mm THICK INTERNAL DUCT LINING FOR FIRST 1500 mm ON ALL SUPPLY AND FIRST 1000 mm ON RETURN DUCTWORK ON ALL AIR-SOURCE HEAT PUMPS.
18. PROVIDE CONDENSATE PIPING FOR ALL CEILING CONCEALED HEAT PUMPS AND VRF FAN COIL UNITS. DRAIN TO NEAREST FLOOR DRAIN (SHEET M-802 DETAIL #1) OR LAVATORY P-TRAP (SHEET M-804 DETAIL #2).



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REVISIONS

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PROJECT NORTH

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PROJECT

DURHAM REGION
409 CENTRE ST S SENIORS HOUSING
409 CENTRE ST. SOUTH, WHITBY, ON L1N 4W3
TOWN FILE NUMBER: SP-26-25

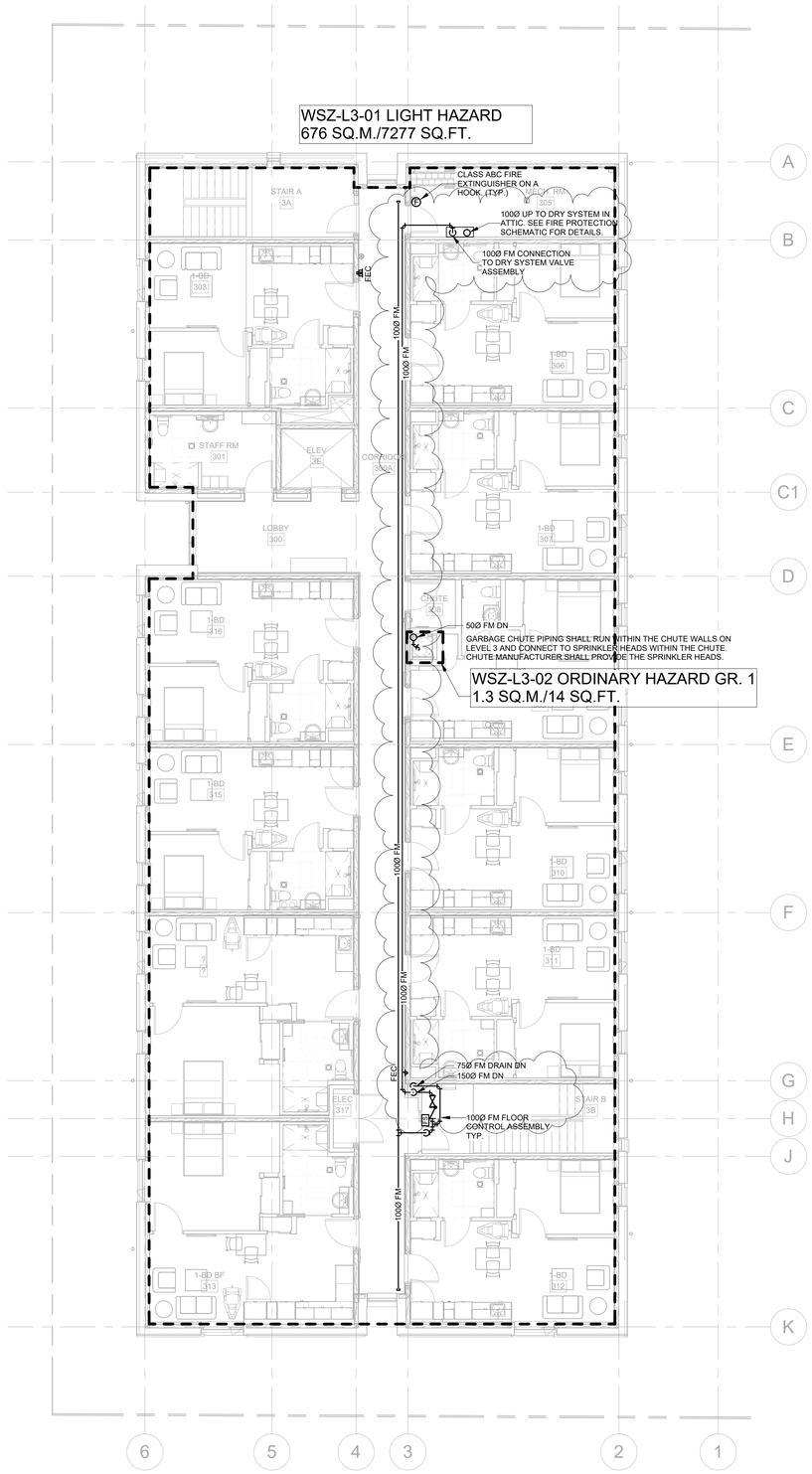
TITLE

HVAC - ROOF

CHECKED	PROJECT No.
SA	RE-24-162
SCALE	DRAWING No.
1:100	M-304

GENERAL NOTES - FIRE PROTECTION

1. SPRINKLER SYSTEM DESIGN AND INSTALLATION TO BE IN ACCORDANCE WITH NFPA 13-2019, THE ONTARIO BUILDING CODE-2024, THE ONTARIO FIRE CODE-2013, AND LOCAL AUTHORITY REQUIREMENTS.
2. ADHERE TO AND OBTAIN ALL PERMITS, LICENSES, AND GOVERNMENT REQUIREMENTS, IF APPLICABLE.
3. CUTTING OF STRUCTURAL AND/OR ARCHITECTURAL MEMBERS TO BE DONE ONLY WITH THE WRITTEN APPROVAL OF THE ARCHITECT AND/OR STRUCTURAL ENGINEER.
4. ALL ELECTRICAL WIRING OF SPRINKLER DEVICES IS BY ELECTRICAL CONTRACTOR. COORDINATE ALL ELECTRICAL ITEMS WITH ELECTRICAL CONTRACTOR AND ENSURE PROPER COORDINATION.
5. PROVIDE STOCK OF EXTRA SPRINKLERS IN ACCORDANCE WITH NFPA-13, 6.2.9.
6. COORDINATION IS TO TAKE PLACE BETWEEN THE SPRINKLER CONTRACTOR AND ALL OTHER TRADES.
7. THE SPRINKLER CONTRACTOR IS TO FIELD SURVEY THE SITE, INCLUDING STRUCTURAL STEEL AND MECHANICAL/ELECTRICAL SERVICES PRIOR TO FABRICATION AND INSTALLATION.
8. CONFLICTS OR DISCREPANCIES ARE TO BE REPORTED IMMEDIATELY TO THE DESIGN CONSULTANTS.
9. INSTALL HIGH TEMPERATURE SPRINKLERS AROUND ALL HEAT SOURCES IN ACCORDANCE WITH NFPA 13-2019.
10. INSTALL GUARDS ON SPRINKLERS IN WAREHOUSE, MECHANICAL, ELECTRICAL AND STORAGE ROOMS.
11. INSTALL LOW POINT DRAINS ON ALL TRAPPED SECTIONS OF PIPING IN ACCORDANCE WITH NFPA 13-2019.
12. PROVIDE TAGS AND SIGNAGE AS PER NFPA 13-2019.
13. SPRINKLER SYSTEMS ARE TO BE HYDROSTATICALLY TESTED IN ACCORDANCE WITH NFPA 13-2019.



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PROJECT

DURHAM REGION
409 CENTRE ST S SENIORS HOUSING
409 CENTRE ST. SOUTH, WHITBY, ON L1N 4W3
TOWN FILE NUMBER: SP-26-25

TITLE

FIRE PROTECTION - LEVEL 3

CHECKED	PROJECT No.
SA	RE-24-162
SCALE	DRAWING No.
1:100	M-403

GENERAL NOTES - FIRE PROTECTION

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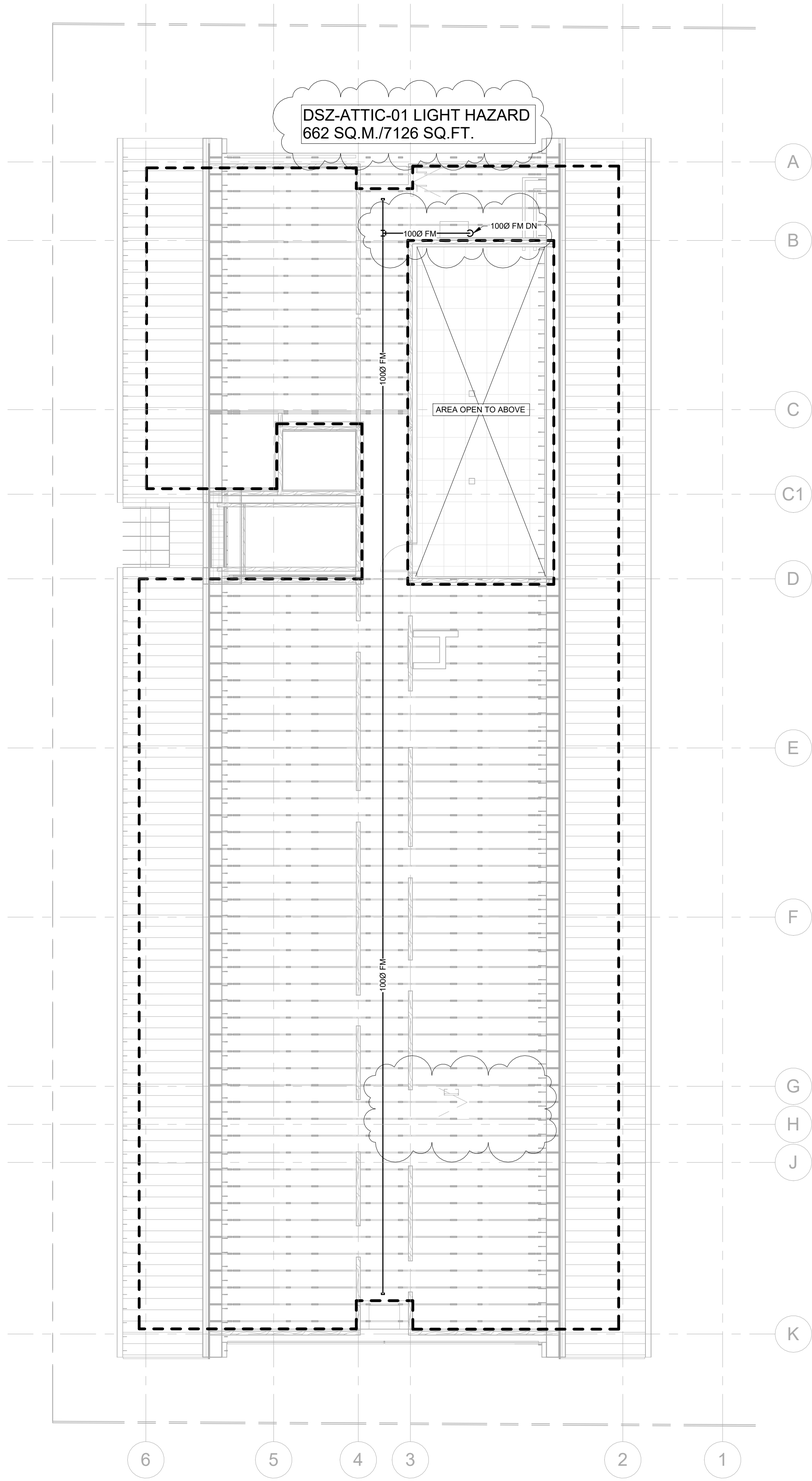

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250 ROWNTREE DAIRY RD, WOODBRIDGE, ON
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PROJECT

DURHAM REGION
409 CENTRE ST S SENIORS HOUSING
409 CENTRE ST. SOUTH, WHITBY, ON L1N 4W3
TOWN FILE NUMBER: SP-26-25

TITLE

FIRE PROTECTION - ROOF

CHECKED	PROJECT No.
SA	RE-24-162
SCALE	DRAWING No.
1:100	M-404

GENERAL NOTES - HVAC

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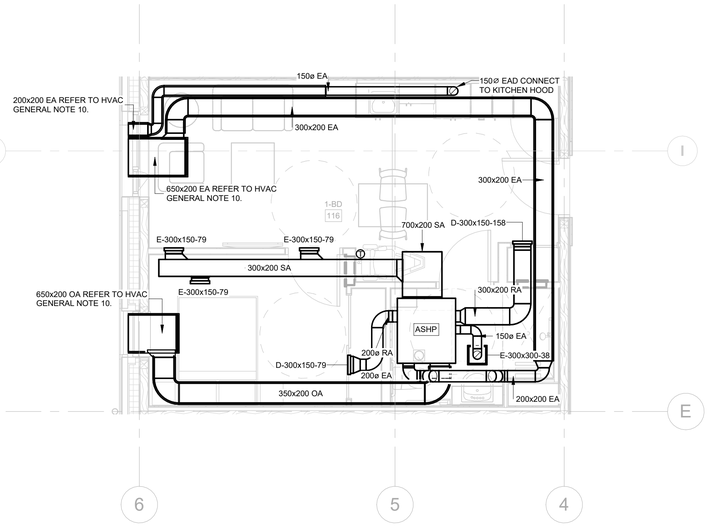
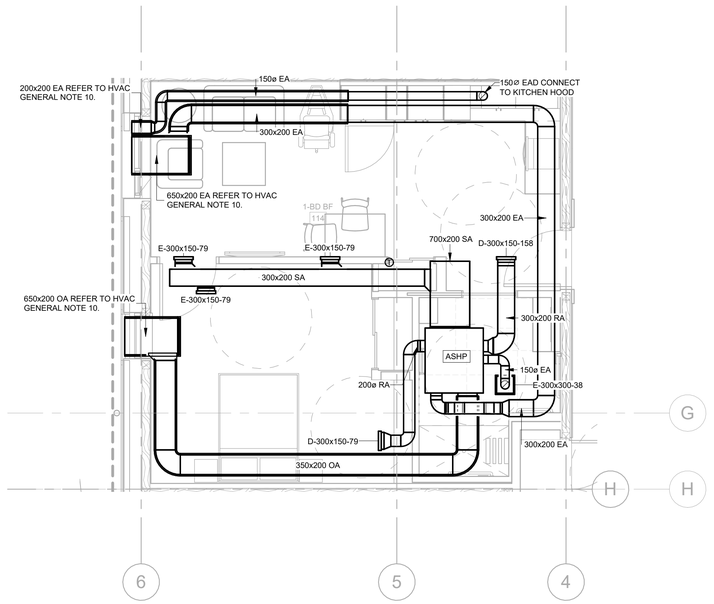


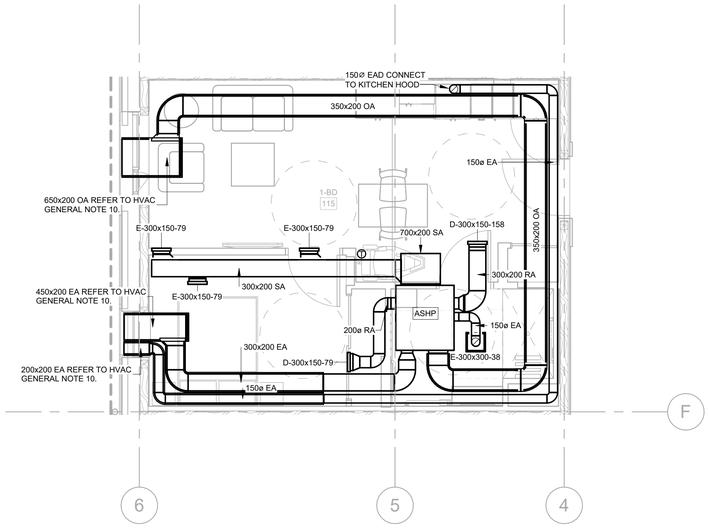
1 HVAC ENLARGED PLAN (1) - TYPICAL SUITE 1BD (116)

SCALE: 1 : 50



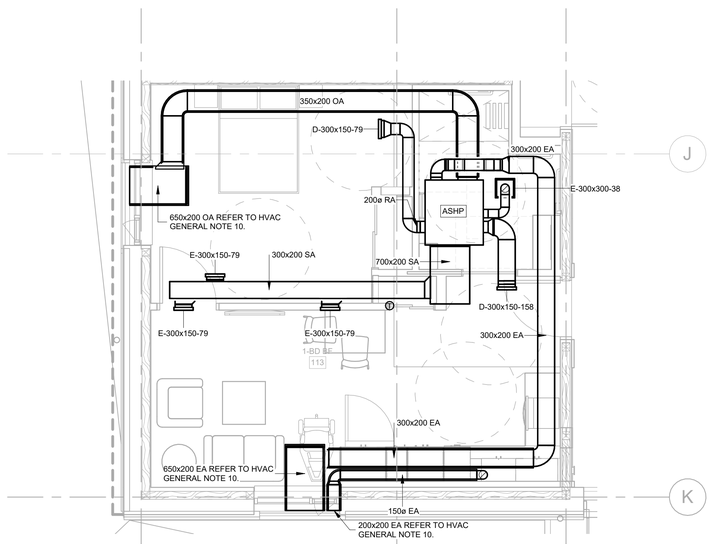
3 HVAC ENLARGED PLAN (3) - TYPICAL SUITE 1BD (114)

SCALE: 1 : 50



2 HVAC ENLARGED PLAN (2) - TYPICAL SUITE 1BD (115)

SCALE: 1 : 50



4 HVAC ENLARGED PLAN (4) - TYPICAL SUITE 1BD (113)

SCALE: 1 : 50

No.	DATE	DESCRIPTION
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REVISIONS

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PROJECT NORTH

QUASAR
250 ROWNTREE DAIRY RD, WOODBRIDGE, ON
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PROJECT

DURHAM REGION
409 CENTRE ST S SENIORS HOUSING
409 CENTRE ST. SOUTH, WHITBY, ON L1N 4W3
TOWN FILE NUMBER: SP-26-25

TITLE

MECHANICAL SECTIONS AND ENLARGED PLANS I

CHECKED	PROJECT No.
SA	RE-24-162
SCALE	DRAWING No.
As indicated	M-501

GENERAL NOTES - HVAC

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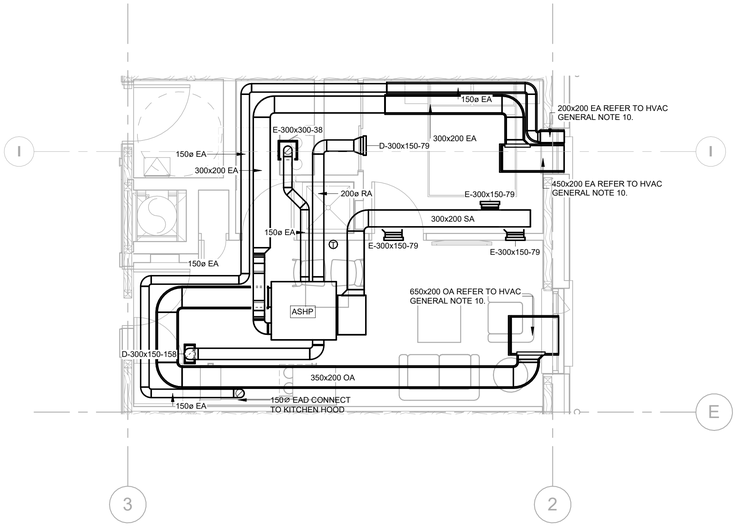
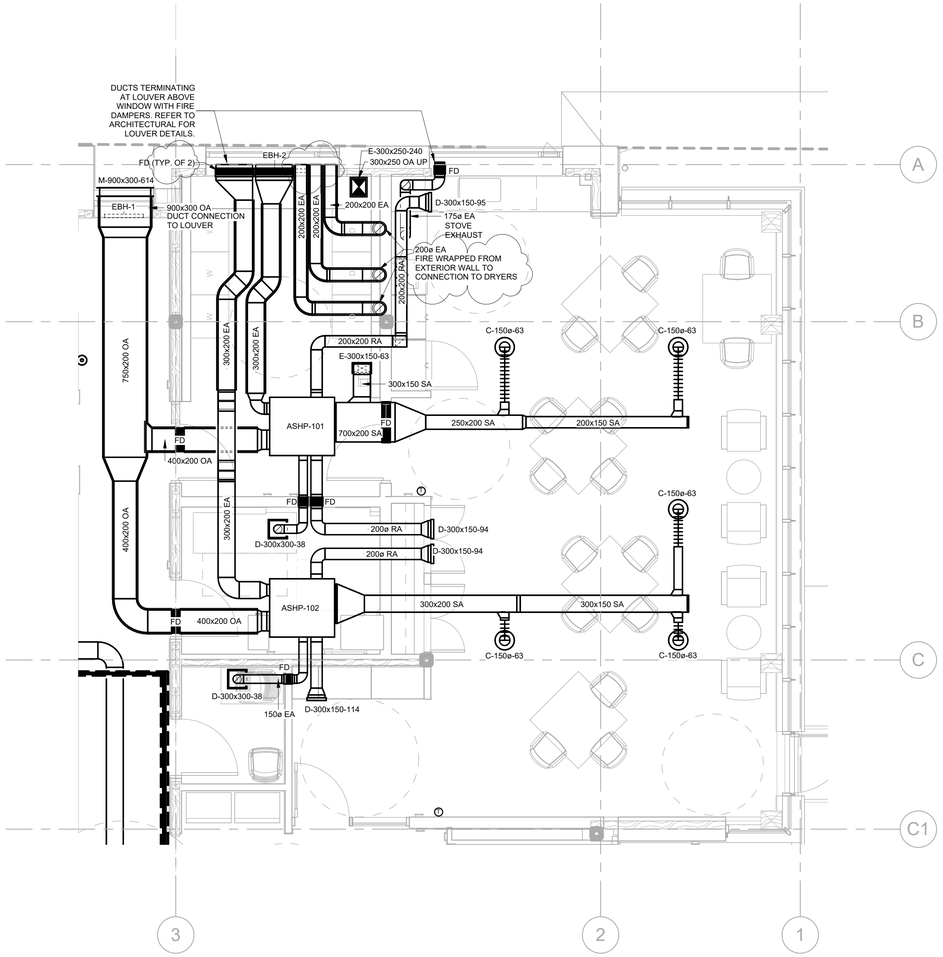
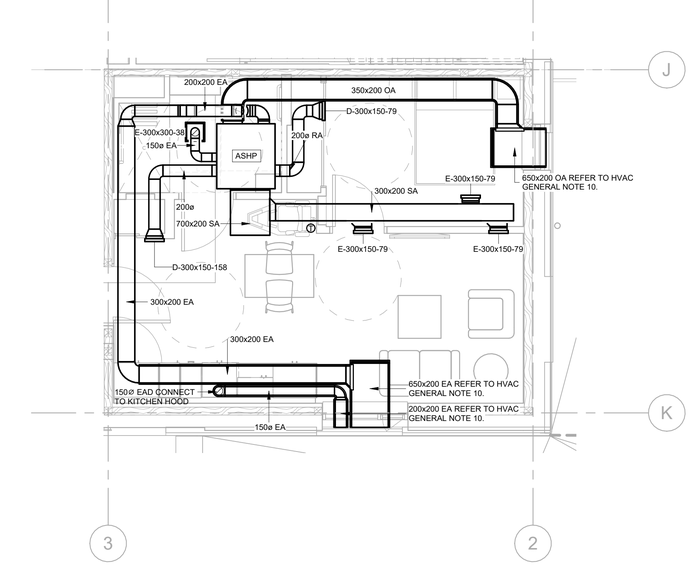


2 HVAC ENLARGED PLAN (6) - TYPICAL SUITE 1BD (209)
SCALE: 1 : 50



1 HVAC ENLARGED PLAN (7) - L1 AMENITY
SCALE: 1 : 50



3 HVAC ENLARGED PLAN (5) - TYPICAL SUITE 1BD (112)
SCALE: 1 : 50

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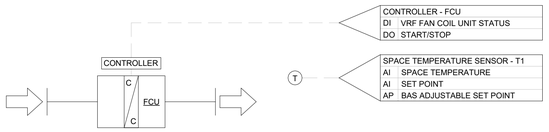
PROJECT

DURHAM REGION
409 CENTRE ST S SENIORS HOUSING
409 CENTRE ST. SOUTH, WHITBY, ON L1N 4W3
TOWN FILE NUMBER: SP-26-25

TITLE

MECHANICAL SECTIONS AND ENLARGED PLANS II

CHECKED	PROJECT No.
SA	RE-24-162
SCALE	DRAWING No.
As indicated	M-502



OPERATING MODE:
1. THE SYSTEM SHALL BE ENABLED ACCORDING TO THE FOLLOWING MODE:
   - SPACE TEMPERATURE CONTROL MODE

INITIAL SET UP:
1. VRF FAN COIL UNIT SHALL BE NORMALLY DISABLED.
2. T1 SHALL MAINTAIN THE SPACE TEMPERATURE SET POINT.

VRF FAN COIL UNIT CONTROL:
1. PROVIDE STAND ALONE PACKAGE DX FAN COIL UNIT AND CONTROL PANEL.
2. VRF SHALL PROVIDE DUAL COOLING AND HEATING FUNCTIONS.

TEMPERATURE CONTROL:
A. IDU-01,02,08 - COOLING ONLY
   - T1 SHALL BE SET TO 27°C IN COOLING MODE.

B. IDU-03,04,05,09,11
   - T1 SHALL BE SET TO 27°C IN COOLING MODE.
   - T1 SHALL BE SET TO 5°C IN HEATING MODE.

C. IDU-06,07,10
   - T1 SHALL BE SET TO 24°C IN COOLING MODE.
   - T1 SHALL BE SET TO 22°C IN HEATING MODE.

1. WHEN IN COOLING MODE:
   - IF THE SPACE TEMPERATURE RISES ABOVE SET POINT, COOLING MODE WILL BE INITIATED AND VRF FAN COIL UNIT SHALL BE ENABLED TO MAINTAIN THE T1 SET POINT.
2. WHEN IN HEATING MODE:
   - IF THE SPACE TEMPERATURE DROPS BELOW SET POINT, HEATING MODE WILL BE INITIATED AND VRF FAN COIL UNIT SHALL BE ENABLED TO MAINTAIN THE T1 SET POINT.

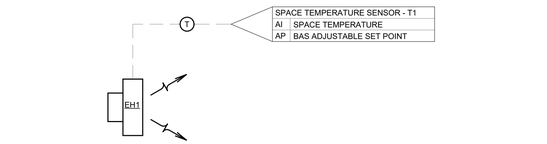
SYSTEM ALARMS & PRIORITY AT BAS:
- VRF FAN COIL UNIT FAILURE: COMMANDED ON/CLOSED FEEDBACK
- SPACE TEMPERATURE OUT OF RANGE (+/- 2°C)

SYSTEM TRENDS AT BAS:
- VRF FAN COIL UNIT STATUS
- SPACE TEMPERATURE
- EQUIPMENT OPERATING HOURS

APPLICABLE SYSTEMS
- IDU-01 TO 11

4 23 81 29.00 - VRF FAN COIL UNIT CONTROL SEQUENCE

SCALE: N.T.S



OPERATING MODE:
1. THE SYSTEM SHALL BE ENABLED ACCORDING TO THE FOLLOWING MODE:
   - SPACE TEMPERATURE CONTROL MODE

INITIAL SET UP:
1. ELECTRIC HEATER SHALL BE NORMALLY DISABLED.
2. T1 SHALL BE A 24V LINE VOLTAGE THERMOSTAT CONTROLLER.
3. T1 SHALL MAINTAIN THE SPACE TEMPERATURE SET POINT.

TEMPERATURE CONTROL:
1. IF THE SPACE TEMPERATURE DROPS BELOW SET POINT UNIT HEATER SHALL BE ENABLED.
2. IF THE SPACE TEMPERATURE RISES ABOVE SET POINT UNIT HEATER SHALL BE DISABLED.

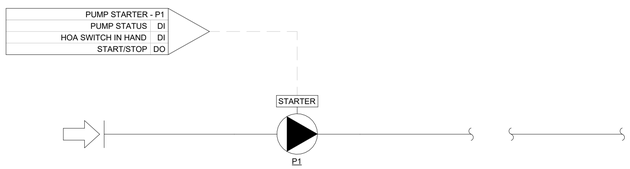
TEMPERATURE SETPOINTS:
A. 4°C
   - ELECTRICAL ROOM
   - INCOMING WATER ROOM
   - WASTE ROOM
   - L2 & L3 MECHANICAL ROOMS
   - ATTIC

B. 21°C
   - ALL OTHER LOCATIONS

NOTE: TEMPERATURE SETPOINT IN INCOMING WATER ROOM, WASTE ROOM, AND MECHANICAL ROOMS IS 1°C LOWER THAN THE SETPOINT OF THE FAN COIL UNITS IN THOSE SPACES. THIS IS TO ENSURE HEATING IS DONE PRIMARILY BY THE HEAT PUMP SYSTEMS INSTEAD OF WITH ELECTRIC RESISTANCE HEAT.

5 23 82 39.02 - ELECTRIC HEATER CONTROL SEQUENCE

SCALE:N.T.S.



OPERATING MODE:
1. THE SYSTEM SHALL BE ENABLED ACCORDING TO THE FOLLOWING MODE:
   - CONTINUOUS OPERATION

INITIAL SET UP:
1. PUMP SHALL BE ENABLED/DISABLED LOCALLY AT THE STARTER OR REMOTELY THROUGH THE BAS.

PUMP CONTROL:
1. ON COMMAND TO START, P1 SHALL BE ENABLED.
2. PUMP SHALL OPERATE AT CONSTANT SPEED.

PUMP FAILURE:
1. N/A

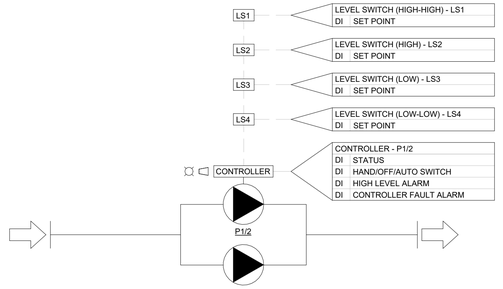
SYSTEM ALARMS & PRIORITY AT BAS:
- PUMP FAILURE : COMMANDED ON/STATUS OFF
- PUMP IN HAND: HOA SWITCH IN HAND

SYSTEM TRENDS AT BAS:
- PUMP STATUS
- EQUIPMENT OPERATING HOURS

APPLICABLE SYSTEMS:
- DRCP-1

3 23 21 00.03 - CONSTANT SPEED PUMP CONTROL SEQUENCE

SCALE: N.T.S.



OPERATING MODE:
1. THE SYSTEM SHALL BE ENABLED ACCORDING TO THE FOLLOWING MODE:
   - LEVEL SWITCH RUN COMMAND

INITIAL SET UP:
1. PUMPS SHALL BE NORMALLY DISABLED AND OPERATE IN A LEAD/LAG CONFIGURATION.
2. P1 SHALL BE THE LEAD PUMP, P2 SHALL BE THE LAG PUMP. PROVIDE AN ALTERNATING CHANGE OVER SEQUENCE FROM A LEAD/LAG OPERATION. THE SEQUENCE SHALL BE ON AN AUTOMATIC CHANGE OVER AND SHALL OCCUR AFTER EACH LEAD PUMP OPERATION TO EQUALIZE EQUIPMENT RUN TIME.
3. LS1 SHALL BE POSITIONED 150MM ABOVE LS2.
4. LS2 SHALL BE POSITIONED 150MM ABOVE LS3.
5. LS3 SHALL BE POSITIONED THREE TIMES A SINGLE PUMP CAPACITY ABOVE LS4.
6. LS4 SHALL BE POSITIONED 150MM ABOVE THE BOTTOM OF THE SUMP PIT.

SUMP PUMP CONTROL CONTROL:
1. PROVIDE STAND ALONE PACKAGE SANITARY SUMP PUMP AND CONTROL PANEL.
2. WHEN LS1 IS MADE, VISUAL INDICATOR/ ALARM SHALL BE ENABLED, LEAD/LAG PUMP SHALL BE ENABLED.
3. WHEN LS2 IS MADE, NO VISUAL INDICATOR/ALARM, LEAD/LAG PUMP SHALL BE ENABLED.
4. WHEN LS3 IS MADE, NO VISUAL INDICATOR/ALARM, LEAD PUMP SHALL BE ENABLED.
5. WHEN LS4 IS MADE, NO VISUAL INDICATOR/ALARM, LEAD/LAG PUMP SHALL BE DISABLED.

SYSTEM ALARMS & PRIORITY AT BAS:
- PUMP FAILURE : COMMANDED ON/STATUS OFF
- PUMP IN HAND: COMMANDED OFF/STATUS ON
- HIGH LEVEL ALARM
- SUMMARY ALARM FROM CONTROL PANEL

SYSTEM TRENDS AT BAS:
- STATUS

APPLICABLE SYSTEMS:
- SP-1/2
- SP-3/4

1 22 30 00.02 - SUMP PUMP CONTROL SEQUENCE

SCALE: N.T.S

No.	DATE	DESCRIPTION
3	2026-10-30	MECHANICAL ADDENDUM 01
2	2026-09-16	ISSUED FOR TENDER
1	2026-03-24	ISSUED FOR 100% CD

REVISIONS

THE SPECIFICATIONS ARE TO BE CONSIDERED AS AN INTEGRAL PART OF THESE DRAWINGS AND NEITHER THE DRAWINGS NOR THE SPECIFICATIONS SHALL BE USED ALONE. REFER TO ARCHITECTURAL DRAWINGS FOR DIMENSIONS. DO NOT SCALE.

PROJECT NORTH

QUASAR
250 ROWNTREE DAIRY RD, WOODBRIDGE, ON
TEL: 905-507-0800
WEB: WWW.QUASARCG.COM

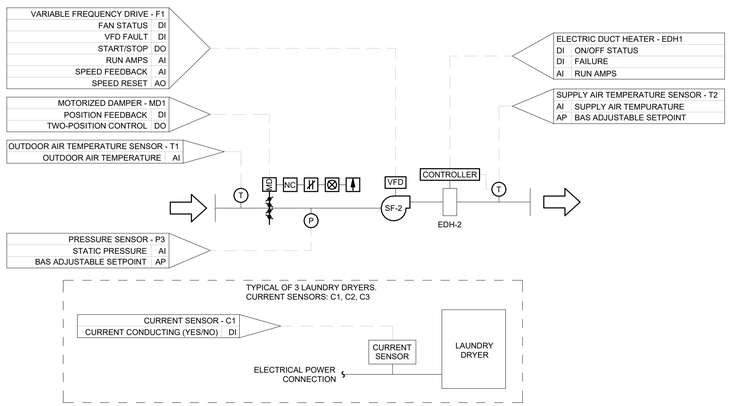
PROJECT

DURHAM REGION
409 CENTRE ST S SENIORS HOUSING
409 CENTRE ST. SOUTH, WHITBY, ON L1N 4W3
TOWN FILE NUMBER: SP-26-25

TITLE

MECHANICAL CONTROL SEQUENCES I

CHECKED	PROJECT No.
SA	RE-24-162
SCALE	DRAWING No.
N.T.S.	M-700



INITIAL SET UP:.
1. MD1 SHALL BE NORMALLY CLOSED.
2. P1 SHALL BE LOCATED APPROXIMATELY TWO-THIRDS UPSTREAM OF THE FAN. FINAL LOCATION SHALL BE COORDINATED WITH THE BALANCING CONTRACTOR AND THE CONTROLS CONTRACTOR.

SEQUENCE OF OPERATIONS:
1. ON COMMAND TO START MD1 SHALL OPEN.
2. UPON PROOF OF MOTORIZED DAMPER POSITION, FAN SHALL BE ENABLED. MD1 SHALL BE HARD WIRED INTERLOCKED WITH SF-2 TO ENSURE CONTROLS OPERATION WHEN FAN IS OPERATING IN "HAND" OR "AUTO" MODE. CONTROLS SIGNAL SHALL BE DUPLICATED AT THE BAS.
3. FAN SPEED SHALL BE MODULATED BASED ON NUMBER OF LAUNDRY DRYERS OPERATING.
   - 1 DRYER: 35% SPEED
   - 2 DRYERS: 65% SPEED
   - 3 DRYERS: 100% SPEED
4. IF OUTDOOR AIR TEMPERATURE IS LESS THAN 16°C, ELECTRIC DUCT HEATER SHALL BE ENABLED UPON CONFIRMATION OF SF-2 BEING ENABLED.
5. SUPPLY AIR TEMPERATURE SETPOINT SHALL BE SET TO 18°C (ADJUSTABLE).

FIRE ALARM MODE:.
1. ELECTRIC DUCT HEATER (EDH-2) SHALL SHUT DOWN
2. FAN SHALL SHUT DOWN.
3. MD1 SHALL REMAIN CLOSED.

FAN FAILURE:
1. UPON FAN FAILURE MD1 SHALL REMAIN CLOSED AND EDH-2 SHALL SHUTDOWN.

ELECTRIC DUCT HEATER FAILURE:
1. UPON FAILURE OF EDH-2, IF OUTDOOR AIR TEMPERATURE IS LESS THAN 4°C, FAN SHALL BE DISABLED AND MD1 SHALL REMAIN CLOSED.

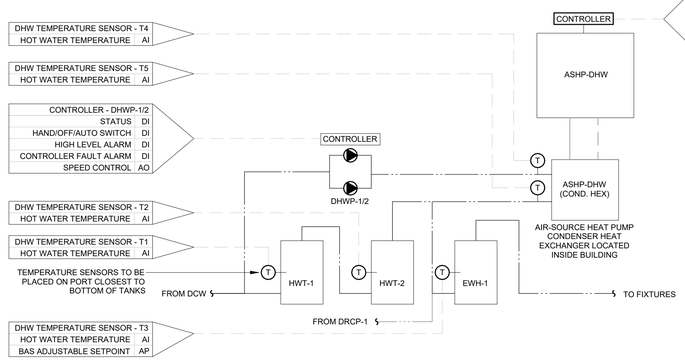
SYSTEM ALARMS & PRIORITY AT BAS:
- FAN FAILURE : COMMANDED ON/STATUS OFF
- FAN IN HAND: COMMANDED OFF/STATUS ON
- VFD FAILURE: FAULT CONTACT
- DAMPER FAILURE: COMMANDED OPEN/CLOSED FEEDBACK
- DAMPER IN HAND: COMMANDED CLOSED/OPEN FEEDBACK
- ELECTRIC DUCT HEATER FAILURE: COMMANDED ON/STATUS OFF
- ELECTRIC DUCT HEATER FAILURE: COMMANDED OFF/STATUS ON
- FREEZE STATUS: T6 IS EQUAL TO LESSER THAN 2°C

SYSTEM TRENDS AT BAS:
- FAN STATUS
- SPEED FEEDBACK
- STATIC PRESSURE
- EQUIPMENT OPERATING HOURS
- MOTORIZED DAMPER POSITION FEEDBACK
- ELECTRIC DUCT HEATER STATUS
- OUTDOOR AIR TEMPERATURE
- SUPPLY AIR TEMPERATURE
- NUMBER OF LAUNDRY DRYERS OPERATING

APPLICABLE SYSTEMS:
- SF-2

4 VARIABLE SPEED FAN SERVING LAUNDRY DRYER MAKEUP AIR - DAMPER INTERLOCK CONTROL SEQUENCE WITH DUCT HEATER

SCALE: N.T.S.



INITIAL SET UP:
1. SYSTEM SHALL BE NORMALLY DISABLED.
2. PUMPS (DHWP-1/2) SHALL OPERATE IN A LEAD/LAG CONFIGURATION. LEAD PUMP SHALL SWITCH ON A PROGRAMMABLE SCHEDULE VIA THE BAS AND SHALL OCCUR EVERY TUESDAY MORNING AT 10 AM (ADJUSTABLE) TO EQUALIZE EQUIPMENT RUN TIME.
3. DHWP-1/2 AND ASHP-DHW SHALL BE ENABLED/DISABLED REMOTELY THROUGH THE BAS.

DHW TEMPERATURE CONTROL:
1. UPON EITHER T1 OR T2 TEMPERATURES MEASURED IN THE STORAGE TANKS BEING BELOW 60°C FOR 10 MINUTES. LEAD PUMP (DHWP-1/2) SHALL BE ENABLED AND OPERATE AT FULL SPEED.
2. 2 MINUTES AFTER DHWP-1/2 IS ENABLED AND OPERATING AT FULL SPEED, ASHP-DHW SHALL BE ENABLED AND SEND SIGNAL TO THE BAS FOR SPEED CONTROL OF THE PUMPS (DHWP-1/2).
3. ONCE BOTH T1 AND T2 STORAGE TANK TEMPERATURES ARE ABOVE 62°C, ASHP-DHW SHALL BE DISABLED. DHWP-1/2 SHALL BE DISABLED 2 MINUTES AFTER ASHP-DHW IS DISABLED.
4. IF PUMP DRCP-1 IS ENABLED AND T3 TEMPERATURE INSIDE EWH-1 IS BELOW 45°C, EWH-1 SHALL BE ENABLED TO HEAT THE WATER TO A SETPOINT OF 55°C. ONCE 55°C IS SENSED BY T3, EHW-1 SHALL BE DIABLED.
5. IF PUMP DRCP-1 IS DISABLED, EWH-1 SHALL HAVE ITS SETPOINT CHANGED TO 60°C

PUMP FAILURE:
1. UPON PUMP FAILURE, LAG PUMP SHALL BE ENABLED.

AIR-SOURCE HEAT PUMP FAILURE:
1. UPON ASHP FAILURE, BAS SHALL NOT ALLOW SYSTEM TO BE ENABLED

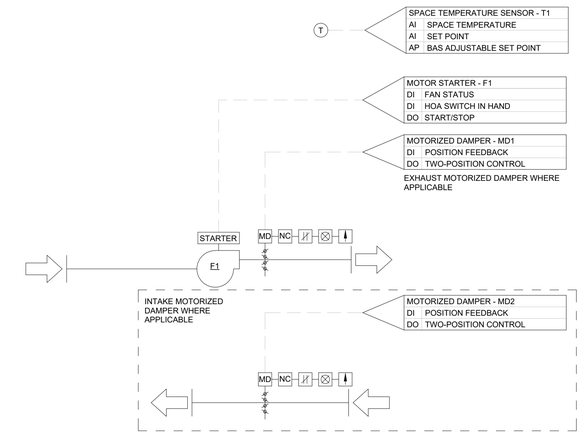
SYSTEM ALARMS & PRIORITY AT BAS:
- ASHP FAILURE : COMMANDED ON/STATUS OFF
- ASHP IN HAND: COMMANDED OFF/STATUS ON
- ASHP CONTROLLER FAULT
- PUMP FAILURE : COMMANDED ON/STATUS OFF
- PUMP IN HAND: COMMANDED OFF/STATUS ON
- PUMP CONTROLLER FAULT

SYSTEM TRENDS AT BAS:
- PUMP STATUS
- ASHP STATUS
- EWH-1 STATUS
- DHW TEMPERATURES (T1, T2, T3)
- ASHP TEMPERATURES (T4 AND T5)
- EQUIPMENT OPERATING ARRANGEMENT
- EQUIPMENT OPERATING HOURS

APPLICABLE SYSTEMS:
- ASHP-DHW
- DHWP-1/2
- EWH-1

2 DOMESTIC WATER HEATING PLANT CONTROL SEQUENCE

SCALE: N.T.S.



OPERATING MODE:
1. THE SYSTEM SHALL BE ENABLED ACCORDING TO THE FOLLOWING MODES:
   - CONTINUOUS OPERATION
   - LIGHT SWITCH
   - TIMED OPERATION
   - TEMPERATURE CONTROL

INITIAL SET UP:
1. MD1 AND MD2 SHALL BE NORMALLY CLOSED.
2. FAN SHALL BE BALANCED TO A FIELD DETERMINED AIRFLOW & STATIC PRESSURE AS PER THE VALUES INDICATED ON THE DRAWINGS DURING SYSTEM BALANCING AND COMMISSIONING.
3. SYSTEM SHALL BE ENABLED BASED ON SCHEDULED RUN TIME
4. FAN SHALL BE ENABLED/DISABLED LOCALLY AT THE STARTER OR REMOTELY THROUGH THE BAS.

FAN CONTROL:
1. ON COMMAND TO START MD1 AND MD2 SHALL OPEN (WHERE APPLICABLE). UPON PROOF OF MOTORIZED DAMPER POSITION, FAN SHALL BE ENABLED. MD1 SHALL BE HARD WIRED INTERLOCKED WITH F1 TO ENSURE CONTROLS OPERATION WHEN FAN IS OPERATING IN "HAND" OR "AUTO" MODE. CONTROLS SIGNAL SHALL BE DUPLICATED AT THE BAS.
2. FAN SHALL OPERATE AT CONSTANT SPEED.

TEMPERATURE CONTROL:
1. SF-1, EF-1
   A. UPON TEMPERATURE RISING ABOVE 26°C IN THE ELECTRICAL ROOM, FANS SHALL BE ENABLED IF OUTDOOR TEMPERATURE IS LESS THAN THE SPACE TEMPERATURE. FAN SHALL BE DISABLED IF EITHER SPACE TEMPERATURE DROPS BELOW 26°C OR OUTDOOR AIR TEMPERATURE RISES ABOVE SPACE TEMPERATURE.
   B. IF FANS HAVE NOT RUN FOR 1 HOUR, FANS SHALL BE ENABLED AND RUN FOR 10 MINUTES THEN BE DISABLED.
2. EF-2, EF-5, EF-6, EF-7
   A. IF SPACE TEMPERATURE RISES ABOVE 24°C AND OUTDOOR TEMPERATURE IS LESS THAN SPACE TEMPERATURE, FAN SHALL BE ENABLED. FAN SHALL BE DISABLED IF EITHER SPACE TEMPERATURE DROPS BELOW 24°C OR OUTDOOR AIR TEMPERATURE RISES ABOVE SPACE TEMPERATURE.
   B. IF SPACE TEMPERATURE DROPS BELOW 5°C AND OUTDOOR TEMPERATURE IS HIGHER THAN SPACE TEMPERATURE, FAN SHALL BE ENABLED. FAN SHALL BE DISABLED IF EITHER SPACE TEMPERATURE RISES ABOVE 5°C OR OUTDOOR AIR TEMPERATURE DROPS BELOW SPACE TEMPERATURE.
   C. IF FANS HAVE NOT RUN FOR 1 HOUR, FANS SHALL BE ENABLED AND RUN FOR 10 MINUTES THEN BE DISABLED.

VENTILATION CONTROL:
1. EF-2 AND EF-4 SHALL BE ENABLED WITH THE LIGHT SWITCH IN THE SPACES SERVED BY EACH FAN.

FIRE ALARM MODE:
1. FAN SHALL SHUT DOWN DURING FIRE ALARM.

FAN FAILURE:
1. UPON FAN FAILURE MD1 AND MD2 SHALL CLOSE.

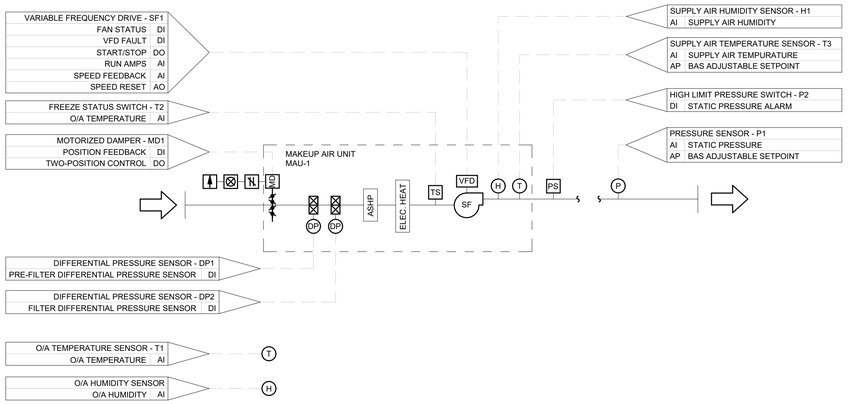
SYSTEM ALARMS & PRIORITY AT BAS:
- FAN FAILURE : COMMANDED ON/STATUS OFF
- FAN IN HAND: HOA SWITCH IN HAND
- DAMPER FAILURE: COMMANDED OPEN/CLOSED FEEDBACK
- DAMPER IN HAND: COMMANDED CLOSED/OPEN FEEDBACK

SYSTEM TRENDS AT BAS:
- FAN STATUS
- EQUIPMENT OPERATING HOURS
- POSITION FEEDBACK

APPLICABLE SYSTEMS:
- EF-1, EF-2, EF-3, EF-4, EF-5, EF-6, EF-7
- SF-1

3 CONSTANT SPEED FAN - DAMPER INTERLOCK CONTROL SEQ.

SCALE:N.T.S.



OPERATING MODE:
- THE SYSTEM SHALL RUN CONTINUOUSLY ACCORDING TO 24 HOUR SCHEDULE

INITIAL SET UP OF MAKEUP AIR UNIT (MUA-1):
1. THE SYSTEM SHALL BE A 100% OUTDOOR AIR SYSTEM.
2. DAMPERS SHALL BE BALANCED AND THEIR MINIMUM POSITIONS RECORDED TO A FIELD DETERMINED AIRFLOW AS PER THE VALUES INDICATED ON THE DRAWINGS DURING SYSTEM BALANCING AND COMMISSIONING.
3. MAKEUP AIR UNIT SHALL BE SUPPLIED WITH A BACNET/MSTP CONTROLLER CAPABLE OF INTERFACING WITH THE BAS.
4. MAKEUP AIR UNIT SHALL BE CAPABLE OF PROVIDING THE FOLLOWING FUNCTIONS:
   - MECHANICAL HEATING AND COOLING VIA AIR-SOURCE HEAT PUMP.
   - MECHANICAL HEATING VIA ELECTRIC HEATER .
5. OUTDOOR AIR MOTORIZED DAMPER (MD1) SHALL BE NORMALLY CLOSED.
6. FAN SHALL BE BALANCED TO A FIELD DETERMINED AIRFLOW & STATIC PRESSURE AS PER THE VALUES INDICATED ON THE DRAWINGS DURING SYSTEM BALANCING AND COMMISSIONING.
7. P1 SHALL BE LOCATED APPROXIMATELY TWO-THIRDS DOWNSTREAM OF THE FAN. FINAL LOCATION SHALL BE COORDINATED WITH THE BALANCING CONTRACTOR AND THE CONTROLS CONTRACTOR.
8. VFD SPEED SHALL BE FIELD OPTIMIZED TO AVOID FAN (TORQUE AND CURRENT) STALL CONDITIONS AND SHALL NOT OPERATE BELOW 30% SPEED.
9. SYSTEM SHALL BE ENABLED BASED ON SCHEDULED OPERATING MODE:
   - MONDAY-SUNDAY - 12:00am-6:00am: 50% OF AIRFLOW
   - MONDAY-SUNDAY - 6:00am-12:00am: 100% OF AIRFLOW
10. FAN SHALL BE ENABLED/DISABLED LOCALLY AT THE VFD OR REMOTELY THROUGH THE BAS.

FAN CONTROL:
1. ON COMMAND TO START OUTDOOR AIR DAMPER SHALL OPEN. UPON PROOF OF MOTORIZED DAMPER POSITION FAN SHALL BE ENABLED.
2. FAN SHALL MODULATE TO MAINTAIN THE STATIC PRESSURE SET POINT DETERMINED BY P1.

FROST PREVENTION CONTROL SEQUENCE:
1. N/A

TEMPERATURE CONTROL:
1. THE AIR HANDLING SYSTEM SHALL MAINTAIN THE FOLLOWING SUPPLY AIR TEMPERATURE SETPOINTS:
   A. COOLING MODE
      - OUTDOOR TEMPERATURE (T1) IS GREATER THAN 22°C
      - 16°C DRY BULB SUPPLY AIR TEMPERATURE
   B. HEATING MODE:
      - OUTDOOR TEMPERATURE (T1) IS LESS THAN 18°C
      - 24°C DRY BULB SUPPLY AIR TEMPERATURE
   C. IF OUTDOOR TEMPERATURE (T1) IS BETWEEN 18°C AND 22°C, SYSTEM SHALL SUPPLY OUTDOOR AIR WITHOUT ANY HEATING OR COOLING

FIRE ALARM MODE:
1. FAN SHALL SHUT DOWN DURING FIRE ALARM.

FAN FAILURE:
1. UPON SUPPLY FAN FAILURE, THE FOLLOW SHALL OCCUR:
   - MOTORIZED DAMPER MD1 SHALL BE CLOSED.

SAFETY SHUT DOWN:
1. HIGH LIMIT DUCT STATIC PRESSURE SENSOR SHALL BE HARDWIRED INTERLOCKED WITH THE SUPPLY FAN AND DISABLE THE FAN WHEN THE PRESSURE EXCEEDS 6 IN/WC.
2. FREEZE STAT T2 SHALL BE HARD WIRED INTERLOCKED WITH THE SUPPLY FAN (SF) AND MOTORIZED DAMPER (MD1). WHEN T2 DROPS BELOW 4°C, SF SHALL BE DISABLED AND MD1 SHALL BE CLOSED. FAN MUST BE MANUALLY RESET PRIOR TO RESTARTING.

SYSTEM ALARMS & PRIORITY AT BAS:
- FAN FAILURE : COMMANDED ON/STATUS OFF
- FAN IN HAND: COMMANDED OFF/STATUS ON
- FAN VFD FAILURE: FAULT CONTACT
- DAMPER FAILURE: COMMANDED ON/CLOSED FEEDBACK
- DAMPER IN HAND: COMMANDED OFF/OPEN FEEDBACK
- HIGH SUPPLY AIR TEMPERATURE: T3 IS GREATER THAN 29°C FOR MORE THAN 30 MINUTES
- LOW SUPPLY AIR TEMPERATURE: T3 IS LOWER THAN 10°C FOR MORE THAN 30 MINUTES
- FREEZE STAT: T2 IS EQUAL TO OR LOWER THAN 4°C

SYSTEM TRENDS AT BAS:
- DIRTY FILTER STATUS
- FAN STATUS
- DIFFERENTIAL PRESSURE
- SUPPLY AIR TEMPERATURE
- SUPPLY AIR HUMIDITY
- MUA CONTROLLER ANALOG VARIABLE (AV) POINTS AS AVAILABLE ON THE CONTROLLER
- MUA CONTROLLER BINARY VARIABLE (BV) POINTS AS AVAILABLE ON THE CONTROLLER

APPLICABLE SYSTEMS:
- MAU-1

1 23 75 00 - AIR-HANDLING UNIT (100% OUTDOOR AIR) - AIR-SOURCE HEAT PUMP

SCALE:N.T.S.

No.	DATE	DESCRIPTION
3	2026-10-30	MECHANICAL ADDENDUM 01
2	2026-09-16	ISSUED FOR TENDER
1	2026-03-24	ISSUED FOR 100% CD

REVISIONS

THE SPECIFICATIONS ARE TO BE CONSIDERED AS AN INTEGRAL PART OF THESE DRAWINGS AND NEITHER THE DRAWINGS NOR THE SPECIFICATIONS SHALL BE USED ALONE. REFER TO ARCHITECTURAL DRAWINGS FOR DIMENSIONS. DO NOT SCALE.

PROJECT NORTH

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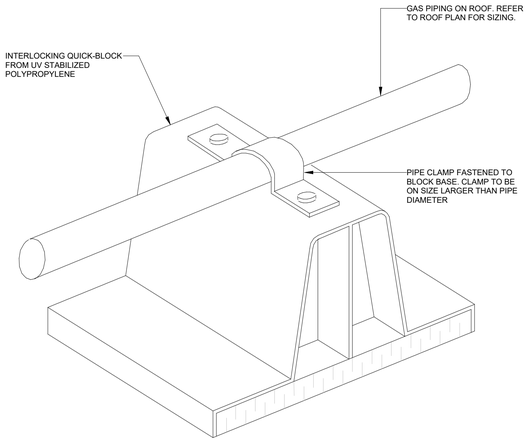
PROJECT

DURHAM REGION
409 CENTRE ST S SENIORS HOUSING
409 CENTRE ST. SOUTH, WHITBY, ON L1N 4W3
TOWN FILE NUMBER: SP-26-25

TITLE

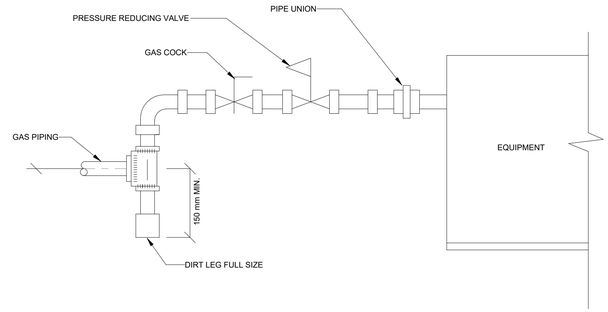
MECHANICAL CONTROL SEQUENCES II

CHECKED	PROJECT No.
SA	RE-24-162
SCALE	DRAWING No.
N.T.S.	M-701



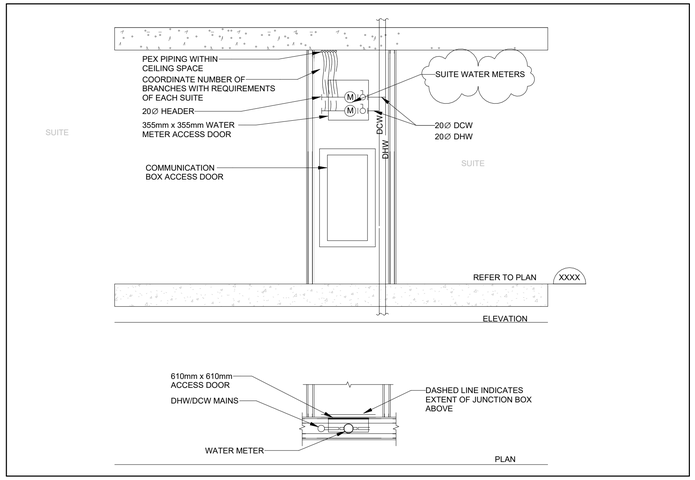
8 GAS PIPING ON ROOF - SUPPORT DETAIL

SCALE:N.T.S.



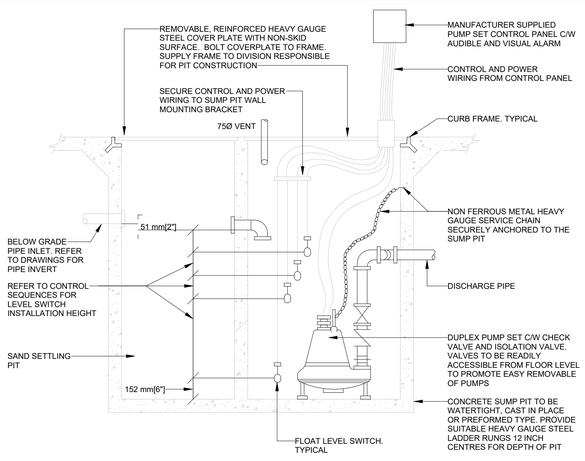
7 GAS PIPE CONNECTION TO EQUIPMENT

SCALE:N.T.S.



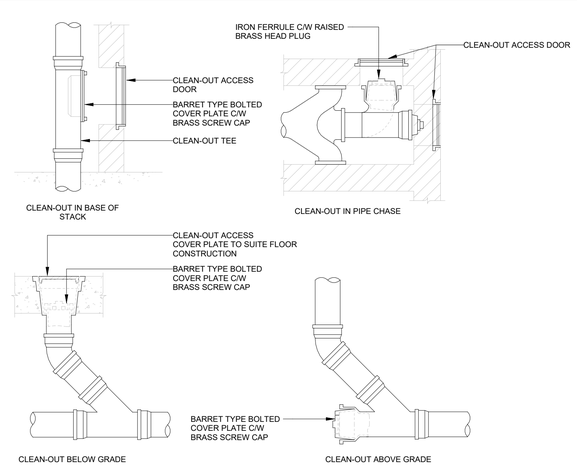
6 IN-SUITE DOMESTIC WATER METER

SCALE:N.T.S.



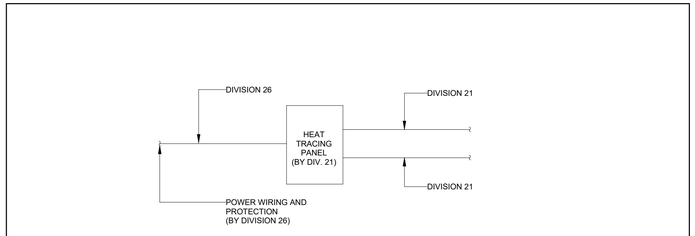
5 22 30 00.03 STORM SUMP PUMP AND SAND SETTLING PIT

SCALE:N.T.S.



9 22 13 00.02 SANITARY & STORM WATER CLEANOUTS

SCALE:N.T.S.

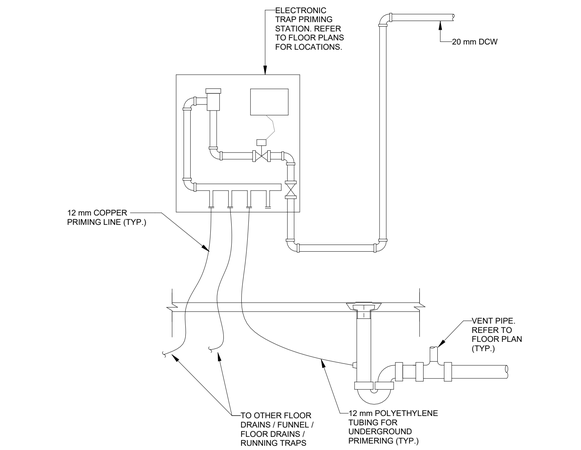


NOTES:

1. DIVISION 21 SHALL PROVIDE ALL NECESSARY DISCONNECT SWITCHES AT THE UNITS.
2. DIVISION 26 SHALL PRVOIDE THE LINE VOLTAGE / FIRE ALARM CONNECTION AT BOTH ENDS OF CABLES WHICH ARE BY DIVISION 26.
3. THE WORD BY UTILIZED ABOVE REFERS TO BOTH SUPPLY & INSTALL.

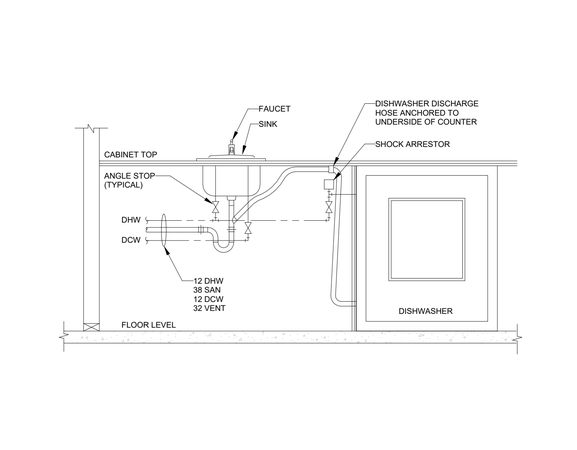
4 ELECTRIC HEAT TRACING

SCALE:N.T.S.



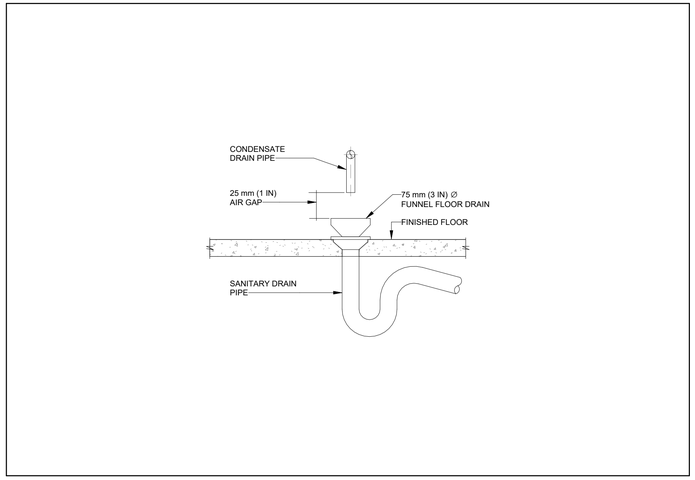
3 ELECTRONIC TRAP PRIMING STATION

SCALE:N.T.S.



2 DISHWASHER PLUMBING

SCALE:N.T.S.



1 CONDENSATE DRAIN TO FUNNEL FLOOR DRAIN

SCALE:N.T.S.

No.	DATE	DESCRIPTION
9	2026-10-30	MECHANICAL ADDENDUM 01
8	2026-09-16	ISSUED FOR TENDER
7	2026-03-24	ISSUED FOR 100% CD
6	2026-02-20	ISSUED FOR 90% CD AND CLASS A COSTING
5	2026-01-30	ISSUED FOR 75%CD
4	2026-01-30	ISSUED FOR SITE SERVICES PERMIT
3	2026-01-30	ISSUED FOR BUILDING PERMIT
2	2025-12-17	ISSUED FOR 30%CD
1	2025-12-01	RE-ISSUED FOR SPA

REVISIONS

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PROJECT NORTH

QUASAR
250 ROWNTREE DAIRY RD, WOODBRIDGE, ON
TEL: 905-507-0800
WEB: WWW.QUASARCG.COM

PROJECT

DURHAM REGION
409 CENTRE ST S SENIORS HOUSING
409 CENTRE ST. SOUTH, WHITBY, ON L1N 4W3
TOWN FILE NUMBER: SP-26-25

TITLE

PLUMBING TYPICAL DETAILS II

CHECKED	PROJECT No.
SA	RE-24-162
SCALE	DRAWING No.
N.T.S.	M-802

Project Name:	409 Centre Street	**Date Issued:**	October 30, 2026
Quasar Project #:	RE-24-162		

Distribution

Company	Name	Email
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Quasar Consulting Group	Umer Qudrat	umer.qudrat@quasarcg.com
Quasar Consulting Group	Justin Purdy	justin.purdy@quasarcg.com

Addendum #:	E-01
Revision #:	00

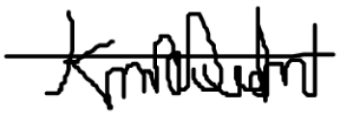
This Addendum forms part of the Contract Specifications and Drawings, and modifies the Bidding Documents, with Amendments and Additions noted below. This Addendum shall be added to the front of the specifications as issued. Bidders shall acknowledge receipt of this Addendum in the space provided in the Bid Form and include in bid amount.

This addendum includes modifications to the drawings as summarized below. Unless otherwise noted, all drawings listed below are attached herewith.

Changes to Drawings:

1. E-303 POWER AND SYSTEMS NEW WORK PLAN – LEVEL 3
    a. Added direct connection for branch selector box.
2. E-304 POWER AND SYSTEMS NEW WORK PLAN - ROOF
    a. Revised location for CU-3 and added CU-4.
3. E-700 SINGLE LINE DIAGRAM
    a. Revised feeder for RP-BC.
4. E-932 ELECTRICAL SCHEDULES – POWER SYSTEMS
    a. Added circuit for branch selector.
5. E-933 ELECTRICAL SCHEDULES – POWER SYSTEMS
    a. Revised RP-2M circuits for CU-3 and added CU-4.

Quasar Consulting Group



Umer Qudrat

Electrical Engineer

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5	2026-02-20	ISSUED FOR 90% CD AND CLASS A COSTING
4	2026-01-30	ISSUED FOR 75%CD
3	2026-01-30	ISSUED FOR SITE SERVICES PERMIT
2	2026-01-30	ISSUED FOR BUILDING PERMIT
1	2025-12-17	ISSUED FOR 30%CD

REVISIONS

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WEB: WWW.QUASARCG.COM

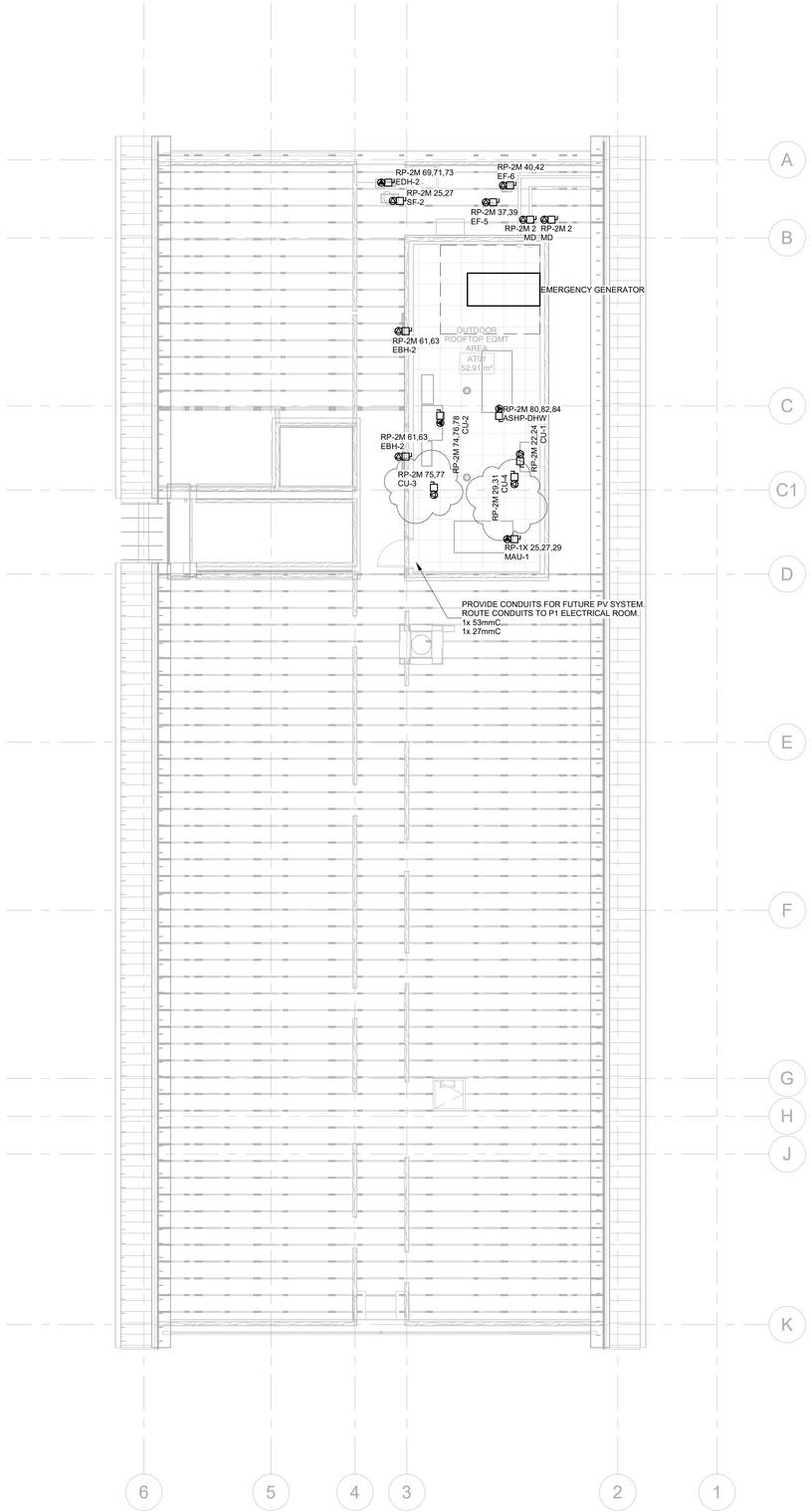
PROJECT

DURHAM REGION
409 CENTRE ST S SENIORS HOUSING
409 CENTRE ST. SOUTH, WHITBY, ON L1N 4W3
TOWN FILE NUMBER: SP-26-25

TITLE

POWER & SYSTEMS NEW WORK PLAN - LEVEL 3

CHECKED	PROJECT No.
EJG	RE-24-162
SCALE	DRAWING No.
1 : 100	E-303



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3	2026-01-30	ISSUED FOR SITE SERVICES PERMIT
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1	2025-12-17	ISSUED FOR 30%CD

REVISIONS

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PROJECT

DURHAM REGION
409 CENTRE ST S SENIORS HOUSING
409 CENTRE ST. SOUTH, WHITBY, ON L1N 4W3
TOWN FILE NUMBER: SP-26-25

TITLE

POWER & SYSTEMS NEW WORK PLAN - ROOF

CHECKED	PROJECT No.
EJG	RE-24-162
SCALE	DRAWING No.
1 : 100	E-304

SLD WIRE FEEDER SCHEDULE

NUMBER OF WIRES	FEEDER ID	# OF RUNS	CONDUCTOR AND GROUNDING SIZE	CONDUIT SIZE	CONDUCTOR TYPE	AMPACITY
3W	A100	1	3 # 1 AWG + # 4 AWG BND	53mm	ALUMINUM Table6A	100
3W	A230	<varies>	<varies>	63mm	ALUMINUM Table6A	<varies>
3W	A270	1	3/C 400 KCMIL + 2 AWG BND	78mm	ALUMINUM Table6A	270
3W	A310	1	3 # 500 MCM + # 1 AWG BND	63mm	ALUMINUM Table6A	310
3W	A385	1	3/C 750 KCMIL + 1 AWG BND	91mm	ALUMINUM Table6A	385
3W	A410	2	3/C 250 KCMIL + 2 AWG BND	63mm	ALUMINUM Table6A	410
3W	A445	1	3/C 1000 KCMIL + 1/0 AWG BND	103mm	ALUMINUM Table6A	445
3W	A615	3	3/C 250 KCMIL + 2 AWG BND	63mm	ALUMINUM Table6A	615
3W	A820	4	3/C 250 KCMIL + 2 AWG BND	63mm	ALUMINUM Table6A	820
3W	A1080	4	3/C 400 KCMIL + 2 AWG BND	78mm	ALUMINUM Table6A	1080
3W	A1240	4	3/C 500 KCMIL + 1 AWG BND	78mm	ALUMINUM Table6A	1240
3W	C20	1	3 # 12 AWG + # 12 AWG BND	21mm	COPPER Table6A	20
3W	C25	1	3/C 12 AWG + 12 AWG BND	16mm	COPPER Table6A	25
3W	C35	1	3/C 10 AWG + 10 AWG BND	16mm	COPPER Table6A	35
3W	C40	1	<varies>	<varies>	<varies>	<varies>
3W	C50	1	3/C 8 AWG + 10 AWG BND	21mm	COPPER Table6A	50
3W	C65	1	3/C 6 AWG + 8 AWG BND	27mm	COPPER Table6A	65
3W	C70	1	3 # 4 AWG + # 8 AWG BND	35mm	COPPER Table6A	07
3W	C85	1	3 # 3 AWG + # 8 AWG BND	41mm	COPPER Table6A	85
3W	C100	1	3/C 3 AWG + 8 AWG BND	35mm	COPPER Table6A	100
3W	C115	1	3/C 2 AWG + 6 AWG BND	35mm	COPPER Table6A	115
3W	C130	1	3/C 1 AWG + 6 AWG BND	41mm	COPPER Table6A	130
3W	C150	1	3/C 1/0 AWG + 6 AWG BND	41mm	COPPER Table6A	150
3W	C175	1	3/C 2/0 AWG + 6 AWG BND	53mm	COPPER Table6A	175
3W	C200	1	3/C 3/0 AWG + 6 AWG BND	53mm	COPPER Table6A	200
3W	C230	1	3/C 4/0 AWG + 4 AWG BND	53mm	COPPER Table6A	230
3W	C255	1	3/C 250 KCMIL + 4 AWG BND	53mm	COPPER Table6A	255
3W	C285	1	3/C 300 KCMIL + 4 AWG BND	63mm	COPPER Table6A	285
3W	C310	1	3/C 350 KCMIL + 3 AWG BND	63mm	COPPER Table6A	310
3W	C335	1	3/C 400 KCMIL + 3 AWG BND	78mm	COPPER Table6A	335
3W	C350	2	3/C 2/0 AWG + 6 AWG BND	53mm	COPPER Table6A	350
3W	C380	1	3/C 500 KCMIL + 3 AWG BND	78mm	COPPER Table6A	380
3W	C400	2	3/C 3/0 AWG + 6 AWG BND	53mm	COPPER Table6A	400
3W	C420	1	3/C 600 KCMIL + 2 AWG BND	78mm	COPPER Table6A	420
3W	C460	2	3/C 4/0 AWG + 4 AWG BND	53mm	COPPER Table6A	460
3W	C510	2	3/C 250 KCMIL + 4 AWG BND	53mm	COPPER Table6A	510
3W	C570	2	3/C 300 KCMIL + 4 AWG BND	63mm	COPPER Table6A	570
3W	C620	2	3/C 350 KCMIL + 3 AWG BND	63mm	COPPER Table6A	620
3W	C670	2	3/C 400 KCMIL + 3 AWG BND	78mm	COPPER Table6A	670
3W	C690	3	3/C 4/0 AWG + 4 AWG BND	53mm	COPPER Table6A	690
3W	C765	3	3/C 250 KCMIL + 4 AWG BND	53mm	COPPER Table6A	765
3W	C920	4	3/C 4/0 AWG + 4 AWG BND	53mm	COPPER Table6A	920
3W	C1005	3	3/C 400 KCMIL + 3 AWG BND	78mm	COPPER Table6A	1005
3W	C1020	4	3/C 250 KCMIL + 4 AWG BND	53mm	COPPER Table6A	1020
3W	C1275	5	3/C 250 KCMIL + 4 AWG BND	53mm	COPPER Table6A	1275

SLD WIRE FEEDER SCHEDULE

NUMBER OF WIRES	FEEDER ID	# OF RUNS	CONDUCTOR AND GROUNDING SIZE	CONDUIT SIZE	CONDUCTOR TYPE	AMPACITY
3W	C1340	4	3/C 400 KCMIL + 3 AWG BND	78mm	COPPER Table6A	1340
3W	C1520	4	3/C 500 KCMIL + 3 AWG BND	78mm	COPPER Table6A	1520
4W	A100	<varies>	<varies>	<varies>	ALUMINUM Table6A	<varies>
4W	A120	1	<varies>	<varies>	ALUMINUM Table6A	<varies>
4W	A180	1	4 #4/0 KCMIL + #1 AWG BND	53mm	ALUMINUM Table6A	180
4W	A230	1	<varies>	<varies>	ALUMINUM Table6A	<varies>
4W	A270	1	4/C 400 KCMIL + 2 AWG BND	78mm	ALUMINUM Table6A	270
4W	A385	1	4/C 750 KCMIL + 1 AWG BND	103mm	ALUMINUM Table6A	385
4W	A410	2	4/C 250 KCMIL + 2 AWG BND	63mm	ALUMINUM Table6A	410
4W	A445	1	4/C 1000 KCMIL + 1/0 AWG BND	129mm	ALUMINUM Table6A	445
4W	A615	3	4/C 250 KCMIL + 2 AWG BND	63mm	ALUMINUM Table6A	615
4W	A620	2	4 # 500 MCM + # 1 AWG BND	103mm	ALUMINUM Table6A	620
4W	A820	4	4/C 250 KCMIL + 2 AWG BND	63mm	ALUMINUM Table6A	820
4W	A1080	4	4/C 400 KCMIL + 2 AWG BND	78mm	ALUMINUM Table6A	1080
4W	A1240	4	4/C 500 KCMIL + 1 AWG BND	91mm	ALUMINUM Table6A	1240
4W	C25	1	4/C 12 AWG + 12 AWG BND	16mm	COPPER Table6A	25
4W	C35	1	4/C 10 AWG + 10 AWG BND	21mm	COPPER Table6A	35
4W	C50	1	4/C 8 AWG + 10 AWG BND	27mm	COPPER Table6A	50
4W	C65	1	4/C 6 AWG + 8 AWG BND	27mm	COPPER Table6A	65
4W	C70	1	4 # 4 AWG + # 8 AWG BND	35mm	COPPER Table6A	70
4W	C85	1	4/C 4 AWG + 8 AWG BND	35mm	COPPER Table6A	85
4W	C100	1	4/C 3 AWG + 8 AWG BND	35mm	COPPER Table6A	100
4W	C115	1	4/C 2 AWG + 6 AWG BND	41mm	COPPER Table6A	115
4W	C130	1	4/C 1 AWG + 6 AWG BND	41mm	COPPER Table6A	130
4W	C150	1	4/C 1/0 AWG + 6 AWG BND	53mm	COPPER Table6A	150

SLD WIRE FEEDER SCHEDULE

NUMBER OF WIRES	FEEDER ID	# OF RUNS	CONDUCTOR AND GROUNDING SIZE	CONDUIT SIZE	CONDUCTOR TYPE	AMPACITY
4W	C175	1	4/C 2/0 AWG + 6 AWG BND	53mm	COPPER Table6A	175
4W	C200	1	4/C 3/0 AWG + 6 AWG BND	53mm	COPPER Table6A	200
4W	C230	1	4/C 4/0 AWG + 4 AWG BND	63mm	COPPER Table6A	230
4W	C255	1	4/C 250 KCMIL + 4 AWG BND	63mm	COPPER Table6A	255
4W	C285	1	4/C 300 KCMIL + 4 AWG BND	78mm	COPPER Table6A	285
4W	C310	1	4/C 350 KCMIL + 3 AWG BND	78mm	COPPER Table6A	310
4W	C335	1	4/C 400 KCMIL + 3 AWG BND	78mm	COPPER Table6A	335
4W	C350	2	4/C 2/0 AWG + 6 AWG BND	53mm	COPPER Table6A	350
4W	C380	1	4/C 500 KCMIL + 3 AWG BND	91mm	COPPER Table6A	380
4W	C400	2	4/C 3/0 AWG + 6 AWG BND	53mm	COPPER Table6A	400
4W	C420	1	4/C 600 KCMIL + 2 AWG BND	91mm	COPPER Table6A	420
4W	C460	2	4/C 4/0 AWG + 4 AWG BND	63mm	COPPER Table6A	460
4W	C510	2	4/C 250 KCMIL + 4 AWG BND	63mm	COPPER Table6A	510
4W	C570	2	4/C 300 KCMIL + 4 AWG BND	78mm	COPPER Table6A	570
4W	C620	2	4/C 350 KCMIL + 3 AWG BND	78mm	COPPER Table6A	620
4W	C670	2	4/C 400 KCMIL + 3 AWG BND	78mm	COPPER Table6A	670
4W	C690	3	4/C 4/0 AWG + 4 AWG BND	63mm	COPPER Table6A	690
4W	C765	3	4/C 250 KCMIL + 4 AWG BND	63mm	COPPER Table6A	765
4W	C920	4	4/C 4/0 AWG + 4 AWG BND	63mm	COPPER Table6A	920
4W	C1005	3	4/C 400 KCMIL + 3 AWG BND	78mm	COPPER Table6A	1005
4W	C1020	4	4/C 250 KCMIL + 4 AWG BND	63mm	COPPER Table6A	1020
4W	C1275	5	4/C 250 KCMIL + 4 AWG BND	63mm	COPPER Table6A	1275
4W	C1340	4	4/C 400 KCMIL + 3 AWG BND	78mm	COPPER Table6A	1340
4W	C1520	4	4/C 500 KCMIL + 3 AWG BND	91mm	COPPER Table6A	1520

NUMBER OF WIRES
3W = 3 WIRES
4W = 4 WIRES

3W A100

FEEDER ID

CONDUCTOR TYPE
C = COPPER
A = ALUMINIUM

GENERAL NOTES:

1. EACH RESIDENTIAL UNITIS TO BE EQUIPPED WITH ITS OWN ELECTRICAL SUB-METERING. ELECTRICAL SUBMETERING FOR THE UNITS TO BE LOCATED IN CENTRAL METERING CABINETS, AND NOT INSIDE EACH UNIT.
2. ELECTRICAL CONTRACTOR IS TO PROVIDE METERING SYSTEM SHOP DRAWING FOR REGION OF DURHAM AND QUASARS REVIEW INCLUDING A METERING PLAN LISTING ALL METERS ALONG WITH TYPE, ENERGY SOURCE METERED, DIAGRAMS, AND/OR REFERENCES TO DESIGN DOCUMENTATION.

NOTE: THE EMERGENCY BACKUP GENERATOR WILL NOT BE USED FOR PARALLEL OPERATION WITH THE GRID OR THAT IT WILL NOT BACKFEED INTO THE GRID IN CASE OF GRID FAILURE.

BASIS OF DESIGN: LMR
PROVIDE GENERATOR WITH FLOATED NEUTRAL POINT. PROVIDE GROUNDING JUMPERS FOR INTERCONNECTION TO THE GROUNDING SYSTEM AS REQUIRED.

DRAWING KEY NOTES:

1. PROVIDE 2 HOUR FIRE RATED CONDUCTORS FOR ALL LIFE-SAFETY LOADS.
   4#300KCMIL + 1#4GND IN 63mmC
   2x (4#300KCMIL + 1#4GND IN 63mmC)
2. BREAKER SIZE SHALL BE CONFIRMED BY DIVISION 16 TO SUIT THE DATA SUPPLIED BY THE SELECTED ELEVATOR CONTRACTOR.
3. INSTANTANEOUS PROTECTION SETTING SHALL BE COORDINATED WITH TRANSFORMER INRUSH CURRENT.

ELECTRICAL DISTRIBUTION DESIGNATION DIAGRAM

DP-1XA-11.1

FLOOR LEVEL
P1->P6 DENOTE PARKING LEVELS
PH DENOTES PENTHOUSE

LIGHTING CIRCUIT NUMBER/ZONE IDENTIFIER

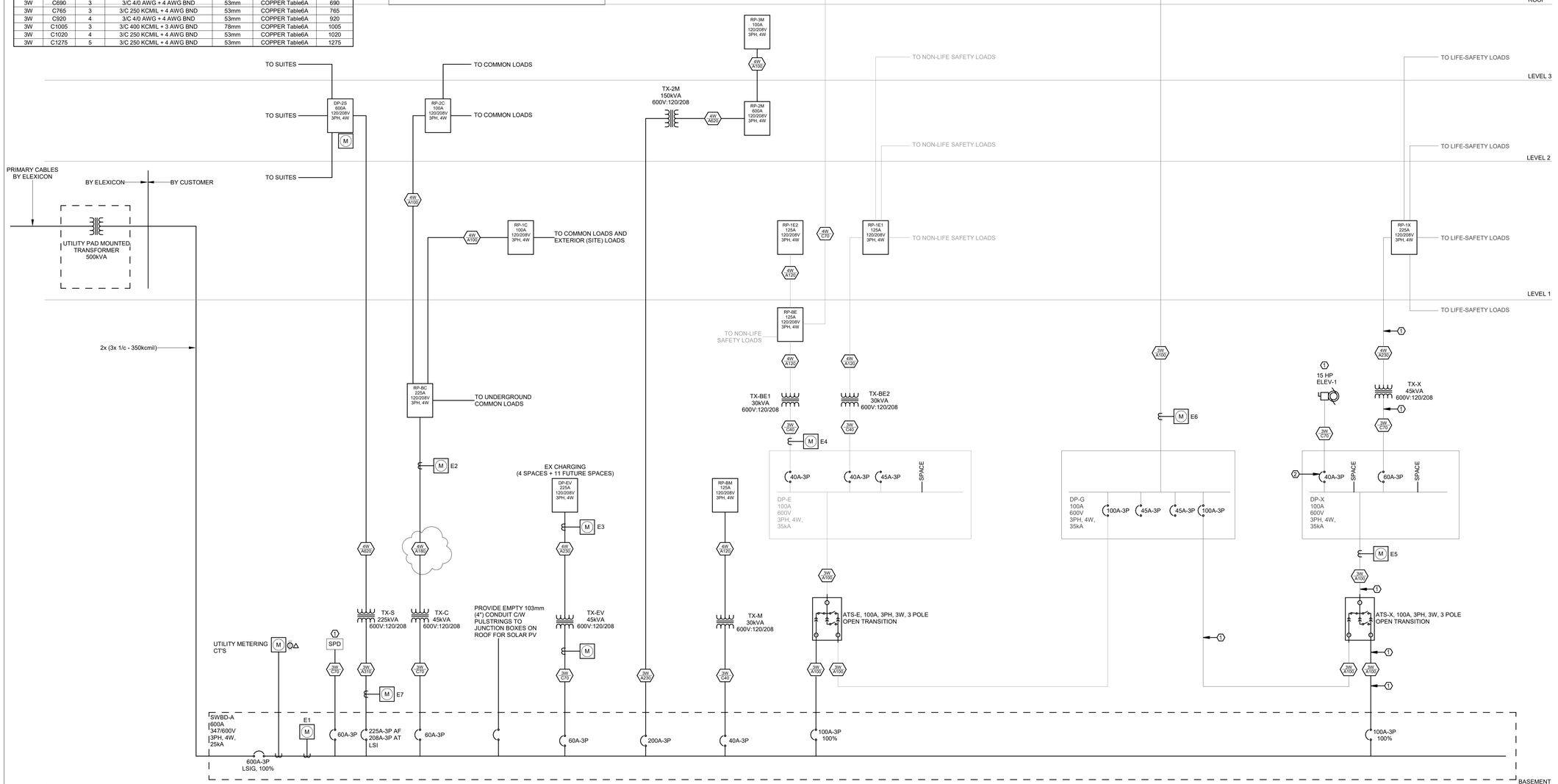
CIRCUIT NUMBER

PANEL NUMBER (SEQUENTIALLY LETTERED).
"O" RESERVED FOR EXTERIOR "OUTSIDE" LIGHTING.

SOURCE TYPE
BLANK = NORMAL (UTILITY) POWER
G = GENERATOR (ONLY LIVE WHEN GENERATOR RUNNING)
X = LIFE SAFETY EMERGENCY POWER
E = NON-LIFE SAFETY STANDBY POWER
M = MECHANICAL POWER

PANEL OR EQUIPMENT TYPE:
ATS - AUTOMATIC TRANSFER SWITCH
DP - 600 V DISTRIBUTION PANELBOARD
G - GENERATOR
LP - 120/208 V LIGHTING PANELBOARD
PP - 120/208 V DISTRIBUTION PANELBOARD
RP - 120/208 V RECEPTACLE PANELBOARD
SP - SPLITTER
SPD - SURGE PROTECTIVE DEVICE
SWBD6 - 600 V SWITCHBOARD
SWBD2 - 120/208 V SWITCHBOARD
SWGR - 13.8 kV SWITCHGEAR
TX - TRANSFORMER

NOTE: "DP", "LP", "RP" MAY BE OMITTED FROM CIRCUIT NUMBERS ON PLANS.



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PROJECT

DURHAM REGION
409 CENTRE ST S SENIORS HOUSING
409 CENTRE ST. SOUTH, WHITBY, ON L1N 4W3
TOWN FILE NUMBER: SP-26-25

TITLE

SINGLE LINE DIAGRAM

CHECKED	PROJECT No.
EJG	RE-24-162
SCALE	DRAWING No.
1 : 1	E-700

Branch Panel: RP-1E1

Location: ELEC 117
Supply From:
Mounting: Surface
Enclosure: Type 1

Volts: 120/208 Wye
Phases: 3
Wires: 4

A.I.C. Rating:
Mains Type:
Mains Rating: 125 A
MCB Rating: 100 A

Notes:

CKT	Circuit Description	Trip	Poles	A	B	C	A	B	C	Poles	Trip	Circuit Description	CKT
1	R: MECHANICAL RM L2	15 A	1	540 VA			180 VA			1	15 A	R: ELEC CLOSET L2	2
3	R: STAIRS A L2 & L3	15 A	1		720 VA			360 VA		1	15 A	R: ELEV CONTROL RM L2	4
5	R: STAIRS B L2 & L3	15 A	1			720 VA			180 VA	1	15 A	R: ELEC CLOSET L3	6
7							540 VA			1	15 A	R: MECHANICAL RM L3	8
9													10
11									420 VA	1	15 A	P: ADO SUITES - 203,206,207 L2	12
13	P: ADO SUITES - 110,111,112 L1	15 A	1	420 VA			420 VA			1	15 A	P: ADO SUITES - 209,210,211 L2	14
15	P: ADO SUITES - 113,114,115,116...	15 A	1		560 VA			420 VA		1	15 A	P: ADO SUITES - 212,213,214 L2	16
17	P: ADO SUITES - 303,306,307 L3	15 A	1			420 VA			280 VA	1	15 A	P: ADO SUITES - 215,216 L2	18
19	P: ADO SUITES - 309,310,311 L3	15 A	1	420 VA			0 VA			1	20 A	SPARE	20
21	P: ADO SUITES - 312,313,314 L3	15 A	1		420 VA			0 VA		1	20 A	SPARE	22
23	P: ADO SUITES - 315,316 L3	15 A	1			280 VA			0 VA	1	20 A	SPARE	24
25	SPARE	20 A	1	0 VA			0 VA			2	20 A	SPARE	26
27	SPARE	20 A	1		0 VA			0 VA					28
29	R: COMM REC - ELEC CLST L3	15 A	1			360 VA			360 VA	1	15 A	R: COMM REC - ELEC CLST L3	30
31	P: ADO - L1 UNI WR & STAIRS A	15 A	1	500 VA			1000 VA			1	15 A	P: ADO WASTE & AMENITY L1	32
33	P: ADO - L1 CORRIDOR & STAIR...	15 A	1		500 VA			1000 VA		1	15 A	P: ADO - L1 VEST. & LOBBY	34
35	R: STAIRS B & ELEC CLOSET L1	15 A	1			540 VA							36
37							0 VA						38
39								0 VA		3	30 A	SPD	40
41									0 VA				42
		Total Load:		4020 VA			3980 VA		3560 VA				
		Total Amps:		34 A			34 A		30 A				

Legend:

Load Classification	Connected Load	Demand Factor	Estimated Demand	Panel Totals	
Other	3000 VA	100.00%	3000 VA		
Spare	4060 VA	100.00%	4060 VA	**Total Conn. Load:**	11560 VA
Receptacle	4500 VA	100.00%	4500 VA	**Total Est. Demand:**	11560 VA
				Total Conn.:	32 A
				Total Est. Demand:	32 A

Notes:

Branch Panel: RP-1E2

Location: STORAGE 106
Supply From: RP-BE
Mounting: Surface
Enclosure: Type 1

Volts: 120/208 Wye
Phases: 3
Wires: 4

A.I.C. Rating:
Mains Type:
Mains Rating: 125 A
MCB Rating: 100 A

Notes:

CKT	Circuit Description	Trip	Poles	A	B	C	A	B	C	Poles	Trip	Circuit Description	CKT
1	R: RACK - AMENITY L1	15 A	1	180 VA			750 VA			2	20 A	P: WALL OVEN AMENITY 107 L1	2
3	R: AMENITY 107 - L1	15 A	1		720 VA			750 VA					4
5	RECEPTACLE	20 A	1			180 VA			180 VA	1	15 A	R: FRIDGE AMENITY 107 - L1	6
7	R: TV AMENITY 107 - L1	15 A	1	900 VA			1250 VA			2	35 A	P: ASHP-2 L1 UNIVERSAL WR	8
9	P: ADO - L1 LAUNDRY, AMENITY,...	20 A	1		1000 VA			1250 VA					10
11	R: MOTORIZED CURTAINS - L1...	15 A	1			900 VA			1250 VA	2	35 A	P: ASHP-1 L1 LAUNDRY RM	12
13	P: COOKTOP - L1 AMENITY 107	15 A	2	750 VA			1250 VA						14
15					750 VA			180 VA		1	15 A	R: KITCHEN COUNTER - AMENIT...	16
17	R: RANGEHOOD - L1 AMENITY 107	15 A	1			180 VA			180 VA	1	20 A	R: KITCHEN COUNTER GFCI -...	18
19	P: ADO - L3 WASHROOM	15 A	1	160 VA									20
21	P: 120VC FOR SECURITY PANELS	15 A	1		500 VA								22
23	P: 120VC FOR SECURITY PANELS	15 A	1			500 VA							24
25													26
27								0 VA					28
29									0 VA	3	30 A	SPD	30
31							0 VA						32
33													34
35													36
37													38
39													40
41													42
		Total Load:		5240 VA			5150 VA		3370 VA				
		Total Amps:		46 A			45 A		28 A				

Legend:

Load Classification	Connected Load	Demand Factor	Estimated Demand	Panel Totals	
Other	10160 VA	100.00%	10160 VA		
Receptacle	3600 VA	100.00%	3600 VA	**Total Conn. Load:**	13760 VA
				Total Est. Demand:	13760 VA
				Total Conn.:	38 A
				Total Est. Demand:	38 A

Notes:

Branch Panel: RP-3M

Location: MECH. RM 305
Supply From: RP-2M
Mounting: Surface
Enclosure: Type 1

Volts: 120/208 Wye
Phases: 3
Wires: 4

A.I.C. Rating:
Mains Type:
Mains Rating: 100 A
MCB Rating:

Notes:

CKT	Circuit Description	Trip	Poles	A	B	C	A	B	C	Poles	Trip	Circuit Description	CKT
1	P: IDU-11 MECHANICAL RM L3	15 A	2	33 VA			150 VA			2	15 A	P: DRCP-1 MECH RM L3	2
3					33 VA			150 VA					4
5	P; DHWP-2 MECH RM L3	15 A	2			187 VA			500 VA	2	15 A	P: EBH - 2 MECH RM L3	6
7				187 VA			500 VA						8
9	P; DHWP-1 MECH RM L3	15 A	2		187 VA			500 VA		2	15 A	P: EBH-1 LOBBY 300 L3	10
11						187 VA			500 VA				12
13	P: EBH - 2 MECH RM L3	15 A	2	500 VA			250 VA			2	15 A	P: EBH-1 CORRIDOR L3	14
15					500 VA			250 VA					16
17	P: EBH-1 CORRIDOR L3	15 A	2			250 VA			1000 VA	1	15 A	P: MOTORIZED DAMPER - L3	18
19				250 VA									20
21	P: EBH-1 STAFF RM L3	15 A	2		250 VA								22
23						250 VA							24
25													26
27													28
29													30
31													32
33													34
35													36
37													38
39													40
41													42
		Total Load:		1869 VA			1869 VA		2873 VA				
		Total Amps:		16 A			16 A		24 A				

Legend: P: POWER R: RECEPTACLE L: LIGHTING

Load Classification	Connected Load	Demand Factor	Estimated Demand	Panel Totals	
Other	6611 VA	100.00%	6611 VA		
				Total Conn. Load:	6611 VA
				Total Est. Demand:	6611 VA
				Total Conn.:	18 A
				Total Est. Demand:	18 A

Notes:

Branch Panel: RP-2C

Location: Space 107
Supply From: RP-BC
Mounting: Surface
Enclosure: Type 1

Volts: 120/208 Wye
Phases: 3
Wires: 4

A.I.C. Rating:
Mains Type:
Mains Rating: 100 A
MCB Rating: 100 A

Notes:

CKT	Circuit Description	Trip	Poles	A	B	C	A	B	C	Poles	Trip	Circuit Description	CKT
1	L: L2 MECHANICAL RM	15 A	1	18 VA			92 VA			1	15 A	L: L3 CORRIDOR	2
3	L: L2 CORRIDOR	15 A	1		200 VA			50 VA		1	15 A	L: L3 STAFF WASHROOM	4
5	L: L2 HOUSEKEEPING	15 A	1			18 VA			50 VA	1	15 A	L: L3 STAFF LOCKER	6
7	L: L3 MECHANICAL RM	15 A	1	18 VA			0 VA			1	20 A	SPARE	8
9	R: L2 LOBBY, CORRIDOR	15 A	1		1080 VA			0 VA		1	20 A	SPARE	10
11	R: L3 LOBBY, CORRIDOR	15 A	1			900 VA			0 VA	1	20 A	SPARE	12
13	SPARE	15 A	1	0 VA			0 VA			1	20 A	SPARE	14
15	SPARE	15 A	1		0 VA			0 VA		1	20 A	SPARE	16
17	SPARE	15 A	1			0 VA			135 VA	1	20 A	LIGHTING	18
19	SPARE	15 A	1	0 VA									20
21	SPARE	15 A	1		0 VA								22
23	BRANCH SELECTOR	15 A	2			80 VA							24
25				80 VA									26
27													28
29						0 VA							30
31	SPD	30 A	3	0 VA									32
33					0 VA								34
35													36
37													38
39													40
41													42
		Total Load:		207 VA			1330 VA		1183 VA				
		Total Amps:		2 A			12 A		11 A				

Legend:

Load Classification	Connected Load	Demand Factor	Estimated Demand	Panel Totals	
Other	160 VA	100.00%	160 VA		
Receptacle	1980 VA	100.00%	1980 VA	**Total Conn. Load:**	2719 VA
				Total Est. Demand:	2719 VA
				Total Conn.:	8 A
				Total Est. Demand:	8 A

Notes:

No.	DATE	DESCRIPTION
10	2026-10-30	ELECTRICAL ADDENDUM 01
9	2026-09-16	ISSUED FOR TENDER
8	2026-06-12	ISSUED FOR TENDER PROGRESS
7	2026-03-24	ISSUED FOR 100% CD
6	2026-02-27	ISSUED FOR ESA PLAN REVIEW
5	2026-02-20	ISSUED FOR 90% CD AND CLASS A COSTING
4	2026-01-30	ISSUED FOR 75%CD
3	2026-01-30	ISSUED FOR SITE SERVICES PERMIT
2	2026-01-30	ISSUED FOR BUILDING PERMIT
1	2025-12-17	ISSUED FOR 30%CD

REVISIONS

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PROJECT NORTH

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PROJECT

DURHAM REGION
409 CENTRE ST S SENIORS HOUSING
409 CENTRE ST. SOUTH, WHITBY, ON L1N 4W3
TOWN FILE NUMBER: SP-26-25

TITLE

ELECTRICAL SCHEDULES - POWER & SYSTEMS

CHECKED	PROJECT No.
EJG	RE-24-162
SCALE	DRAWING No.
N.T.S.	E-932

Branch Panel: DP-2S

Location: Space 107
Supply From:
Mounting: Surface
Enclosure: Type 1

Volts: 120/208 Wye
Phases: 3
Wires: 4

A.I.C. Rating:
Mains Type:
Mains Rating: 600 A
MCB Rating: 600 A

Notes:

CKT	Circuit Description	Trip	Poles	A	B	C	Poles	Trip	Circuit Description	CKT
1	SUITE - 103 - L1	100 A	2	0 VA			2	100 A	SUITE - 110 - L1	2
3					0 VA					4
5	SUITE - 111 - L1	100 A	2			0 VA	2	100 A	SUITE - 112 - L1	6
7				0 VA						8
9	SUITE - 113 - L1	100 A	2		0 VA		2	100 A	SUITE - 114 - L1	10
11						0 VA				12
13	SUITE - 115 - L1	100 A	2	0 VA			2	100 A	SUITE - 116 - L1	14
15					0 VA					16
17	SUITE - 203 - L2	100 A	2			0 VA	2	100 A	SUITE - 206 - L2	18
19				0 VA						20
21	SUITE - 207 - L2	100 A	2		0 VA		2	100 A	SUITE - 209 - L2	22
23						0 VA				24
25	SUITE - 210 - L2	100 A	2	0 VA			2	100 A	SUITE - 211 - L2	26
27					0 VA					28
29	SUITE - 212 - L2	100 A	2			0 VA	2	100 A	SUITE - 213 - L2	30
31				0 VA						32
33	SUITE - 214 - L2	100 A	2		0 VA		2	100 A	SUITE - 215 - L2	34
35						0 VA				36
37	SUITE - 216 - L2	100 A	2	0 VA			2	100 A	SUITE - 306 - L3	38
39					0 VA					40
41	SUITE - 303 - L3	100 A	2			0 VA	2	100 A	SUITE - 307 - L3	42
43				0 VA						44
45	SUITE - 309 - L3	100 A	2		0 VA		2	100 A	SUITE - 310 - L3	46
47						0 VA				48
49	SUITE - 311 - L3	100 A	2	0 VA			2	100 A	SUITE - 312 - L3	50
51					0 VA					52
53	SUITE - 313 - L3	100 A	2			0 VA	3	15 A	METER	54
55				0 VA						56
57	SUITE - 314 - L3	100 A	2		0 VA					58
59						0 VA				60
61	SUITE - 315 - L3	100 A	2	0 VA						62
63					0 VA					64
65	SUITE - 316 - L3	100 A	2			0 VA				66
67				0 VA						68
69	SPARE	20 A	2		0 VA					70
71						0 VA				72
73	SPARE	20 A	2	0 VA			3	30 A	SPD	74
75					0 VA					76
77	SPARE	15 A	2			0 VA				78
79				0 VA						80
81	SPARE	15 A	2		0 VA					82
83						0 VA				84
		Total Load:		0 VA	0 VA	0 VA				
		Total Amps:		0 A	0 A	0 A				

Legend:

Load Classification	Connected Load	Demand Factor	Estimated Demand	Panel Totals	
				Total Conn. Load:	0 VA
				Total Est. Demand:	0 VA
				Total Conn.:	0 A
				Total Est. Demand:	0 A

Notes:

Branch Panel: RP-2M

Location: Space 136
Supply From:
Mounting: Surface
Enclosure: Type 1

Volts: 120/208 Wye
Phases: 3
Wires: 4

A.I.C. Rating:
Mains Type:
Mains Rating: 600 A
MCB Rating: 500 A

Notes:

CKT	Circuit Description	Trip	Poles	A	B	C	Poles	Trip	Circuit Description	CKT
1	P: IDU-06 L1 OFFICE	15 A	2	18 VA	1000...		1	15 A	P: MOTORIZED DAMPERS - ROOF	2
3					18 VA	18 VA				4
5	P: EF-3 L2	15 A	2			33 VA	2	15 A	P: IDU-10 STAFF RM L3	6
7				33 VA	33 VA					8
9	P: MOTORIZED DAMPERS - L2	15 A	1		500 VA	33 VA	2	15 A	P: IDU-09 MECHANICAL RM L2	10
11	P: IDU-07,08 HSKP & ELEV CONTROL L2	15 A	2			50 VA				12
13				50 VA	95 VA		2	15 A	P: EFH-1 L1 WASTE	14
15	P: MOTORIZED DAMPERS - L1	15 A	1		380 VA	380 VA	1	15 A	P: MOTORIZED DAMPERS - L1	16
17	P: IDU-04, 05 L1 WASTE	15 A	2			105 VA	2	15 A	P: EBH - 2 MECH RM L2	18
19				105 VA	500 VA					20
21	P: EF-4 STAFF RM L3	15 A	2		750 VA	1603...	2	40 A	CU-1 ROOF	22
23						750 VA				24
25	P: SF-2 ROOF	15 A	2	4000...	1869...					26
27					4000...	1869...	3	100 A	RP-2M2	28
29	CU-4 ROOF	20 A	2			1603...				30
31				1603...	250 VA		2	15 A	P: EBH-1 CORRIDOR L2	32
33	P: EBH-1 L1 CORRIDOR	15 A	2		250 VA	250 VA				34
35						250 VA	2	15 A	P: EBH-1 LEVEL 2 HSKP	36
37	EF-5 ROOF	15 A	2	250 VA	250 VA					38
39					250 VA	250 VA	2	15 A	EF-6 ROOF	40
41	P: EBH-1 CORRIDOR L2	15 A	2			250 VA				42
43				250 VA	500 VA		2	15 A	P: EBH - 2 MECH RM L2	44
45	P: EBH-1 LOBBY 200 L2	15 A	2		500 VA	500 VA				46
47						500 VA	2	15 A	P; EBH-2 LEVEL 1	48
49	P: FFH-1 L1 VEST 108	15 A	2	1000...	500 VA					50
51					1000...	1000...	2	15 A	P: FFH-1 L1 VEST. 101	52
53	P: FFH-1 STAIRS A L1	15 A	2			1000...				54
55				1000...	1000...		2	15 A	P: EFH-1 L1 STAIRS B	56
57	P: FFH-1 L1 STAIRS B	15 A	2		1000...	1000...				58
59						1000...	2	15 A	P: FFH-1 CORRIDOR L1	60
61	P: EBH-2 ROOF	20 A	2	1000...	1000...					62
63					1000...	1000...	2	15 A	P: EFH-1 L1 CORRIDOR	64
65	P: EFH-1 L1 CORRIDOR 118	15 A	2			1000...				66
67				1000...	2650...					68
69	P: EDH-2 ROOF	30 A	3		2667...	2650...	3	30 A	P: EWH-1 MECH ROOM L2	70
71						2667...				72
73	CU-3	40 A	2	2667...	4333...					74
75					1603...	4333...	3	90 A	P: CU-2 ROOF	76
77						1603...				78
79				0 VA	1213...					80
81	SPD	30 A	3		0 VA	1213...	3	220 A	P: ASHP- DHW - ROOF	82
83						0 VA				84
		Total Load:		39087 VA	40935 VA	39014 VA				
		Total Amps:		326 A	341 A	325 A				

Legend: P: POWER R: RECEPTACLE L: LIGHTING

Load Classification	Connected Load	Demand Factor	Estimated Demand	Panel Totals	
Other	119036 VA	100.00%	119036 VA		
				Total Conn. Load:	119036 VA
				Total Est. Demand:	119036 VA
				Total Conn.:	330 A
				Total Est. Demand:	330 A

Notes:

No.	DATE	DESCRIPTION
10	2026-10-30	ELECTRICAL ADDENDUM 01
9	2026-09-16	ISSUED FOR TENDER
8	2026-06-12	ISSUED FOR TENDER PROGRESS
7	2026-03-24	ISSUED FOR 100% CD
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5	2026-02-20	ISSUED FOR 90% CD AND CLASS A COSTING
4	2026-01-30	ISSUED FOR 75%CD
3	2026-01-30	ISSUED FOR SITE SERVICES PERMIT
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1	2025-12-17	ISSUED FOR 30%CD

REVISIONS

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PROJECT

DURHAM REGION
409 CENTRE ST S SENIORS HOUSING
409 CENTRE ST. SOUTH, WHITBY, ON L1N 4W3
TOWN FILE NUMBER: SP-26-25

TITLE

ELECTRICAL SCHEDULES - POWER & SYSTEMS

CHECKED	PROJECT No.
EJG	RE-24-162
SCALE	DRAWING No.
N.T.S.	E-933

409 CENTRE ST S SENIORS HOUSING

409 CENTRE ST. SOUTH, WHITBY, ON L1N 4W3
TOWN FILE NUMBER: SP-26-25
ICAT SET

RE-24-162

QUASAR

NORTH BAY, OTTAWA, TORONTO, VANCOUVER, WOODBRIDGE

ICAT LEGEND	
SYMBOL	DESCRIPTION
LINETYPES	
———	NEW WORK
———	WORK TO BE DEMOLISHED, OR REMOVED
———	EXISTING MATERIAL/EQUIPMENT/SERVICES TO REMAIN
– – – –	FUTURE WORK (NOT IN SCOPE)
———	EXTENTS OF FIRE ALARM ZONE, WET LOCATION, OR OTHER AREA AS NOTED ON PLANS
ABBREVIATIONS	
E	EXISTING TO REMAIN
R	EXISTING TO BE DEMOLISHED/REMOVED
ER	EXISTING IN RELOCATED POSITION
RR	REMOVE AND RELOCATE
C	CEILING MOUNTED CONNECTION
W	WALL MOUNTED CONNECTION
F	FLOOR MOUNTED CONNECTION
℄	CENTRE LINE
AFF	ABOVE FINISHED FLOOR
AFG	ABOVE FINISHED GRADE
O/C	OVER COUNTER
U/C	UNDER CABINET
U/F	UNDER RAISED FLOOR
CCT	CIRCUIT
CTE	CONNECT TO EXISTING
AFCI	ARC FAULT CIRCUIT INTERRUPTER
GFCI	GROUND FAULT CIRCUIT INTERRUPTER
GFI	GROUND FAULT INTERRUPTER
IG	ISOLATED GROUND
TL	TWIST LOCK
TR	TAMPER RESISTANT
WG	WIRE GUARD
WP	WEATHER PROOF
EX	EXPLOSION PROOF
HZ	HAZARDOUS LOCATION
R/I	ROUGH-IN ONLY
NIC	NOT IN CONTRACT
SIM.	SIMILAR TO
TYP.	TYPICAL
ABBREVIATIONS - CODES AND STANDARDS	
OBC	ONTARIO BUILDING CODE
OESC	ONTARIO ELECTRICAL SAFETY CODE
OFC	ONTARIO FIRE CODE
ABBREVIATIONS - CEILING TYPES	
ACT	ACOUSTIC CEILING TILE (T-BAR)
EXP	EXPOSED CEILING
OWSJ	OPEN WEB STEEL JOISTS
PCC	PAINTED OR POPCORN CEILING ON EXPOSED CONCRETE
WD	WOOD CEILING
ANNOTATIONS	
CL	CLOSET
WR	WASHROOM
CONDUIT AND BOXES	
—⊐	CONDUIT WITH END BUSHING
—○	CONDUIT UP
—◌	CONDUIT DOWN
—⊣	CONDUIT CONTINUES
JB	JUNCTION BOX
PB	PULL BOX
HH	HAND HOLE
DETAIL REFERENCES	
1	SHEET KEYNOTE
1	REVISION NUMBER

ICAT LEGEND	
SYMBOL	DESCRIPTION
	TELECOMMUNICATIONS
	SYSTEM FURNITURE FEED.
W	ADJACENT TO SYSTEM FURNITURE FEED, DENOTES WALL SYSTEM FURNITURE FEED FOR COMMUNICATIONS.
F	ADJACENT TO SYSTEM FURNITURE FEED, DENOTES FLOOR SYSTEM FURNITURE FEED FOR COMMUNICATIONS.
C	ADJACENT TO SYSTEM FURNITURE FEED, DENOTES CEILING SYSTEM FURNITURE FEED FOR COMMUNICATIONS (SERVICE POLE OR DROP CORD AS NOTED).
	CABLE TRAY (LADDER TYPE)
	CABLE TRAY (BASKET TYPE)
	WALL MOUNTED DATA (D) OR VOICE (V) OUTLET. PROVIDE 1V AND 1D UNLESS NOTED OTHERWISE.
	WALL MOUNTED VOICE (TELEPHONE) OUTLET. PROVIDE 1V UNLESS NOTED OTHERWISE.
	WALL MOUNTED DATA OUTLET. PROVIDE 1D UNLESS NOTED OTHERWISE.
	WALL MOUNTED TELEVISION OUTLET.
	VOICE, DATA, OR TV OUTLET AS DESCRIBED ABOVE, MOUNTED ABOVE COUNTER TOP OR AS INSTRUCTED ON SITE.
B	ADJACENT TO COMMUNICATIONS OUTLET, INDICATES BLANK-OFF PLATE.
HDMI	HDMI OUTLET.
AV	AUDIO VIDEO GANG, AS NOTED.
WAP	WIRELESS ACCESS POINT (WIFI)
A	AUDIO VISUAL SYSTEM SPEAKER, CEILING MOUNTED.
A	AUDIO VISUAL SYSTEM SPEAKER, WALL MOUNTED.
PA	PUBLIC ADDRESS SYSTEM SPEAKER, CEILING MOUNTED.
PA	PUBLIC ADDRESS SYSTEM SPEAKER, WALL MOUNTED.
	PUBLIC ADDRESS SPEAKER, WALL MOUNTED HORN SPEAKER.
HS	PUBLIC ADDRESS SYSTEM HANDSET
ACC	PUBLIC ADDRESS SYSTEM ADMIN CONTROL CONSOLE
VOL	PUBLIC ADDRESS SPEAKER VOLUME CONTROL SWITCH.
	INTERCOM
IDC	INSULATION DISPLACEMENT CONNECTION
	CLOCK
MTX	GPS CLOCK SYSTEM MASTER TRANSMITTER
	GPS CLOCK SYSTEM GPS RECEIVER
RTX	GPS CLOCK SSTEM SETTELITE TRANSMITTER (RECEIVER)
	GPS CLOCK SYSTEM RECEIVER SWITCH
	ACCESS CONTROL AND DOOR HARDWARE
CR	CARD READER
DA	DOOR ALARM SOUNDER
DC	DOOR CONTACT
ODC	OVERHEAD DOOR CONTACT
EL	ELECTRIC LATCH
ES	ELECTRIC STRIKE
PTC	ELECTRIC POWER TRANSFER CABLE
PTH	POWER TRANSFER HINGE
KS	KEY SWITCH
ML	ELECTROMAGNETIC LOCK
MLR	MOTORIZED LATCH RETRACTION. PROVIDE 120 V.
REX	REQUEST TO EXIT SENSOR
	MUSHROOM HEAD PUSH BUTTON FOR 'REQUEST TO EXIT' MAGLOCK RELEASE, OR OTHER PUSH BUTTON AS INDICATED
RF	RF NODE
	BARRIER FREE DOOR OPERATOR PUSH BUTTON
WS	TOUCHLESS "WAVE SWITCH" FOR DOOR OPERATOR CONTROL
DB	DOOR BELL C/W SOUNDER AND STROBE
DB	DOOR BELL (SOUNDER ONLY)
INTRUSION DETECTION	
	GLASS BREAK (GB)
MD	MOTION DETECTOR (MD)
KP	KEYPAD (KP)
VIDEO SURVEILLANCE	
	CCTV CAMERA
C/P	CCTV CAMERA, CEILING OR POLE MOUNTED
	CCTV CAMERA, WALL MOUNTED
PTZ	PAN-TILT-ZOOM
DURESS SYSTEM	
DB	DURESS BUTTON (MOUNTED ON UNDERSIDE OF TABLETOP)
DB-W	WALL MOUNTED DURESS BUTTON WITH POLYCARBONATE ANTI-TAMPER COVER
DB	DURESS SYSTEM STROBE LIGHT

THIS LEGEND IS GENERIC. ALL SYMBOLS LISTED MAY NOT BE APPLICABLE FOR THIS PROJECT. REFER TO FLOOR PLANS TO DETERMINE USED DEVICES AND EQUIPMENT.

ICAT DRAWING LIST

DRAWING #	DRAWING NAME
T-000	COVER PAGE
T-001	ICAT LEGEND & DRAWING LIST
T-100	ICAT SITE PLAN
T-101	ICAT SITE DETAILS 1
T-300	ICAT NEW WORK PLAN - BASEMENT
T-301	ICAT NEW WORK PLAN - LEVEL 1
T-302	ICAT NEW WORK PLAN - LEVEL 2
T-303	ICAT NEW WORK PLAN - LEVEL 3
T-304	ICAT NEW WORK PLAN - ROOF
T-600	ENLARGED SUITES AND IT SPACES - ICAT
T-700	COMMUNICATIONS RISER AND SCHEMATICS
T-701	SECURITY RISER DIAGRAM
T-702	AUDIO/VISUAL RISER AND SCHEMATICS
T-703	CABINET ELEVATIONS AND NETWORK DIAGRAM
T-800	ICAT TYPICAL DETAILS 1
T-801	ICAT TYPICAL DETAILS 2

ICAT RESPONSIBILITY MATRIX				
ITEM	SUPPLY	INSTALL	TERMINATE	COMMISSIONING
STRUCTURED CABLING (HORIZONTAL AND BACKBONE)	DIV 27	DIV 27	DIV 27	DIV 27
JACKS	DIV 27	DIV 27	DIV 27	DIV 27
OUTLETS	DIV 27	DIV 27	DIV 27	DIV 27
PATCH PANELS	DIV 27	DIV 27	DIV 27	DIV 27
EQUIPMENT RACKS	DIV 27	DIV 27	DIV 27	DIV 27
FIREWALL	CLIENT	DIV 27	N/A	CLIENT
NETWORK SERVICES	CLIENT	CLIENT	CLIENT	CLIENT
WIRELESS NETWORK	CLIENT	DIV 27	N/A	CLIENT
SERVERS	CLIENT	DIV 27	N/A	CLIENT
TELEPHONES	CLIENT	DIV 27	N/A	CLIENT
PATHWAYS, CONDUITS, BACKBOXES	DIV 26	DIV 26	N/A	N/A

4 1 ICAT RESPONSIBILITY MATRIX
SCALE:N.T.S

No.	DATE	DESCRIPTION
4	2026-10-30	ICAT ADDENDUM 02
3	2026-09-16	ISSUED FOR TENDER
2	2026-03-24	ISSUED FOR 100% CD
1	2026-02-20	ISSUED FOR 90% CD AND CLASS A COSTING

REVISIONS

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PROJECT NORTH

QUASAR
250 ROWNTREE DAIRY RD, WOODBRIDGE, ON
TEL: 905-507-0800
WEB: WWW.QUASARCG.COM

PROJECT

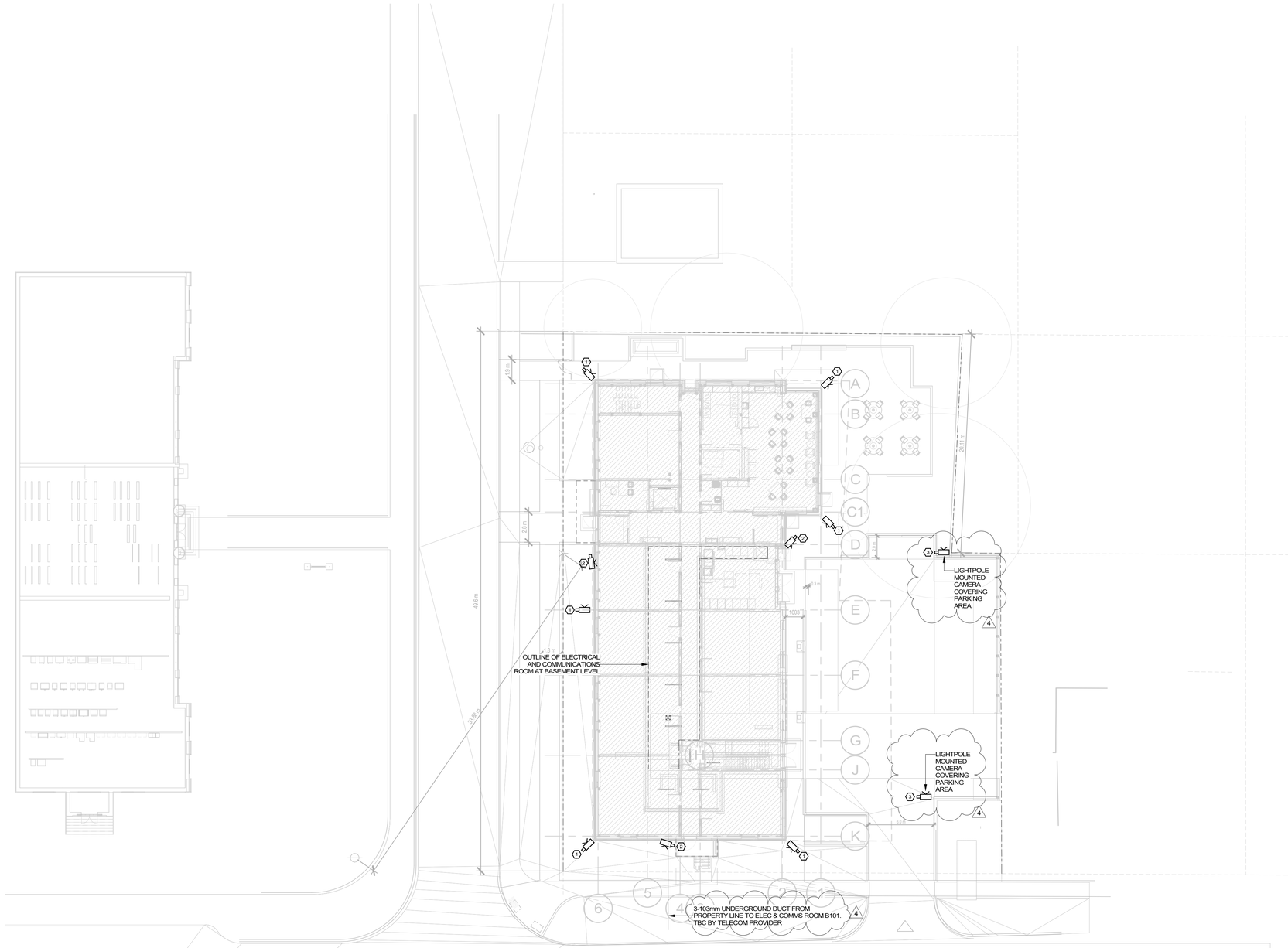
DURHAM REGION
409 CENTRE ST S SENIORS HOUSING
409 CENTRE ST. SOUTH, WHITBY, ON L1N 4W3
TOWN FILE NUMBER: SP-26-25

TITLE

ICAT LEGEND & DRAWING LIST

CHECKED	PROJECT No.
JP	RE-24-162
SCALE	DRAWING No.
N.T.S.	T-001

4



KEYNOTE LEGEND	
Key Value	Keynote Text
1	360° CORNER MOUNT CAMERA MOUNTED AT 3600mm
2	FIXED WALL MOUNT CAMERA MOUNTED AT 2750mm.
3	360° POLE MOUNT CAMERA MOUNTED AT 3600MM. PROVIDE MINIMUM 27MM PVC CONDUIT IN COMMON TRENCH WITH POWER TO BASEMENT TELECOM ROOM.

No.	DATE	DESCRIPTION
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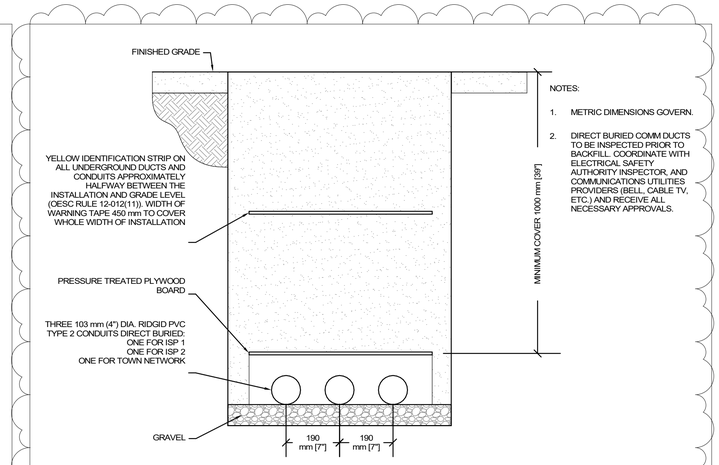
PROJECT

DURHAM REGION
409 CENTRE ST S SENIORS HOUSING
409 CENTRE ST. SOUTH, WHITBY, ON L1N 4W3
TOWN FILE NUMBER: SP-26-25

TITLE

ICAT SITE PLAN

CHECKED	PROJECT No.
JP	RE-24-162
SCALE	DRAWING No.
1 : 200	T-100



1 DIRECT BURIED COMMUNICATIONS THREE DUCTS 4

SCALE:N.T.S

No.	DATE	DESCRIPTION
4	2026-10-30	ICAT ADDENDUM 02
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PROJECT

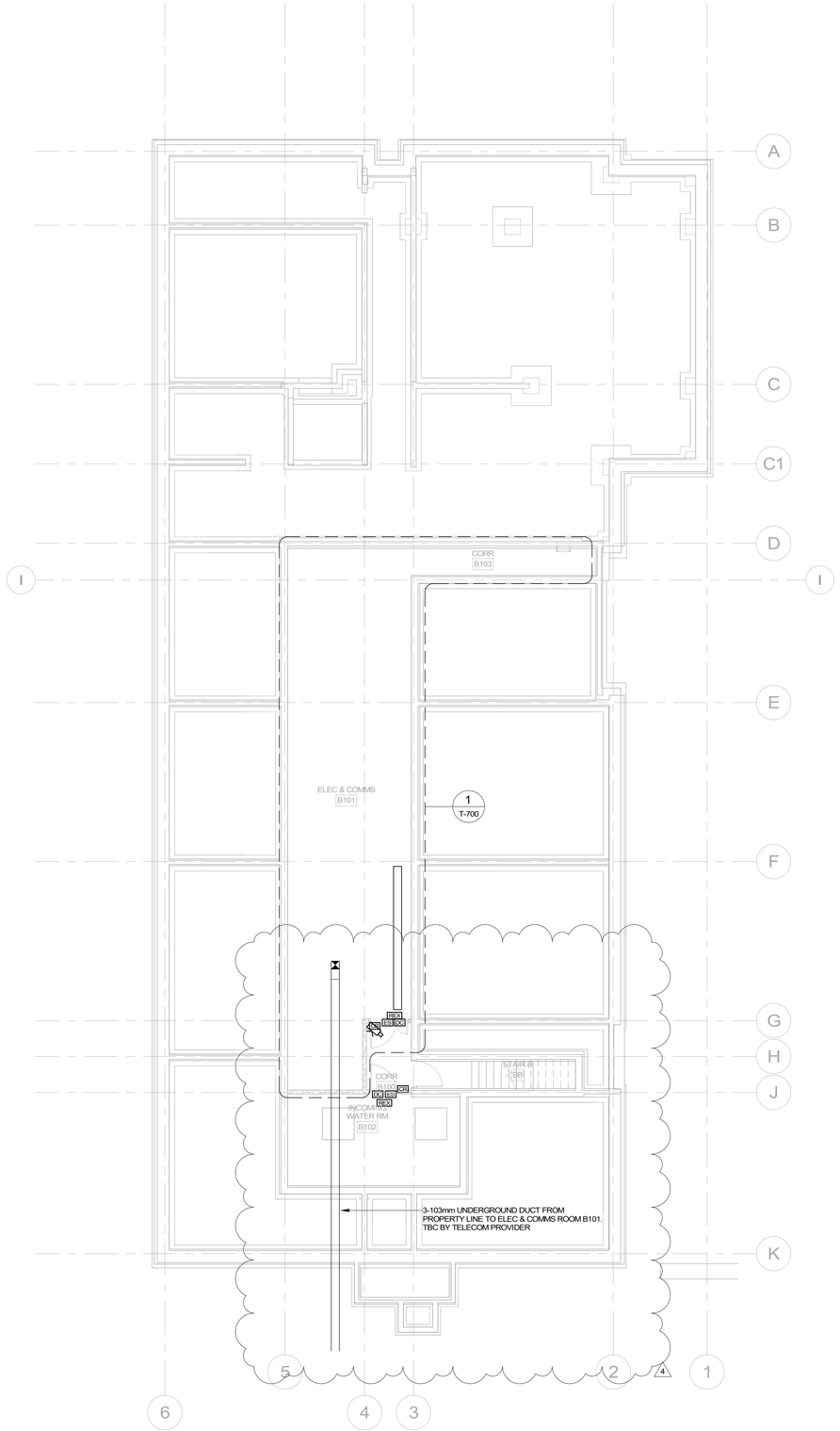
DURHAM REGION
409 CENTRE ST S SENIORS HOUSING
409 CENTRE ST. SOUTH, WHITBY, ON L1N 4W3
TOWN FILE NUMBER: SP-26-25

TITLE

ICAT SITE DETAILS 1

CHECKED	PROJECT No.
JP	RE-24-162
SCALE	DRAWING No.
N.T.S.	T-101

4



KEYNOTE LEGEND

Key Value	Keynote Text

4

No.	DATE	DESCRIPTION
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PROJECT

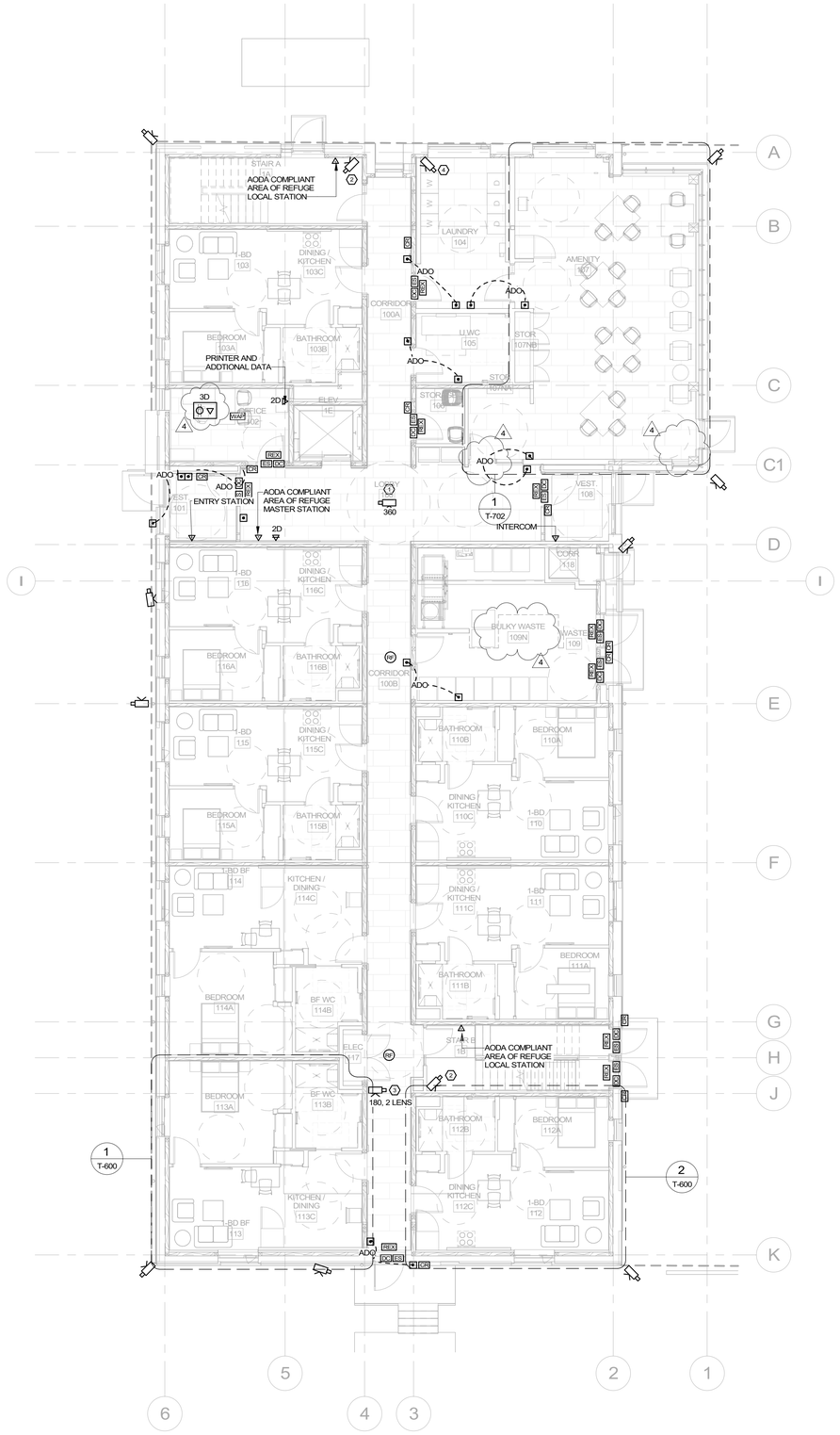
DURHAM REGION
409 CENTRE ST S SENIORS HOUSING
409 CENTRE ST. SOUTH, WHITBY, ON L1N 4W3
TOWN FILE NUMBER: SP-26-25

TITLE

ICAT NEW WORK PLAN - BASEMENT

CHECKED	PROJECT No.
JP	RE-24-162
SCALE	DRAWING No.
1 : 100	T-300

4



KEYNOTE LEGEND

Key Value	Keynote Text
1	360° CEILING MOUNTED CAMERA.
2	FIXED WALL MOUNT CAMERA MOUNTED AT 2750mm
3	180° WALL MOUNT CAMERA MOUNTED AT 2750mm
4	FIXED CEILING MOUNTED CAMERA

No.	DATE	DESCRIPTION
4	2026-10-30	ICAT ADDENDUM 02
3	2026-09-16	ISSUED FOR TENDER
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PROJECT

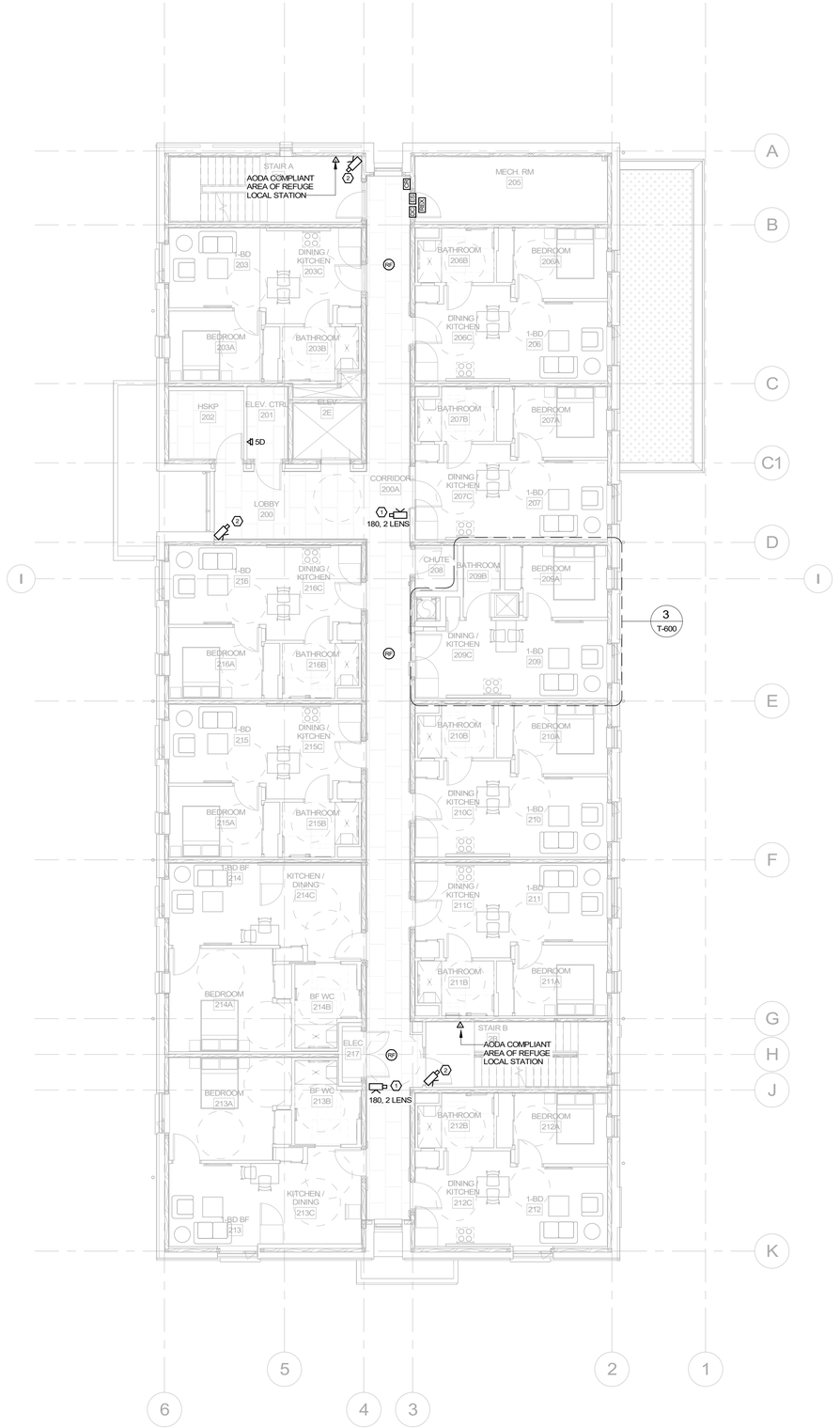
DURHAM REGION
409 CENTRE ST S SENIORS HOUSING
409 CENTRE ST. SOUTH, WHITBY, ON L1N 4W3
TOWN FILE NUMBER: SP-26-25

TITLE

ICAT NEW WORK PLAN - LEVEL 1

CHECKED	PROJECT No.
JP	RE-24-162
SCALE	DRAWING No.
1 : 100	T-301

4



KEYNOTE LEGEND

Key Value	Keynote Text
1	180° WALL MOUNT CAMERA MOUNTED AT 2750mm
2	FIXED WALL MOUNT CAMERA MOUNTED AT 2750mm

No.	DATE	DESCRIPTION
4	2026-10-30	ICAT ADDENDUM 02
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PROJECT

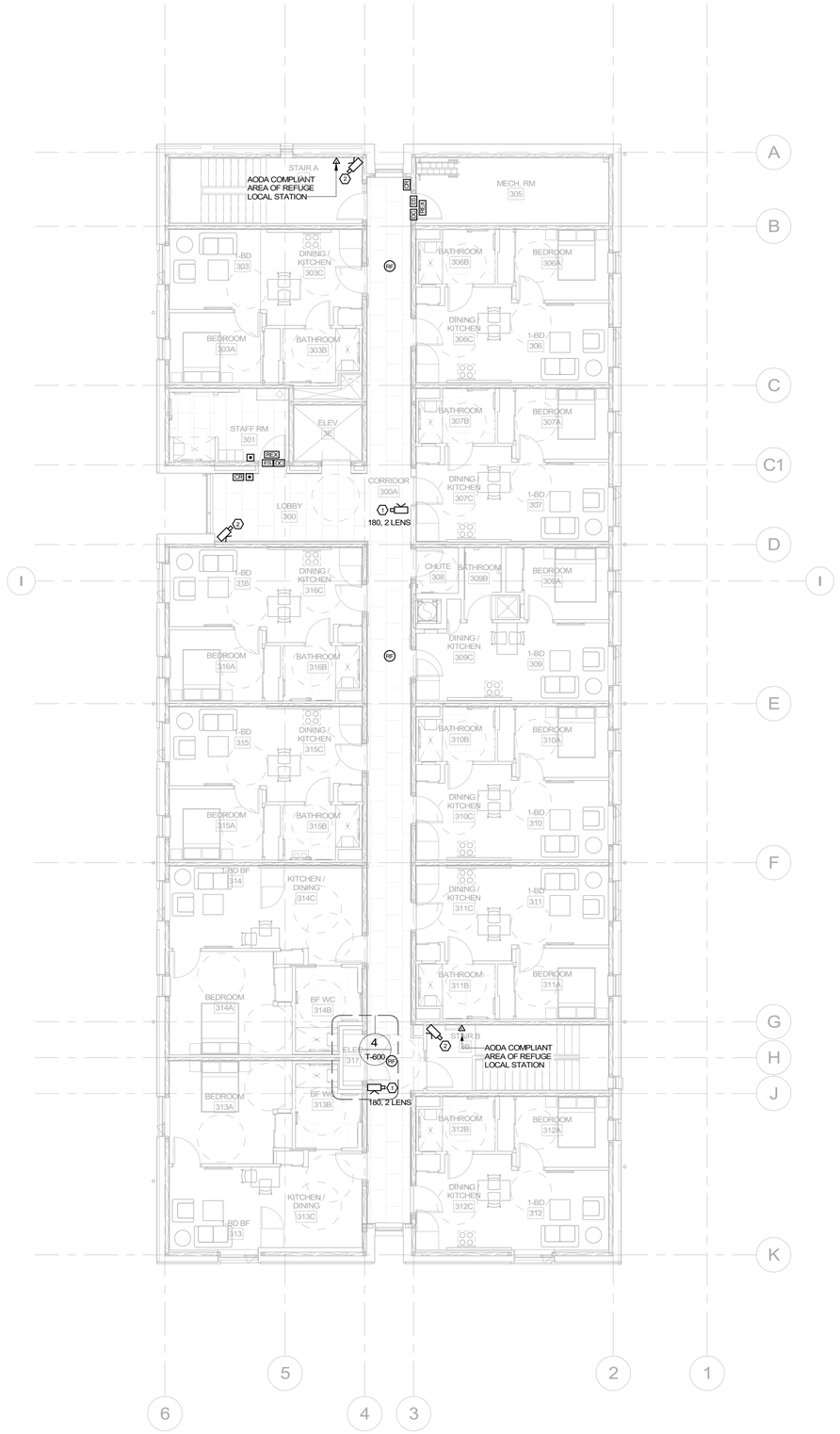
DURHAM REGION
409 CENTRE ST S SENIORS HOUSING
409 CENTRE ST. SOUTH, WHITBY, ON L1N 4W3
TOWN FILE NUMBER: SP-26-25

TITLE

ICAT NEW WORK PLAN - LEVEL 2

CHECKED	PROJECT No.
JP	RE-24-162
SCALE	DRAWING No.
1 : 100	T-302

4



KEYNOTE LEGEND	
Key Value	Keynote Text
1	180° WALL MOUNT CAMERA MOUNTED AT 2750mm
2	FIXED WALL MOUNT CAMERA MOUNTED AT 2750mm

No.	DATE	DESCRIPTION
4	2026-10-30	ICAT ADDENDUM 02
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PROJECT

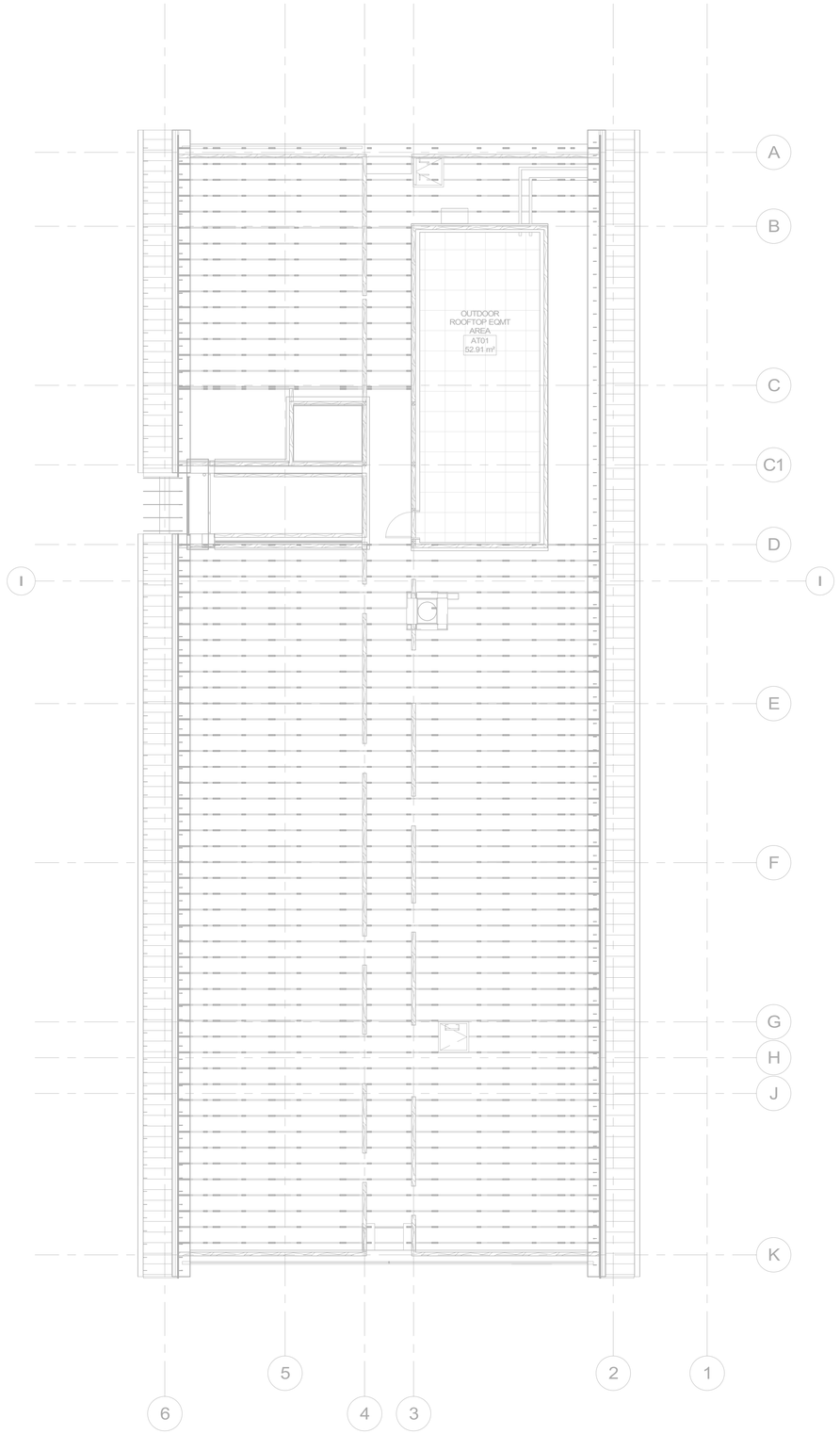
DURHAM REGION
409 CENTRE ST S SENIORS HOUSING
409 CENTRE ST. SOUTH, WHITBY, ON L1N 4W3
TOWN FILE NUMBER: SP-26-25

TITLE

ICAT NEW WORK PLAN - LEVEL 3

CHECKED	PROJECT No.
JP	RE-24-162
SCALE	DRAWING No.
1 : 100	T-303

4



KEYNOTE LEGEND

Key Value	Keynote Text

4

No.	DATE	DESCRIPTION
4	2026-10-30	ICAT ADDENDUM 02
3	2026-09-16	ISSUED FOR TENDER
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PROJECT

DURHAM REGION
409 CENTRE ST S SENIORS HOUSING
409 CENTRE ST. SOUTH, WHITBY, ON L1N 4W3
TOWN FILE NUMBER: SP-26-25

TITLE

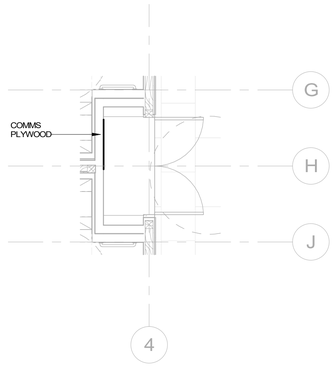
ICAT NEW WORK PLAN - ROOF

CHECKED	PROJECT No.
JP	RE-24-162
SCALE	DRAWING No.
1 : 100	T-304

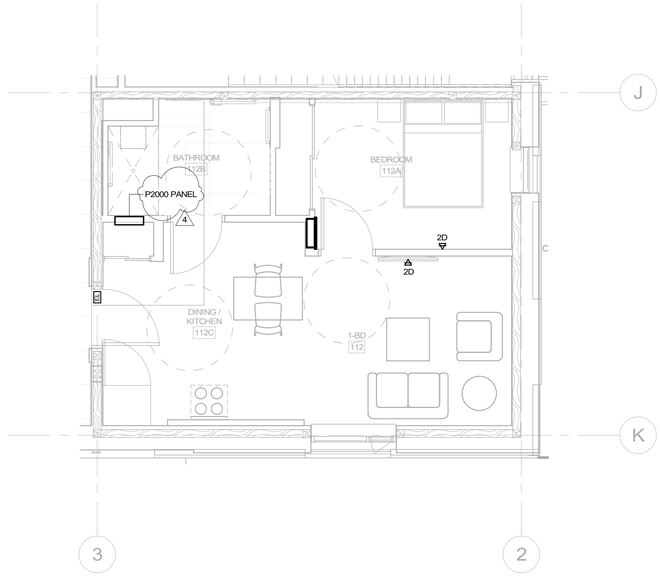
4

GENERAL NOTES

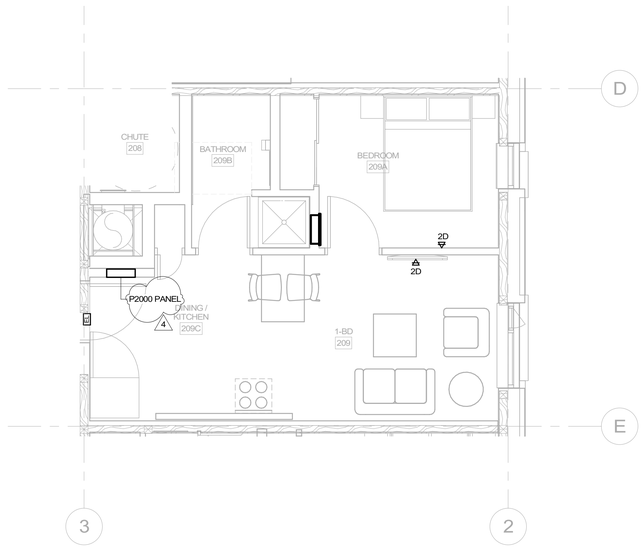
1. ROUGH IN FOR ADO SHOWN ON ELECTRIC DRAWINGS



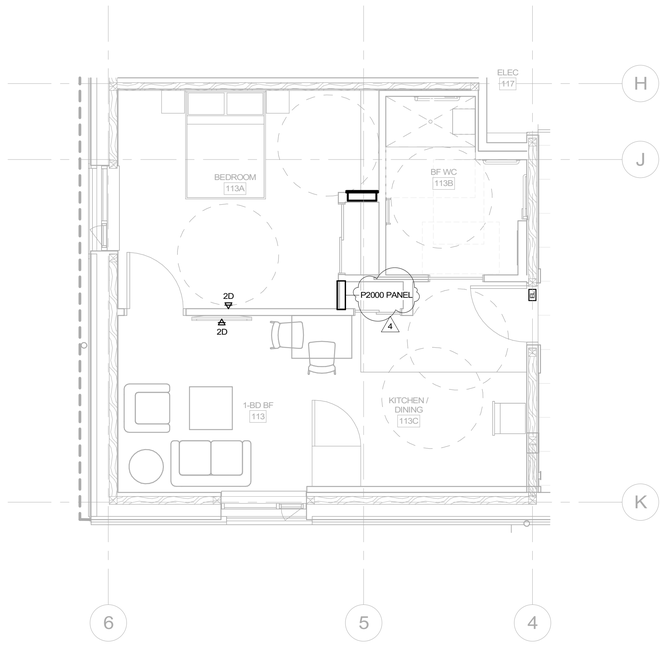
4 ICAT - ELEC CLOSET 317
SCALE: 1 : 50



2 ICAT - SUITE TYPE - 1-BD
SCALE: 1 : 50



3 ICAT - SUITE TYPE - 1-BD-ALTERNATE
SCALE: 1 : 50



1 ICAT - SUITE TYPE - 1-BD BF
SCALE: 1 : 50

No.	DATE	DESCRIPTION
4	2026-10-30	ICAT ADDENDUM 02
3	2026-09-16	ISSUED FOR TENDER
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PROJECT NORTH

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TEL: 905-507-0800
WEB: WWW.QUASARCG.COM

PROJECT

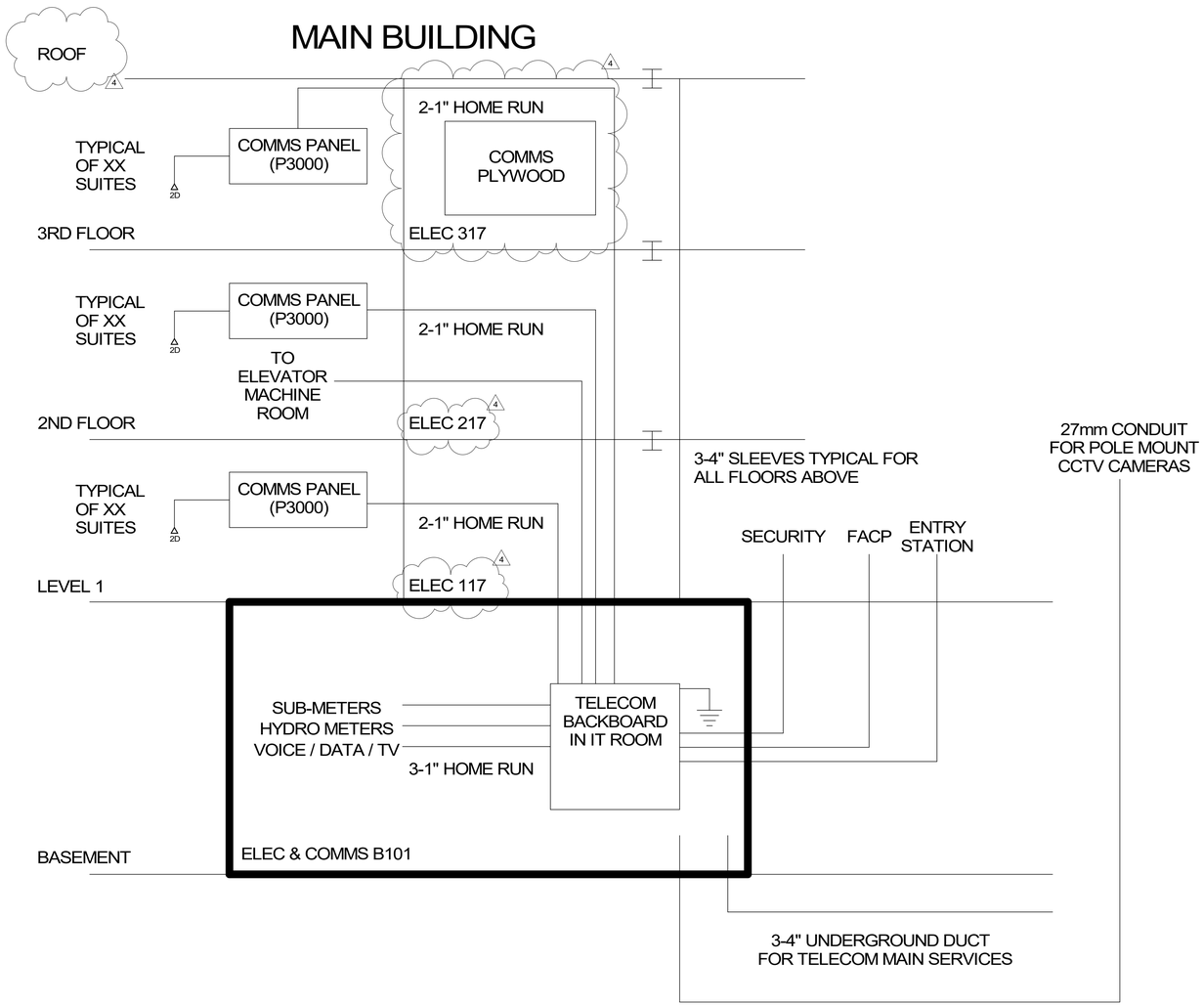
DURHAM REGION
409 CENTRE ST S SENIORS HOUSING
409 CENTRE ST. SOUTH, WHITBY, ON L1N 4W3
TOWN FILE NUMBER: SP-26-25

TITLE
ENLARGED SUITES AND IT SPACES - ICAT

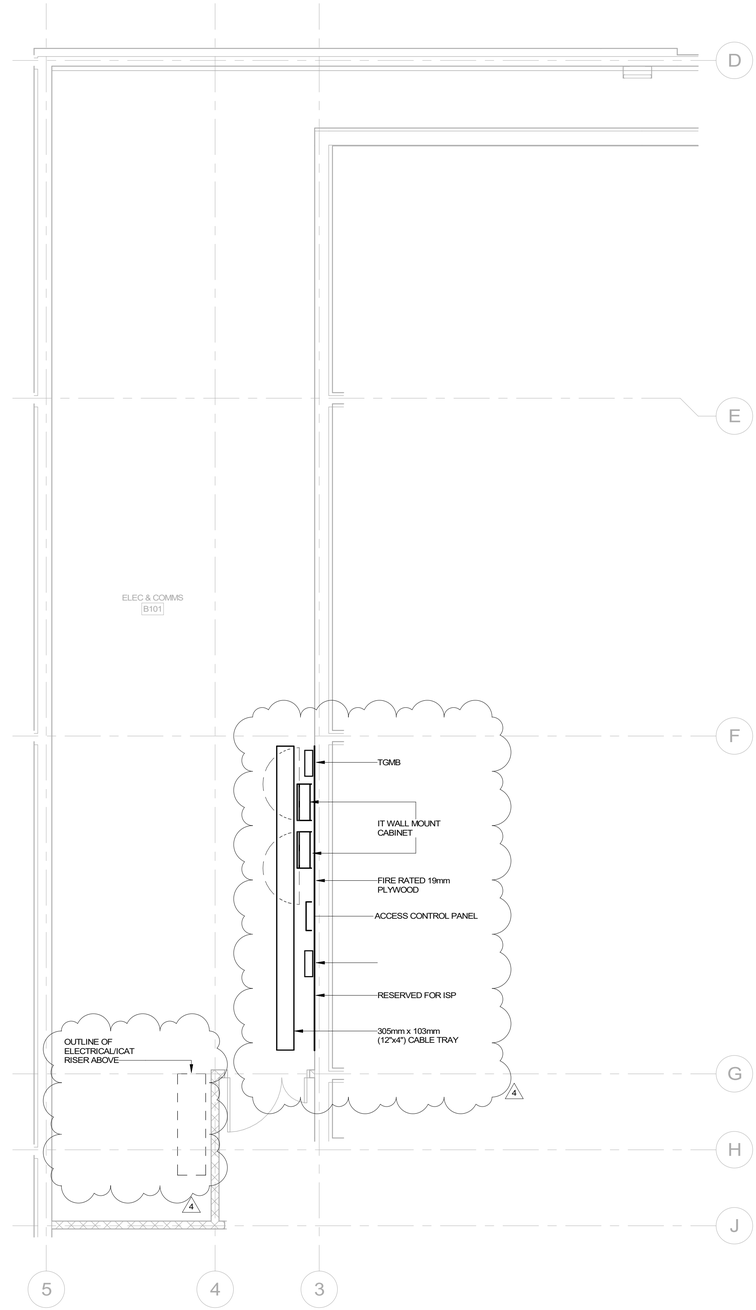
CHECKED	PROJECT No.
JP	RE-24-162
SCALE	DRAWING No.
1 : 50	T-600

GENERAL SHEET NOTES

1.	4 THESE SCHEMATICS ARE FOR REFERENCE ONLY, ELECTRICAL CONTRACTOR IS TO COORDINATE WITH CLIENT SECURITY CONTRACTOR'S DETAILED DESIGN AND FLOOR PLANS FOR EXACT SECURITY SYSTEM EQUIPMENT LOCATION, DEVICES QUANTITY AND INSTALLATION REQUIREMENT PRIOR TO ROUGH-IN.
2.	PROVIDE EACH CAMERA A JUNCTION BOX IN CLOSE PROXIMITY TO FINAL INSTALLATION LOCATION USING 27mmC CONDUIT FOR THE FINAL RUN TO THE CAMERA. THIS SCHEMATIC ASSUME ALL CAMERAS ARE POE TYPE. LOCATION NOT SHOWN ON DRAWING, SHALL BE SUBJECT TO SECURITY CONTRACTOR'S DESIGN.
3.	PROVIDE MINIMUM ONE SINGLE GANG BACK BOX FOR EACH DEVICE. RECESS MOUNT BOXES IN FINISHED AREAS AND EXTERIOR AREAS. SURFACE MOUNTED BOXES ARE ALLOWED IN UNFINISHED AREAS AND CEILING SPACES.
4.	#6 INSULATED COPPER GROUND IN 21mmC TO BUILDING MAIN GROUNDING BUS.
5.	PROVIDE ROUGH-IN EMPTY CONDUITS WITH PULL WIRES AND PATHWAY ONLY FOR TELECOM SYSTEM. REFER TO FLOOR LAYOUTS FOR EXACT QUANTITY AND LOCATION OF TELEPHONE AND TV OUTLETS.
6.	PROVIDE HOME RUN OF 3/4"C FROM EACH TELEPHONE AND/OR TV OUTLET, CONNECTED BACK TO JUNCTION BOX IN SUITE, AND/OR BACK TO NEAREST TEL./TV CLOSET IN COMMON AREA.
7.	PROVIDE 16"X16" BOX SPRINKLER PROOF TYPE ON 4'(W)X8'(H)X3/4"(D) FIRE RATED PLYWOOD BACKBOARD. TYPICAL FOR ALL COMMUNICATION CLOSETS.
8.	PROVIDE DUPLEX RECEPTACLE ON 4'(W)X8'(H)X3/4"(D) FIRE RATED PLYWOOD BACKBOARD. TYPICAL FOR ALL COMMUNICATION CLOSETS.



2 TELECOM RISER
SCALE:N.T.S



1 ICAT - ELEC & COMMS ROOM B101
SCALE: 1 : 50

No.	DATE	DESCRIPTION
4	2026-10-30	ICAT ADDENDUM 02
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PROJECT

DURHAM REGION
409 CENTRE ST S SENIORS HOUSING
409 CENTRE ST. SOUTH, WHITBY, ON L1N 4W3
TOWN FILE NUMBER: SP-26-25

TITLE

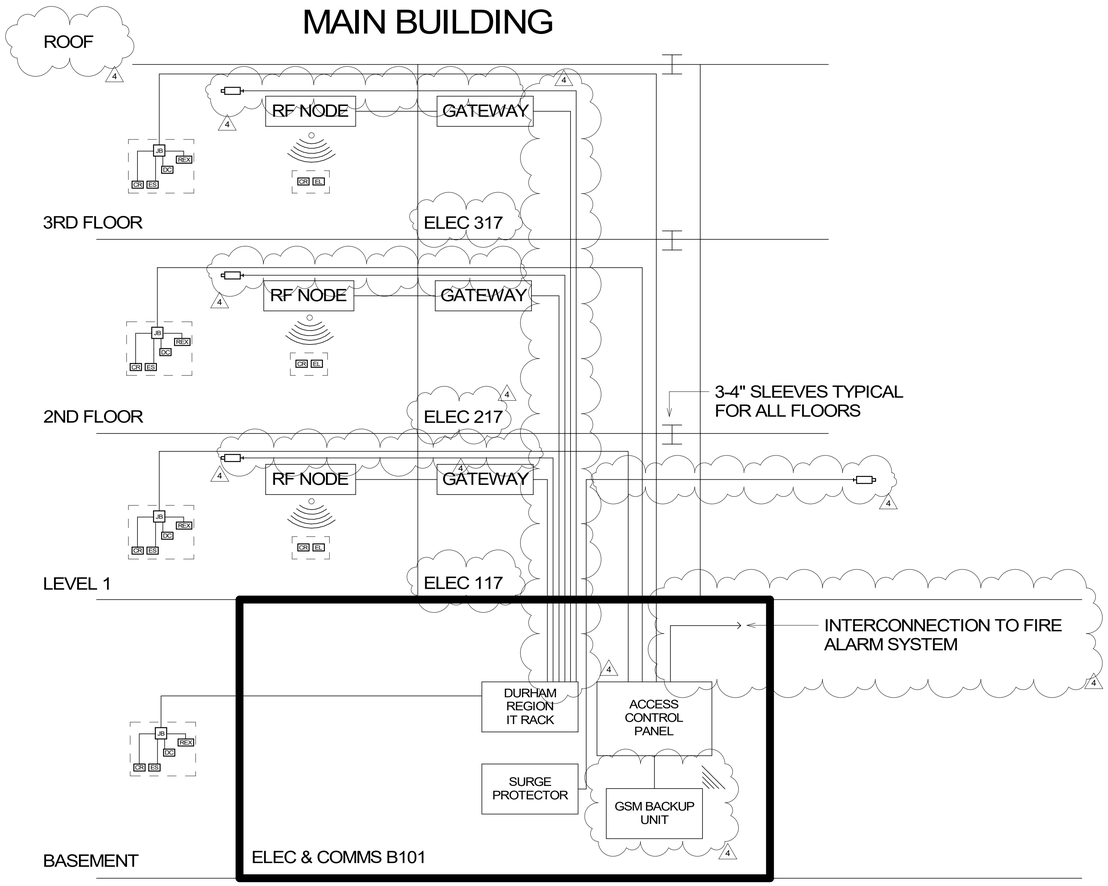
COMMUNICATIONS RISER AND SCHEMATICS

CHECKED	PROJECT No.
JP	RE-24-162
SCALE	DRAWING No.
As indicated	T-700

GENERAL SHEET NOTES

1.	RISERS ARE DIAGRAMITIC ONLY AND DO NOT INCLUDE ALL DEVICES AND CONNECTION, CONTRACTOR IS RESPONCIBLE FOR A COMPLETE AND FUNCTIONING SYSTEM.
4.	PROVIDE #6 INSULATED COPPER GROUND TO TMGB FOR EACH TELECOM RACK.
6.	PROVIDE HOME RUN OF 19mm C FROM EACH TELEPHONE AND/OR TV OUTLET, CONNECTED BACK TO JUNCTION BOX IN SUITE, AND/OR BACK TO NEAREST TEL./TV CLOSET IN COMMON AREA.
7.	PROVIDE PRIMARY PROTECTOR (LIGHTENING/SURGE PROTECTOR AXIS TU8001 OR EQUIVALENT) FOR EACH EXTERNAL CCTV/SURVEILANCE CAMERA. PROVIDE COPPER GROUND TO TMGB FOR EACH PROTECTOR AS PER MANUFACTURES INSTRUCTIONS.
8.	PROVIDE DUPLEX RECEPTACLE ON 4'(W)X8'(H)X3/4"(D) FIRE RATED PLYWOOD BACKBOARD. TYPICAL FOR ALL COMMUNICATION CLOSETS.

MAIN BUILDING



1 SECURITY RISER
SCALE:N.T.S

No.	DATE	DESCRIPTION
4	2026-10-30	ICAT ADDENDUM 02
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PROJECT

DURHAM REGION
409 CENTRE ST S SENIORS HOUSING
409 CENTRE ST. SOUTH, WHITBY, ON L1N 4W3
TOWN FILE NUMBER: SP-26-25

TITLE
SECURITY RISER DIAGRAM

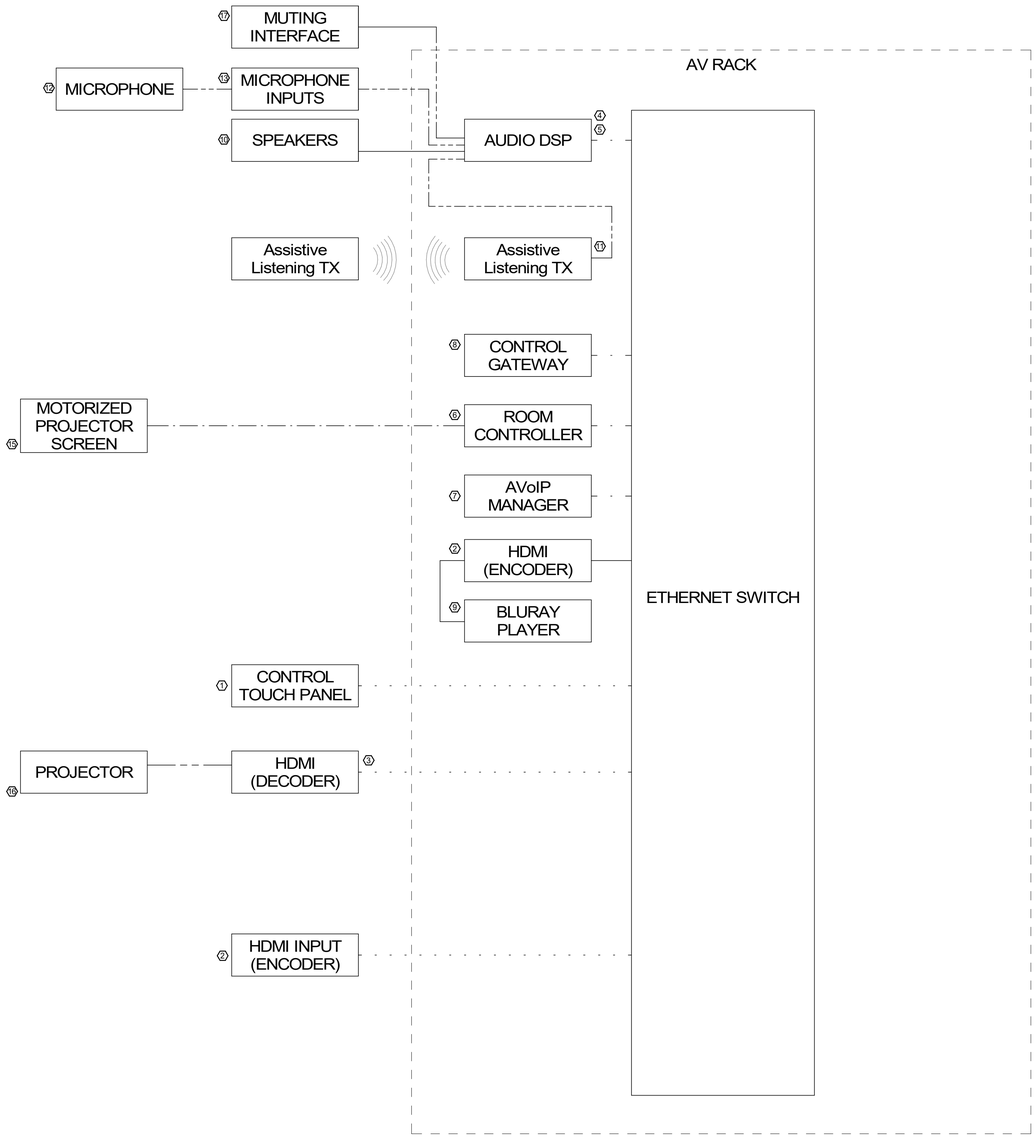
CHECKED	PROJECT No.
JP	RE-24-162
SCALE	DRAWING No.
N.T.S	T-701

DETAIL LEGEND

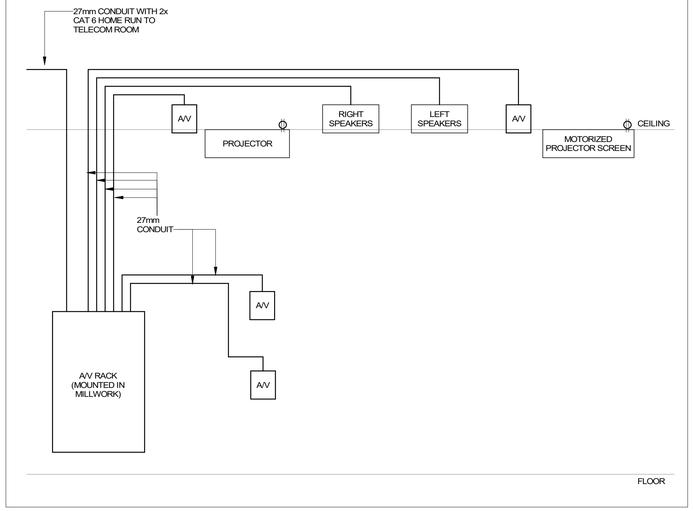
1.	KT 2010, TOUCH SCREEN	9.	PANSONIC (DPUB420K), SONY, TOSHIBA, OTHERS AS APPROVED
2.	WP-SW2-EN7, AVoIP ENCODER	10.	PN-6P, 6.5" PENDANT SPEAKER
3.	WP-DEC7, AVoIP DECODER	11.	FM-557, ASSISTIVE LISTENING SYSTEM
4.	PA-120Net, NETWORK POWER AMP	12.	CVG12D (SHURE) CORDED GOOSE NECK MICROPHONE
5.	AFM-20DSP-AEC, AUDIO MATRIX	13.	MP-WP-2XLRO-1G, CORDED MICROPHONE INPUT PLATES
6.	KC-Virtual Brain5, ROOM CONTROLLER	14.	S3400-24T4FP, PoE ETHERNET SWITCH
7.	KDS-7-MNGER AVoIP MANAGER	15.	94" DRAPER ACCESS V ELECTRIC PROJECTOR SCREEN "140111"
8.	FC-6, 2-PORT MULTI FUNCTION SERIAL/IR CONTROL GATEWAY	16.	BACRO F80-Q12 12000 LUMEN PROJECTOR
		17.	RADIO DESIGN LABS ST-SSR1 AUDIO SWITCH

CABLING LEGEND

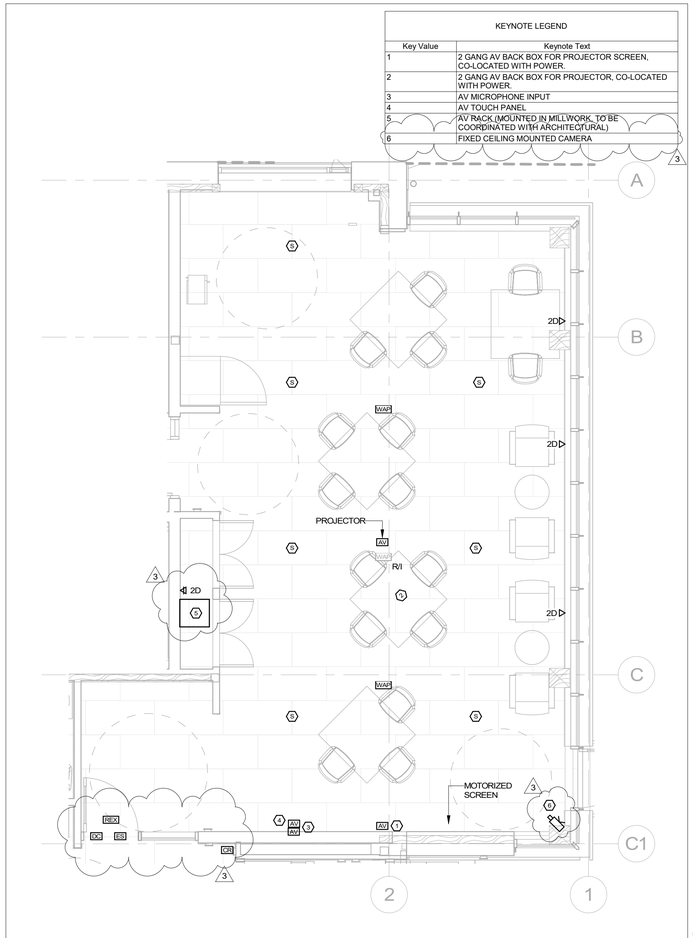
———————	14 AWG 2 CONDUCTOR SPEAKER CABLE
—— · —— · ——	RS-232, 2 CONDUCTOR SHIELDED
— · · · · —	CAT-6 ETHERNET
—— – – ——	HDMI CABLE, VERSION 2.2 COMPLIANT
– – —— – – –	ANALOG AUDIO, 2 CONDUCTOR SHIELDED



2 AMENITY ROOM AV LOGIC DIAGRAM
SCALE:N.T.S



3 A/V CONDUIT RISER DIAGRAM
SCALE:N.T.S



1 ICAT - LEVEL 1 - AMENITY - 107
SCALE: 1 : 50

KEYNOTE LEGEND

Key Value	Keynote Text
1	2 GANG AV BACK BOX FOR PROJECTOR SCREEN, CO-LOCATED WITH POWER.
2	2 GANG AV BACK BOX FOR PROJECTOR, CO-LOCATED WITH POWER.
3	AV MICROPHONE INPUT
4	AV TOUCH PANEL
5	AV RACK (MOUNTED IN MILLWORK, TO BE COORDINATED WITH ARCHITECTURAL)
6	FIXED CEILING MOUNTED CAMERA

No.	DATE	DESCRIPTION
3	2026-10-30	ICAT ADDENDUM 02
2	2026-09-16	ISSUED FOR TENDER
1	2026-03-24	ISSUED FOR 100% CD

REVISIONS

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PROJECT NORTH

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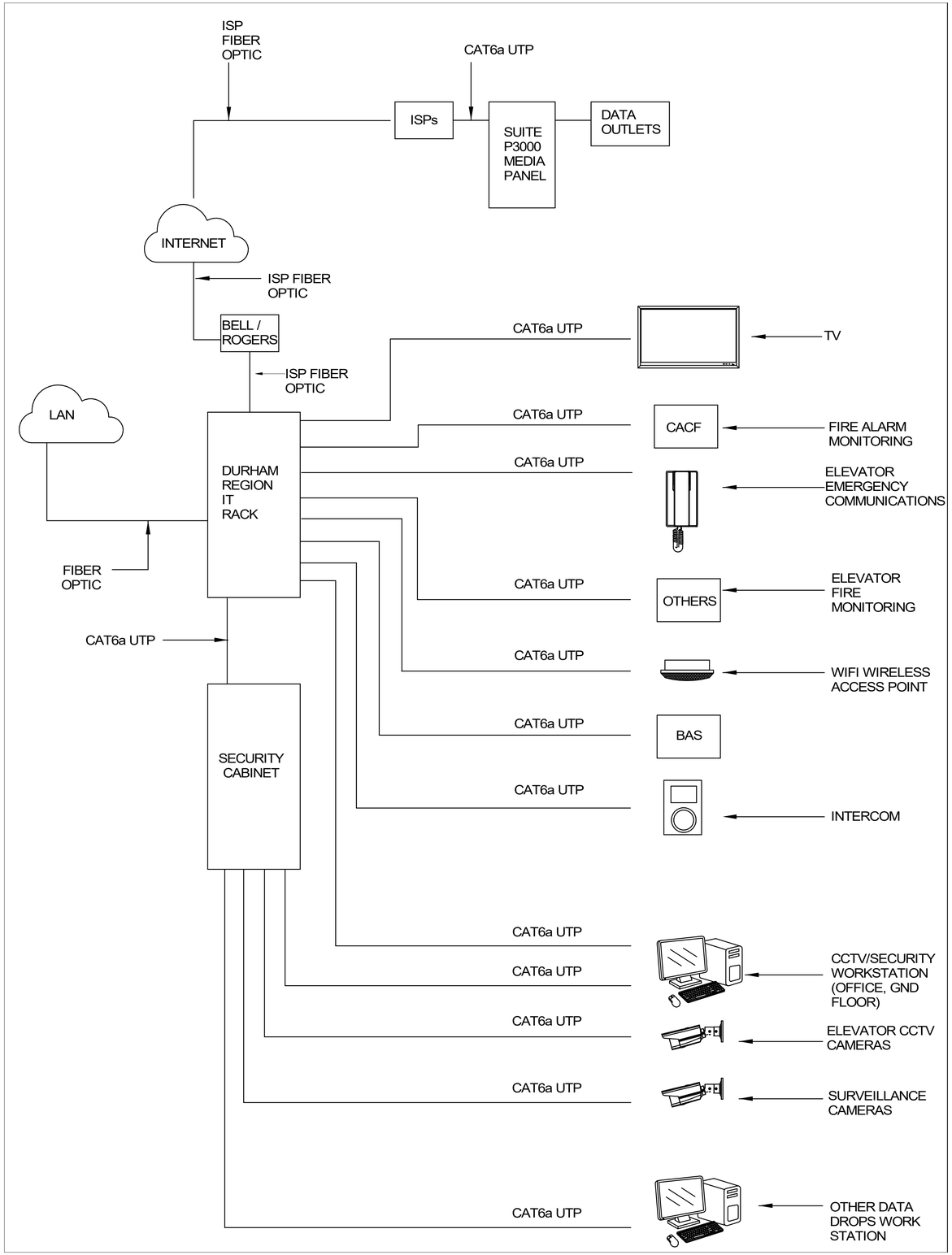
PROJECT

DURHAM REGION
409 CENTRE ST S SENIORS HOUSING
409 CENTRE ST. SOUTH, WHITBY, ON L1N 4W3
TOWN FILE NUMBER: SP-26-25

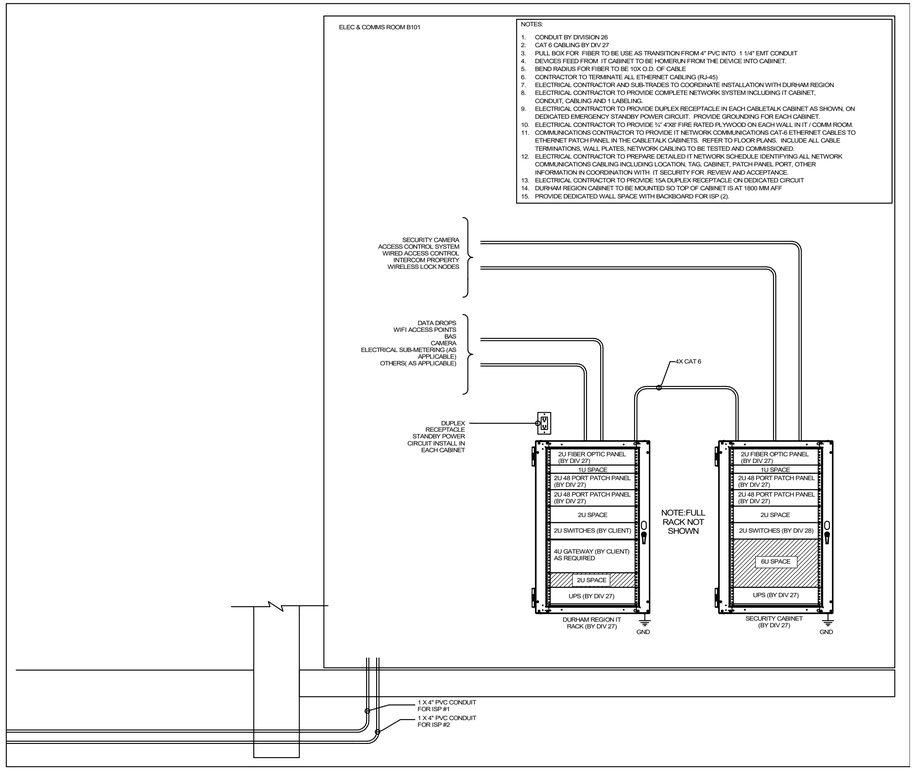
TITLE

AUDIO/VISUAL RISER AND SCHEMATICS

CHECKED	PROJECT No.
JP	RE-24-162
SCALE	DRAWING No.
As indicated	T-702



2 NETWORK DIAGRAM
SCALE:N.T.S



1 CABINET ELEVATION
SCALE:N.T.S

No.	DATE	DESCRIPTION
4	2026-10-30	ICAT ADDENDUM 02
3	2026-09-16	ISSUED FOR TENDER
2	2026-03-24	ISSUED FOR 100% CD
1	2026-02-20	ISSUED FOR 90% CD AND CLASS A COSTING

REVISIONS

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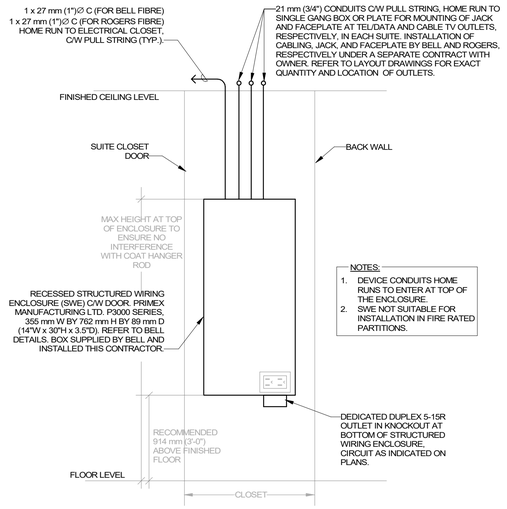
TOWN FILE NUMBER: SP-26-25

TITLE

CABINET ELEVATIONS AND NETWORK DIAGRAM

CHECKED	PROJECT No.
JP	RE-24-162
SCALE	DRAWING No.
N.T.S.	T-703

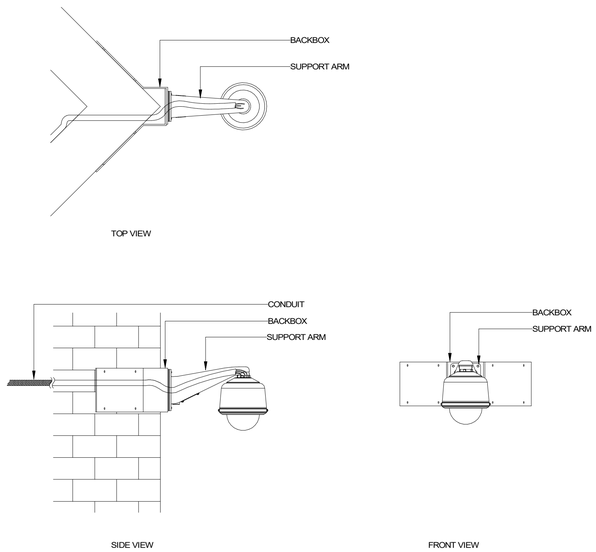


BELL/ROGERS STRUCTURED WIRING ENCLOSURE IN SUITE CLOSET WALL

N.T.S.

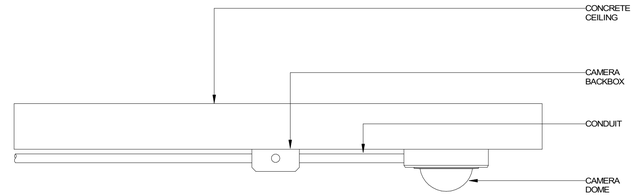
6 RESIDENTIAL SUITE STRUCTURED WIRING ENCLOSURE

SCALE:N.T.S.



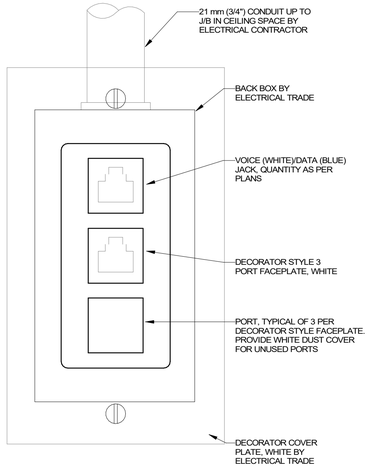
4 CCTV DETAIL - EXTERNAL CORNER MOUNT

SCALE:N.T.S



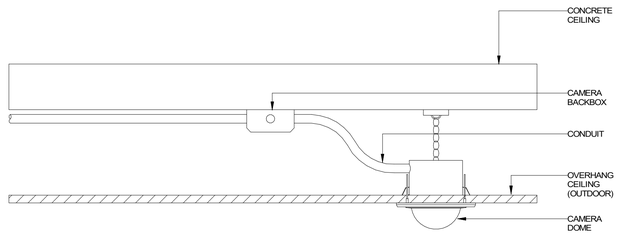
2 CCTV DETAIL - SURFACE MOUNT

SCALE:N.T.S



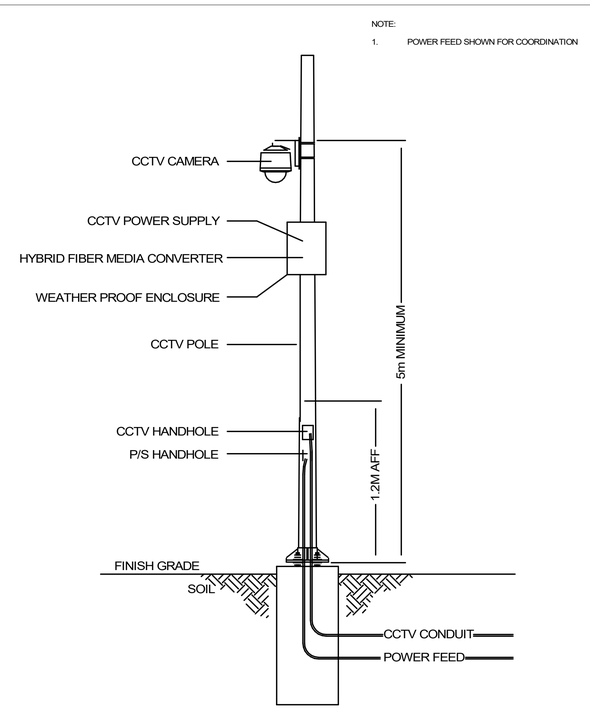
5 TYPICAL VOICE AND DATA OUTLET

SCALE:N.T.S.



3 CCTV DETAIL - RECESSED MOUNT

SCALE:N.T.S



1 CCTV DETAIL - POLE MOUNT

SCALE:N.T.S

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4	2026-10-30	ICAT ADDENDUM 02
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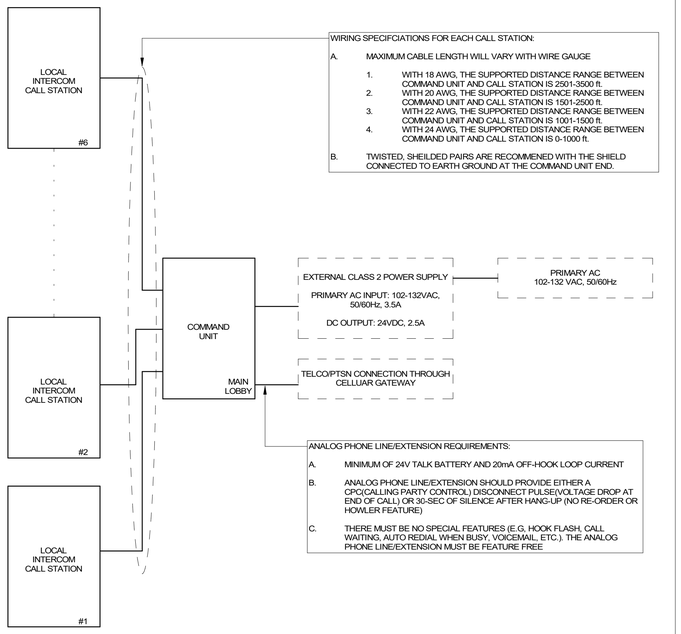
409 CENTRE ST. SOUTH, WHITBY, ON L1N 4W3
TOWN FILE NUMBER: SP-26-25

TITLE

ICAT TYPICAL DETAILS 1

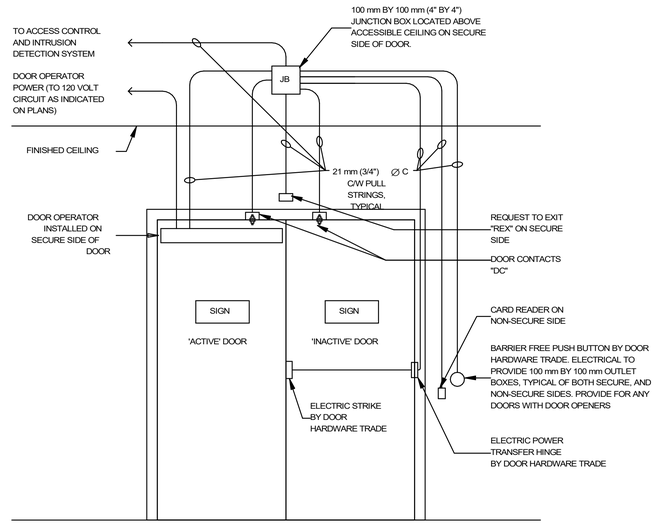
CHECKED	PROJECT No.
JP	RE-24-162
SCALE	DRAWING No.
N.T.S.	T-800

4



5 6-STATION AREA OF REFUGE SYSTEMS RISER DIAGRAM

SCALE:N.T.S

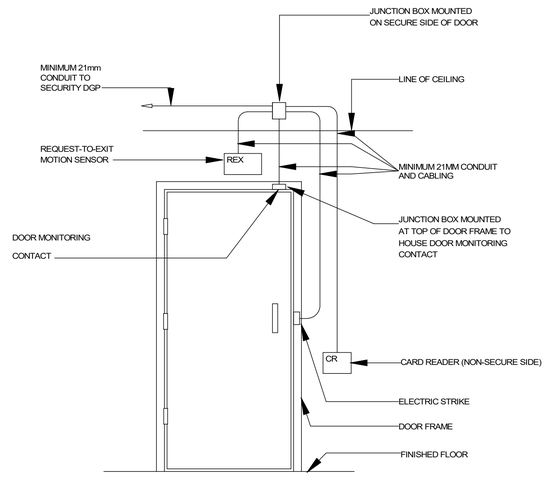


NOTES:

1. REFER TO DETAIL "MOUNTING HEIGHTS FOR ACCESSIBLE DESIGN" FOR EXACT MOUNTING HEIGHTS OF DEVICES AND APPLICABLE CODE REFERENCES.
2. REFER TO DOOR HARDWARE SCHEDULE FOR EXACT SCHEDULE OF DEVICES AT DOOR. COORDINATE EXACT ROUGH-IN DETAILS WITH DOOR HARDWARE TRADE.
3. INTERLOCK OPERATION OF DOOR OPERATOR WITH CARD READER ON THE NON-SECURE SIDE.

4 TYPICAL DOUBLE DOOR WITH ACCESS CONTROL AND OPERATOR

SCALE:N.T.S

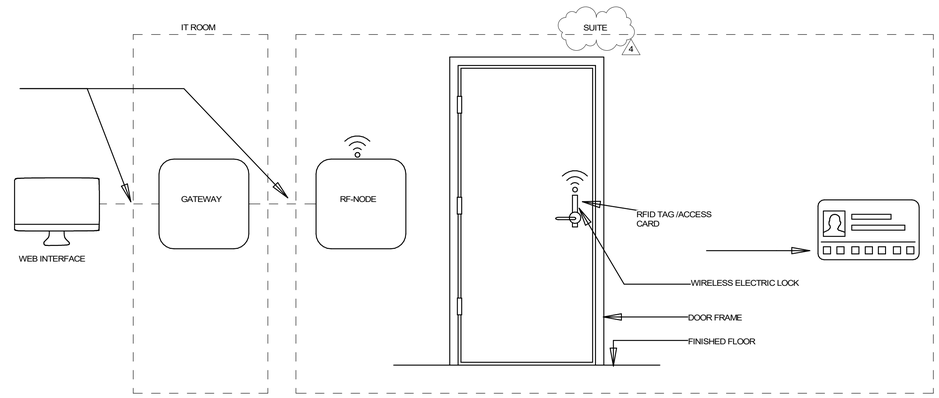


NOTES

1. REFER TO PLAN DRAWINGS FOR EXACT QUANTITY AND TYPE OF DEVICE AT EACH DOOR.
2. MINIMUM CONDUIT SIZE SHALL BE 3/4" (21mm) UNLESS OTHERWISE NOTED.
3. CONFIRM EXACT LOCATION AND MOUNTING HEIGHT OF ALL BACK BOXES ON SITE PRIOR TO INSTALLATION.
4. WIRING SHALL BE COORDINATED AND CONFIRMED WITH SUPPLIER.
5. TYPE AND QUANTITY OF WIRING AND SIZE AND QUANTITY OF CONDUIT SHALL BE TO THE SPECIFIC MANUFACTURERS REQUIREMENTS.
6. ALL SECURITY DEVICE WIRING SHALL BE IN CONDUIT UNLESS OTHERWISE NOTED.
7. THESE DETAILS ARE TYPICAL AND ARE INTENDED TO SHOW GENERAL ARRANGEMENT ONLY. REFER TO DOOR HARDWARE SCHEDULE AND ELEVATION SCHEMATICS FOR EXACT ROUGH-IN DETAILS.

2 TYPICAL SINGLE DOOR WITH ACCESS CONTROL

SCALE:N.T.S

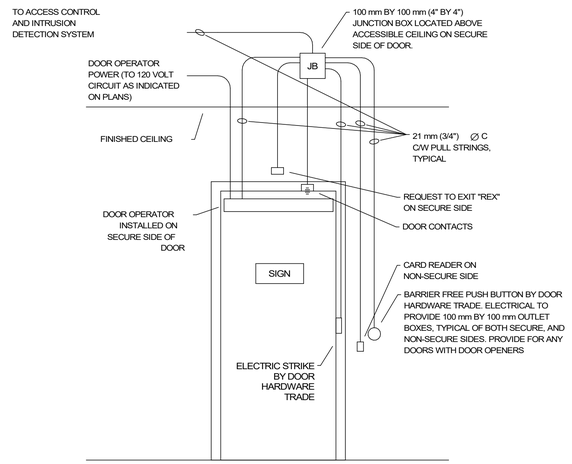


NOTES

1. REFER TO PLAN DRAWINGS FOR EXACT QUANTITY AND TYPE OF DEVICE AT EACH DOOR.
2. CONFIRM EXACT LOCATION AND MOUNTING HEIGHT OF ALL BACK BOXES ON SITE PRIOR TO INSTALLATION.
3. TYPE AND QUANTITY OF WIRING AND SIZE AND QUANTITY OF CONDUIT SHALL BE TO THE SPECIFIC MANUFACTURERS REQUIREMENTS
4. ALL SECURITY DEVICE WIRING SHALL BE IN CONDUIT UNLESS OTHERWISE NOTED.
5. THESE DETAILS ARE TYPICAL AND ARE INTENDED TO SHOW GENERAL ARRANGEMENT ONLY. REFER TO DOOR HARDWARE SCHEDULE AND ELEVATION SCHEMATICS FOR EXACT ROUGH-IN DETAILS.

3 TYPICAL WIRELESS LOCK LOGIC DIAGRAM

SCALE:N.T.S



NOTES:

1. REFER TO DETAIL "MOUNTING HEIGHTS FOR ACCESSIBLE DESIGN" FOR EXACT MOUNTING HEIGHTS OF DEVICES AND APPLICABLE CODE REFERENCES.
2. REFER TO DOOR HARDWARE SCHEDULE FOR EXACT SCHEDULE OF DEVICES AT DOOR. COORDINATE EXACT ROUGH-IN DETAILS WITH DOOR HARDWARE TRADE.
3. INTERLOCK OPERATION OF DOOR OPERATOR WITH CARD READER ON THE NON-SECURE SIDE.

1 TYPICAL SINGLE DOOR WITH ACCESS CONTROL AND OPERATOR

SCALE:N.T.S

No.	DATE	DESCRIPTION
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TOWN FILE NUMBER: SP-26-25

TITLE

ICAT TYPICAL DETAILS 2

CHECKED	PROJECT No.
JP	RE-24-162
SCALE	DRAWING No.
N.T.S.	T-801

4